MSKTD

Purdue University College of Engineering

ME-RAIL-HAMP Study

June 30, 2015 (Rev)

PURDUE UNIVERSITY

MSKTD & Associates, Inc.

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Project Scope

The scope of the ME-RAIL-HAMP study for the College of Engineering includes a programming and conceptual design exercise to review and reconfigure existing spaces and program functions within the Mechanical Engineering Building (ME), American Railway Building (RAIL) and Hampton Hall (HAMP) on Purdue University's West Lafayette campus. A primary goal of the study is to maintain core academic functions within the central part of Purdue's campus. These core academic functions are supported by specific space types, including (1) classrooms, (2) instructional/teaching labs, and (3) offices and open-office workstations for faculty, staff and graduate assistants.

As part of recent efforts to guide the College of Engineering growth plan, the SLAM Collaborative was commissioned to develop a master plan for the College, and although the ME-RAIL-HAMP study and space allocation priorities within each building are based on discussions with each department, it is also based on a conclusion originally developed as part of the College's master plan document. Page 1-10 of the master plan states:

> *"Potential for spatial growth on the North Campus is limited due to the site density that hinders the ability to add new buildings and program space. Due to these restrictions it was determined that the North Campus will remain the academic core and functions that do not support this mission should be relocated away from this campus. A prime opportunity for additional square footage are new building sites located on the South Campus in Discovery Park. The location of Discovery Park is not ideal for undergraduate student activities, so all space added at this location will support research activity."*

The ME-RAIL-HAMP study includes two parts that were developed concurrently for each building. Part 1 includes elements that directly impact the College of Engineering's growth plan, including increased office and open-office workstation quantities, as well as teaching lab configurations and priorities, which are limited to specific areas within each building. Part 2 of the study includes additional considerations for the existing buildings to identify overall building use and how future projects may impact each building in the long term.

As part of the master planning process, conceptual plans were developed for each building, including ME, RAIL and HAMP, to accommodate various program functions, as well as growth for the College. To accommodate the needs, the master plan approach was to work additional program elements into existing space configurations where feasible, and this included reusing existing spaces and layout, as appropriate. Based on recent input from the College of Engineering, the ME-RAIL-HAMP study takes a different approach and reviews overall building utilization, space adjacencies and related program functions to develop options to reorganize and improve space relationships to increase efficiency for program elements and function. Larger blocks of related spaces were reviewed and discussed to develop flexible configurations, including options for office suite configurations and concepts recently developed and implemented in recent projects at Wang and Grissom Halls.

The ME-RAIL-HAMP study included reviewing the layout of the existing buildings, as well as systems and components related to the impacted spaces, and providing overall space concepts to improve quantities, flexibility and efficiencies. As each building was considered, space functions and adjacencies were observed, discussed and reviewed with building committee team members from the affected departments within each respective building. A primary goal was to develop an approach to space organization and building layout to provide a long-term conceptual plan for each building, while accommodating as

much growth as feasible for spaces that support the core academic functions within each building and considering other relevant and existing program functions (e.g., research labs) that are necessary to remain within each building.

The study process included multiple meetings and review of the existing building conditions, as well as developing, reviewing and discussing proposed concepts to accommodate growth and program needs. Although this report does not directly include all of the considerations, conversations, diagrams and space needs analysis that were reviewed, our team, including each building committee, spent time assessing the program components and proposed layouts to ensure that the proposed solutions considered space function, space proximities, and overall building space relationships. Although not all of each department's space needs, growth goals and program elements can be met within the existing constraints of each building, multiple options and potential solutions were reviewed for increasing program efficiencies and developing functional relationships to address needs (current and future), improve efficiencies and adjacencies, and consider allocation of resources for each overall building renovation.

Within the ME building, the scope of the study comprises nearly 123,800 gross square feet, including the 1932 (north) wing, the 1929 (central) wing, and the 1942 and 1948 east wing. The original study scope as defined by Purdue University excludes the recent Gatewood Addition. The American Railway building (RAIL) is immediately adjacent to the ME building and Gatewood Addition, and although they are not connected with interior circulation, RAIL program functions are often considered as part of ME. RAIL is a two-story building that is approximately 2,500 gross square feet and was built in 1926.

Hampton Hall (HAMP) comprises approximately 284,000 gross square feet, although the scope of the study is limited to specific areas within the building that are assigned to the College of Engineering. Although other departments have offices and lab spaces within HAMP, those areas were not included as part of the study and project considerations.

Limitations of the study project scope areas are graphically identified in the following color-coded existing floor plan diagrams for each building. The heavy, red-dashed lines indicate areas originally excluded from the project scope.

ME-RAIL-HAMP
Square Footage Summary Info

BUILDING SQUARE FOOTAGE SUMMARY

AMERICAN RAILWAY BUILDING

LEVEL	GSF	STUDY SCOPE	PROPOSED RENO.	DISPLACED AREA
FIRST FLOOR	1,250	1,250	1,150	0
SECOND FLOOR	1,250	1,250	1,248	0
TOTALS SF	2,500	2,500	2,398	0

MECHANICAL ENGINEERING BUILDING (ME) GATEWOOD EXCLUDED

LEVEL	GSF	STUDY SCOPE	PROPOSED RENO.	DISPLACED AREA
BASEMENT	10,430	0	0	0
GROUND FLOOR	31,380	31,380	14,829	7,753
FIRST FLOOR	31,180	31,180	11,513	2,102
SECOND FLOOR	23,775	23,775	15,179	0
THIRD FLOOR	13,500	13,500	12,797	0
FOURTH FLOOR	13,530	13,530	148	0
TOTALS SF	123,795	113,365	54,466	9,855

HAMPTON HALL

LEVEL	GSF	STUDY SCOPE	PROPOSED RENO.	DISPLACED AREA
SUB-BASEMENT	9,110	0	0	0
BASEMENT	54,942	46,340	10,210	0
GROUND FLOOR	49,415	43,585	22,904	6,885
FIRST FLOOR	41,472	25,485	21,376	0
SECOND FLOOR	44,862	19,465	6,027	1,527
THIRD FLOOR	44,509	19,470	17,765	9,333
FOURTH FLOOR	39,678	15,675	13,673	0
TOTALS SF	283,988	170,020	91,955	17,745

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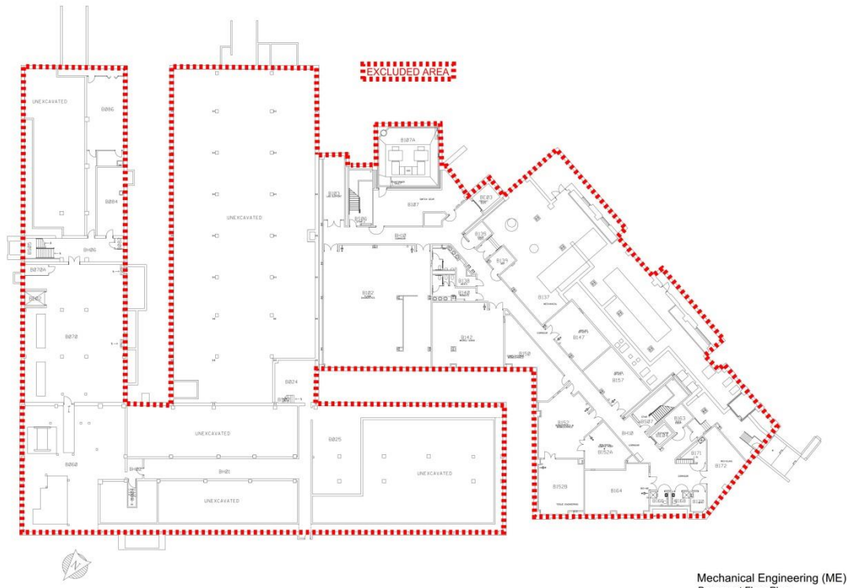
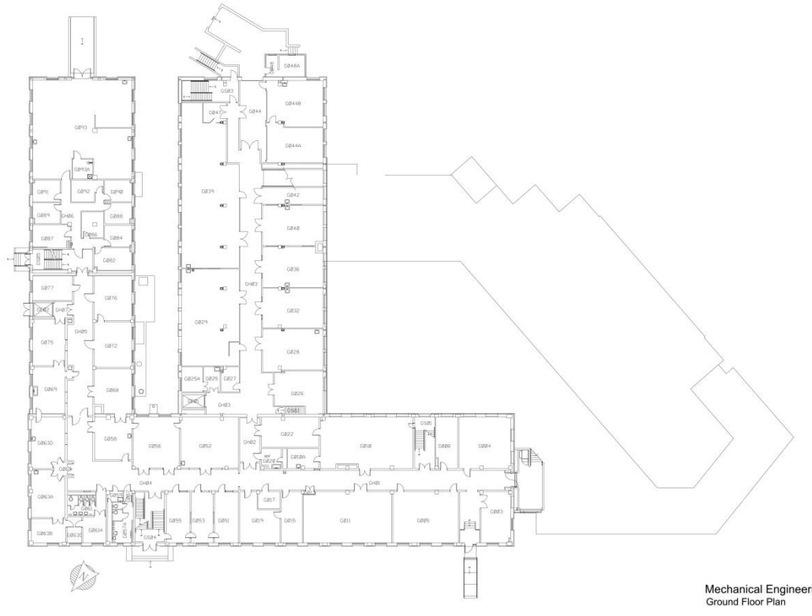
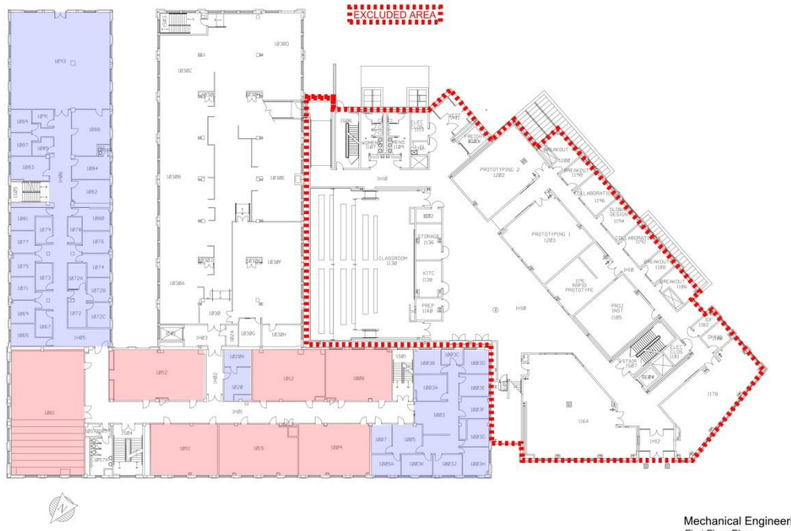


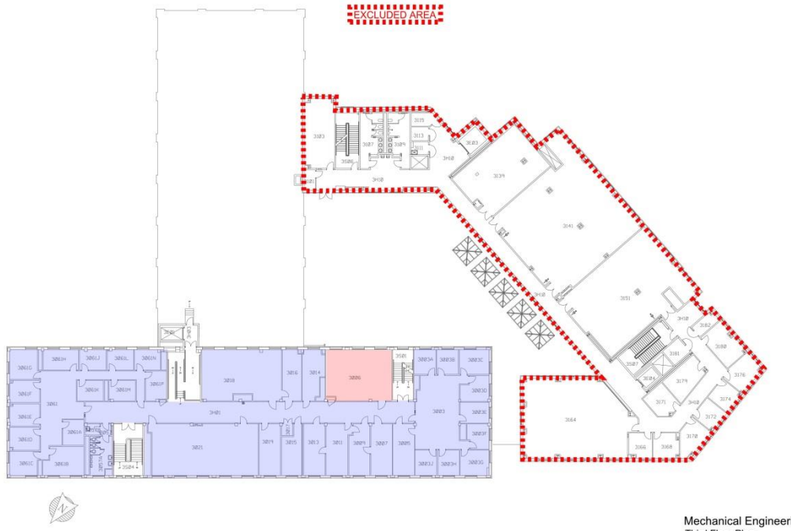
Mechanical Engineering (ME)
Basement Floor Plan



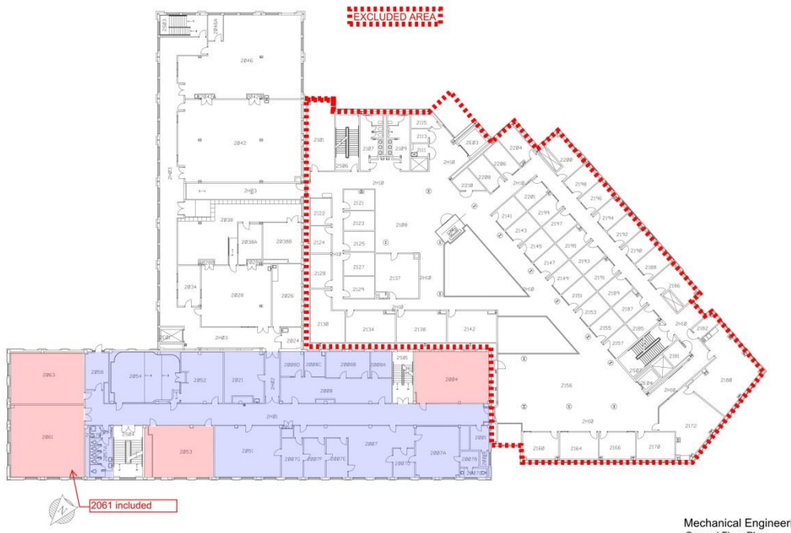
Mechanical Engineering (ME)
Ground Floor Plan



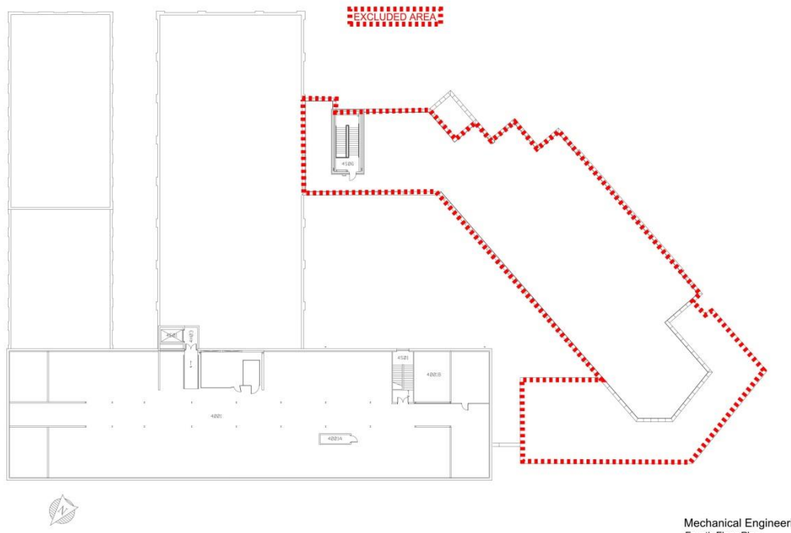
Mechanical Engineering (ME)
First Floor Plan



Mechanical Engineering (ME)
Third Floor Plan



Mechanical Engineering (ME)
Second Floor Plan



Mechanical Engineering (ME)
Fourth Floor Plan

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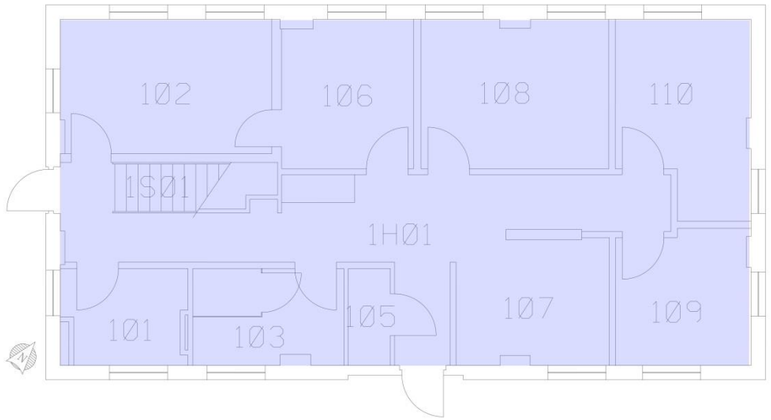
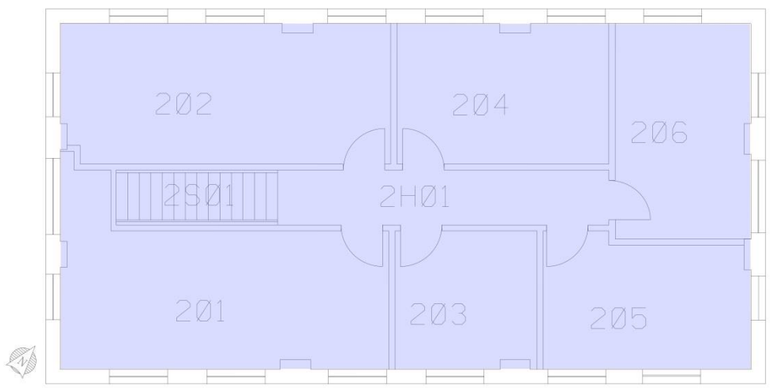
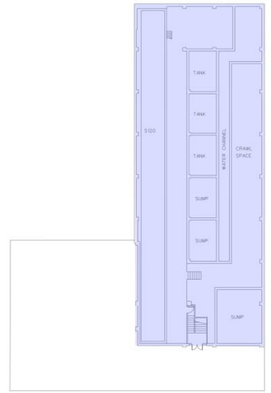


American Railway Building (RAIL)
First Floor Plan



American Railway Building (RAIL)
Second Floor Plan



Hampton Hall (HAMP)
Sub-Basement Floor Plan

Hampton Hall (HAMP)
Basement Floor Plan

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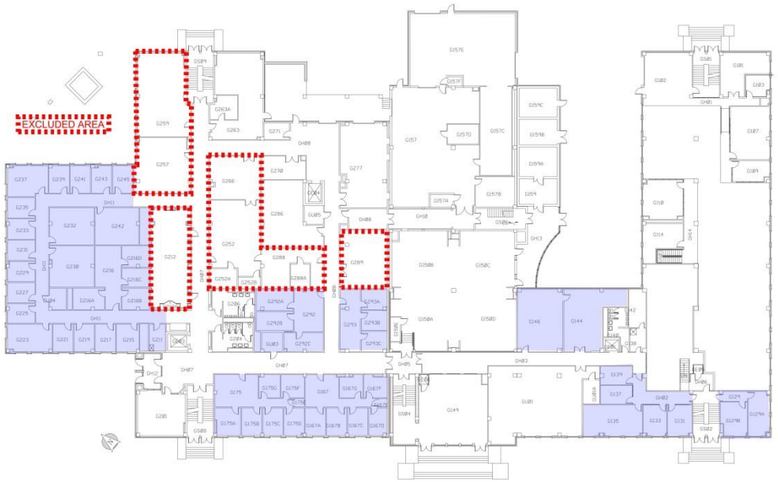
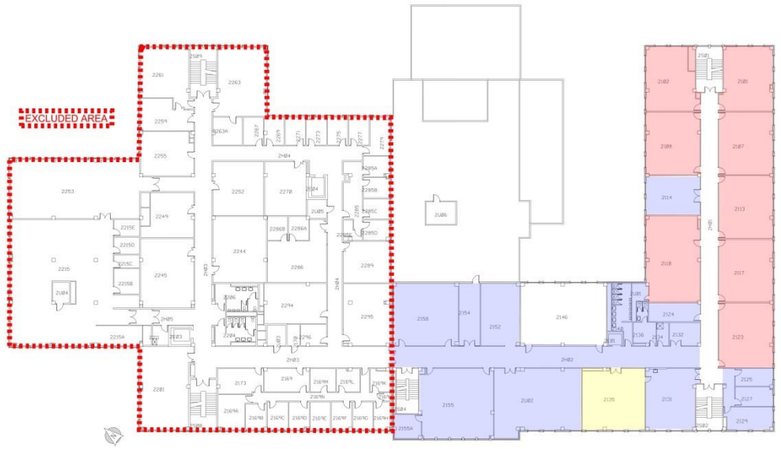
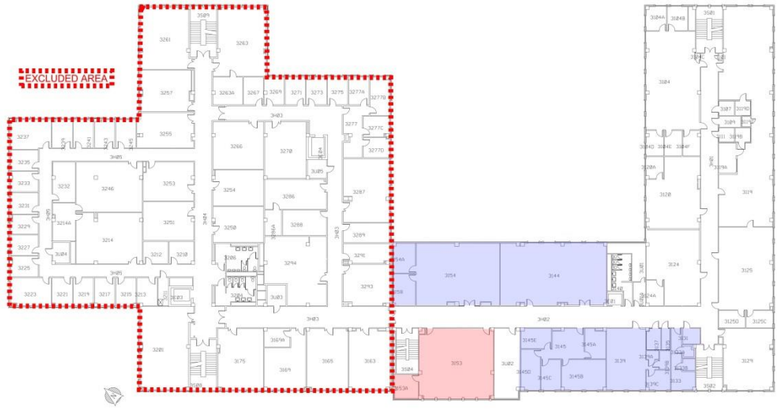


Hampton Hall (HAMP)
Ground Floor Plan



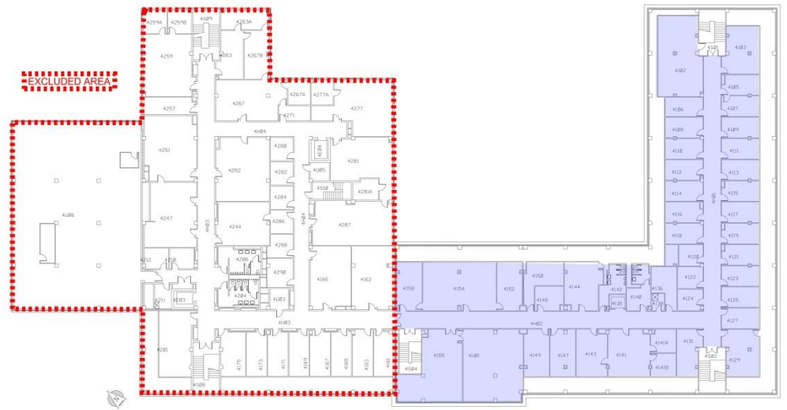
Hampton Hall (HAMP)
Second Floor Plan

Hampton Hall (HAMP)
First Floor Plan



Hampton Hall (HAMP)
Third Floor Plan

PURDUE
UNIVERSITY



Hampton Hall (HAMP)
Fourth Floor Plan

Location

The ME, RAIL and HAMP buildings are within core academic areas at the heart of Purdue University's West Lafayette campus. Because of the central campus locations, space utilization and efficiencies are important considerations when reviewing program functions and departmental needs, and these are considered in the narratives that follow for each building.

ME, RAIL and HAMPTON location

Evaluation

As part of the study, our team reviewed each building from various perspectives. Following are some of the major criteria we utilized in developing the proposed direction and final study documentation:

- BUILDING FUNCTION AND CURRENT USE: Review of the existing buildings and drawings indicate that ME and HAMP were originally designed for multiple educational functions, including offices, classrooms, instructional/teaching labs and research labs, and these are common functions at the core of the study. However, one aspect not specifically considered in the original designs was flexibility. As part of the ME-RAIL-HAMP study, we explore options for flexibility as part of the final study documentation.
- CIRCULATION: Existing circulation considerations for each building include building entrances, horizontal circulation on each individual floor or level within a building, and vertical circulation within each building to effectively connect multiple floors or levels. As part of the study documentation, we include recommendations to modify existing circulation patterns within a building with the goal to improve circulation, space efficiencies or to address exiting concerns.
- BUILDING ENVELOPE AND SITE: While the building envelope and site were not included as part of the project scope and considerations, we make general recommendations as part of the study narratives and drawings to consider additional enhancements to improve ADA access and/or bring natural daylighting into additional interior spaces. However, in general, the existing exterior walls, roofs and windows are to remain for each building.
- ELEVATORS: Review of the existing elevators is an important consideration when discussing and addressing circulation within a building, as well as accessibility of different floors and intermediate levels. Specific review and evaluation of the efficiencies and operation of the existing elevator systems and controls was not included in the study scope, but the narrative for each building discusses options and considerations related to vertical circulation, based on proposed layouts.
- MECHANICAL AND PLUMBING SYSTEMS: Modifications to the existing mechanical, plumbing and fire protection systems are major considerations in understanding the overall scope and modifications required within each building to accommodate program requirements and proposed renovations. In addition, plumbing systems are to be modified or replaced to better serve the building occupants, including more restrooms and better access in more locations. Refer to the mechanical portions of this study for more detail and information.
- ELECTRICAL SYSTEMS: Existing electrical systems and service for each building was reviewed to understand the extent of work required to accommodate the proposed work and understand the implications and complexities for phasing. Refer to the electrical portions of this study for more detail and information.
- TECHNOLOGY SYSTEMS: Similar to the mechanical, electrical and plumbing systems, technology and telecommunications systems infrastructure are important considerations, including upgrades and reorganization of core telecommunications spaces. Existing spaces in some buildings are not located to provide adequate services for renovations into the future and will be reorganized to improve layout and services. More detail and additional information is included in the technology portions of the study.

Phasing

In general, the parts for the ME-RAIL-HAMP study correspond with the proposed phasing for construction implementation. Part 1 (and Phase 1) includes spaces that directly impact the College of Engineering's growth plan, including office and open-office workstation and suites, as well as instructional / teaching lab configurations that are impacted by the reorganization and additional office quantities. Part 2 (and Phase 2) of the study includes additional considerations for each of the existing buildings that impacts overall building use and future flexibility such as relocation of instructional / teaching labs, research labs and support spaces. Phasing is discussed in more detail in the 'Systems Narratives' section for each building.

PROGRAM OVERVIEW AND MASTER PLAN

Master Plan and Program Overview:

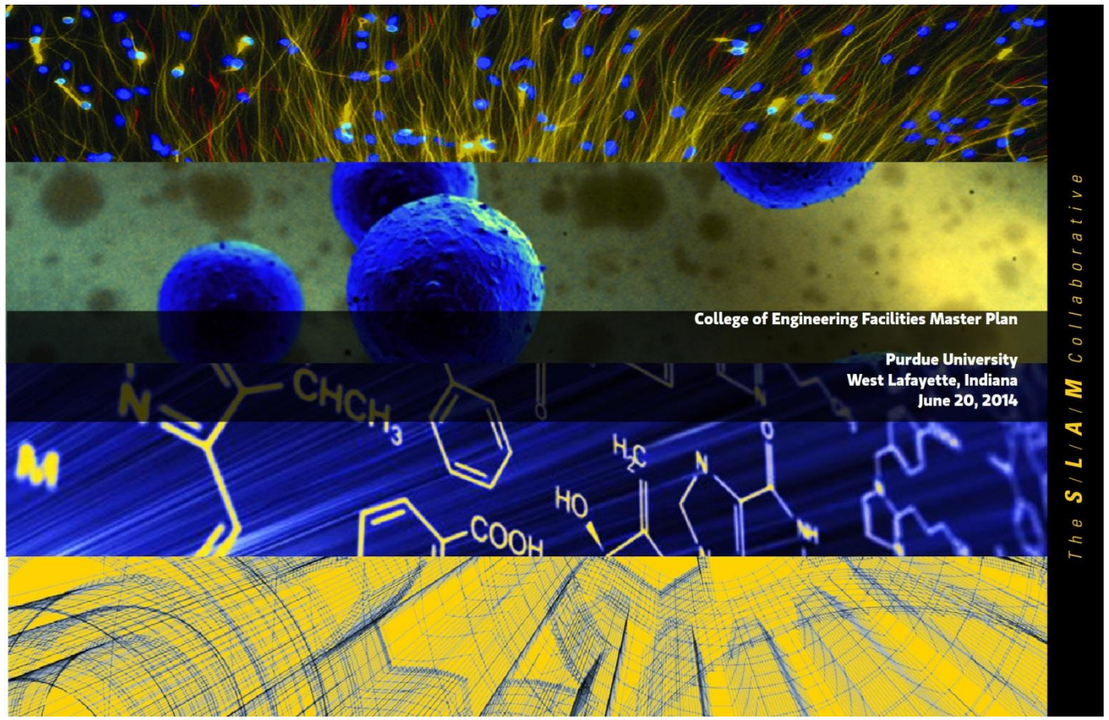
As noted in other parts of this document, the original study for ME, RAIL and HAMP began with a focus on the program of spaces developed as part of the College of Engineering master planning process. The program requirements, including square footage guidelines for each space type, were developed as part of previous master planning efforts to accommodate the office and student support spaces for each building and associated department(s), as well as instructional and research labs, and other related support services.

As discussed and included as part of the master planning project, the SLAM Collaborative identified *quantitative* and *qualitative* space needs and guidelines. The ME-RAIL-HAMP study approaches each building in a similar way. Quantitatively, we address and accommodate anticipated growth and establish equity in space delineation among building users to align with the College of Engineering's space standards that were developed as part of the master plan. And qualitatively, we explore opportunities to establish integrated, experiential and collaborative environments for the core academic spaces within each building, including the private and open-office environments, to meet the College of Engineering's goals within the growth plan.

The following images represent a few select pages from the SLAM Collaborative Master Plan document dated June 20, 2014, and the page 1-2 includes a general statement of the purpose of the facility master plan for the College of Engineering. The ultimate goal for the master plan and this study for ME, RAIL and HAMP is to support and fulfill the strategic growth initiative of the College of Engineering, along with the renewal and replacement of physical equipment or facilities within each building.

As part of the master plan document conclusion, it was recommended that research lab functions at the core of the campus be displaced within each building to accommodate academic functions such as classrooms, instructional/teaching labs, and offices. And to address space requirements for the research labs displacement, there were recommendations that these program functions be relocated to a Flex Lab research building on the South Campus (page 1-10).

> *"With the need for more space on the North Campus, a solution was developed to accommodate core academic growth on the North Campus. The recommended solution is to build a Flex Lab research building on the South Campus. This building will allow research to be moved out of the older buildings on the North Academic Campus freeing space within those existing buildings for personnel worksetting and teaching lab growth. The Flex Lab will also promote multi- / inter-disciplinary research by co-locating research from all schools and units. This building needs to support many types of research while remaining flexible enough to evolve with changing space needs over time."*



Introduction

COLLEGE OF ENGINEERING

VISION

"We will be known for our impact on the world."

MISSION

"To advance engineering learning, discovery, and engagement in fulfillment of the Land Grant promise and the evolving responsibility of a global university."

GOALS

- **GRADUATES EFFECTIVE IN A GLOBAL CONTEXT**

  Purdue Engineers will be prepared for leadership roles in responding to the global technological, economic, and societal challenges of the 21st century

- **RESEARCH OF GLOBAL SIGNIFICANCE**

  We will focus our talent and facilities on research with great potential for expanding the boundaries of science and technology and addressing the global challenges and opportunities of the 21st century

- **EMPOWERING OUR PEOPLE, ENRICHING OUR CULTURE**

  We will create a leadership culture and environment where the people of Purdue Engineering can dream their boldest dreams and collaborate in a place where those dreams can become reality

PURDUE UNIVERSITY

College of Engineering
Facility Master Plan *2014*

1—2

Introduction

Purdue University's College of Engineering is among the largest and most highly ranked in the United States and is committed to leveraging its strength in undergraduate education, to further expanding and strengthening its faculty, to increasing its excellence in research and growing creative discovery, to improving the quality of its graduate programs, to completing improvements to its facilities and environment, and to continuing its deep commitment to increasing diversity among its students and faculty. The College is comprised of fourteen academic schools and divisions and nine programs as well as the Dean's Office and Engineering Computer Network. Engineering is crucial to innovation, economic development, jobs creation and to addressing the grand challenges facing our world in the 21st century. And the College is accelerating the speed at which it progresses to bring itself from excellence to preeminence.

In 2009, the College of Engineering announced its strategic plan, *Extraordinary People, Global Impact*. This plan sets aside business as usual and calls on the creativity and drive of its people—faculty, students, staff, alumni, and friends—pursuing individual passions and shared goals to make Purdue Engineering *known for its impact on the world*.

Since 2009, some remarkable and game-changing opportunities for strategic growth and expansion have occurred:

- In 2012, Purdue's Board of Trustees endorsed a five-year plan for a landmark investment in the College of Engineering, prompted by significant growth in its undergraduate and graduate enrollments and in its research enterprise, and spurred by the national call for the U.S. to graduate 10,000 more engineers a year. The College expects to increase the size of its faculty by 30% and staff by 28% by 2017.

- In 2013, President Mitch Daniels launched the Purdue Moves initiatives, a set of big ideas that will drive innovation, achievement, and growth across the University and differentiate Purdue as one of the true STEM capitals of higher education. A key Purdue Moves initiative is the expansion of the College of Engineering—a move that builds on and broadens the College's strategic plan.

- Growth on this scale is an opportunity for transformational change and must be supported by appropriate physical resources to accommodate personnel, instructional, research, and collaboration activities. In 2013 the College commissioned The SLAM Collaborative to develop a facilities master plan to align the College's physical resources with its Strategic Plan.

Current College of Engineering space associated with the West Lafayette campus is located on the core academic North Campus, on the South Campus, and off-campus and is comprised of approximately 950,000 net square feet. Certain University initiatives including the Active Learning Center (classrooms) which is led by Libraries, and certain College of Engineering initiatives including the Innovation Design Center (student organization space), Zucrow Test Cell Facility, and Jischke Addition are already underway and informed the master plan space program.

Existing space varies in quality depending on building and status of renewal activity. Space as currently occupied is highly utilized and does not allow for growth or expansion of individual worksettings, instruction, research, and collaboration.

Master Plan Objectives

The purpose of the facilities master plan is to:

- Create a comprehensive physical facilities plan to support the College's Strategic Plan and to align with the University Campus Plan
- Define the physical resources required to sustain and advance the College's mission, goals, objectives, and priorities
- Identify the College's immediate needs and establish a long-term strategy for additions to and renewal and/or replacement of physical facilities
- Identify a long-term physical vision for the College, reflective of its position as one of the best colleges of engineering in the nation and the world
- Provide the College with a sense of place that proclaims its purpose, distinction, and domain

The facilities master plan specifically studies and resolves the following issues:

- Determine individual group current space needs and projected over a 10-year timeframe
- Accommodate the current program of space requirements and projected growth
- Establish appropriate space assignments and a consistency in space quality
- Understand and develop existing building condition assessments in order to assure the best and highest use for existing facilities
- Understand space use and optimize utilization of available space
- Maintain strategic and/or desired adjacencies
- Maintain strategic and/or desired north, south, and off-campus locations
- Enable classrooms upgrades
- Provide space for unknown future initiatives and/or aspirational growth
- Develop a prioritized plan for major renovations and new construction projects with appropriate phasing, costs, and schedules for integration into Purdue's ongoing capital planning budget process.

Project Description

This document provides a facilities master plan to:

- Accommodate the addition of 107 faculty (28% growth), 100 staff (30% growth), 300 paid graduate students, and 500-600 unpaid graduate students in 5 years.
- Improve the quality of outdated instructional and research space.
- Provide worksettings and research space for personnel growth, and to increase efficiency of space utilization.

Growth in personnel is anticipated to be completed by the fall of 2017 with facility alignment completed in 2021. The master plan also takes longer range growth into account by identifying opportunities for additional growth of physical resources.

This study defines programmatic space needs, assesses existing conditions, defines planning criteria, looks at conceptual planning options, develops the selected master plan option, and develops a budget and schedule model. This master plan meets the anticipated needs of the College of Engineering through continued use of existing space, new construction and remodeling/repurposing of existing space. The study identifies needs that can be accommodated in repurposed space and those that can only be met by construction of new space. The study has taken into account prioritized needs and has developed phasing strategies for implementation.

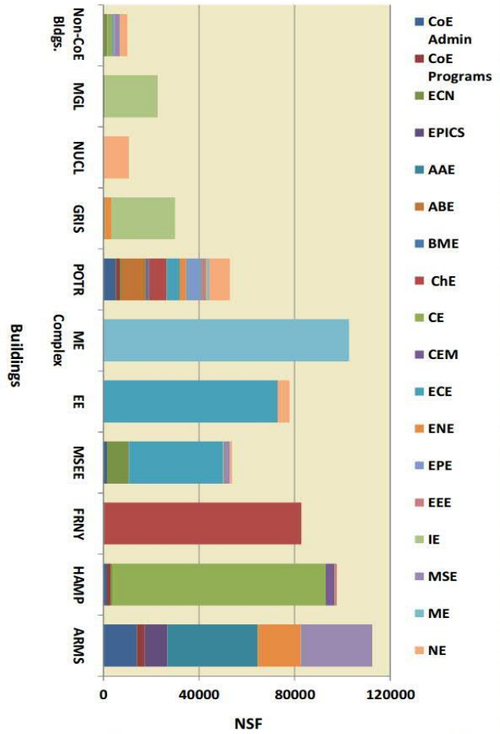


Chart 1.3.a showing CoE space in North Campus buildings

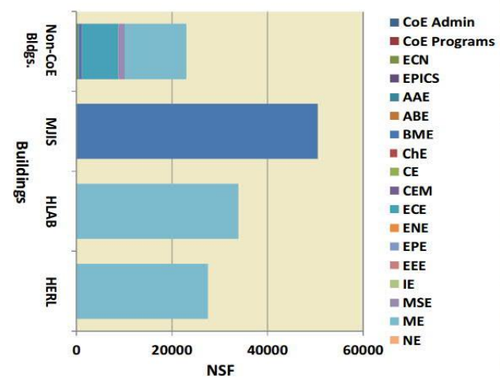
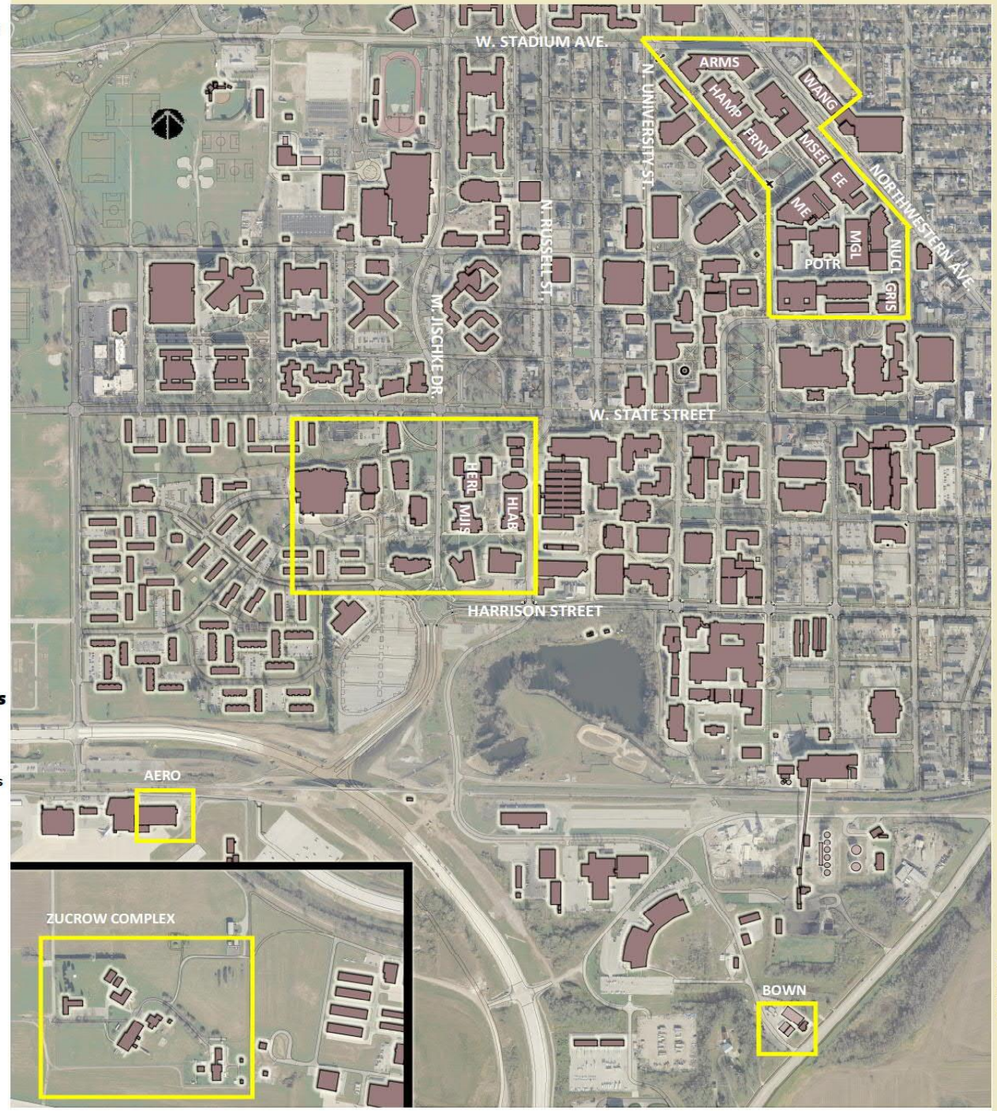


Chart 1.3.b showing CoE space in South Campus buildings

Aerial view of campus:

Existing College of Engineering facilities are located within the highlighted areas.



The College of Engineering is comprised of the following Schools, Divisions, Programs, and Units

- School of Aeronautics and Astronautics
- School of Agricultural and Biological Engineering
- Weldon School of Biomedical Engineering
- School of Chemical Engineering
- Lyles School of Civil Engineering
- Division of Construction Engineering and Management
- School of Electrical and Computer Engineering
- School of Engineering Education
- Engineering Professional Education
- Division of Environmental and Ecological Engineering
- School of Industrial Engineering
- School of Materials Engineering
- School of Mechanical Engineering
- School of Nuclear Engineering
- System of Systems
- Dean's Office
- Engineering Computer Network
- Engineering Projects in Community Service (EPICS)
- Global Engineering Program
- College of Engineering Honors Program
- Minority Engineering Program
- Office of Future Engineers
- Office of Professional Practice
- Women in Engineering Program

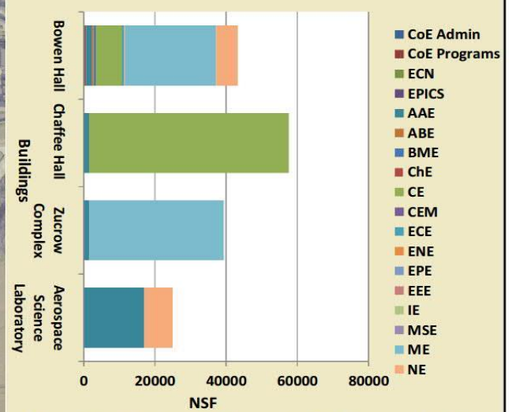


Chart 1.3.c showing CoE space in Off-Campus buildings

The S/L/A/M Collaborative

Introduction

INTRODUCTION

Goals and Objectives

A program of space requirements has been developed to accommodate the needs of the College of Engineering through the year 2017. The goal of the space program is to document the types, quantity, and quality of spaces that are required to accommodate the activities that will meet the Strategic Plan of the College. Through meetings with the Core Committee, Steering Committee, representatives of the College of Engineering Schools and Divisions, the Facilities Department, and other stakeholders a clear understanding of what the facility master plan program should provide was developed.

The resulting program encompasses a broad vision for facilities, both new and existing, that:

- Meets current and future needs and provides space to grow
- Develops a sense of community and place
- Creates collaborative facilities that catalyze people
- Enables organization of space on an interdisciplinary and synergistic basis rather than only departmental
- Develops a "road map" for future capital growth that can be phased as funding becomes available

Space needs are both quantitative and qualitative. Quantitative space needs are perhaps the easiest to understand as these address accommodating anticipated growth and achieving parity of space among all users in alignment with Purdue University space guidelines. Qualitative space requirements address the need to create an experiential, integrated, and collaborative learning and research environment that have a tremendous impact on productivity and success in learning and research at the College of Engineering.

Thus, the facility master plan program has been developed to accomplish the following.

- Provide appropriate office and office support space for the anticipated number of personnel
- Create a state-of-the-art teaching and learning environment including:
  - Experiential teaching lab spaces
  - Large teaching spaces
  - Opportunities for informal learning
  - Enable inter– and trans-disciplinary learning
  - Student project space
  - Access to more equipment and instrumentation
- Create a state-of-the-art research environment including:
  - Appropriate heating, ventilation, air conditioning, electrical, and plumbing systems
  - Grouping of disciplines regardless of department
  - Flexible research and support spaces to accommodate evolving research targets, group sizes, and equipment and instrumentation.
  - Core and specialized support facilities that are consolidated and avoid duplication where appropriate
- Incorporate state-of-the-art technology:
  - Increased data usage and storage space to enable collaborations
  - Enabling use of technology to share knowledge beyond Purdue borders
- Create collaboration and interaction environments:
  - Opportunities for spontaneous interactions
  - Space for student/faculty and student/grad/postdoc engagement
  - Surge office and social space for visitors that is not isolated
- Enhance Purdue's close relationship with state and industry partners:
  - Space to meet specific industry needs
  - Space for after-hours courses and meetings
- Create a community with a sense of place that proclaims the stature of the College of Engineering and that:
  - Provides compelling and exciting space
  - Enhances faculty and student recruitment
  - Provides clarity, wayfinding, and organization
  - Fosters transparency and accessibility
  - Creates density and buzz
  - Puts science on display

And finally, the facility master plan program for the College of Engineering must align with the Campus Master Plan in order to add to a cohesive and comprehensible campus.

Program Development Process and Methodology

The program was developed based on Purdue space guidelines, scientific requirements, utilization, and the requirement to create community and enhance interaction. All of these were informed by the Core Committee, Steering Committee, Departmental, and other stakeholder meetings.

Purdue space guidelines delineate the amount of space allocated for an individual worksetting for each category of personnel. Office support space and collaboration space is added to achieve a state-of-the-art workplace.

Space Program

LYLES SCHOOL OF CIVIL ENGINEERING (CE)

Personnel Change

- 13% faculty and staff increases to support the College's growth initiative
- 15% faculty increase
- 13% increase of staff

Table 2.28.a—Master Plan space program

Space Type	PERSONNEL ALL CAMPUSES: Total Number of Existing Personnel	Projected Growth	Total Master Plan Personnel	EXISTING SPACE NORTH CAMPUS: Area	Worksettings or Seats	SOUTH CAMPUS: Area	Worksettings or Seats	OFF - CAMPUS: Area	Worksettings or Seats	TOTAL: TOTAL EXISTING AREA (NSF)	TOTAL EXISTING WORKSETTINGS	MASTER PLAN SPACE PROGRAM NORTH CAMPUS Worksettings: Located on North Campus	North Campus Adjustment/ Growth	Total Worksettings	Required Seats: Instr. Teaching Lab Seats	Collaboration: Personnel Collaboration	Collaboration: Student Collaboration	Res. Research seats	North Campus Totals: cap. Seats	per room: NSF / Seat	per room: NSF / station or Room	per room: No. of stations or Rooms	TOTALS: Total Area (NSF)	SOUTH CAMPUS TOTALS	OFF-CAMPUS TOTALS	WANG TOTALS	FLEX LAB TOTALS	GRAND TOTAL	Comments
Offices	**385**	**51**	**436**	**22,314**	**317**	**0**		**7,500**	**81**	**29,814**	**401**	**321**	**80**	**401**				**203.5**					**23,169**	**0**	**7,500**	**0**	**0**	**30,669**	
Dean																					300	0	0						
Academic Department Head	1		1	195	1						1	1		1				1			180	1	180						
Academic Assoc Head, Asst. Head																					150	0	0						
Tenured/TT Faculty	51	8	59	7,534	47			1,565	11		58	47	8	55				44			120	55	6,600						
Joint Appointment Faculty																					60	0	0						
Visiting Scholars (Total space allocation is equal to 7% of above total faculty space allocation.)				472	8						8	8		8				8			475	1	475						
Research or Visiting Faculty	1		1	295	4						4	4		4				4			60	4	240						
Instructor or Lecturer	1	3	4									0	4	4							120	4	480						
Emeritus/Emeritas Office				119	1						2	1		1							120	1	120						
Post-Doctoral Assistants - Workstation	30		30	935	20			213	3		23	20	10	30				30			90	30	2,700						
Grad - Thesis (See Table 2.7b)	239	35	274	7,621	198			2,972	42		240	198	35	233				116			40	163	6,524						*MP provides grad desks at ratio of .7 desk/grad
Professional Staff - Office	23	1	24	2,103	16			925	7		23	16	6	22							120	22	2,640						
Business Manager	1		1									0	1	1							120	1	120						
Business Staff	5		5									0	5	5							90	5	450						
ECN IT Staff	2		2								2	0	2	2							120	2	240						
Management	2		2									0	2	2							120	2	240						
Admin/Operations Staff - Workstation	7	2	9	2,024	12			1,825	16		28	7	2	9							90	9	810						
Clerical - Workstation	9	2	11						2		2	9	2	11							90	11	990						
Service Staff - Workstation	3		3										3	3							120	3	360						
Hoteling offices																					40	0	0						
JTRP - Professional Staff	3		3	1,016	3						3	3		3							120	3	360						
JTRP - Admin.	7		7		7						7	7		7							90	7	630						
Office Support				**3,030**		**0**		**2,156**		**5,186**													**1,960**	**0**	**2,156**	**0**	**0**	**4,116**	
Department Reception																					300	1	400						
Files/Storage																					300	1	400						
Admin Workroom																					300	1	400						
Business Workroom																					300	1	400						
JTRP - Department Reception																					120	1	120						
JTRP - Files/Storage																					120	1	120						
JTRP - Admin Workroom																					120	1	120						
Research Labs & Support				**41,959**		**0**		**49,189**		**91,148**								**203.5**					**41,536**	**0**	**49,189**	**0**	**9,339**	**100,064**	
Wet Bench																													
Engineering																		152.6	153	325	49,595	1	49,603						
Dry Comp Collaboration																		51	51	25	1,272	1	1,272						
Relocate to Flex Lab																							-9,339				9,339		
Specialized Shared Support				**6,912**		**0**		**2,686**		**9,598**													**6,600**	**0**	**2,686**	**0**	**0**	**9,286**	
Data Visualization Lab				459																	459	1	459						
Laser Lab				622																	622	1	622						
Hydr. Tow. Tank				4,227																	4,227	1	4,227						
Fieldwork Prep Room																					400	1	400						
Shop				892				2,375													892	1	892						
Central Storage				712				311																					

LYLES SCHOOL OF CIVIL ENGINEERING (CE)

Table 2.29.b—Master Plan space program

Space Type	EXISTING SPACE: NORTH CAMPUS Area	NORTH CAMPUS Worksettings or Seats	SOUTH CAMPUS Area	SOUTH CAMPUS Worksettings or Seats	OFF - CAMPUS Area	OFF - CAMPUS Worksettings or Seats	TOTAL: TOTAL EXISTING AREA (NSF)	TOTAL EXISTING WORKSETTINGS	MASTER PLAN SPACE PROGRAM: NORTH CAMPUS Worksettings: Located on North Campus	North Campus Adjustment / Growth	Total Worksettings	Required Seats: Instr. Teaching Lab Seats	Collaboration: Personnel Collaboration	Student Collaboration	Res. Research seats	North Campus Totals: cap. Seats	per room: NSF / Seat	NSF / station or Room	No. of stations or Rooms	TOTALS: Total Area (NSF)	SOUTH CAMPUS TOTALS	OFF-CAMPUS TOTALS	WANG TOTALS	FLEX LAB TOTALS	GRAND TOTAL	Comments
Teaching Labs & Support	**9,531**		**0**		**0**		**9,531**					**276**								**16,191**	**0**	**0**	**0**	**0**	**16,191**	
Teaching Lab - I2I Lab NW	1,582											36				36	50	1,800	1	1,582						HAMP G150A, G150B
Teaching Lab - I2I Lab SW	1,649											36				36	50	1,800	1	1,649						HAMP G150C, G150D
Teaching Lab -Hydromechic Room	1,371											24				24	75	1,800	1	1,371						HAMP B115G
Teaching Lab - Hydraulics Lab	847											24				24	50	1,200	1	847						HAMP B115F
Teaching Lab - D-CL Grad TL	791											14				14	50	700	1	791						HAMP B268
Teaching Lab - Structures	870											24				24	50	1,200	1	1,200						HAMP 2158
Teaching Lab - Environmental	577											40				40	50	2,000	1	2,000						HAMP 2146
Teaching Lab/Open Lab - Geomatics	790											24				24	75	1,800	1	1,800						HAMP 1115
Teaching Lab - Arch Eng												24				24	75	1,800	1	1,800						
Teaching Lab - Senior Capstone												30				30	50	1,500	1	1,457						HAMP 3154, 3154A, 3158
Teaching Lab Support - 25%	1,054																	3,260	1	1,694						HAMP 1119, 1121, G159C
Collaboration Space	**7,235**		**0**		**3,517**		**10,752**		**0**	**0**	**0**		**216**	**194**						**11,982**	**0**	**3,517**	**0**	**0**	**15,499**	
Personnel Collaboration																										
Conference Room	1,887												48			48	25	1,200	1	1,200						divisible 4144, 4158
Conference Room													72			36	25	900	2	1,800						G212
Conference Room													32			16	25	400	2	800						1101E, 1141C
JTRP - Conference Room													12			12	25	300	1	300						
Focus/TA Consulting/Help/Advising													16			4	25	100	4	400						
Mediascapes													24			6	25	150	4	600						
Break/Coffee Bar	681												12					400	1	400						
Student Collaboration																										
Open Lab - D-CL Computer	1,661													40		40	30	1,200	1	1,200						ex HAMP 1212, rel HAMP 1252, 1266
Open Lab - Project	1,165													24		24	75	1,800	1	1,800						HAMP 3154
Student Organizations	290													18		6	25	150	3	450						HAMP 2154 w/storage lockers
Student Clubs																				0						concrete canoe, steel bridge currently in Hydraulics Lab B115
Group Study (quiet)														32		4	25	100	8	800						
Break-out / Commons	1,551													80		40	25	1,116	2	2,232						
CIVIL ENGINEERING TOTAL	**90,981**	**317**	**0**	**0**	**65,048**	**81**	**156,029**	**401**	**321**	**80**	**401**	**276**	**216**	**194**	**407**					**101,437**	**0**	**65,048**	**0**	**9,339**	**175,824**	

See Page 2-30 for Program summary by location and space type, and Change in program space type Charts.

	MP Total: NORTH CAMPUS	SOUTH CAMPUS	OFF CAMPUS	WANG	FLEX	Total
Total	**101,438**	**0**	**65,048**	**0**	**9,339**	**175,825**
Offices & Support	25,129		9,656			34,785
Research & Support	41,536		49,189		9,339	100,064
Specialized Shared Support	6,600		2,686			9,286
Teaching Labs & Support	16,191					16,191
Collaboration	11,982		3,517			15,499

Table 2.29.a -Program summary by location and space type

Space Program

The CE graduate program is ranked 6th in the nation and the undergraduate program is ranked 10th in the nation. Specialty groups include:

- Architectural Engineering
- Construction Engineering
- Environmental Engineering
- Geomatics Engineering
- Geotechnical Engineering
- Hydraulic and Hydrologic Engineering
- Materials Engineering
- Structural Engineering
- Transportation Engineering

Space Drivers

- To efficiently accommodate faculty and staff growth
- To support growing student enrollment

Program Changes

Offices and Support

Despite a 13% growth in personnel office space was been reduced by 1% due to new more efficient space guidelines.

Research Labs and Support

Total required research space exceeds the available space on the North Campus. Therefore, some research will be located in the South Campus Flex Lab. It is recommended that the research best served by intra-disciplinary actives should be at the Flex Lab.

Teaching Labs and Support

The current facilities are overcrowded and more space will be required to add new teaching labs and student work space.

Collaboration

This space category increased slightly with added personnel and students.

LYLES SCHOOL OF CIVIL ENGINEERING (CE) - CONTIINUED

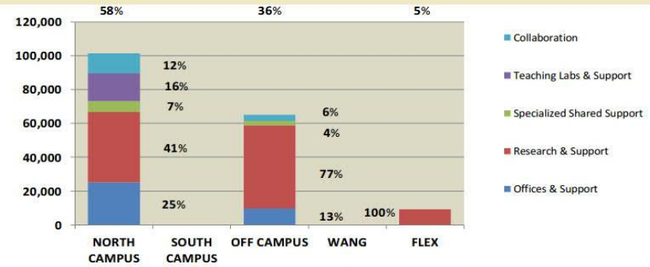


Chart 2.30.a –CE Program summary by location and space type

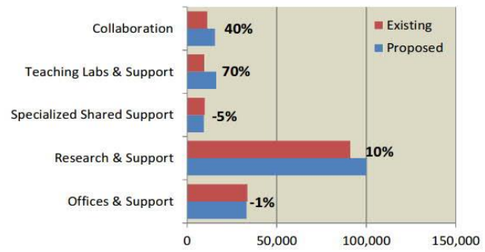


Chart 2.30.b—CE Change in program space type

Space Program

DIVISION OF CONSTRUCTION ENGINEERING AND MANAGEMENT (CEM)

Table 2.30.a—CEM Master Plan space program

Space Type	PERSONNEL ALL CAMPUSES: Total Number of Existing Personnel	Projected Growth	Total Master Plan Personnel	EXISTING SPACE NORTH CAMPUS: Area	Worksettings or Seats	SOUTH CAMPUS: Area	Worksettings or Seats	OFF - CAMPUS: Area	Worksettings or Seats	TOTAL: TOTAL EXISTING AREA (NSF)	TOTAL EXISTING WORKSETTINGS	MASTER PLAN SPACE PROGRAM NORTH CAMPUS Worksettings: Located on North Campus	North Campus Adjustment / Growth	Total Worksettings	Required Seats: Instr. Teaching Lab Seats	Collaboration: Personnel Collaboration	Student Collaboration	Res. Research seats	North Campus Totals: cap. Seats	per room: NSF / Seat	NSF / station or Room	No. of stations or Rooms	TOTALS: Total Area (NSF)	SOUTH CAMPUS TOTALS	OFF-CAMPUS TOTALS	WANG TOTALS	FLEX LAB TOTALS	GRAND TOTAL	Comments
Offices	14	48	62	3,147	32					3,147	32	26	36	62	0	0	0	35			300	0	3,993	0	0	0	0	3,993	
Dean																					300	0	0						
Academic Department Head	1		1	252	1						1	1		1				1			180	1	180						
Academic Assoc Head, Asst. Head																					150	0	0						
Tenured/TT Faculty	6	5	11	625	5						5	5	5	10				10			120	10	1,200						
Adjuncts or Joint Appointments																					60	0	0						
Visiting Scholars (Total space allocation is equal to 7% of above total faculty space allocation.)				264	5																97	1	97						
Research or Visiting Faculty																					60	0	0						
Instructor or Lecturer	1		1	126	1						1	1		1							120	1	120						
Emeritus/Emeritas Office																					0	0	0						
Post-Doctoral Assistants - Workstation		8	8										8	8				8			90	8	720						
Grad - Thesis (See Table 2.7b)		32	32	664	13						13	13	19	32				16			40	22	896						*MP provides grad desks at ratio of .7 desk/grad
Professional Staff - Office	2	1	3	546	4						4	2	1	3							120	3	360						
Business Manager																					120	0	0						
Business Staff	1		1								0	1		1							90	1	90						
Management																					120	0	0						
Admin/Operations Staff - Workstation	3	1	4	512	3						3	3	1	4							90	1	90						
Clerical - Workstation																					90	0	0						
Service Staff - Workstation		1	1										2	2							120	2	240						
Hoteling offices																					40	0	0						

Personnel Change

- 342% faculty and staff increases to support the College's growth initiative
- 71% faculty increase
- 614% increase of staff

DIVISION OF CONSTRUCTION ENGINEERING AND MANAGEMENT (CEM)

Table 2.31.b—CEM Master Plan space program

Space Type	EXISTING SPACE: NORTH CAMPUS Area	NORTH CAMPUS Worksettings or Seats	SOUTH CAMPUS Area	SOUTH CAMPUS Worksettings or Seats	OFF - CAMPUS Area	OFF - CAMPUS Worksettings or Seats	TOTAL: TOTAL EXISTING AREA (NSF)	TOTAL: TOTAL EXISTING WORKSETTINGS	MASTER PLAN SPACE PROGRAM — NORTH CAMPUS — Worksettings: Located on North Campus	North Campus Adjustment / Growth	Total Worksettings	Required Seats: Instr. Teaching Lab Seats	Collaboration: Personnel Collaboration	Collaboration: Student Collaboration	Res. Research seats	North Campus Totals: cap. Seats	per room: NSF / Seat	NSF / station or Room	No. of stations or Rooms	TOTALS: Total Area (NSF)	SOUTH CAMPUS TOTALS	OFF-CAMPUS TOTALS	WANG TOTALS	FLEX LAB TOTALS	GRAND TOTAL	Comments
Office Support	**267**						**267**		**0**	**0**	**0**									**420**	**0**	**0**	**0**	**0**	**420**	
Department Reception																		300	1	300						
Files/Storage																		120	1	120						
Admin Workroom																		0	0	0						
IT Workroom																		0	0	0						
Business Workroom																		0	0	0						
Research Labs & Support	**0**						**0**		**0**	**0**	**0**									**0**	**0**	**0**	**0**	**875**	**875**	
Wet Bench																										
Dry Comp Collaboration																35	25	875	1	875						
Relocate to Flex Lab																				-875				875		
Specialized Shared Support	**0**						**0**		**0**	**0**	**0**									**0**	**0**	**0**	**0**	**800**	**800**	
Simulation Labs																4	100	400	2	800						
Central Storage																										
Relocate to Flex Lab																				-800				800		
Teaching Labs & Support	**0**						**0**		**0**	**0**	**0**	**24**								**1,800**	**0**	**0**	**0**	**0**	**1,800**	
Teaching Lab												24				24	75	1,800	1	1,800						High-bay
Collaboration Space	**275**						**275**		**0**	**0**	**0**		**16**	**20**						**1,100**	**0**	**0**	**0**	**0**	**1,100**	
Personnel Collaboration																										
Conference Room	275												16			16	25	400	1	400						
Focus/TA Consulting/Help/Advising																										
Mediascapes																										See Simulation Lab
Break/Coffee Bar																										
Student Collaboration																										
Open Lab - Computer														0		0	30	0	1	0						Share w/CE
Open Lab - Project														4		4	75	300	1	300						
Student Organizations																		400	1	400						
Group Study (quiet)														4		4	25	100	1	100						
Break-out / Commons														12		12	25	300	1	300						
CONSTRUCTION ENG AND MGMT	**3,689**	**32**	**0**	**0**	**0**	**0**	**3,689**	**32**	**26**	**36**	**62**	**24**	**16**	**20**	**35**					**7,313**	**0**	**0**	**0**	**1,675**	**8,988**	

Space Program

CEM began as part of the School of Civil Engineering and there are close synergies between these programs.

Space Drivers

- To efficiently accommodate faculty and staff growth
- To support growing student enrollment

Program Changes

Offices and Support

This category is experiencing only a 30% growth in space while the personnel growth is over 300%. The new space standards help to minimize the growth in this category.

Research Labs and Support

Total required research space exceeds the available space on the North Campus. Therefore, some research will be located in the South Campus Flex Lab. It is recommended that the research best served by intra-disciplinary actives should be at the Flex Lab.

Teaching Labs and Support

There is no existing space currently dedicated for this function; however with the expansion of this program a teaching lab will be required to support growing student enrollment.

Collaboration

More collaboration space is required to support the faculty and staff growth as well as the student growth.

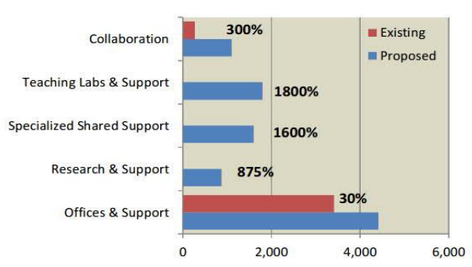


Chart 2.31.b -Change in program space type

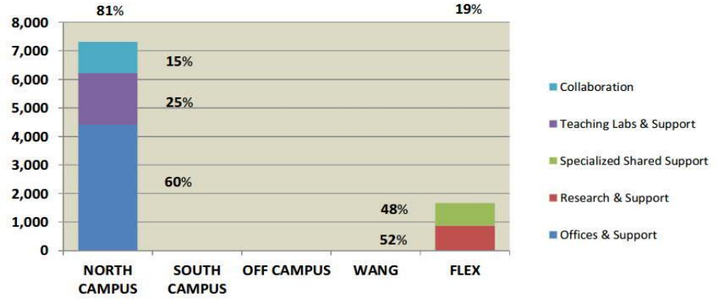


Chart 2.31.a -Program summary by location and space type

Table 2.31.a -Program summary by location and space type

	MP Total: NORTH CAMPUS	SOUTH CAMPUS	OFF CAMPUS	WANG	FLEX	Total
Total	**7,313**	**0**	**0**	**0**	**1,675**	**8,988**
Offices & Support	4,413					4,413
Research & Support					875	875
Specialized Shared Support					800	800
Teaching Labs & Support	1,800					1,800
Collaboration	1,100					1,100

Master Plan Program

Space Program

SCHOOL OF MECHANICAL ENGINEERING (ME)

Table 2.46.a—Master Plan space program

Space Type	PERSONNEL ALL CAMPUSES: Total Number of Existing Personnel	Projected Growth	Total Master Plan Personnel	EXISTING SPACE NORTH CAMPUS: Area	Worksettings or Seats	SOUTH CAMPUS: Area	Worksettings or Seats	OFF - CAMPUS: Area	Worksettings or Seats	TOTAL: TOTAL EXISTING AREA (NSF)	TOTAL EXISTING WORKSETTINGS	MASTER PLAN SPACE SUMMARY — NORTH CAMPUS: Worksettings: Located on North Campus	North Campus Adjustment / Growth	Total Worksettings	Required Seats: Instr. Teaching Lab Seats	Collaboration: Personnel Collaboration	Collaboration: Student Collaboration	Res. Research seats	North Campus Totals: cap. Seats	per room: NSF / Seat	per room: NSF / station or Room	per room: No. of stations or Rooms	TOTALS: Total Area (NSF)	SOUTH CAMPUS TOTALS	OFF-CAMPUS TOTALS	WANG TOTALS	FLEX LAB TOTALS	GRAND TOTAL	Comments
Offices	**421**	**105**	**526**	**25,014**	**299**	**15,546**	**218**	**10,990**	**105**	**51,550**	**622**	**299**	**102**	**401**				**190**						**24,386**	**15,546**	**10,990**	**0**	**0**	**50,922**	
Dean																					300	0	0						
Academic Department Head	1		1	462	1						1	1		1				1			180	1	180						
Academic Assoc Head, Asst. Head	1	1	2	139	1						1	1	1	2				2			150	2	300						
Tenured/TT Faculty	50	16	66	7,484	46	5,639	19	1,769	13		78	46	20	66				34			120	66	7,920						+4 fac offices at Zucrow
Adjuncts or Joint Appointments	10		10	1,834	13						13	13	-3	10				10			60	10	600						
Visiting Scholars (Total space allocation is equal to 7% of above total faculty space allocation.)																					630	1	630						
Research or Visiting Faculty	10		10	249	3	263	2				5	3	5	8				8			60	8	480						
Instructor or Lecturer	2	2	4	765	10						10	10	-6	4							120	4	480						
Emeritus/Emeritas Office				151	1						1	1		1							120	1	120						
Post-Doctoral Assistants - Workstation	14		14	988	15	352	6	864	8		29	15	-1	14				14			90	14	1,260						
Grad - Thesis (See Table 2.7b)	269	80	349	6,963	162	7,092	181	4,046	77		420	162	80	242				121			40	169	6,776						* MP provides grad desks at ratio of .7 desk/grad. +46 grad desks at Zucrow
Professional Staff - Office	16	2	18	1,690	10		6	195			16	10	2	12							120	12	1,440						
Business Manager	1		1		1						1	1		1							120	1	120						
Business Staff	9		9		7				2		9	7		7							90	7	630						
ECN IT Staff	2		2		2						2	2		2							120	2	240						
Management	3	1	4		3						3	3	1	4							120	4	480						
Admin/Operations Staff - Workstation	10	1	11	4,565	5	1,888		412	5		10	5	1	6							90	6	540						
Clerical - Workstation	13	2	15		9		4				13	9	2	11							90	11	990						
Service Staff - Workstation	10		10		10						10	10		10							120	10	1,200						
Hoteling offices																					40	0	0						
Office Support				**4,561**		**1,206**		**995**		**6,762**													**1,989**	**1,206**	**995**	**0**	**0**	**4,190**	
Department Reception																					300	1	300						
Files/Storage																					300	1	300						
Admin Workroom																					300	1	300						
Business Workroom																					300	1	300						
Shipping & Receiving																							789						
Research Labs & Support				**32,458**		**41,529**		**45,550**		**119,537**								**190**	190				**14,305**	**41,529**	**55,450**	**0**	**23,295**	**134,579**	
Wet Bench																													
Engineering																			190	250	47,500	1	47,500						
Dry Comp Collaboration																													
Relocated to Zucrow addition																							-9,900		9,900				
Relocate to Flex Lab																							-23,295				23,295		
Specialized Shared Support				**10,198**		**2,376**		**600**		**13,174**													**13,014**	**2,376**	**600**	**0**	**0**	**15,990**	
Electronics shop				973																			973						
UG Machine shop				2,970																			2,970						G029, G039
Research shop				539																			539						use ADM
Manufacturing shop				1,235																			4,200						G010, G028
Central Storage and Computer				4,332				600															4,332		600				

Personnel Change

- 25% faculty and staff increases to support the College's growth initiative
- 31% faculty increase
- 24% increase of staff

SCHOOL OF MECHANICAL ENGINEERING (ME)

Table 2.47.a—Master Plan space program

Space Type	EXISTING SPACE: NORTH CAMPUS Area	NORTH CAMPUS Worksettings or Seats	SOUTH CAMPUS Area	SOUTH CAMPUS Worksettings or Seats	OFF - CAMPUS Area	OFF - CAMPUS Worksettings or Seats	TOTAL: TOTAL EXISTING AREA (NSF)	TOTAL: TOTAL EXISTING WORKSETTINGS	MASTER PLAN SPACE SUMMARY: NORTH CAMPUS Worksettings: Located on North Campus	North Campus Adjustment / Growth	Total Worksettings	Required Seats: Teaching Lab Seats (Instr.)	Collaboration: Personnel Collaboration	Collaboration: Student Collaboration	Res.: Research seats	North Campus Totals: cap. Seats	per room: NSF / Seat	per room: NSF / station or Room	per room: No. of stations or Rooms	TOTALS: Total Area (NSF)	SOUTH CAMPUS TOTALS	OFF-CAMPUS TOTALS	WANG TOTALS	FLEX LAB TOTALS	GRAND TOTAL	Comments
Teaching Labs & Support	**14,888**		**0**		**0**		**14,888**					**332**								**21,255**	**0**	**0**	**0**	**0**	**21,255**	
ME 475 Automatic Control Systems	912											16				16	50	800	1	912						ME 1030E
ME 586/588 Micro- processors	1,432											20				20	75	1,500	1	1,432						ME 1030D
ME 557 Design for Manufacturability/ME 463	1,146											24				24	50	1,200	1	1,146						ME 1178
ME 315 Heat and Mass Transfer	1,576											30				30	50	1,500	1	1,576						ME 1030B
Project Instru/ ME 297 How Stuff Works	774											30				30	50	1,500	1	774						ME 1185
ME 365/ Systems & Measurement	1,102											24				24	50	1,200	1	1,102						ME 1030A
D-CL PC/ ME444 CAD & Prototyping												30				30	50	1,500	1	1,500						ME 3021, sim. to ME1178
ME 263 Sophomore Cornerstone/ Intro to Mech Eng Design												24				24	50	1,200	1	1,200						sim. to ME 1178 in lieu of cr. ME2063
ME 309 Fluid Mechanics	1,470											30				30	50	1,500	1	1,470						ME 1030F
ME 352 Machine Design												16				16	75	1,200	1	1,200						in lieu of cr ME1006, ME1009, ME1012
ME 363 Manufacturing	316											8				8	75	600	1	316						ME G032
ME 497 Finite Element Analysis												0				16	50	800	0	0						Not required per P. Meckl
Senior Capstone ME 463												24				24	75	1,800	1	1,800						1 per P. Meckl
Combustion IC Engines												16				16	50	800	1	788						Rooms G051, G053, G055
Computer Lab	1,216											40				40	30	1,200	1	1,216						ME 2028
Shops																										
Electronics	866																			900						ME 2043
Rapid Prototype	584																			600						ME 1191
Teaching Support - 25%	3,323																			3,323						ME 1030, ME 1030H
Collaboration Space	**16,350**	**434**	**13,584**		**5,542**		**35,476**						######	######						**18,884**	**13,584**	**5,542**	**0**	**0**	**38,010**	
Personnel Collaboration		233																								
Conference Room - Existing	3,442		1,195		5,040								138							3,442						+20 seat conf rm at Zuc
Focus/TA Consult/Help/Advising	1,268												51							1,268						
Break/Coffee Bar	456				393								18							456						
Mediascapes													12			6	25	150	2	300						
Break-out / Commons - Grad	1,226				109								25							1,226						
Student Collaboration		201																								
Open Lab - Computer														40		40	30	1,200	1	1,200						
Open Lab - Student Project	7,065													115						7,276						
Open Lab - Student Clubs																										Cars/IDC?
Student Organizations	881		9,321											35						881						
Group Study (quiet)	1,222													49						1,222						
Break-out / Commons - Undergrad	1,613													32						1,613						
MECHANICAL ENGINEERING TOTAL	**103,469**	**733**	**74,241**	**218**	**63,677**	**105**	**241,387**	**622**	**299**	**102**	**401**	**332**	**243**	**271**	**380**					**93,833**	**74,241**	**73,577**	**0**	**23,295**	**264,946**	

See Page 2-49 for Program summary by location and space type, and Change in program space type Charts.

Space Program

The ME graduate program is ranked 10th in the nation. Areas of research include:

- Acoustics and Noise Control
- Bioengineering
- Combustion
- Design
- Fluid Mechanics and Propulsion
- Heat Transfer
- Heating, Ventilation, Air Conditioning and Refrigeration
- Manufacturing and Materials Processing
- Mechanics and Vibration
- Nanotechnology
- Robotics
- Solid Mechanics
- Systems Measurements and Control

Space Drivers

- To efficiently accommodate faculty and staff growth
- To support growing student enrollment

Program Changes

Offices and Support

This category has a 6% reduction in space while the personnel growth is 25%. The new space standards help to minimize the growth in this category.

Research Labs and Support

Due to the increased personnel growth and current overcrowding, additional space will be required to support the research mission of this school.

Total required research space exceeds the available space on the North Campus. Therefore, some research will be located in the South Campus Flex Lab. It is recommended that the research best served by intra-disciplinary actives should be at the Flex Lab.

Teaching Labs and Support

The current facilities are overcrowded and more space will be required to add new teaching labs and student work space to support growing student populations.

Collaboration

Greater collaboration space is required to support the faculty and staff as well as the students.

Master Plan Program

Building Condition Analysis

Delon & Elizabeth Hampton Hall of Civil Engineering (HAMP)

Constructed:	1951, Additions in 1962, 1986
Structural System:	Concrete
Building efficiency:	51.7%
Highest and best use:	Technical (Research & Instructional)
Repair & Rehabilitation Level:	4
Current Replacement Value:	$122,423,121
Deferred Maintenance Value:	$12,430,950
Facility Condition Index:	0.10

This building was built in three phases; the first was completed in 1951 with the first addition completed in 1962. There was a third addition completed in 1986 that included renovations to the existing building. The building houses Civil Engineering, Construction Engineering and Management, Global Engineering Program, Earth, Atmospheric and Planetary Sciences, Libraries, Health Sciences, Pharmacy Practice, and Radiological and Environmental Management (REM).

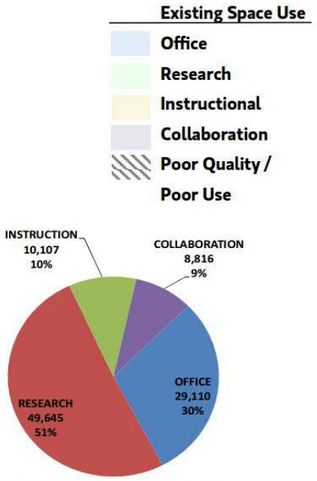
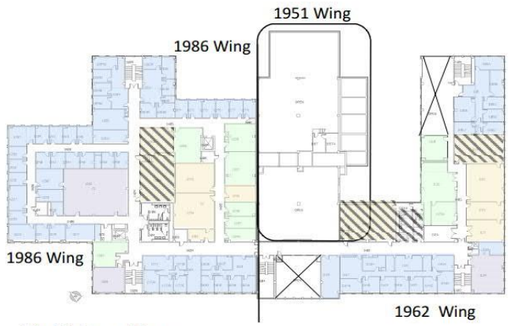


Chart 3.6 -Distribution of NSF by space type

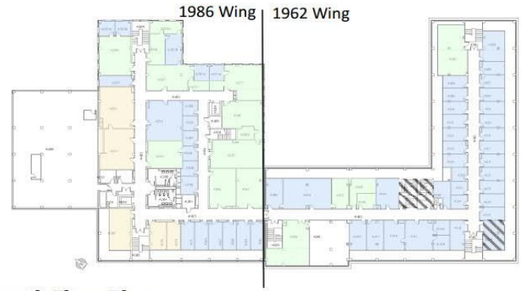
Building	Adjacencies	Appropriate Use	Research Space	Instruction Space	Support Space	Wayfinding	Finishes	Amenities	Accessibility	HVAC	Electrical Systems	Roof	Windows/ Doors	Elevator
	Functionality						Environment			Infrastructure				
HAMPTON HALL														
HAMP Wing 0 - 1951														
First Floor Plan	◒	●	◒		◒	◒	◒	◒	◒	◒	◒	◒	◒	
Ground Floor Plan	◒	●	◒	◒	◒	◒	◒	◒	◒	◒	◒	◒	◒	
Basement Plan	◒	●	◎		◒	◒	◒	◒	◒	◒	◒	◒	◒	
HAMP Wing 1 - 1962														
Fourth Floor Plan	◒	◒	◒		◒	◒	◒	◎	◒	◒	◒	◒	◒	◎
Third Floor Plan	◒	●	◎	◎	◒	◒	◒	◎	◒	◒	◒	◒	◒	◎
Second Floor Plan	◒	●	◒	◎	◒	◒	◒	◎	◒	◒	◒	◒	◒	◎
First Floor Plan	◒	●	◒	◎	◒	◒	◒	◎	◒	◒	◒	◒	◒	◎
Ground Floor Plan	◒	●	◒	◒	◒	◒	◒	◎	◒	◒	◒	◒	◒	◎
Basement Plan	◒	●	◒	◒	◒	◒	◒	◎	◒	◒	◒	◒	◒	◎
HAMP Wing 2 - 1986														
Fourth Floor Plan	●	●	●	●	●	●	●	●	◒	◒	●	●	●	●
Third Floor Plan	●	●	●	●	●	●	●	●	◒	●	●	●	●	●
Second Floor Plan	●	◒	●	●	●	●	●	●	◒	●	●	●	●	●
First Floor Plan	●	●	●	●	●	●	●	●	◒	●	●	●	●	●
Ground Floor Plan	●	●	●	●	●	●	●	●	◒	●	●	●	●	●
Basement Plan	●	◒	●	●	●	●	●	●	◒	●	●	●	●	●

Legend: ◎ Poor ◒ Fair ● Good

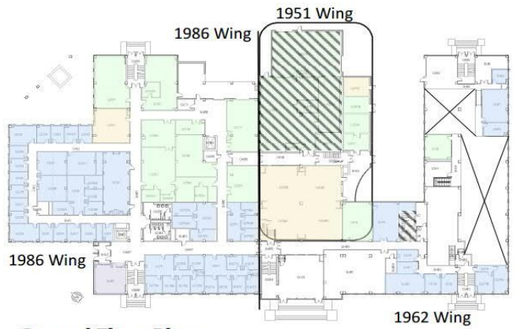
Table 3.6 -Hampton Hall existing space analysis



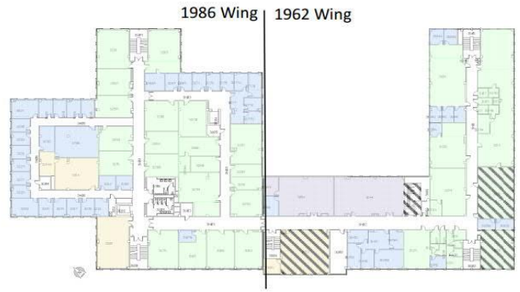
First Floor Plan



Fourth Floor Plan



Ground Floor Plan



Third Floor Plan

Basement Plan

Second Floor Plan

Building Condition Analysis

Mechanical Engineering Building (ME)

Constructed:	1929, Additions in 1932,1942,1948, and 2011
Structural System:	Concrete
Building efficiency:	54.2%
Highest and best use:	Technical (Research & Instructional)
Repair & Rehab. Level:	3
Current Replacement :	$77,913,187
Deferred Maintenance :	$17,107,550
Facility Condition Index:	0.22

The original Mechanical Engineering Building, built in 1929, is a 3 story concrete frame building which is now ringed by additions and housed the School of Mechanical Engineering. The 1932 wing faces the Purdue Mall and serves as the main entrance to the facility. The most recent addition is the Roger B. Gatewood Wing, which was completed in 2012 and is the first Purdue building constructed to LEED certification standards.

American Railway Building (RAIL)

Constructed:	1926
Structural System:	Load Bearing Masonry
Building efficiency:	59.8%
Highest and best use:	Office
Repair & Rehab. Level:	4
Current Replacement :	$963,125
Deferred Maintenance :	$47,000
Facility Condition Index:	0.05

Constructed in 1926, the American Railway building was originally used for research in railway draft gears. The building is now used as office facilities for the School of Mechanical Engineering.

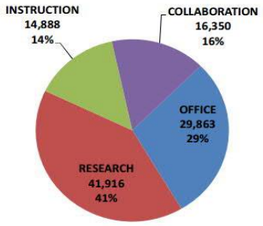
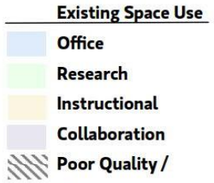


Chart 3.10 -Distribution of NSF by space type

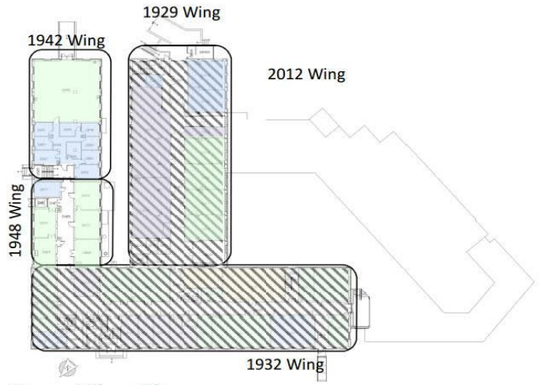


Third Floor Plan

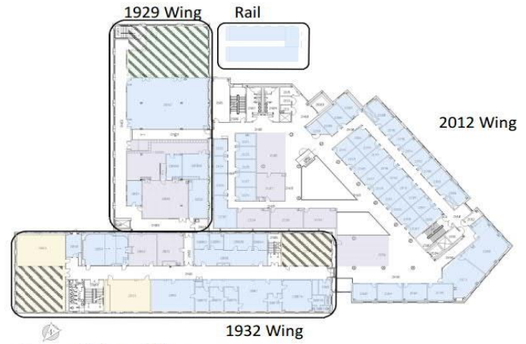
Building	Functionality: Adjacencies	Appropriate Use	Research Space	Instruction Space	Support Space	Wayfinding	Environment: Finishes	Amenities	Accessibility	Infrastructure: HVAC	Electrical Systems	Roof	Windows/ Doors	Elevator
Mechanical Engineering Building														
ME - Wing 0 - 1929														
Second Floor Plan	◒	◒	○		○	◒	○	○	◒	○	○	◒	○	◒
First Floor Plan	◒	◒		○	○	◒	○	○	◒	○	○		○	◒
Ground Floor Plan	◒	○	○		◒	◒	○	○	◒	○	○		○	◒
ME - Wing 1 - 1932														
Third Floor Plan	◒	◒	◒	○	◒	◒	◒	◒	◒	◒	◒	◒	○	
Second Floor Plan	◒	◒	◒	○	◒	◒	◒	◒	◒	◒	◒		○	
First Floor Plan	◒	◒	◒	○	◒	◒	◒	◒	◒	◒	◒		○	
Ground Floor Plan	◒	○	○	○	○	◒	○	○	○	◒	◒		○	
ME - Wing 2 - 1942														
First Floor Plan	◒	◒	○		◒	◒	◒	◒	◒	◒	◒	◒	○	
Ground Floor Plan	◒	◒	◒	◒	◒	◒	◒	◒	◒	◒	◒		○	
Basement Plan	◒	◒			◒	○	○	○	○	○	○			
ME - Wing 3 - 1948														
First Floor Plan	◒	◒			◒	◒	◒	◒	◒	◒	◒	◒	◒	
Ground Floor Plan	◒	◒	◒		◒	◒	◒	◒	◒	◒	◒		◒	
ME - Gatewood Wing - 2011														
Third Floor Plan	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Second Floor Plan	●	●	●	●	●	●	●	●	●	●	●	●	●	●
First Floor Plan	●	●	●	●	●	●	●	●	●	●	●		●	●
Ground Floor Plan	●	●	●	●	●	●	●	●	●	●	●		●	●
Basement Plan	●	●	●	●	●	●	●	●	●	●	●		●	●
ME - RAIL- 1926	◒	◒			◒	◒	◒	◒	○	◒	◒	◒	◒	

Legend: ◎ Poor, ◒ Fair, ● Good

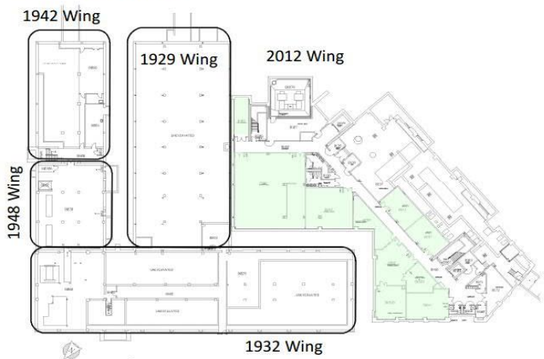
Table 3.10 -ME complex existing space analysis



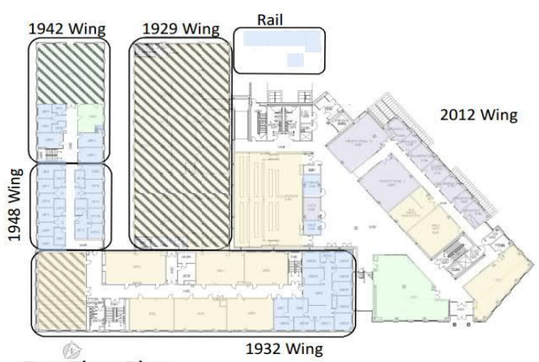
Ground Floor Plan



Second Floor Plan



Basement Plan



First Floor Plan

Cost and Time Line

Cost Estimate Methodology

Cost estimates have been developed using Purdue Construction Cost Estimating Planning Index data adjusted to the 3rd quarter 2013. Total costs for each renovated building have been weighted depending on the condition of the building being renovated and the extent of infrastructure upgrades that can be expected. ARMS has been weighted at 50% of new construction cost. EE, HAMP, MSEE, HAMP, and RAIL have been weighted at 75% of new construction. POTR, and all proposed new buildings have been weighted at 100%

All costs include a 30% mark up to cover soft costs including fees, contingency, and fit up. All costs have been escalated at 4% per year.

Time Line Explanation

Costs for each building have been broken down by Phase and indicated in the year in which they can be expected to be accrued. Total costs for each year are indicated on the rightmost column. Total costs per building are listed at the bottom of the table on the facing page

Sequential activities are indicated by like colors. Activities that rely on the completion of the Flex Lab and the vacated space that will result are colored Olive. Activities that rely on the relocation of research on the 3rd floor of MSEE are colored gold. Activities that rely on the completion of the new Classroom building are light blue.

		ALC	ARMS	EE	FRNY	GRIS	HAMP	NUCL	ME	MSEE	POTR	RAIL	WANG	South Campus Flex Lab	Classroom Bldg	Total Cost per Year
2014	**Activity**	JTRP to WANG 4th				IE moves to WANG Renovation begins				Design 3rd flr for ECN, Design 1st for ECE	EPE, ENE to WANG - IE from GRIS		IE from GRIS, EPE from POTR, ENE from ARMS, ECE from EE, JTRP from ENAD	Predesign		
	Cost															
2015	**Activity**	Demo ENAD HPN Start Construction		ECE swings 100 to 115 seats to WANG, Efficiency Renovations	Construction	Occupy - IE from WANG & POTR			Design Renovations for ME	3rd flr research to TBD, Renovate for ECN - ECE swings to WANG, Efficiency renovation Base.	IE to GRIS		IE to GRIS, ECE from EE & MSEE	Design		
	Cost			2,440,000		11,700,000				6,000,000				6,170,000		**26,310,000**
2016	**Activity**	Construction		Design 1st flr teaching labs, ECE swings back from WANG - Phase 1 Complete	Occupied by ChE		Design renovations for CE		ME swings to WANG, Efficiency Renovations - Phase 1 Complete	ECN to MSEE 3rd flr, Renovate 1st ECE Office and stud. Ctr. - Phase 1 Complete	Predesign	Efficiency renovations - Phase 1 Complete	ECE to EE, ME from ME	Construction		
	Cost			3,660,000					12,200,000	9,000,000		1,200,000		37,020,000		**63,080,000**
2017	**Activity**	Construction	Artisan Lab to IDC, Renovations for EPICS, AAE - Phase 1 Complete	ECE 1st flr to MSEE 1st flr - Renovate for ECE Teaching Labs			CE swing to WANG Efficiency Renovations G, 1st, & 4th		ME swings back from WANG	ECE from EE occupies 1st flr - Phase 2 Complete	Design Renovated POTR		ME to ME, CE from HAMP	Construction		
	Cost		1,500,000	2,040,000			1,260,000			16,600,000				55,530,000		**76,930,000**
2018	**Activity**	Occupy - Libraries move from HAMP, POTR	Research to FLEX Lab, Renovations for AAE - Phase 2 Complete	Research to Flex Lab, Renovate for ECE Teaching Labs - Phase 2 Complete			Research to Flex Lab, Libraries to ALC, REM to TBD, Renovate for CE growth - Phase 1 Complete	Predesign, Research to Flex Lab	Research to Flex Lab, Renovate for ME teaching Labs, Classrooms, growth		BME to TBD, Libraries to ALC, Research to FLEX Lab, EEE 10 seats & OPP 10 seats to WANG		OPP, EEE from POTR	Occupy -		
	Cost		1,800,000	3,060,000			2,940,000		9,000,000					24,680,000		**41,480,000**

Time Line Key

	Activity related to completion of the South Campus Flex Lab
	Activity related to the completion of renovations in MSEE
	Activity related to the completion of a Classroom Building

		ALC	ARMS	EE	FRNY	GRIS	HAMP	NUCL	ME	MSEE	POTR	RAIL	WANG	South Campus Flex Lab	Classroom Bldg	Total Cost per Year
2019	**Activity**		Design renovations	Renovation for Flex Teaching Labs on 1st floor - Phase 3 Complete			Occupy - CE / JTRP from WANG - Phase 2 Complete	Design new NUCL building, NE to WANG - Demo NUCL	Occupy teaching labs & Classrooms - Phase 2 Complete		Under Renovation		CE / JTRP to HAMP			
	Cost			5,700,000			14,000,000	3,145,000	6,000,000		59,760,000					**88,605,000**
2020	**Activity**		Programs & ENE to POTR, Renovate for MSE, AAE, EPICS growth - Phase 3 Complete				GEP, Honors to POTR - Phase 3 Complete	Construction			Occupy - ENE, Programs from ARMS, EEE from WANG, GEP Honors from HAMP		OPP and EEE to POTR, NE from NUCL		Predesign	
	Cost		5,100,000				3,300,000	18,870,000			14,940,000					**42,210,000**
2021	**Activity**		Occupy EPICS, MSE, AAE, Renovate for EPICS Computer Lab - Phase 4 Complete					Construction					ENE Research growth		Design -	
	Cost		600,000					31,450,000							1,446,825	**33,496,825**
2022	**Activity**							Occupy - IE from MGL S., SOS, NE from WANG					NE to new NUCL building		Construction	
	Cost							9,435,000							14,910,000	**24,345,000**
2023	**Activity**			Design classroom back fill			Design classroom back fill		Design classroom back fill	Design classroom back fill					Construction	
	Cost														22,365,000	**22,365,000**
2024	**Activity**			Back fill SMAS Classrms - Phase 4 Complete			Back fill SMAS Classrms - Phase 4 Complete		Back fill SMAS Classrms - Phase 3 Complete	Back fill SMAS Classrms - Phase 3 Complete					Occupy - SMAS	
	Cost			4,500,000			3,300,000		4,200,000	1,500,000					9,940,000	**23,440,000**
Total Cost per Building		**0**	**9,000,000**	**21,400,000**	**0**	**11,700,000**	**24,800,000**	**62,900,000**	**31,400,000**	**33,100,000**	**74,700,000**	**1,200,000**	**0**	**123,400,000**	**48,661,825**	**442,261,825**

Preliminary Code Information:

A preliminary code study is included to help guide future development of projects associated with the ME-RAIL-HAMP study, as follows:

PRELIMINARY CODE INFORMATION

RALPH GERDES CONSULTANTS
BUILDING CODES • FIRE SAFETY

June 10, 2015

PURDUE UNIVERSITY
MECHANICAL ENGINEERING, AMERICAN RAIL
AND HAMPTON HALL BUILDINGS
STUDY PHASE

The purpose of this report is to document the code review of the study phase of these similar projects. The review and comments are based on a preliminary programmatic study plans, phone calls, and one meeting on April 16, 2015. The following is offered:

Scope of Work

Existing academic buildings on a college campus are having a renovation study conducted.

The intent of the study is to review existing conditions and provide overall space concepts for improving efficiencies. The study also includes evaluations of how to increase work spaces quantities and provide environments similar to previous project.

Although the proposed functions of spaces may change, it is not anticipated there will be any code defined 'change of use' of the spaces. The use of building will remain the same.

Mechanical Engineering

The original building was built in 1929, with additions in 1932, 1942, 1948 and 2012. With some very minor exceptions, the building is protected throughout with an automatic fire suppression system (partial), standpipe system, and fire alarm system. It is anticipated that those areas not currently protected by the sprinkler system, will be provided with sprinklers as the project moves forward.

There may be a small addition to the building in the form of a stairway.

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American Rail

The building was built in 1926. The building is assumed to be protected with an automatic fire suppression system (may be partial) and fire alarm system. It is anticipated that the areas that may not be currently protected by the sprinkler system or other life safety system, will be provided with them as the project moves forward.

It is anticipated that the entire existing stair will be moved and reconstructed.

Hampton Hall

The original building was built in 1951, with additions in 1962 and 1986. With some very minor exceptions, the building is protected with an automatic fire suppression system (partial), standpipe system, and fire alarm system. It is anticipated that those areas not currently protected by the sprinkler system, will be provided with sprinklers as the project moves forward.

There is anticipated to be a substantial addition to the building in the form of a proposed large entry space located in the 1951 wing courtyard.

General Code Approach

Since the building, and the proposed scope of work will remain the same occupancy (A-3, Assembly >50; B, Offices and Education over the 12th grade) the existing building and systems are allowed to remain "as is" (i.e.: Grandfathered). The existing building and systems are not required to be brought up to today's current code. However, General Administrative Rules (GAR) require existing systems to be maintained as installed and, if added to, shall not cause the system to become non-compliant with current code requirements.

All new work shall comply with current codes and standards as if a new building. Anticipated new work may include:

- Additional sq ft added to the building (stair enclosure, new entry space, infill of floors, exteriors)
- New shafts
- New duct penetrations (vertically or horizontally)
- Work through and around existing rated walls (area separations/fire walls, use group/occupancy, etc.)
- New openings in existing interior walls, some fire walls
- Establish new restroom locations
- Relocation of an existing stairway

Only new work must comply with current code, even though it may be built into existing conditions.

Although it is encouraged to install complete new systems, unless other code sections or trade-offs are dependent on a complete installation, partial installation of some systems and subsequent code compliance is acceptable.

Although, some existing conditions may not be technically code compliant when installed, they are still grandfathered, due to their age. These conditions do however become the University's liability.

Wet labs will maintain hazardous materials and chemicals below the exempt amounts that they nor the building become a 'H' occupancy.

It is the desire of the University to maintain, if not improve, the overall fire and life safety of the building. Field verification of existing condition is needed.

Applicable Codes and Standards

Indiana Building Code – 2014	(2012 International Building Code & State Amendments)
Indiana Fire Code – 2014	(2012 International Fire Code & State Amendments)
Indiana Electrical Code – 2009	(2008 National Electrical Code & State Amendments)
Indiana Mechanical Code - 2014	(2012 International Mechanical Code & State Amendments)
Indiana Plumbing Code – 2012	(2006 International Plumbing Code & State Amendments)
Indiana Elevator Code - 2011	(ASME A17.1 - 2007 with State Amendments)
Indiana Energy Conservation Code-2010	(American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc. (ASHRAE) – (Standard 90.1 – 2007 with Amendments)

Applicable Standards

NFPA 10 – 2010 Edition, Portable Fire Extinguishers
NFPA 13 – 2010 Edition, Installation of Sprinkler Systems
NFPA 14 – 2000 Edition, Installation of Standpipes
NFPA 20 – 1999 Edition, Installation of Centrifugal Fire Pumps
NFPA 25 – 2002 Edition, inspection Testing and Maintenance of Water Based Fire Protection Systems
NFPA 37 – 2002 Edition, Installation and Use of Stationary Combustible Engines and Gas Turbines
NFPA 72 – 2002 Edition, National Fire Alarm Code
NFPA 70 – 2008 Edition with Indiana Amendments as Indiana Electric Code 2009

Other Rules and Regulations

Indiana General Administrative Rules (675 IAC 12), (Indiana Administrative Code)
American with Disabilities Act (Federal Law), July 26, 1992

Mechanical Engineering

- Existing Building

1929	Three (3) story with a basement
1932	Five (5) story with a basement
1942	Two (2) story with a basement
1948	Two (2) story with a basement
2012	Three (3) story with a basement

- Construction Type

1929	Field verification needed
1932	Field verification needed
1942	Field verification needed
1948	Field verification needed
2012	Field verification needed

- Fire Wall Locations and Ratings

  1929 Field verification needed
  1932 Field verification needed
  1942 Field verification needed
  1948 Between 1932 wing, rating unknown, two (2) story with a basement
  2012 Between 1929 wing, rating unknown, three (3) story with a basement

- Fire Barrier Locations and Ratings

  1929 Field verification needed
  1932 Field verification needed
  1942 Field verification needed
  1948 Field verification needed
  2012 Field verification needed

- Interior Partition Locations and Ratings

  Typically masonry, glazed and clay tile and non-combustible construction with some wood wall construction (field verification needed).

- Automatic fire suppression system throughout , with minor exceptions

- Standpipe system located in exit stair enclosures.

Proposed New Work

- New program and space relocations on multiple floors requiring new interior wall and corridor construction.
- A new stair enclosure added to the south end of 1942 wing of _____sq ft. (unknown sq.ft. at this time)
- Establish new men's and women's restrooms on multiple floors.
- New building systems (electrical, data, plumbing, HVAC, emergency generator) as necessary.
- Upgrade renovated/remodeled areas to ANSI/ADA compliance.

Code Issues

A. Adding Additional Square Footage

Issue
- Either in the building (i.e.: floor infill) or an addition outside the existing building will cause the existing building square footage and any new square footage to meet the allowable height and area of the current code.

Solution
- Separate new square footage from existing with a 4 hour wall
- Although these calculations can be done, most likely it will be easier to apply for a General Administrative Rule variance vs figuring out construction types, open frontage increases, fire wall locations, etc. of a 80 year old building with several additions, and applicable code over the decades.

B. New Shafts

Issue
- New shafts could create a challenge. Today's codes require shafts to be supported by construction affording the same rating as the shaft.

Solution
- There are some exceptions that we may be able to use, but the existing potential fire resistive rating of the floor and its supporting structure must match the required rating of the new shaft requirements. Ratings and exceptions will depend on construction type of building and number of floors the duct passes.

C. Removal of Existing Hose Cabinet/Hoses

- Can be removed, if today's code is followed [GAR 675 IAC 12-4-9(f)]

Issue
- Installation of a Class I standpipe system in each stairwell with outlets at intermediate landings and on the roof (Sec. 905.4)
- Additional outlets may be required on each floor, if coverage can not be met within 200 feet of the stairwell connections (Sec 905.4, Item 6)

Solution
- Installation of sprinklered riser(s) in each stairwell can also become the Class I standpipe system (Sec. 905.3.1)
- If location of standpipe outlets at the Intermediate landings in the existing stairwell is an issue, a variance can be attempted with the Local Fire Department input to locate outlets at the main landings (Sec. 905.4) Purdue Fire Department may prefer outlets at main floor landings.

American Rail

- Existing Building

  1926 Two (2) story building

- Construction Type

  1926 Field verification needed

- Fire Wall Locations and Ratings

  1926 Field verification needed

- Fire Barrier Locations and Ratings

  1926 Field verification needed

- Interior Partition Locations and Ratings

  Typically masonry, glazed and clay tile and non-combustible construction with some wood wall construction (field verification needed).

- Automatic fire suppression system throughout , with minor exceptions

- Standpipe system located in exit stair enclosures. (Field verification needed)

Proposed New Work

- New program and space relocations on two floors requiring new interior wall and corridor construction.
- Remove and construct new stair in new location between first and second floors.
- Establish new men's and women's restrooms on two floors.
- New building systems (electrical, data, plumbing, HVAC, emergency generator) as necessary.
- Upgrade renovated/remodeled areas to ANSI/ADA compliance.

Code Issues

A. Relocation of Existing Stair

Issue
- It is assumed that the stair can remain open due to the small footprint of the building, small occupant load and limited common path of travel/travel distance. Review of new code requirements based on proposed design with sprinklers is essential to avoid an enclosed stairway.

Solution
- Provide sprinklers and fire alarm system if not already provided.
- Apply for a variance if straight code cannot be achieved through design.

B. New Shafts

Issue
- New shafts could create a challenge. Today's codes require shafts to be supported by construction affording the same rating as the shaft.

Solution
- There are some exceptions that we may be able to use, but the existing potential fire resistive rating of the floor and its supporting structure must match the required rating of the new shaft requirements. Ratings and exceptions will depend on construction type of building and number of floors the duct passes.

C. Removal of Existing Hose Cabinet/Hoses
- Can be removed, if today's code is followed [GAR675 IAC 12-4-9(f)]

Issue
- Installation of a Class I standpipe system in each stairwell with outlets at intermediate landings and on the roof (Sec. 905.4)
- Additional outlets may be required on each floor, if coverage can not be met within 200 feet of the stairwell connections (Sec 905.4, Item 6)

Solution
- Installation of sprinklered riser(s) in each stairwell can also become the Class I standpipe system (Sec. 905.3.1)
- If location of standpipe outlets at the Intermediate landings in the existing stairwell is an issue, a variance can be attempted with the Local Fire Department input to locate outlets at the main landings (Sec. 905.4) Purdue Fire Department may prefer outlets at main floor landings.

Hampton Hall

- Existing Building

  1951 Three (3) story with a basement
  1962 Five (5) story with a basement
  1986 Five (5) story with a basement

- Construction Types

  1951 Field verification needed
  1962 Field verification needed
  1986 Field verification needed

- Fire Wall Locations and Types

  1951 Field verification needed
  1962 Field verification needed
  1986 Field verification needed

- Fire Barrier Locations and Types

  1951 Field verification needed
  1962 Field verification needed
  1986 Field verification needed

- Interior Partition Locations and Types

  Typically masonry, glazed and clay tile and non-combustible construction with some wood wall construction (field verification needed).

- Automatic fire suppression system throughout , with minor exceptions

- Standpipe system located in exit stair enclosures.

Proposed New Work

- New program and space relocations on multiple floors requiring new interior wall and corridor construction.
- Possibly a new large entry space added to the courtyard of the 1951 wing of unknown square footage at this time
- Establish new men's and women's restrooms on multiple floors.
- New building systems (electrical, data, plumbing, HVAC, emergency generator) as necessary.
- Upgrade renovated/remodeled areas to ANSI/ADA compliance.

Code Issues

A. Adding Additional Square Footage

Issue
- Either in the building (i.e.: floor infill) or an addition outside the existing building will cause the existing building square footage and any new square footage to meet the allowable height and area of the current code.

Solution
- Separate new square footage from existing with a 4 hour wall
- Although these calculations can be done, most likely it will be easier to apply for a General Administrative Rule variance vs figuring out construction types, open frontage increases, fire wall locations, etc. of a 60 year old building with several additions, and applicable code over the decades.

B. New Shafts

Issue
- New shafts could create a challenge. Today's codes require shafts to be supported by construction affording the same rating as the shaft.

Solution
- There are some exceptions that we may be able to use, but the existing potential fire resistive rating of the floor and its supporting structure must match the required rating of the new shaft requirements. Ratings and exceptions will depend on construction type of building and number of floors the duct passes.

C. Removal of Existing Hose Cabinet/Hoses

- Can be removed, if today's code is followed [GAR675 IAC 12-4-9(f)]

Issue
- Installation of a Class I standpipe system in each stairwell with outlets at intermediate landings and on the roof (Sec. 905.4)
- Additional outlets may be required on each floor, if coverage can not be met within 200 feet of the stairwell connections (Sec 905.4, Item 6)

Solution
- Installation of sprinklered riser(s) in each stairwell can also become the Class I standpipe system (Sec. 905.3.1)
- If location of standpipe outlets at the Intermediate landings in the existing stairwell is an issue, a variance can be attempted with the Local Fire Department input to locate outlets at the main landings (Sec. 905.4) Purdue Fire Department may prefer outlets at main floor landings.

Existing Conditions vs Current Code

For new construction to meet current code, and with the desire to improve the overall fire and life safety of the building, a list of items has been developed to field verify. Verification of existing conditions are required in most situations in order to assist in determining code requirements for new construction. Current code requirements of these items have been provided for both new construction and consideration to improve the fire and life safety of the building.

General Program

	Existing Conditions	Indiana 2014 Codes
GENERAL PROGRAM		
Use Group Classification:		Business (Education Above) the 12th grade assembly <50 (B) Storage Occupancies (S-1)
Allowable construction types:	Varies per building; field verification is needed	
Construction type used for this report		Varies per building area; as the project progresses, the reports will be modified to individual buildings
BUILDING PROGRAM		
Mechanical Engineering	Actual	Maximum Allowed
Construction Type	1929 field verification needed 1932 field verification needed 1942 field verification needed 1948 field verification needed 2012 field verification needed	Unknown at this time
American Rail		
Construction Type	1926 field verification needed	Unknown at this time
Sprinkler system	Provided NFPA 13	For various sections or by Owner's choice

Hampton Hall

Construction Type	1951 field verification needed 1962 field verification needed 1986 field verification needed	Unknown at this time

OCCUPANCY CLASSIFICATIONS

	Existing Conditions	2014 Codes
B	Business (education above the 12th grade)	B Business (education above the 12th grade)
A-3	Assembly > 50 occupants	A-3 Assembly > 50 occupants Assembly > 50 occupants and accessory to another occupancy shall be included as part of that occupancy
S-1	Storage (Sec. 311.2)	S-1 Storage (Sec. 311.2)

STRUCTURAL REQUIREMENTS

Category	Existing Conditions	2014 Codes Minimum Fire Resistance (based on fire resistance rating required for each building. Field verification needed to match existing requirements)
Beams:		
Supporting roof only		Unknown at this time (based on several factors)
Supporting floors		Unknown at this time (based on several factors)
Columns:		
Supporting roof only		Unknown at this time (based on several factors)
Supporting floors		Unknown at this time (based on several factors)
Exterior bearing walls		Unknown at this time (based on several factors)
Exterior non-bearing walls		Unknown at this time (based on several factors)
Interior bearing walls		Unknown at this time (based on several factors)

STRUCTURAL REQUIREMENTS

Category (continued)	Existing Conditions	2014 Codes
Interior non-bearing walls		Unknown at this time (based on several factors)
Exit access corridors		0 hour (B, A) (Table 1017.1)
Vertical exit stair enclosure		2 hour (connecting ≥ 4 stories, including basement) (Sec 1009.3.1.2. and 1022.2) 1 hour (connecting less than 4 stories including basements Exceptions for up to 50% required stairs serving one adjacent floor (Sec. 1009.3.1.2 No requirement (connecting 1st and 2nd floors and not interconnected to other floors, with at least 2 additional means of egress) (Sec. 1009.3, Exception 9) No requirement for stairways within dwelling units 1009.3, Except to follow requirements of Sec. 705 (Sec. 1009.3.1.8)
Vertical exit stair enclosure (exterior walls)		Follow requirements of Sec. 705 (Sec. 1009.3.1.8)
Exit passageways		Not less than 2 hours (Sec. 1023)
Stairway exit discharge		Continuous to the exterior maintaining the same rating as the stair (Sec. 1024) Exc. 1 Max 50% of the number and capacity of the enclosures may exit through the level of exit discharge if: 1.1 Unobstructed path to exterior 1.2 Separation from areas below by construction conforming to the enclosure

STRUCTURAL REQUIREMENTS

	Existing Conditions	2014 Codes
Category (continued)		
Stairway exit discharge (continued)		Exc. 2 Max of 50% of the number and capacity through a vestibule 2.1 Vestibule is protected from below by the same as the enclosure 2.2 Vestibule does not exceed 10' 2.3 Vestibule is separated by wire glass equivalent 2.4 Vestibule is used as exit and is unobstructed (Sec. 1027,Exc 2)
Fire walls between maximum areas allowed by code		2 or 3 hour (Table 706.4)
Horizontal Exit Fire Rating		2 hour (Sec. 1025)
Fire barriers		As required for their individual use as exit passageways, vertical exit and shaft enclosure, incidental use areas, exit passageways, etc. (Sec 707)
Shafts		2 hour (connecting ≥ four (4) stories, including basement levels) (Sec. 713) 1 hour (connecting < four (4) stories, including basement levels) (Sec. 713) 1 hour shafts, building <420 feet other than exit enclosures and elevator hoistways where sprinkler heads are installed within the shafts and alternating floor levels (Sec 403.2.1.2)

STRUCTURAL REQUIREMENTS

	Existing Conditions	2014 Codes
Category (continued)		
Shafts (continued)		Supporting construction of shafts shall afford the same fire resistance rating as the shaft requirements (Sec. 707.5) See exceptions: (5) 4 stories or less containing sheet metal ductwork or non-combustible piping (Sec 707.5.1) (6) 4 stories or less with 25 inches of light weight concrete or 2.7 inches of sand - light weight concrete (Sec 707.5.1)
Floors below level of exit discharge		No floor below exit discharge used for non residential use shall have unprotected openings to other residential floors
Structural Continuity		
Fire Barriers		Supporting construction of shafts shall afford the same fire resistance rating as the shaft requirements (Sec. 707.5)
Fire Partitions		Supporting construction of fire partition; shall afford the same fire resistance as the fire partition (Sec 708.4)
Elevator shafts		
Enclosure		2 hour separation required when omitting sprinklers within hoistways and machine rooms (Sec. 903.3.1.1.1) Minimum Fire Resistance (Based on fire-resistance rating reduction for high-rise from IA to IB)
Venting		Required (Sec. 3004.1)

STRUCTURAL REQUIREMENTS

	Existing Conditions	2014 Codes
Category (continued)		
Elevator shafts (continued)		
Drainage		Not required (Indiana Elevator Safety Code 2011, Indiana Amendments Sec. 2.2.2.5)
Elevator lobby vestibules		Not required (Sec. 713.14.1) Not required where the elevator shafts do not connect to floors over 75' in B, M, A-2, A-3, S-2)
Protection of wall openings		
Openings in exterior walls (sprinklered building)		0 to 3' Not permitted 3' to 5' 15" unprotected 1 hour rated elsewhere 5' to 10' 25% 10' to 15' 45% 15' to 20' 75% 20' to 25' No limit > 25' No limit > 30' Not required (Table 705.8, Sec. 705.8.2 & 705.10)
Openings in exterior walls first floor		Unlimited when >15' facing a street or unoccupied space on same lot or dedicated for public use not less than 30' (Sec. 705.8.1, Exc 1)
Openings in interior bearing walls		0 hour (Table 716.5)
Openings in interior non-bearing walls (partitions)		0 hour (Table 716.5)
Openings in occupancy		3/4 hour in 1 hour fire barriers

STRUCTURAL REQUIREMENTS

	Existing Conditions	2014 Codes
Category (continued)		
Protection of wall openings (continued)		
Separations (Fire barriers)		(Table 716.5) 1 ½ hour in 2 hour fire barriers (Table 716.5)
Openings in exit corridors (Fire partitions)		0 hour (B, A & S-1) Table 716.5 when sprinklered
Openings in exit passageway (Fire barrier)		1 ½ hour (Table 716.5)
Openings in stair enclosures (Fire barrier)		1 ½ hour (Table 716.5)
Openings in vertical shafts (Fire barrier)		1 ½ hour (Table 716.5)
Openings in incidental-use (smoke resistant construction)		Doors with closers; air transfer grilles to have fire dampers. Ducted penetrations have no requirements (Sec. 509.4.2)
Vertical Openings		
Protection of vertical openings of fire resistive rated horizontal assemblies		Penetrations of a floor, floor/ceiling assembly or the ceiling membrane or the membrane or the roof/ceiling assembly shall be protected in accordance with Sec. 713. Penetrations of fire resistive rate horizontal assemblies shall comply with Sec.714.4.1 thru 714.4.2.2) Penetrations by steel, ferrous or copper conduits, pipes, tubes, vents, concrete or masonry through a single fire-resistive floor assembly must have the annular space (between the floor and the penetrating item) filled with non-combustible material to prevent the passage of flame and hot gases (Sec. 714.4.1.1 Exc 1)

STRUCTURAL REQUIREMENTS

	Existing Conditions	2014 Codes
Category (continued)		
Vertical Openings (continued)		
Protection of vertical openings for fire resistive rated floors		Penetrations through a single concrete floor by steel, ferrous or copper conduits, pipes, tubes and vents with a maximum 6 inches nominal diameter are allowed without a shaft provided concrete, grout or mortar is the floor to maintain the fire resistive rating (Sec. 714.4.1.1 Exc 2) Duct penetrations of more than one (1) fire-rated floor/ceiling assembly shall be located within a shaft per Section 717.6 Duct penetrations of fire rated floor/ceiling assembly that connects not more than two (2) stories are permitted without a shaft provided a fire damper is installed at the floor line. (Sec. 717.6.1) see exceptions Through penetrations of only one (1) fire-rated floor/ceiling assemblies or three (3) stories or more by other items shall be located within a shaft or comply with Exception 1, 2 or 3 of Section 714.4.1.1 Electrical outlet boxes of any material are permitted provided such boxes are tested for use in fire-resistive rated assemblies (Sec 714.4.1.1)

STRUCTURAL REQUIREMENTS

	Existing Conditions	2014 Codes
Category (continued)		
Seismic Requirements		Seismic Design Category ? Unknown at this time
Site Classification		Class C ? (from Geotech Report)
Occupancy/Category Risk		Category II ? (Sec 1604.5, Table 1604.5)
Sprinkler Bracing		Unknown (ASCE7-10, 13.6.8.2)
Mechanical Bracing		Unknown (ASCE7-10,13.6.3)
Interior Finish		Maximum Flame Spread and Smoke Developed
		Sprinklered (Sec. 803.7) (minimum requirements)

Occupancy

	Vertical exits & passageway	Exit access corridors & other exitways	Rooms and enclosed spaces	Vertical exits & passageways	Exit access corridors & other exitways	Rooms and enclosed spaces
A-3	B	B	C	B	B	C
B	B	C	C	B	C	C
S	C	C	C	C	C	C

Class A	Flame Spread of 0-25 Smoke Development limits of 450 (Sec 803.1.1)
Class B	Flame Spread of 26-75 Smoke Development limits of 450 (Sec. 803.1.1)
Class C	Flame Spread of 76-200 Smoke Development limits of 450 (Sec. 803.1.1)
Elevator Finishes	Class B (See Elevator Code 2.14.2.1)

STRUCTURAL REQUIREMENTS

	Existing Conditions	2014 Codes
Category (continued)		
Interior Finish (continued)		
Interior floor finish		
Required vertical exits & passageways		Class II or DOC F-1 "Pill Test" (Sec. 804.4.2, Exc.)
Corridors providing exit access		Class II or DOC FF-1 "Pill Test" (Sec. 804.4.2. Exc.)
Rooms or enclosed spaces		Class II or DOC FF-1 "Pill Test" (Sec. 804.4.3 Exc.)
Elevator Cab		Class I per ASTMD 648 .45 w/cm^2 (Elevator Code, Sec. 2.14.2.1.5)
Insulations considerations		
Concealed		25 flame spread (Sec. 720.2) 450 smoke developed
Exposed		25 flame spread (Sec. 720.3) 450 smoke developed
Within plenum/ ducts		25 flame spread (IMC, Sec 604.3) 50 smoke developed
Piping		25 flame spread (IMC, Sec. 1204.1) 450 smoke developed

USE GROUP SEPARATIONS

	Existing Conditions	Indiana 2014 Codes
Mixed occupancy separations		Not required, built to the most restrictive height and area of non-separated uses except dwelling unit separations (Sec. 508.3)

Incidental Use/Hazard Areas	
Incidental use definition	<10% of the story (Sec. 509.3)
Furnace and boiler rooms over 400,000 BTU/hour input	1 hour or smoke resistant construction with automatic fire suppression system (Sec. 509)
Refrigerant machinery rooms	1 hour or smoke resistant construction with automatic fire suppression system (Sec. 508.2)
Storage rooms > 100 sq.ft.	Unseparated is allowed: if <10% of floor area (Sec. 509.3) or using non-separated uses (Sec. 508.3.2) or automatic Sprinkler system (Sec. 903.2.9)
OTHER SEPARATIONS	
Exit access corridors	0 hour (B, A and S-1) (Table 1017.1)
Maintenance Shops	NR Maintain flammable combustible liquids below the exempt amounts
Elevator equipment rooms and hoistways (to avoid shunt trip devices)	2 hour fire barrier (Sec. 903.3.1.1.1)
Vertical exit enclosures	2 hour fire barrier (Sec. 1022.2) (Sec. 1009.3.1.2)
Electrical "Rooms"	
Electrical vaults > 600 Volts	3 hour rating [IEC, Sec. 110.31 (A)]
Electrical "Rooms"	
Transformers Dry Type < 112 ½ kVA	Not required [IEC 450.2(A)]
< 112 ½ kVA	1 hour [IEC 450.2(A)] See exceptions
Vaults	See Section IEC 450.41 for for exceptions
Switch Gear - unenclosed switchgear over 35,000 volts	In a vault following IEC 450.41 through, 450.48 Sec IEC 230.212

EXIT REQUIREMENTS

Exit Sizes

Exit size measured in units of feet and inches.

		Existing Conditions		Indiana 2014 Codes	
		Minimum	Maximum	Minimum	Maximum
Doors	-width			32" (Sec. 1008.1.1)	4'-0" (Sec.1008.1.1)
	-height			6'-8" (Sec. 1008.1.1)	NR
Stairs	-risers			4" (Sec. 1009.7.2)	7" (Sec. 1009.7.2)
	-treads			11" (Sec. 1009.7.2)	NR
	-maximum height between landings			NR (Sec. 1009.3)	12' vertically
	-width			44" (36" < 50 occ.) (Sec 1009.4)	NR
	-headroom			6'-8" (Sec. 1009.5)	NR
	-landing width			Equal to stair (Sec. 1009.8)	NR
	-reduction of landing width by door			NR	½ of required width 7" when fully open (Sec.1009.8)
	-clear space between handrail & wall			2 1/4" (Sec. 1012.7)	NR
	-handrail projection into required width			NR	4-1/2" (Sec 1012.7)
	-handrail height			34" (Sec.1012.2)	38" (Sec.1012.2)

EXIT REQUIREMENTS (continued)

Exit Sizes

		Existing Conditions		Indiana 2014 Codes	
		Minimum	Maximum	Minimum	Maximum
Stairs	-required number			Each side of stair (Sec. 1009.10)	N/A
	-graspability			1.1/4" dia (Sec. 1012.3)	2" dia (Sec. 1012.3)
	-extensions			Required when not continuous	Both sides beyond the top riser, and to slope for a depth of one tread beyond the bottom riser (Sec. 1012.6)
	-intermediate handrails			Stairs exceeding 6'-3" shall have a handrail within 30' of all portions of required egress width (Sec 1012.9)	
Corridors	-width			44" (Sec. 1018.1 Table 1018.2	NR
	-height			7'-6" (Sec. 1003.2)	NR
	-projections			NR	7" reduction of corridor width when door is fully open. Door cannot reduce corridor width width less than ½ during swing (Sec 1003.3)

EXIT REQUIREMENTS (continued)

Exit Sizes

	Existing Conditions		Indiana 2014 Codes	
	Minimum	Maximum	Minimum	Maximum
Egress width per occupant served				
Stairways			0.3	Table 1005.3.1
Other egress components (doors, corridors, horizontal exits)			0.2	Table 1005.3.1
Occupancy load factors			Square Feet/Person	
Classroom			20 Gross Table 1004.1.2	
Laboratory			50 Gross Table 1004.1	
Storage/Mechanical			300 Gross Table 1004.1.2	
Office			100 Gross Table 1004.1.2	
Assembly (Chairs only)			7 sq.ft. Table 1004.1.2	
Assembly (Tables & chairs)			15 sq.ft. Table 1004.12	
Number of Exits				
Two exits or exit access doorways from any space provided (Sec.1015)			(1) Occupant exceed: A, B 49 R 10 S 29 Table 1014.3	
			(2) Common path of travel exceeds: A 75 B, S 100 Table 1014.3	

EXIT REQUIREMENTS (continued)

	Existing Conditions	Indiana 2014 Codes
Egress width per occupant served		
	Existing Conditions	Indiana 2014 Codes
		(3) Where required for special rooms: Boilers, incinerator, furnace (Sec. 1015.3) Refrigeration machinery rooms (Sec. 1015.4) Refrigerated rooms (Sec. 1015.5)

Where one exit is allowed

Occupancy	Max Ht. Stories Above or Below Grade	Max. Allowances
A, B	1	50 occupants (Table 1019.2) 75' travel distance Table 1022.2 (2)
B, S	2	30 occupants (Table 1019.2) 75' path of travel (Table 1021.2)
S	1	30 occupants (Table 1019.2) 100' of travel Table 1021.2 (2)
Mechanical spaces that are not rooms		No building code requirement Review mechanical code access (IMC 306 and OSHA requirements)

Multiple exits

	Exits	Occupant Load	Min. Number of exits
1-500	2	1-500	2 (Sec 1021.2)
501-1000	3	501-1000	3 (Sec 1021.2.3)
More than 1000	4	More than 1000	4 (Sec 1021.2.4)

EXIT REQUIREMENTS (continued)

Existing Conditions	Indiana 2014 Codes
Means of egress continuity	
Egress width maintained	Required (Sec.1005.4)
Multiple means of egress	Shall be sized that loss of any one (1) means of egress shall not reduce the available capacity to less than 50% of the required capacity (Sec. 1005.5)
Assembly exiting < 300	
Main exit	
Well defined	Not required, <300 (Sec 1028.2)
Not well defined	N/A
Side exits	NR
Other exits	Not required (Sec 1028.3)
Aisle width	Required (Sec. 1028.6)
Seats in a row	14 seats maximum with 12" aisle (Sec. 1028.10.2)
Assembly exiting > 300	
Main exit Well defined	Required, 1/2 occupant load (Sec. 1028.3)
Not well defined	100% capacity distributed around the perimeter (Sec 10028.2)
Assembly exiting > 300	
Side exits	Not required
Other exits	Required ½ occupant load (Sec. 1028.3)
Aisle width	Required (Sec. 1028.9)
Seat in a row	Required (Sec. 1028.10.2)

EXIT REQUIREMENTS (continued)

Existing Conditions	Indiana 2014 Codes
Assembly exiting > 300 (continued)	
Aisle termination	Each aisle shall terminate at cross aisle or other means of egress (Sec. 1028.9.5)
Special exiting situations	
Usable space under stairs	Permitted with 1 hour fire resistance rated construction or same rating of stair enclosure and sprinklers and entrance is not from within the exit enclosure (Sec. 1009.9.3)
Basement stair interruption (Discharge identification)	Required (Sec. 1022.8)
Stair access to roof	Required,when >4 connected stories including basements (Sec. 1009.16)
Roof access	Through a penthouse except unoccupied roofs may be through a roof hatch >16 sq.ft. (Sec. 1009.16.1)
Alternating tread stair	Limited use (Sec. 1009.13)
Changes in elevation	By sloped surfaces when less 12" (Sec. 1003.5)
Guards/rails	Required on open sides of stairs ramps, landings >30" above surface below (Sec. 1013.1)
Boiler and fuel equipment rooms over 400,00 BTU's and 500 sq.ft.	2 exits required (Sec. 1015.3)
Refrigeration machinery rooms over 1,000 sq.ft.	2 exits required (Sec. 1015.4)
Refrigerated Rooms >1000 sq.ft. >20 degrees	See IEC 110.26 (C) for situations (Sec. 1015.4)
Electrical rooms	See IEC 110.26 (C) for situations

EXIT REQUIREMENTS (continued)

	Existing Conditions	Indiana 2014 Codes
General		
Panic hardware required		Each door is a means of egress from an occupancy of Group A & E having an occupant load of 50 or more from rooms, corridors or the building (Sec. 1008.1.10)
Door swing in direction of travel		Occupant load of 50 or more (Sec. 1008.1.2)
Panic hardware		Electrical rooms 1200 amps or more and over 6 ft. wide (Sec. 1008.1.10)
Door swing		Electrical rooms 1200 amps or more and over 6 ft. wide (Sec. 1008.1.10)
Separate atmosphere egress systems		NR
Maximum travel distance to exits		
A		250' with sprinklers (Table 1016.2)
B		300' with sprinklers (Table 1016.2)
Common path of travel		
A		75' with sprinklers (Table 1014.3)
B, S		100' with sprinklers (Sec. 1014.3, Exc. 1)
General		
Maximum dead end corridor		20' (Sec. 1018.4)
		50' B occupancies with sprinklers (Sec. 1018.4, Exc.2)
		Length of dead end is less than 2.5 times the width (Sec. 1018.4, Exc. 3)
Air movement in corridors		Shall not be used as supply, return, exhaust, relief or ventilation
		See exceptions
		(Sec. 1018.5)

EXIT REQUIREMENTS (continued)

	Existing Conditions	Indiana 2014 Codes
General (continued)		
Corridor ceiling		Use as a plenum See conditions (Sec.1018.6)
Grade level stair discharge		Enclosed until exiting the building (Sec. 1027.1) 1. Maximum 50% of number & capacity of enclosed is permitted .1 Clear path to exterior from exit enclosure .2 Area if exit discharge is separated from area below with the same rating as the exit enclosure .3 Protected by sprinklers (Sec. 1027. Exc. 2) 2. Maximum 50% of number and capacity of the enclosures is permitted to egress through a vestibule. .1 Area of exit discharge is separated from area below with the same rating as the exit enclosure .2 Depth from exterior or building is not greater than 30 ft. .3 Area is separated from the remainder of level by at least wired glass and steel frame. (Sec. 1027, Exc.2) .4 The area is only used of exit discharge (Sec. 1024, Exc. 2)
Exit Signs		Required if rooms requires two (2) or more means of egress (Sec. 1011.1);
Emergency lighting		Required for not less than 90 minutes (Sec. 1011.6.3)

EXIT REQUIREMENTS (continued)

	Existing Conditions	Indiana 2014 Codes
Egress miscellaneous		
Delayed egress locks		Allowed for B occupancies following criteria of Sec. 1008.1.9.7,15 seconds maximum
Access-controlled egress doors		Allowed, where permitted in special occupancies, following criteria of Sec. 1008.1.9.8
Electromagnetically Locked egress doors		Allowed following criteria of Sec.1008.1.9.9
Security grilles		Permitted in B, F, M & S at the main exit. Operable from the inside without the use of key or special knowledge (Sec 1008.1.4.4)
Stair enclosure door locks		> 4 stories operable from both sides (Exception for stairway discharge doors) (Sec. 1008.1.9.11) Allowed to be locked if arranged to 403.12 (Sec. 1008.1.8.7 Exc. 3) other allowances, see Indiana Amendments
Ship's ladder		Not allowed (Sec. 1009.14) Note: Since mechanical space penthouses are considered unoccupied, the building code has no requirement. Review mechanical code access (IMC 306 and OSHA requirements)
Alternating tread stairs		Allowed in specific conditions (Sec. 1009.13)

EXIT REQUIREMENTS (continued)

	Existing Conditions	Indiana 2014 Codes
Egress miscellaneous		
Outdoor conditions		
Ramps		Designed not to accumulate water w/o ice and snow and to minimize the accumulation of ice and snow (Sec. 1010.8.2) (Exception: if facility has a removal maintenance program)
Stairs		Designed not to accumulate water w/o ice and snow and to minimize the accumulation of ice and snow (Sec. 1009.9.2) (Exception: if facility has a removal maintenance program)
Posting of occupant loads		Required in assembly occupancies (Sec.1004.3)
Storage under stairs		Allowed if following conditions are met: (IFC, Sec. 1009.9.3 and IBC, Sec. 1009.9.3) Walls and soffits enclosing usable space are protected by 1 hour fire fire resistive construction or the rating of the stair enclosure. Enclosed space is protected by an automatic sprinkler system as the rest of the building It is advised to install a rated door into the space Access to provided from outside the stair enclosure. (IFC Sec. 1009.9.3)
Vehicle ramps as exits access ramps		Not allowed unless pedestrian facilities are provided (Sec. 1021.4)

FIRE PROTECTION SYSTEMS

Existing Conditions	Indiana 2014 Codes
Automatic Sprinklers	Height & area (Sec. 509.4) Required for other code trade-offs (Sec. 903.2.1)
Standpipe and Hoses	Required Class I
	Every required stairway [Sec. 905.4(1)]
	Hose connection located at an intermediate landing between floors [Sec. 905.4(1)]
Standpipe and Hoses	Roof connection when slope is < 4 ½ or the highest landing with stair access to roof [Sec. 905.4(4)]
	Additional hose connections when the most remote portion of the building is more than 200 ft. from a hose connection [Sec. 905.4(6)]
Portable Fire Extinguishers	
Building	Required (IFC Sec. 9061 and IMCFC Sec. 591-401)
Maintenance	2-A, 10BC <75 travel distance
Elevator machine room	Required in elevator machine room (Elevator Code, Sec. 8.6.1.6.5)
Fire Alarm Systems	Required (Sec. 907.1)
Assembly < 300	Manual pull stations not required (Sec. 907.2.1)
B Occupancy	Manual pull stations required (Sec. 907.2.2) Manuals pulls not required if notification devices activate upon sprinkler water flow (Exc. 9072.2)

FIRE PROTECTION SYSTEMS (continued)

Existing Conditions	Indiana 2014 Codes
Fire Alarm Systems	
A,B,S Occupancies	Manual pull stations required (Sec.907.2.7) Manual pulls not required if notification devices activate upon sprinkler water flow (Exc. 907.2.7)
EMERGENCY OPERATIONS	
Voice/alarm communication	Not required
Smoke detection	In air handling units over 2000 CFM to shut down the air distribution system (IMC, Sec 606.2.1 IBC)
	Elevator automatic recall by smoke or heat detection in the hoistway and smoke detection in the elevator lobbies and machine room (ANSI/ASME A 17.1, Sec. 2.27.3.2.1 and NFPA 72 Sections 3-8. 14.3 & 3-8, 14.4)
Annunciation	
Sprinkler water flow	General building alarm
	Signal to a constantly attended location capable of contacting the local fire department
Trouble alarm	Constantly attended location; no fire department notification
Tamper switch	Constantly attended location; no fire department notification

EMERGENCY OPERATIONS (continued)

Smoke detectors	
Part of elevator operations	Recall elevators Constantly attended location; no fire department notification (NFPA 72, Sec. 3-18.14.15)
Part of HVAC units	Shut down HVAC units Constantly attended location; no fire department notification (Sec. 907.11)

ACCESSIBILITY

All public and common areas in building and site are required to be accessible (except mechanical rooms, janitor closets, tunnels, etc.)	ADA IBC, Chapter 11
Elevator	Required (Title III, Sec. 36.401, IBC Chapter 11, Sec. 4.1.3)
Unisex bathing and toilet rooms	Required in Group A and M occupancies (Sec. 2912.1)
Exits (accessible)	
Number required	1 required (Sec. 1007.1)
First floor	Required from each room/area equal to the number of exits required by the building code [ADA Sec.4.1.3. (9) IBC 1007.1]
Above and below the first floor	Same as number of exits Areas of Refuge Assistance [ADA Sec. 4.1.3(9)] (Exception: supervised automatic fire suppression system through-out (ADA Sec. 4.1.3 (9)]
Elevators	Not required for egress due to enclosed stairs and sprinklers (Sec. 1007.4, Exc. 2)

ACCESSIBILITY (continued)

Areas of refuge	No requirement at exit stairs (Sec. 1007.3, Exc. 4)
Horn/strobe units	Required – common use area including all classrooms restrooms, meeting rooms, etc.

I trust this communicates the Study Phase Code Review and other items discussed during this phase. If you have any further questions or comments, please do not hesitate to contact me.

We look forward to working with you during the next phase.

Very truly yours,

David Cook, RA

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EXISTING PLANS

Existing Plans

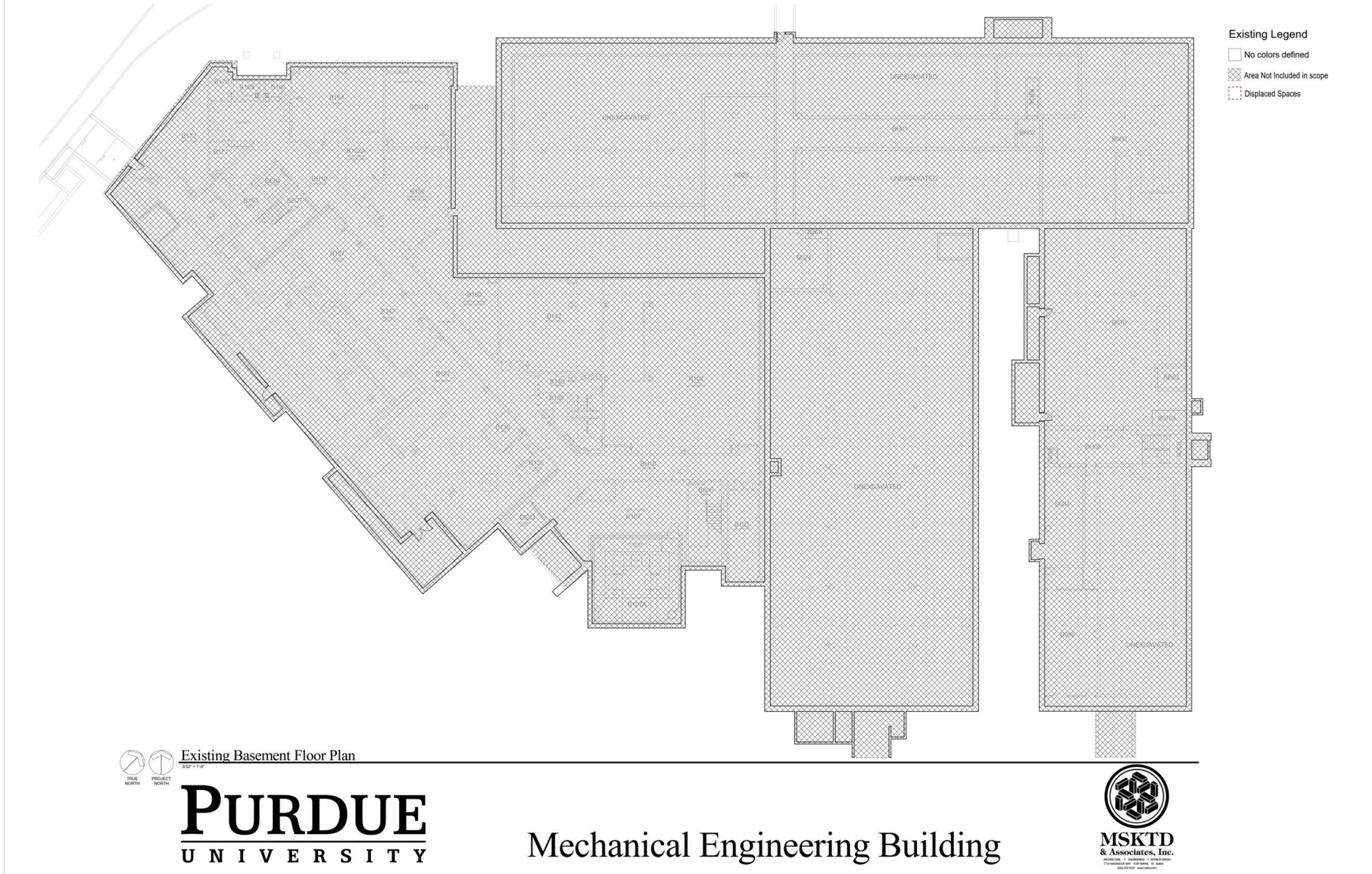
The study encompasses only specific areas within the Mechanical Engineering (ME) building, as indicated on the following color-coded floor plans of the existing building. Based on the study scope originally defined by Purdue University, study considerations exclude the recent Gatewood Addition and related spaces.

The total gross square footage of the renovated space is approximately 54,466 GSF, with the following breakdown by floor.

Basement .....0 GSF
Ground Floor ..... 14,829 GSF
First Floor .....11,513 GSF
Second Floor.....15,179 GSF
Third Floor .....12,797 GSF
Fourth Floor.....148 GSF

As a brief summary, we are showing approximately 24,330 NSF of proposed instructional lab space, including existing labs to remain; 26,080 NSF of proposed office space, including existing office spaces to remain; and 5,343 NSF of proposed research lab space. These NSF calculations do not include support, restrooms or circulation (existing and new).

Existing Plans – ME Basement Floor



Existing Plans – ME Ground Floor

Existing Ground Floor Plan

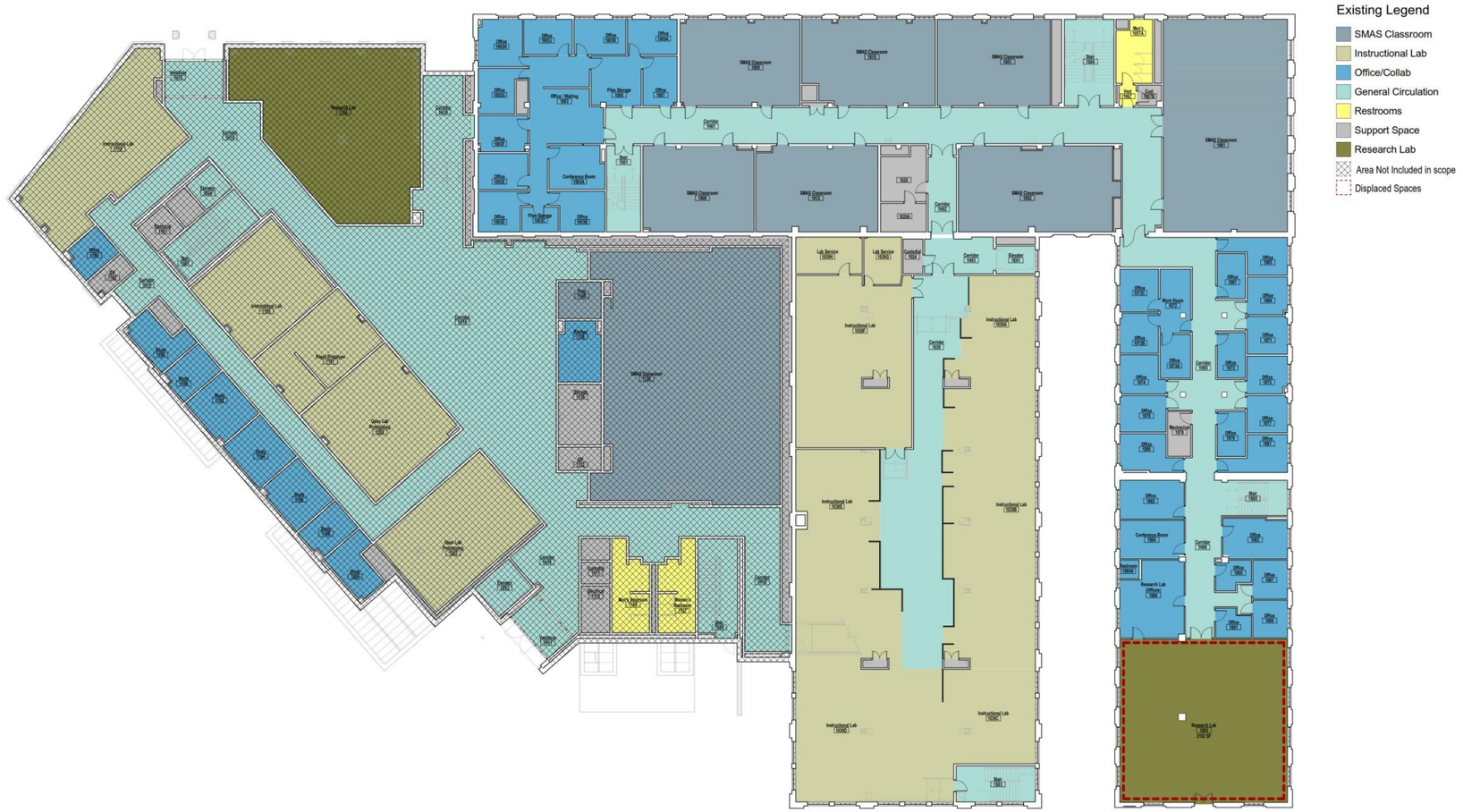
3/32" = 1'-0"

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Existing Plans – ME First Floor



Existing First Floor Plan

3/32" = 1'-0"

TRUE NORTH

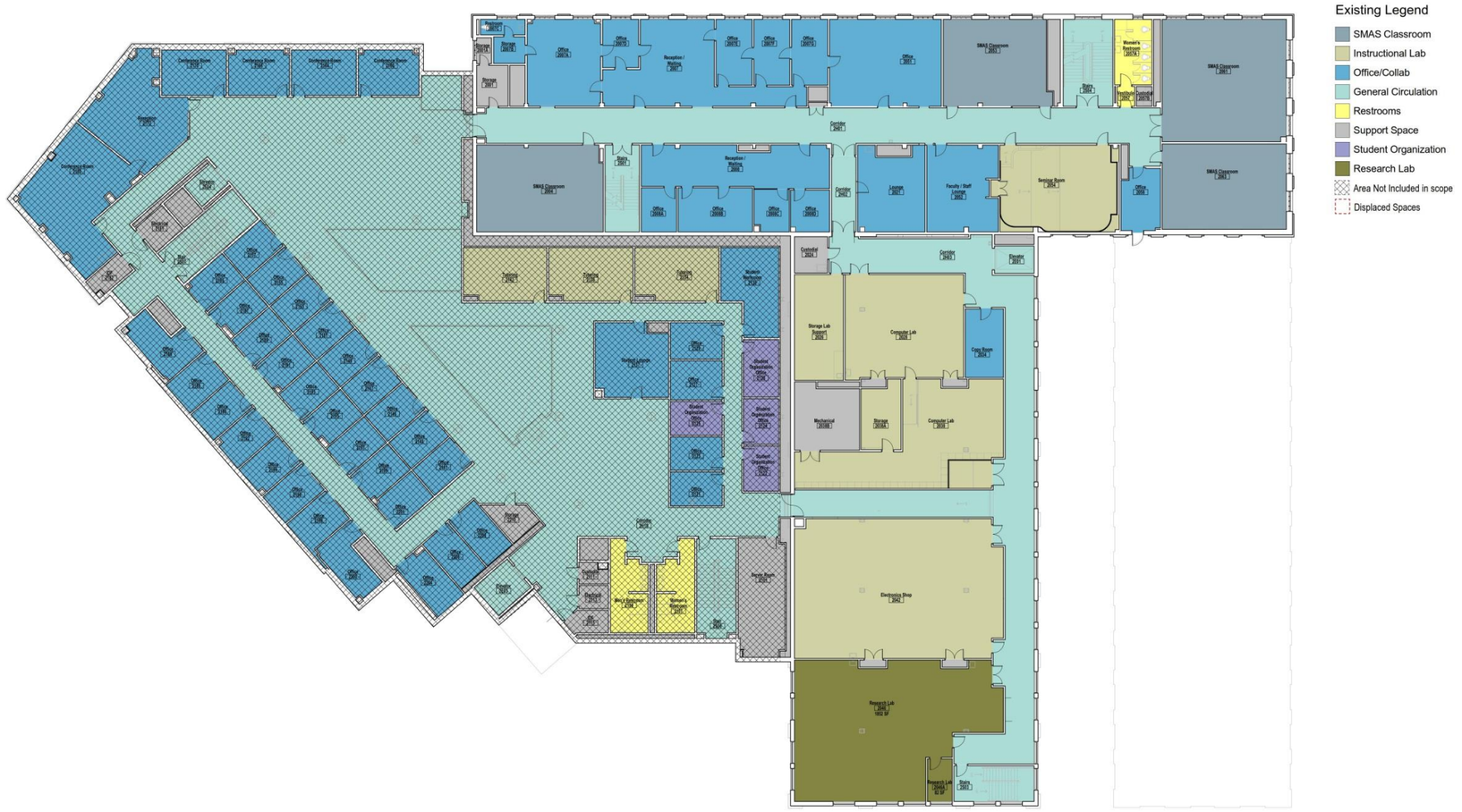
PROJECT NORTH

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Existing Plans – ME Second Floor



Existing Second Floor Plan

3/32" = 1'-0"

TRUE NORTH PROJECT NORTH

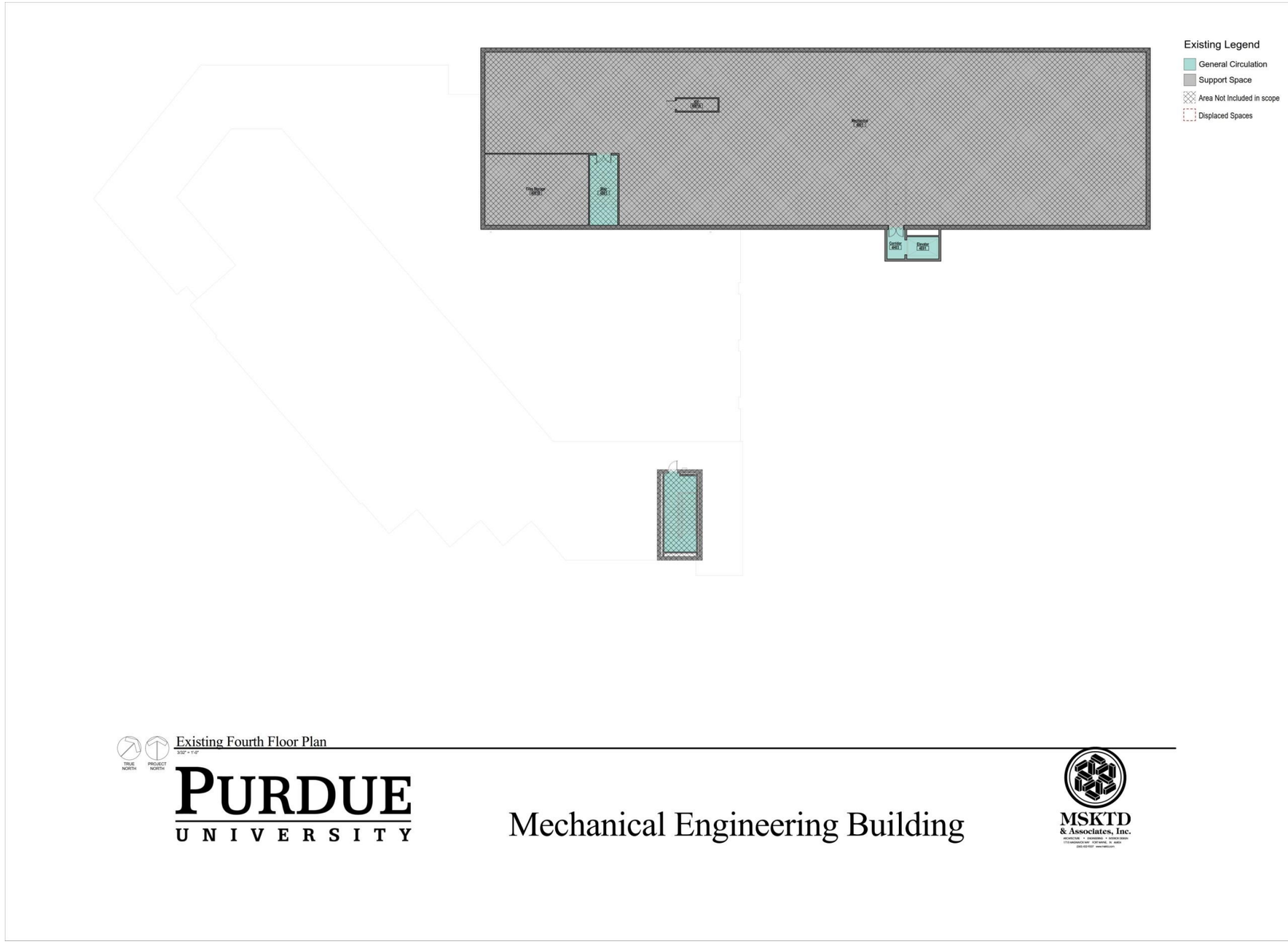
PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Existing Plans – ME Third Floor

Existing Plans – ME Fourth Floor



Existing Space Use

The following spreadsheets are similar to standard forms used by architects and engineers to quantify and describe space use in a typical project. These sheets identify current/existing room names, room numbers, area or assignable square footage of rooms, departmental assignment, quantities of faculty, staff or graduate assistants accommodated, and other comments regarding space usage and adjacencies.

The room data information originated from the Space Management department (SMAS) at Purdue University, and the following sheets contain modified information that was confirmed with the Mechanical Engineering department to validate current space usage and existing quantities. The following information on the existing spaces provides a snapshot in time to understand current faculty, staff and graduate student offices and open-office workstations accommodated in the current ME portion of the building. During our team meetings and discussions, it was observed that growth within the Mechanical Engineering department had already begun (as compared to the master plan document, which identified projected future growth). In essence, the goals identified as part of the master planning process have already been impacted by the initial departmental growth that is summarized in the following pages.

In order to develop the final program of spaces for the design team, quantities, spatial relationships and important adjacencies will need to be confirmed. At the next phase of the project, more detailed, individual room data sheets will provide additional space requirements and other necessary information. These room data sheets will also include specific requirements for instructional / teaching labs and reconfigured / relocated research labs. Sample room data summary sheets are included, but relevant lab information data sheets were not available during the review process for the study.

Purdue University ME/RAIL/HAMP Study

Mechanical Engineering (ME) and American Railway Building (RAIL)

EXISTING SQUARE FOOTAGE SUMMARY OF TEACHING AND RESEARCH LAB SPACES									
	Basement	**Ground Floor**	**First Floor**	**Second Floor**	**Third Floor**	**Fourth Floor**		**Current Quantities**	
TEACHING LABS	0	2,841	7,409	0	1,216	0		**11,466**	
RESEARCH LABS	0	11,483	2,409	1,989	0	0		**15,881**	

Purdue University ME/RAIL/HAMP Study

Mechanical Engineering (ME) and American Railway Building (RAIL)

EXISTING PROGRAM SUMMARY OF FACULTY, STAFF AND SPACE COUNTS

	Basement	Ground Floor	First Floor	Second Floor	Third Floor	Fourth Floor		Current Quantities	*GATE*	*TOTAL*	
TEACHING FACULTY											
Mechanical Engineering (ME)	N/A	9	13	2	22	0		**46**	31	77	
American Railway (RAIL)	N/A	N/A	N/A	N/A	N/A	N/A		**N/A**			
COLLEGE/DEPARTMENT STAFF											
Mechanical Engineering (ME)	N/A	6	15	25	3	0		**49**	10	59	
American Railway (RAIL)	N/A	N/A	N/A	N/A	N/A	N/A		**N/A**			
GRADUATE ASSISTANTS											
Mechanical Engineering (ME)	N/A	20	26	0	61	0		**107**	60	167	
American Railway (RAIL)	N/A	N/A	7	19	N/A	N/A		**26**	---	---	

PROPOSED SUMMARY OF FACULTY, STAFF AND SPACE COUNTS (Based on CoE Master Plan Document) - Mechanical Engineering

	Mechanical Engineering (ME)						
	Quantity	*Workstation Size (SF)*	*SF Info*				*Comments*
FACULTY	**74**		**9,000**				
Academic Department Head	1	180	180				
Academic Associate Head / Asst. Head	2	150	300				
Tenured/TT Faculty	66	120	7,920				
Instructor/Lecturer	4	120	480				
Emeritus/Emeritas	1	120	120				
OTHER *(Wkstns)*	**19**		**1,710**				
Adjuncts or Joint Appointments	10	60	600				
Visiting Scholars	1	630	630				
Research / Visiting Faculty	8	60	480				
STAFF *(Offices)*	**19**		**10,320**				
Professional Staff	12	120	1,440				
Business Manager	1	120	120				
ECN IT Staff	2	120	240				
Management	4	120	480				
STAFF *(Wkstns)*	**48**		**3,420**				
Post Doctoral Assistants	14	90	1,260				
Business Staff	7	90	630				
Admin/Operations Staff	6	90	540				
Clerical	11	90	990				
Service Staff	10	120	1,200				
GRADUATE ASSISTANTS	**169**		**6,760**				
Grad - Thesis	169	40	6,760				0.7 desk / grad
Subtotal	**329**		**31,210**				

Room Description / Name	Room Number	Area (SF)	Faculty / Stat	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
BASEMENT															
Mechanical	B024			Gen Building	Gen Building	MECHANICAL									Not included in study
Mechanical	B025			Gen Building	Gen Building	MECHANICAL							Circulating Water Tank		Not included in study
Mechanical / Storage	B060			Mech Engr	Mech Engr	CENT STORAGE							Air Compressors + Storage		Not included in study
Mechanical	B070			Mech Engr	Mech Engr	CENT STORAGE									Not included in study
Mechanical	B070			Gen Building	Gen Building	MECHANICAL							Ventilation Equip		Not included in study
Mechanical	B070A			Gen Building	Gen Building	MECHANICAL							Utility Room		Not included in study
Custodial	B082			Gen Building	Gen Building	BLDG SERV									Not included in study
Storage	B084		Building Storage - R. DuVall	Mech Engr	Mech Engr	CENT STORAGE									Not included in study
Storage	B086		Building Storage - R. DuVall	Mech Engr	Mech Engr	CENT STORAGE									Not included in study
Research Lab	B102		R. Lucht + G. King	Mech Engr	Mech Engr	NONCLAS LAB							Flame Diagnostics Lab		Not included in study
Research Lab Support	B103		Lab Support - DuVall	Mech Engr	Mech Engr	NONCLS SV							Lab Support		Not included in study
Electrical	B107			Gen Building	Gen Building	MECHANICAL							Switchgear Room		Not included in study
Electrical	B107A			Gen Building	Gen Building	MECHANICAL							Transformer Vault		Not included in study
Elevator Equipment	B135			Gen Building	Gen Building	MECHANICAL									Not included in study
Mechanical	B137			Gen Building	Gen Building	MECHANICAL									Not included in study
Men's Restroom	B138			Gen Building	Gen Building	REST ROOM							MEN'S RESTROOM-1WC,2U,3L-HCA		Not included in study
BDF	B139			Gen Building	Gen Building	MECHANICAL									Not included in study
Women's Restroom	B140			Gen Building	Gen Building	REST ROOM							WOMEN'S RESTROOM-3WC,3L-HCA		Not included in study
Research Lab	B142		P. Vlachos	Mech Engr	Mech Engr	NONCLAS LAB									Not included in study
Research Lab	B147		R. Kramer	Mech Engr	Mech Engr	NONCLAS LAB									Not included in study
Office	B150		E. Nauman; Orig: Computational Biomechanic Lab"	Mech Engr	Mech Engr	OFFICE			10				Graduate Assistants		Not included in study
Research Lab	B152		E. Nauman	Mech Engr	Mech Engr	NONCLAS LAB							Biomaterials and Biomechanics		Not included in study
Research Lab	B152A		E. Nauman	Mech Engr	Mech Engr	NONCLS SV							Biological Sample Prep		Not included in study
Research Lab	B152B		E. Nauman	Mech Engr	Mech Engr	NONCLAS LAB							Tissue Engineering Lab		Not included in study
Research Lab	B157		T. Siegmund	Mech Engr	Mech Engr	NONCLAS LAB							Applied Optics Lab		Not included in study
Elevator Equipment	B163			Gen Building	Gen Building	MECHANICAL									Not included in study
Research Lab	B164		B. Han	Mech Engr	Mech Engr	NONCLAS LAB							Bio-ME Lab		Not included in study
Unisex Restroom	B166			Gen Building	Gen Building	REST ROOM									Not included in study
Unisex Restroom	B168			Gen Building	Gen Building	REST ROOM									Not included in study
IDF	B170			Gen Building	Gen Building	MECHANICAL									Not included in study
Custodial	B171			Gen Building	Gen Building	BLDG SERV									Not included in study
Recycling Room	B172			Gen Building	Gen Building	BLDG SERV									Not included in study
Elevator	BE02			Gen Building	Gen Building	CIRCULATE									Not included in study
Elevator	BE03			Gen Building	Gen Building	CIRCULATE									Not included in study
Elevator	BE04			Gen Building	Gen Building	CIRCULATE									Not included in study

Room Description / Name	Room Number	Area (SF)	Faculty / Stat	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
BASEMENT															
Corridor	BH01			Gen Building	Gen Building	CIRCULATE									Not included in study
Corridor	BH02			Gen Building	Gen Building	CIRCULATE									Not included in study
Corridor	BH06			Gen Building	Gen Building	CIRCULATE									Not included in study
Corridor	BH10			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	BS02			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	BS04			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	BS05			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	BS06			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	BS07			Gen Building	Gen Building	CIRCULATE									Not included in study
Subtotals		**0**	**SF**				0	0	10	0	0	0			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Ground Floor															
Research Lab	G003	332	J. Panchal	Mech Engr	Mech Engr	NONCLAS LAB							Grad student offices only		
Research Lab	G004	588	X. Deng	Mech Engr	Mech Engr	NONCLAS LAB						588			
Research Lab / Office	G005	329	F. Sadeghi	Mech Engr	Mech Engr	NONCLAS LAB				12		329	Mixed use - Design Research Lab and Grad Assts Office	*G011*	
Research Lab Office	G005	402		Mech Engr	Mech Engr	R LAB OFFICE			14				Graduate Assistants		
SAE Projects	G006	247		Mech Engr	Mech Engr	NONCLAS SV						247	Storage		
Machine Shop	G010	695	M. Sherwood	Mech Engr	Mech Engr	CLASS LAB					695				
Teaching Lab / Manuf. Lab	G010	358		Mech Engr	Mech Engr	CLASS LAB					358		Teaching Lab Manuf. Facility - *New*	*G028*	
Machine Shop Storage	G010A	89		Mech Engr	Mech Engr	CLS LB SV					89				
Research Lab Office	G010A	46	sherwood	Mech Engr	Mech Engr	R LAB OFFICE		2					Currently unused; Machinists		
Research Lab	G011	968	F. Sadeghi	Mech Engr	Mech Engr	NONCLAS LAB						968	Design Research	*G005*	
Electrical	G013	6		Gen Building	Gen Building	MECHANICAL									
Research Lab	G015	247	T. Reid	Mech Engr	Mech Engr	NONCLAS LAB						247	Mixed use		
Electrical	G017	84		Gen Building	Gen Building	MECHANICAL									
Teaching Lab	G019	371		Mech Engr	Mech Engr	OPEN LAB					371		Internal Combustion Engine - Tear Down		
BDF	G020	101		Gen Building	Gen Building	MECHANICAL									
Machine Shop	G022	250		Mech Engr	Mech Engr	NONCLS SV							Welding?		
Machine Shop	G022	129		Mech Engr	Mech Engr	CLS LB SV									
Custodial	G025	114		Gen Building	Gen Building	BLDG SERV									
Elevator Equipment	G025A	96		Gen Building	Gen Building	MECHANICAL									
Electrical	G026	402		Gen Building	Gen Building	MECHANICAL									
Office	G027	99	Duvall OR S. Florence?	Mech Engr	Mech Engr	OFFICE		1					Staff		
Teaching Lab	G028	540	M. Sherwood	Mech Engr	Mech Engr	NONCLAS LAB					540		Nano/Microfabrication Lab	*G010*	
Student Shop and Project Lab	G029	1192		Mech Engr	Mech Engr	OPEN LAB							Project Machine Shop		
Research Lab	G032	316	Y. Shin	Mech Engr	Mech Engr	NONCLAS LAB						316	Manufacturing		
Research Lab	G032	211	Y. Shin	Mech Engr	Mech Engr	OPEN LAB						211	Manufacturing		
Research Lab	G036	525	Y. Shin	Mech Engr	Mech Engr	NONCLAS LAB						525	Manufacturing		
Research Lab / Student Machine Shop	G039	1778	A. Krichbaum + J. Wheele	Mech Engr	Mech Engr	OPEN LAB		1				1,778	Student Shop + Projects Lab		
Research Lab	G040	544	Y. Shin	Mech Engr	Mech Engr	NONCLAS LAB						544	Manufacturing		
Student Shop Storage	G042	221		Mech Engr	Mech Engr	OPEN SERV									
Receiving	G044	248	R. Duvall + A. Matlock	Gen Building	Gen Building	RECEIVING							Loading Dock		
Office	G044A	228	building deputy - r. duvall	Mech Engr	Mech Engr	OFFICE		2						*Shipping/Receiving*	
Shipping and Receiving	G044A	229		Mech Engr	Mech Engr	OFF SERV									
Storage	G044B	560	R. Duvall	Mech Engr	Mech Engr	OFF SERV							Shipping + Storage Fork Lift + Carts		
Office	G047	102	Krichbaum + Wheeler	Mech Engr	Mech Engr	OFFICE		1					Shop Office		
Storage	G048	55		Gen Building	Gen Building	BLDG SERV							Shipping + Receiving		
Mechanical	G048A	120	R. Duvall	Mech Engr	Mech Engr	NONCLS SV							Air Compressor		
Teaching Lab	G051	269		Mech Engr	Mech Engr	OPEN SERV					269		Internal Combustion Test Cell		
SAE Projects	G052	648	"Car Groups"	Mech Engr	Mech Engr	NONCLS SV							Student Vehicle Lab		
Control Room	G053	250		Mech Engr	Mech Engr	OPEN LAB					250				
Teaching Lab	G055	269		Mech Engr	Mech Engr	OPEN SERV					269		Internal Combustion Test Cell		
Research Lab	G056	534	Y. Shin	Mech Engr	Mech Engr	NONCLAS LAB						534	Manufacturing		
Vestibule	G057	19		Gen Building	Gen Building	REST ROOM							Men's Restroom		
Men's Restroom	G057A	177		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,2U,2L-HCA		
Research Lab	G058	349	B. Wu	Mech Engr	Mech Engr	NONCLAS LAB						349			
Custodial	G059	36		Gen Building	Gen Building	BLDG SERV									
Women's Restroom	G061	174		Gen Building	Gen Building	REST ROOM							Women's Restroom - 4WC, 3L-HCA		

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Ground Floor															
Mechanical	G061A	111		Gen Building	Gen Building	MECHANICAL									
Vestibule	G063	30		Mech Engr	Mech Engr	NONCLS SV							Research Lab Entrance		
Research Lab	G063A	331	S. Garimella	Mech Engr	Mech Engr	NONCLAS LAB						331	Heat Transfer Lab		
Research Lab	G063B	178	S. Garimella	Mech Engr	Mech Engr	NONCLAS LAB						178			
Research Lab	G063C	76		Mech Engr	Mech Engr	NONCLAS LAB						76			
Research Lab	G063D	363	S. Garimella	Mech Engr	Mech Engr	NONCLAS LAB						363	Heat Transfer Lab		
Research Lab	G068	372	F. Zhao	Mech Engr	Mech Engr	NONCLAS LAB						372	Heat Transfer Lab		
Research Lab	G069	363	S. Garimella	Mech Engr	Mech Engr	NONCLAS LAB						363	Thermal Microsystems		
Research Lab	G072	375	G. Subbarayan + M. Sher	Mech Engr	Mech Engr	NONCLAS LAB						375			
Research Lab	G075	350	A. Bae + S. Garimella	Mech Engr	Mech Engr	NONCLAS LAB						350	Applied Optics		
Research Lab	G076	370	G. Subbarayan	Mech Engr	Mech Engr	NONCLAS LAB						370			
Office	G077	213		Mech Engr	Mech Engr	OFFICE			6				Graduate Assts + Visitor		
Office	G082	191	Y. Shin	Mech Engr	Mech Engr	OFFICE	1								
Office	G084	127	P. Meckl	Mech Engr	Mech Engr	OFFICE	1								
Office	G086	146	Secretary - D. Akers	Mech Engr	Mech Engr	OFFICE		1							
Office	G087	185	G. King	Mech Engr	Mech Engr	OFFICE	1								
Office	G088	125	S. Zhang	Mech Engr	Mech Engr	OFFICE	1						Post Docs		
Office	G089	129	G. Shaver + Paniagua	Mech Engr	Mech Engr	OFFICE	2								
Office	G090	127	K. Ariyur	Mech Engr	Mech Engr	OFFICE	1						Post Docs		
Office	G091	131	H. Mongia	Mech Engr	Mech Engr	OFFICE	1								
Office	G092	137	S. Naik	Mech Engr	Mech Engr	OFFICE	1								
Research Lab	G093	2069	X. Xu + Vlachos	Mech Engr	Mech Engr	NONCLAS LAB						2,069	Heat Transfer		
Elevator	GE01	83		Gen Building	Gen Building	CIRCULATE									
Elevator	GE02	64		Gen Building	Gen Building	CIRCULATE									
Corridor	GH01	1442		Gen Building	Gen Building	CIRCULATE									
Corridor	GH02	206		Gen Building	Gen Building	CIRCULATE									
Corridor	GH03	1901		Gen Building	Gen Building	CIRCULATE									
Corridor	GH04	1117		Gen Building	Gen Building	CIRCULATE									
Corridor	GH05	708		Gen Building	Gen Building	CIRCULATE									
Corridor	GH06	287		Gen Building	Gen Building	CIRCULATE									
Elevator Lobby	GH07	62		Gen Building	Gen Building	CIRCULATE									
Stair	GS01	234		Gen Building	Gen Building	CIRCULATE									
Stair	GS02	34		Gen Building	Gen Building	CIRCULATE									
Stair	GS03	264		Gen Building	Gen Building	CIRCULATE									
Stair	GS04	333		Gen Building	Gen Building	CIRCULATE									
Stair	GS05	277		Gen Building	Gen Building	CIRCULATE									
Subtotals		**29058**	**SF**				9	8	20	12	2,841	11,483			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignmen	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor															
Office	1003	338	Receptionist/Empty	Mech Engr	Mech Engr	OFFICE		1					Reception		
Waiting	1003	199		Mech Engr	Mech Engr	OFF SERV							Internal Corridor + Waiting Area		
Conference Room	1003A	184		Mech Engr	Mech Engr	CONFERENCE				10			Conference Room		
Office	1003B	149	S. Banfield	PRF Devlpmnt	Mech Engr	OFFICE		1					Development Director	*Development Office Suite*	
Files Storage	1003C	67		Mech Engr	Mech Engr	OFF SERV							Development	*Development Office Suite*	
Office	1003D	139	T. Stoutenborough	PRF Devlpmnt	Mech Engr	OFFICE		1					Development Director	*Development Office Suite*	
Office	1003E	140	T. Finkbiner	Mech Engr	Mech Engr	OFFICE		1					Development	*Development Office Suite*	
Office	1003F	135	M. Anderson	Mech Engr	Mech Engr	OFFICE		1					Development	*Development Office Suite*	
Office	1003G	140	B. Shriner	Mech Engr	Mech Engr	OFFICE		1					Development	*Development Office Suite*	
Office	1003H	164	D. Anderson OR Subbara	Mech Engr	Mech Engr	OFFICE	1	0					Graduate Head	*Graduate Office Suite*	
Office	1003J	130	C. Glotzbach	Mech Engr	Mech Engr	OFFICE		1					Graduate	*Graduate Office Suite*	
Office	1003K	139	J. Moser	Mech Engr	Mech Engr	OFFICE		1					Graduate	*Graduate Office Suite*	
Office	1005	97	Grad Office Secretary	Mech Engr	Mech Engr	OFFICE		1					Graduate	*Graduate Office Suite*	
Files Storage	1005	98		Mech Engr	Mech Engr	OFF SERV							Graduate	*Graduate Office Suite*	
Office	1005A	132	C. Elwell	Mech Engr	Mech Engr	OFFICE		1					Graduate	*Graduate Office Suite*	
SMAS Classroom	1006	730		Gen Acad Cls	Gen Acad Cls	CLASSROOM				30			Tables + Chairs		
Office	1007	133	Secretary - K. Elwell	Mech Engr	Mech Engr	OFFICE		1					Graduate	*Graduate Office Suite*	
SMAS Classroom	1009	800		Gen Acad Cls	Gen Acad Cls	CLASSROOM				54			Tablet Arm Chairs		
SMAS Classroom	1012	794		Gen Acad Cls	Gen Acad Cls	CLASSROOM				54			Tablet Arm Chairs		
Electrical	1013	7		Gen Building	Gen Building	MECHANICAL									
SMAS Classroom	1015	856		Gen Acad Cls	Gen Acad Cls	CLASSROOM				44			Tables + Chairs		
Research Lab	1020	198	Logan	Mech Engr	Mech Engr	NONCLAS LAB						198	Research Lab		
Lab Service	1020A	100		Mech Engr	Mech Engr	NONCLS SV						100	Lab Service		
Custodial	1024	51		Gen Building	Gen Building	BLDG SERV									
Corridor	1030	2214		Mech Engr	Mech Engr	CLS LB SV							Internal Corridor + Computers		
Teaching Lab	1030A	1102	M. Logan	Mech Engr	Mech Engr	CLASS LAB					1,102		ME 365 Laboratory		*Existing raised floor*
Teaching Lab	1030B	1576	M. Logan	Mech Engr	Mech Engr	CLASS LAB					1,576		ME 315 Laboratory		*Existing raised floor*
Study	1030C	680	M. Logan	Mech Engr	Mech Engr	CLS LB SV					680		Study Tables		*Existing raised floor*
Teaching Lab	1030D	1432	M. Logan	Mech Engr	Mech Engr	CLASS LAB					1,432		ME 586/588 Laboratory		*Existing raised floor*
Teaching Lab	1030E	912	M. Logan	Mech Engr	Mech Engr	CLASS LAB					912		ME 475 Laboratory		*Existing raised floor*
Teaching Lab	1030F	1470	M. Logan	Mech Engr	Mech Engr	CLASS LAB					1,470		ME 309 Laboratory		
Lab Service	1030G	99	M. Logan	Mech Engr	Mech Engr	CLS LB SV					99		Department Firewalls		
Lab Service	1030H	138	M. Logan	Mech Engr	Mech Engr	CLS LB SV					138		ME 309 Laboratory piv		
Chase	1030J	16		Gen Building	Gen Building	MECHANICAL							pipe chase		*Existing raised floor (adjacent)*
Chase	1030K	16		Gen Building	Gen Building	MECHANICAL							pipe chase		*Existing raised floor (adjacent)*
Chase	1030L	16		Gen Building	Gen Building	MECHANICAL							pipe chase		*Existing raised floor (adjacent)*
Chase	1030M	16		Gen Building	Gen Building	MECHANICAL							pipe chase		*Existing raised floor (adjacent)*
SMAS Classroom	1051	756		Gen Acad Cls	Gen Acad Cls	CLASSROOM				50			Tablet Arm Chairs		
SMAS Classroom	1052	1010		Gen Acad Cls	Gen Acad Cls	CLASSROOM				66			Tablet Arm Chairs		
Vestibule	1057	26		Gen Building	Gen Building	REST ROOM							Men's Restroom Entrance		
Men's Restroom	1057A	194		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,3U,2L-HCA		
Custodial	1057B	26		Gen Building	Gen Building	BLDG SERV									
SMAS Classroom	1061	2007		Gen Acad Cls	Gen Acad Cls	CLASSROOM				174			Tablet Arm Chairs		
Office	1065	152	S. Fleeter	Mech Engr	Mech Engr	OFFICE	1								
Office	1067	99	Bilel?	Mech Engr	Mech Engr	OFFICE	1	0					Post Docs		

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							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor															
Office	1069	118	I. Bilionis	Mech Engr	Mech Engr	OFFICE	1								
Office	1071	118	J. Weibel + E. Bae	Mech Engr	Mech Engr	OFFICE	0	2							
Work Room	1072	152		Mech Engr	Mech Engr	OFF SERV									
Office	1072A	104	Secretary	Mech Engr	Mech Engr	OFFICE		1							
Office	1072B	119	P. Sojka	Mech Engr	Mech Engr	OFFICE	1								
Office	1072C	127	J. Abraham	Mech Engr	Mech Engr	OFFICE	1								
Office	1073	100		Mech Engr	Mech Engr	OFFICE			4				Graduate Assistants		
Office	1074	125	K. Zhao	Mech Engr	Mech Engr	OFFICE	1								
Office	1075	118		Mech Engr	Mech Engr	OFFICE	2						Flex Office - Graduate Assistant; Faculty; Visiting Professor		
Office	1076	131	B. Wu	Mech Engr	Mech Engr	OFFICE	1								
Office	1077	115	S. Wereley	Mech Engr	Mech Engr	OFFICE	1								
Mechanical	1078	80		Gen Building	Gen Building	MECHANICAL									
Office	1079	99		Mech Engr	Mech Engr	OFFICE			4				Graduate Assistants		
Office	1080	148		Mech Engr	Mech Engr	OFFICE			4				Graduate Assistants		
Office	1081	151	emeritus - R. Viskanta	Mech Engr	Mech Engr	OFFICE	1						Emeritus		
Office	1082	193		Mech Engr	Mech Engr	OFFICE			5				Graduate Assistants		
Office	1083	202		Mech Engr	Mech Engr	OFFICE			5				Graduate Assistants		
Conference Room	1084	192		Mech Engr	Mech Engr	CONFERENCE				10			Conference Room		
Unisex Restroom	1084A	28		Mech Engr	Mech Engr	CONF SERV							unisex restroom - 1wc,1l		
Office	1085	77		Mech Engr	Mech Engr	OFFICE			2				Graduate Assistants		
Office	1087	125	C. Savran	Mech Engr	Mech Engr	OFFICE	1								
Research Lab	1088	352	I. Mudawar	Mech Engr	Mech Engr	NONCLAS LAB							Heat Transfer; Grad stdnts offices	*1093*	
Office	1089	125	K. Ariyur OR E. Robinson	Mech Engr	Mech Engr	OFFICE		1							
Office	1091	75		Mech Engr	Mech Engr	OFFICE			2				Graduate Assistants		
Research Lab	1093	2111	I. Mudawar	Mech Engr	Mech Engr	NONCLAS LAB						2,111	Heat Transfer	*1088*	
Women's Restroom	1107			Gen Building	Gen Building	REST ROOM							WOMEN'S RESTROOM-3WC,3L-HCA		*Not included in study*
Men's Restroom	1109			Gen Building	Gen Building	REST ROOM							MEN'S RESTROOM-1WC,2U,3L-HCA		*Not included in study*
Custodial	1111			Gen Building	Gen Building	BLDG SERV									*Not included in study*
Electrical	1113			Gen Building	Gen Building	MECHANICAL									*Not included in study*
SMAS Classroom	1130			Gen Acad Cls	Gen Acad Cls	CLASSROOM				120			Strip Tables		*Not included in study*
IDF	1132			Gen Building	Gen Building	MECHANICAL									*Not included in study*
Storage	1136			Mech Engr	Mech Engr	CLS LB SV							Project Storage		*Not included in study*
Kitchenette	1138			Mech Engr	Mech Engr	OFF SERV							Kitchenette		*Not included in study*
Prep Room	1140			Mech Engr	Mech Engr	CLSRM SRV							Lecture Prep Room		*Not included in study*
Research Lab	1164		K. Ariyur + G. Chiu + D. C	Mech Engr	Mech Engr	NONCLAS LAB							SPIRA Lab for Electro-Mech Sy		*Not included in study*
Teaching Lab	1178			Mech Engr	Mech Engr	CLASS LAB							Conceptualization and Design		*Not included in study*
Office	1180		M. Sherwood	Mech Engr	Mech Engr	OFFICE							Lab Manager		*Not included in study*
Electrical	1181			Gen Building	Gen Building	MECHANICAL									*Not included in study*
IDF	1182			Gen Building	Gen Building	MECHANICAL									*Not included in study*
Teaching Lab	1185			Mech Engr	Mech Engr	CLASS LAB									*Not included in study*
Study	1186			Mech Engr	Mech Engr	STUDY							Breakout Room		*Not included in study*
Study	1188			Mech Engr	Mech Engr	STUDY							Breakout Room		*Not included in study*
Lab Service	1191		Rapid Prototype Fab Equ	Mech Engr	Mech Engr	CLS LB SV							Rapid Prototype Fab Equipment		*Not included in study*
Study	1192			Mech Engr	Mech Engr	STUDY							Breakout Room		*Not included in study*
Study	1194			Mech Engr	Mech Engr	STUDY							Breakout Room		*Not included in study*
Study	1196			Mech Engr	Mech Engr	STUDY							Breakout Room		*Not included in study*
Study	1198			Mech Engr	Mech Engr	STUDY							Breakout Room		*Not included in study*

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignmen	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor															
Study	1200			Mech Engr	Mech Engr	STUDY							Breakout Room		Not included in study
Open Lab	1202			Mech Engr	Mech Engr	OPEN LAB							Prototyping and Assembly		Not included in study
Open Lab	1203			Mech Engr	Mech Engr	OPEN LAB							Prototyping and Assembly		Not included in study
Elevator	1E01	83		Gen Building	Gen Building	CIRCULATE									
Elevator	1E03			Gen Building	Gen Building	CIRCULATE									Not included in study
Elevator	1E04			Gen Building	Gen Building	CIRCULATE									Not included in study
Lift	1E05			Gen Building	Gen Building	CIRCULATE									Not included in study
Corridor	1H01	1857		Gen Building	Gen Building	CIRCULATE									
Corridor	1H02	187		Gen Building	Gen Building	CIRCULATE									
Corridor	1H03	190		Gen Building	Gen Building	CIRCULATE									
Corridor	1H05	821		Gen Building	Gen Building	CIRCULATE									
Corridor	1H06	450		Gen Building	Gen Building	CIRCULATE									
Corridor	1H10			Gen Building	Gen Building	CIRCULATE									Not included in study
Vestibule	1H11			Gen Building	Gen Building	CIRCULATE									Not included in study
Vestibule	1H12			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	1S01	228		Gen Building	Gen Building	CIRCULATE									
Stair	1S03	231		Gen Building	Gen Building	CIRCULATE									
Stair	1S04	331		Gen Building	Gen Building	CIRCULATE									
Stair	1S05	161		Gen Building	Gen Building	CIRCULATE									
Stair	1S06			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	1S07			Gen Building	Gen Building	CIRCULATE									Not included in study
Subtotals		**29701**	**SF**				13	15	26	612	7,409	2,409			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor															
Storage	2001	127		Mech Engr	Mech Engr	OFF SERV							Records Storage		
Storage	2001A	35		Mech Engr	Mech Engr	OFF SERV									
SMAS Classroom	2004	831		Gen Acad Cls	Gen Acad Cls	CLASSROOM				48			Tablet Arm Chairs		
Office	2007	517		Mech Engr	Mech Engr	OFFICE		0					ME Main Office		
Waiting	2007	222		Mech Engr	Mech Engr	OFF SERV				4					
Office	2007A	462	A. Bajaj	Mech Engr	Mech Engr	OFFICE	1						Department Head Office	*ME Main Office Suite*	
Storage	2007B	69		Mech Engr	Mech Engr	OFF SERV								*ME Main Office Suite*	
Unisex Restroom	2007C	52		Mech Engr	Mech Engr	OFF SERV							unisex restroom - 1wc,1l		
Office	2007D	137	J. Baumgardt	Mech Engr	Mech Engr	OFFICE		1						*ME Main Office Suite*	
Office	2007E	181	C. Dalton + D. Akers	Engineering	Mech Engr	OFFICE		2					Divided space	*ME Main Office Suite*	
Office	2007F	178	J. Butler + S. Whaley	Engineering	Mech Engr	OFFICE		2					Business Office	*Business Office Suite*	
Office	2007G	181	C. Bluett OR C. Cornell?	Engineering	Mech Engr	OFFICE		1					Business Manager	*Business Office Suite*	
Office	2008	153	h. kendig	Mech Engr	Mech Engr	OFFICE		1					Undergrad Office	*Undergrad Office Suite*	
~~Conference Room~~	~~2008~~	~~214~~		~~Mech Engr~~	~~Mech Engr~~	~~CONFERENCE~~				~~0~~			~~Conference Room~~	~~*Undergrad Office Suite*~~	
Waiting	2008	243	H. Kendig	Mech Engr	Mech Engr	OFF SERV		1		6				*Undergrad Office Suite*	
Office	2008A	127	T. Brickler	Mech Engr	Mech Engr	OFFICE		1						*Undergrad Office Suite*	
Office	2008B	245	J. Jones	Mech Engr	Mech Engr	OFFICE	1						Assoc. Head for Undergrad. Affairs	*Undergrad Office Suite*	
Office	2008C	111	J. Haller	Mech Engr	Mech Engr	OFFICE		1						*Undergrad Office Suite*	
Office	2008D	113	K. Deckard	Mech Engr	Mech Engr	OFFICE		1						*Undergrad Office Suite*	
Electrical	2013	5		Gen Building	Gen Building	MECHANICAL									
Lounge	2021	444		Mech Engr	Mech Engr	DEPT COMMONS				12					
Custodial	2024	97		Gen Building	Gen Building	BLDG SERV									
Storage - Lab Support	2026	400	Storage - Logan	Mech Engr	Mech Engr	OFF SERV							Storage for electronic equipment		
Computer Lab	2028	918		Mech Engr	Mech Engr	OPEN LAB							Teaching Lab; MECL	*2038*	
Chase	2028A	16		Gen Building	Gen Building	MECHANICAL							Pipe Chase		
Chase	2028B	16		Gen Building	Gen Building	MECHANICAL							Pipe Chase		
Copy Room	2034	199		Mech Engr	Mech Engr	OFF SERV							Work room, supply, copy		
Computer Lab	2038	1108		Mech Engr	Mech Engr	OPEN LAB							D-CL Open Computer Lab	*2028*	
Storage	2038A	187	File Storage - C. Bluett	Mech Engr	Mech Engr	OFF SERV									
Mech HVAC	2038B	324		Mech Engr	Mech Engr	OPEN SERV									
Electronics Shop	2042	973		Mech Engr	Mech Engr	NONCLS SV							Instructional equipment for labs; services UG and G labs below. Equipment and Technicians		
Electronics Shop	2042	866		Mech Engr	Mech Engr	CLS LB SV									
Office	2042	325	T. Armstrong + M. Black +	Mech Engr	Mech Engr	OFFICE		6					Technicians		
Chase	2042A	16		Gen Building	Gen Building	MECHANICAL							pipe chase		
Chase	2042B	16		Gen Building	Gen Building	MECHANICAL							pipe chase		
Research Lab	2046	1905	J. Seipel + P. Meckl + R. (	Mech Engr	Mech Engr	NONCLAS LAB						1,905	Spiya lab of electro-mech sys		
Research Lab	2046A	84	Research Lab - Raman +	Mech Engr	Mech Engr	NONCLAS LAB						84			
Office	2051	743	S. Bordner + J. Clauson +	Mech Engr	Mech Engr	OFFICE		7					Business Office (Bullpen)	*Business Office Suite*	
Faculty/Staff Lounge	2052	456		Mech Engr	Mech Engr	DEPT COMMONS				12					
SMAS Classroom	2053	692		Gen Acad Cls	Gen Acad Cls	CLASSROOM				48			Tablet Arm Chairs		
Seminar Room	2054	711		Mech Engr	Mech Engr	CONFERENCE				37			Ford Seminar Room		
Vestibule	2057	26		Gen Building	Gen Building	REST ROOM							Women's Restroom Entrance		
Women's Restroom	2057A	194		Gen Building	Gen Building	REST ROOM							Women's Restroom - 5WC,3L-HCA		
Custodial	2057B	25		Gen Building	Gen Building	BLDG SERV									

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor															
Office / Files Storage	2058	182	J. Matthews	Mech Engr	Mech Engr	OFF SERV		1							
SMAS Classroom	2061	1170		Gen Acad Cls	Gen Acad Cls	CLASSROOM				78			Tablet Arm Chairs; R&R		
SMAS Classroom	2063	817		Gen Acad Cls	Gen Acad Cls	CLASSROOM				30			Tables + Chairs		
ECN Server Room	2101			Mech Engr	Mech Engr	CEN COMPUTER							Mixed Server Room, incl ECN, accss ctl and video headend		*Not included in study*
Women's Restroom	2107			Gen Building	Gen Building	REST ROOM							WOMEN'S RESTROOM-3WC,3L-HCA		*Not included in study*
Student Commons	2108			Mech Engr	Mech Engr	BLDG COMMONS									*Not included in study*
Men's Restroom	2109			Gen Building	Gen Building	REST ROOM									*Not included in study*
Custodial	2111			Gen Building	Gen Building	BLDG SERV									*Not included in study*
Electrical	2113			Gen Building	Gen Building	MECHANICAL									*Not included in study*
IDF	2115			Gen Building	Gen Building	MECHANICAL									*Not included in study*
Office	2121			Mech Engr	Mech Engr	OFFICE			3				Post Doc		*Not included in study*
Student Organization Office	2122			Mech Engr	Mech Engr	OFFICE							ASME		*Not included in study*
Office	2123			Mech Engr	Mech Engr	OFFICE			3				Post Doc		*Not included in study*
Student Organization Office	2124			Mech Engr	Mech Engr	OFFICE							Ambassadors		*Not included in study*
Student Organization Office	2125			Mech Engr	Mech Engr	OFFICE							Graduate Student Organization		*Not included in study*
Office	2127			Mech Engr	Mech Engr	OFFICE			4				Graduate Assistants		*Not included in study*
Student Organization Office	2128			Mech Engr	Mech Engr	OFFICE							Pi Tau Sigma		*Not included in study*
Office	2129			Mech Engr	Mech Engr	OFFICE			4				Graduate Assistants		*Not included in study*
Student Workroom	2130			Mech Engr	Mech Engr	OFF SERV							Project Workroom		*Not included in study*
Tutoring	2134			Mech Engr	Mech Engr	STUDY									*Not included in study*
Student Lounge	2137			Mech Engr	Mech Engr	BLDG COMMONS									*Not included in study*
Tutoring	2138			Mech Engr	Mech Engr	STUDY									*Not included in study*
Office	2141		S. Dyke	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Tutoring	2142			Mech Engr	Mech Engr	STUDY									*Not included in study*
Office	2143		J. Choi	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2145		J. Chen	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2147		R. Kramer	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2149		N. Key	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2151		A. Marconnet	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2153		X. Ruan	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2155		X. Xu	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Student Lounge	2156			Mech Engr	Mech Engr	BLDG COMMONS									*Not included in study*
Office	2157		B. Han	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Conference Room	2160			Mech Engr	Mech Engr	CONFERENCE							Conference Room		*Not included in study*
Conference Room	2164			Mech Engr	Mech Engr	CONFERENCE							Conference Room		*Not included in study*
Conference Room	2166			Mech Engr	Mech Engr	CONFERENCE							Conference Room		*Not included in study*
Conference Room	2170			Mech Engr	Mech Engr	CONFERENCE							Conference Room		*Not included in study*
Reception	2172		Clerical + Reception - E. I	Mech Engr	Mech Engr	OFFICE		2							*Not included in study*
Conference Room	2180			Mech Engr	Mech Engr	CONFERENCE							Conference Room		*Not included in study*
Electrical	2181			Gen Building	Gen Building	MECHANICAL									*Not included in study*
IDF	2182			Gen Building	Gen Building	MECHANICAL									*Not included in study*
Office	2185		L. Pan	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2186		T. Siegmund	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2187		CONSULTATION ROOM	Mech Engr	Mech Engr	INTERVIEW	1								*Not included in study*
Office	2188		J. Starkey	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2189		D. Cappelleri	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor															
Office	2190		X. Deng	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2191		P. Vlachos	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2192		J. Panchal	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2193		SHARED FACULTY	Mech Engr	Mech Engr	OFFICE	2						Shared Faculty		*Not included in study*
Office	2194		F. Zhao	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2195		CONSULTATION ROOM	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2196		M. Koslowski	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2197		T. Fisher	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2198		T. Reid	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2199		A. Vacca	Mech Engr	Mech Engr	OFFICE	2								*Not included in study*
Office	2200		G. Subbarayan	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2201		J. Nolfi	Mech Engr	Mech Engr	OFFICE		1							*Not included in study*
Office	2204		B. Lucht	Mech Engr	Mech Engr	OFFICE	1								*Not included in study*
Office	2206			Mech Engr	Mech Engr	OFFICE			3				Post Doc		*Not included in study*
Office	2208			Mech Engr	Mech Engr	OFFICE			3				Post Doc		*Not included in study*
Storage	2210		Storage - DuVall	Mech Engr	Mech Engr	OFF SERV									*Not included in study*
Elevator	2E01	83		Gen Building	Gen Building	CIRCULATE									
Elevator	2E03			Gen Building	Gen Building	CIRCULATE									*Not included in study*
Elevator	2E04			Gen Building	Gen Building	CIRCULATE									*Not included in study*
Corridor	2H01	1940		Gen Building	Gen Building	CIRCULATE									
Corridor	2H02	161		Gen Building	Gen Building	CIRCULATE									
Corridor	2H03	2293		Gen Building	Gen Building	CIRCULATE									
Corridor	2H10			Gen Building	Gen Building	CIRCULATE									*Not included in study*
Stair	2S01	229		Gen Building	Gen Building	CIRCULATE									
Stair	2S03	220		Gen Building	Gen Building	CIRCULATE									
Stair	2S04	335		Gen Building	Gen Building	CIRCULATE									
Stair	2S06			Gen Building	Gen Building	CIRCULATE									*Not included in study*
Stair	2S07			Gen Building	Gen Building	CIRCULATE									*Not included in study*
Subtotals		**22374**	**SF**				30	28	20	275	0	1,989			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignmen	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Third Floor															
Reception	3003	515	Receptionist - M. Morrisor	Mech Engr	Mech Engr	OFFICE		1					Faculty office suite, incl reception		
Waiting	3003	200		Mech Engr	Mech Engr	OFF SERV							Internal Corridor + Waiting Area		
Office	3003A	128	N. Jain	Mech Engr	Mech Engr	OFFICE	1								
Office	3003B	116	D. Atkinson	Mech Engr	Mech Engr	OFFICE		1							
Office	3003C	158	F. Sadeghi	Mech Engr	Mech Engr	OFFICE	1								
Office	3003D	136	S. Son	Mech Engr	Mech Engr	OFFICE	1								
Office	3003E	135	E. Groll	Mech Engr	Mech Engr	OFFICE	1								
Office	3003F	113	G. Pennock	Mech Engr	Mech Engr	OFFICE	1								
Office	3003G	153	Y. Chen	Mech Engr	Mech Engr	OFFICE	1								
Office	3003H	115	J. Braun	Mech Engr	Mech Engr	OFFICE	1								
Office	3003J	117	Wassgren	Mech Engr	Mech Engr	OFFICE	1								
Office	3005	221		Mech Engr	Mech Engr	OFFICE			7				Graduate Assistants		
SMAS Classrooom	3006	720		Gen Acad Cls	Gen Acad Cls	CLASSROOM				40			Tables + Chairs		
Conference Room	3007	235		Mech Engr	Mech Engr	CONFERENCE				8			Conference Room		
Office	3009	217		Mech Engr	Mech Engr	OFFICE			6				Graduate Assistants		
Office	3011	247		Mech Engr	Mech Engr	OFFICE			7				Graduate Assistants		
Office	3013	239		Mech Engr	Mech Engr	OFFICE			7				Graduate Assistants		
Office	3014	216		Mech Engr	Mech Engr	OFFICE			5				Graduate Assistants		
Design Resource Room	3015	204		Mech Engr	Mech Engr	NONCLS SV							TA space, incl office hours		
Office	3016	199		Mech Engr	Mech Engr	OFFICE			6				Post Docs or Graduate Assistants?		
Electrical	3017	6		Gen Building	Gen Building	MECHANICAL									
Office	3018	870		Mech Engr	Mech Engr	OFFICE			16				Graduate Assistants		
Office	3019	243		Mech Engr	Mech Engr	OFFICE			7				Graduate Assistants		
Computer Lab (ME)	3021	1216		Mech Engr	Mech Engr	CLASS LAB				40	1,216		D-CL PC Lab; ME Computer Lab		
Vestibule	3057	27		Gen Building	Gen Building	REST ROOM							men's restroom entrance		
Men's Restroom	3057A	190		Gen Building	Gen Building	REST ROOM							men's restroom - 3wc,3u,2l-hca		
Custodial	3057B	25		Gen Building	Gen Building	BLDG SERV									
Reception	3061	549	receptionist - D. Arihood	Mech Engr	Mech Engr	OFFICE		1					Faculty office suite, incl reception; pods working weel for faculty interaction and community		
Corridor	3061	183		Mech Engr	Mech Engr	OFF SERV									
Office	3061A	120	E. Nauman	Mech Engr	Mech Engr	OFFICE	1								
Conference Room	3061B	241		Mech Engr	Mech Engr	CONFERENCE				12			Conference Room		
Office	3061C	147	G. Chiu	Mech Engr	Mech Engr	OFFICE	1								
Office	3061D	148	J. Seipel	Mech Engr	Mech Engr	OFFICE	1								
Office	3061E	143	A. Raman	Mech Engr	Mech Engr	OFFICE	1								
Office	3061F	143	K. Kokini	Mech Engr	Mech Engr	OFFICE	1								
Office	3061G	173	C. Krousgrill	Mech Engr	Mech Engr	OFFICE	1								
Office	3061H	147	J. Chagdes	Mech Engr	Mech Engr	OFFICE	1								
Office	3061J	120	P. Davies + E. Berger	Mech Engr	Mech Engr	OFFICE	2								
Office	3061K	119	S. Bolton	Mech Engr	Mech Engr	OFFICE	1								
Office	3061L	119	B. Yao	Mech Engr	Mech Engr	OFFICE	1								
Office	3061M	117	M. Gonzalez	Mech Engr	Mech Engr	OFFICE	1								
Office	3061N	122	J. Rhoads	Mech Engr	Mech Engr	OFFICE	1								
Office	3061P	134	K. Li	Mech Engr	Mech Engr	OFFICE	1								
Mechanical	3101	52		Mech Engr	Mech Engr	NONCLS SV							Pre-Action System Room		
Research Lab	3103		S. Zhang - Lab?	Mech Engr	Mech Engr										*Not included in study*

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignmen	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Third Floor															
Women's Restroom	3107			Gen Building	Gen Building	REST ROOM							WOMEN'S RESTROOM-3WC,3L-HCA		Not included in study
Men's Restroom	3109			Gen Building	Gen Building	REST ROOM							MEN'S RESTROOM-1WC,2U,3L-HCA		Not included in study
Custodial	3111			Gen Building	Gen Building	BLDG SERV									Not included in study
Electrical	3113			Gen Building	Gen Building	MECHANICAL									Not included in study
IDF	3115			Gen Building	Gen Building	MECHANICAL									Not included in study
Office	3139		MULTISCALE SIMULATI	Mech Engr	Mech Engr	OFFICE			20				Graduate Assistants		Not included in study
Research Lab	3141		A. Ardekani + M. Gonzale	Mech Engr	Mech Engr	NONCLAS LAB									Not included in study
Research Lab	3151		Research Lab - J. Choi +	Mech Engr	Mech Engr	NONCLAS LAB									Not included in study
Research Lab	3164		Research Lab - K. Ramar	Mech Engr	Mech Engr	NONCLAS LAB									Not included in study
Office	3166		J. Gore	Mech Engr	Mech Engr	OFFICE	1								Not included in study
Office	3168		R. Cipra	Mech Engr	Mech Engr	OFFICE	1								Not included in study
Office	3170		I. Mudawar	Mech Engr	Mech Engr	OFFICE	1								Not included in study
Office	3171			Mech Engr	Mech Engr	OFFICE			4				Graduate Assistants		Not included in study
Office	3172			Mech Engr	Mech Engr	OFFICE	1								Not included in study
Office	3174			Mech Engr	Mech Engr	OFFICE	1						Currently unused		Not included in study
Office	3176		K. Ramani	Mech Engr	Mech Engr	OFFICE	1								Not included in study
Office	3179			Mech Engr	Mech Engr	OFFICE			6				Graduate Assistants		Not included in study
Office	3180		S. Garimella	Mech Engr	Mech Engr	OFFICE	1								Not included in study
Electrical	3181			Gen Building	Gen Building	MECHANICAL									Not included in study
IDF	3182			Gen Building	Gen Building	MECHANICAL									Not included in study
Elevator	3E01	83		Gen Building	Gen Building	CIRCULATE									
Elevator	3E03			Gen Building	Gen Building	CIRCULATE									Not included in study
Elevator	3E04			Gen Building	Gen Building	CIRCULATE									Not included in study
Corridor	3H01	1894		Gen Building	Gen Building	CIRCULATE									
Elevator Lobby	3H03	63		Gen Building	Gen Building	CIRCULATE									
Corridor	3H10			Gen Building	Gen Building	CIRCULATE									Not included in study
Stair	3S01	229		Gen Building	Gen Building	CIRCULATE									
Stair	3S04	335		Gen Building	Gen Building	CIRCULATE									
Stair	3S06			Gen Building	Gen Building	CIRCULATE									
Stair	3S07			Gen Building	Gen Building	CIRCULATE									
Subtotals		**12342**	**SF**				29	3	91	100	1,216	0			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignmen	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Fourth Floor															
Mechanical	4001	12419		Gen Building	Gen Building	MECHANICAL									
IDF	4001A	60		Gen Building	Gen Building	MECHANICAL									
Files Storage	4001B	726		Mech Engr	Mech Engr	OFF SERV									
Elevator	4E01	83		Gen Building	Gen Building	CIRCULATE									
Elevator Lobby	4H03	162		Gen Building	Gen Building	CIRCULATE									
Stair	4S01	270		Gen Building	Gen Building	CIRCULATE									
Subtotals		**13,720**	**SF**				0	0	0	0	0	0			

Sample Room Data Sheets

Room data sheets were provided to each department to summarize important space requirements and adjacencies. At the conclusion of the study, the room data sheets were not complete, but a sample sheet is included for reference.

PURDUE UNIVERSITY

(TL) Room Data Sheets

PURDUE UNIVERSITY - ME/RAIL/HAMP Study - ME | Project Name | Date Last Revised | **1,500** Assignable SF | Room Number | MSKTD Proj

Mechanical Engineering | Department/Group Name | **D-CL OC/ME444 CAD & Prototyping** Room Name | Contact, Phone

Room Function and Type (Laboratory/Lab Support (Dedicated)/Biosafety Level/Clean Room Classification): ME3021, sim. To ME1178

Room Adjacencies - *What other rooms/functions should this room be directly adjacent to, if any? Why? (Includes exterior adjacencies or direct access to exterior.)*

Number of Occupants: | Special Ceiling Height (if any): | Fire Rating (if known) | Special/Hazardous Room Conditions (if any):

Finishes and Materials

Floor/Base:
- ◯ Vinyl Composition Tile/4" Vinyl Base
- ◯ Seamless Sheet Vinyl/Integral Coved Vinyl Base
- ◯ Epoxy/Integral Coved Epoxy Base
- ◯ Carpet/4" Vinyl Base
- ◯ Sealed Concrete/4" Vinyl Base
- ◯ Other :

Ceiling:
- ◯ Standard Lay-In Tile ◯ Mylar Faced Tile
- ◯ Gypsum Board/Painted ◯ Exposed/Non-Painted
- ◯ Exposed/Painted ◯ Other :

Doors
- ◯ Hollow Metal ◯ Wood ◯ Vision Panel ◯ Light Seals ◯ Acoustical Seals ◯ Other:

Security/Access Control:
- ◯ Key Lock ◯ Alarmed Door ◯ Card Access ◯ Motion Sensors ◯ Closed Circuit TV ◯ Other:

Walls:
- ◯ Gypsum Wall Board
- ◯ Cementitous Wall Board
- ◯ Concrete Masonry Units
- ◯ Other :

Wall Finishes:
- ◯ Latex Enamel Paint
- ◯ Epoxy Paint
- ◯ Vinyl Wallcovering
- ◯ Seamless Sheet Vinyl
- ◯ Other :

Wall Cabinets/Shelving:
- ◯ Open ◯ Swing Doors
- ◯ Sliding Doors ◯ Adjustable

Lab Casework:
- ◯ Fixed ◯ Portable
- ◯ Metal ◯ Wood
- ◯ Other :

Lab Benchtop:
- ◯ Chemical-resistant Plastic Laminate
- ◯ Epoxy Resin ◯ Stainless Steel
- ◯ Other :

Structural Considerations:
- ◯ Special Floor Loading Requirements: ____ PSF ◯ Recessed Floor Area Requirements ◯ Ceiling Mounted Equipment

Environmental Considerations:
- ◯ Significant Noise/Vibrational Generator ◯ Visual Privacy Required ◯ Natural Light Desirable ◯ Odor Producing Operations
- ◯ Significant Noise/Vibrational Sensitivity ◯ Views Desirable ◯ Avoid Natural Light ◯ Waste Production : ◯ Low ◯ High

Mechanical, Plumbing and Electrical Considerations:

HVAC Considerations (Brief description of any special HVAC considerations for the space, if any, including temperature/relative humidity, air flow/changes, air quality/exhaust.)

Plumbing Considerations (Brief description of any special plumbing considerations for the space, if any, including fixtures/features, services, waste.)

Electrical Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)

Communications Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)
- ◯ Telephone Connections ◯ Data Connections ◯ Network Service Type: ____ ◯ Clocks ◯ Intercom System ◯ Public Address System

Signatures

User Representative | Date

PU Representative | Date

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PURDUE UNIVERSITY

(RSCH) Room Data Sheets

PURDUE UNIVERSITY - ME/RAIL/HAMP Study - ME | Project Name | Date Last Revised | Assignable SF | Room Number | MSKTD Proj

Mechanical Engineering | Department/Group Name | Room Name | Contact, Phone

Room Function and Type (Laboratory/Lab Support (Dedicated)/Biosafety Level/Clean Room Classification):

Room Adjacencies - *What other rooms/functions should this room be directly adjacent to, if any? Why? (Includes exterior adjacencies or direct access to exterior.)*

Number of Occupants: | Special Ceiling Height (if any): | Fire Rating (if known) | Special/Hazardous Room Conditions (if any):

Finishes and Materials

Floor/Base:
- ◯ Vinyl Composition Tile/4" Vinyl Base
- ◯ Seamless Sheet Vinyl/Integral Coved Vinyl Base
- ◯ Epoxy/Integral Coved Epoxy Base
- ◯ Carpet/4" Vinyl Base
- ◯ Sealed Concrete/4" Vinyl Base
- ◯ Other :

Ceiling:
- ◯ Standard Lay-In Tile ◯ Mylar Faced Tile
- ◯ Gypsum Board/Painted ◯ Exposed/Non-Painted
- ◯ Exposed/Painted ◯ Other :

Doors
- ◯ Hollow Metal ◯ Wood ◯ Vision Panel ◯ Light Seals ◯ Acoustical Seals ◯ Other:

Security/Access Control:
- ◯ Key Lock ◯ Alarmed Door ◯ Card Access ◯ Motion Sensors ◯ Closed Circuit TV ◯ Other:

Walls:
- ◯ Gypsum Wall Board
- ◯ Cementitous Wall Board
- ◯ Concrete Masonry Units
- ◯ Other :

Wall Finishes:
- ◯ Latex Enamel Paint
- ◯ Epoxy Paint
- ◯ Vinyl Wallcovering
- ◯ Seamless Sheet Vinyl
- ◯ Other :

Wall Cabinets/Shelving:
- ◯ Open ◯ Swing Doors
- ◯ Sliding Doors ◯ Adjustable

Lab Casework:
- ◯ Fixed ◯ Portable
- ◯ Metal ◯ Wood
- ◯ Other :

Lab Benchtop:
- ◯ Chemical-resistant Plastic Laminate
- ◯ Epoxy Resin ◯ Stainless Steel
- ◯ Other :

Structural Considerations:
- ◯ Special Floor Loading Requirements: ____ PSF ◯ Recessed Floor Area Requirements ◯ Ceiling Mounted Equipment

Environmental Considerations:
- ◯ Significant Noise/Vibrational Generator ◯ Visual Privacy Required ◯ Natural Light Desirable ◯ Odor Producing Operations
- ◯ Significant Noise/Vibrational Sensitivity ◯ Views Desirable ◯ Avoid Natural Light ◯ Waste Production : ◯ Low ◯ High

Mechanical, Plumbing and Electrical Considerations:

HVAC Considerations (Brief description of any special HVAC considerations for the space, if any, including temperature/relative humidity, air flow/changes, air quality/exhaust.)

Plumbing Considerations (Brief description of any special plumbing considerations for the space, if any, including fixtures/features, services, waste.)

Electrical Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)

Communications Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)
- ◯ Telephone Connections ◯ Data Connections ◯ Network Service Type: ____ ◯ Clocks ◯ Intercom System ◯ Public Address System

Signatures

User Representative | Date

PU Representative | Date

MSKTD Associates, Inc. RSCH-1 Research Lab Page 17

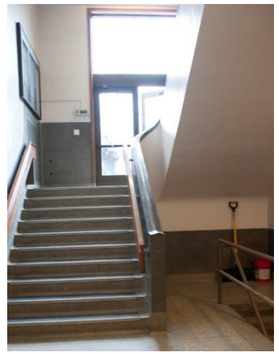
SYSTEMS NARRATIVES
(Existing and Proposed)

Architectural Narrative

The ME building, including the recent Gatewood Addition, was built in multiple phases. The original building was built in 1929 (central wing) with separate projects/additions in 1932 (north wing), 1942 (east wing, south) and 1948 (east wing, north). The Gatewood Addition was built in 2012. The original 1920s, 1930s and 1940s portions of the ME building are in need of renovation to satisfy educational and organizational needs. These proposed renovations will also accommodate the core academic functions initially identified in the College of Engineering master plan, as well as recommended upgrades to the mechanical, electrical, plumbing and telecommunications systems, as identified within this study. As indicated, the Gatewood Addition is not included in the study and related recommendations.

ME Building, NE corner (north wing)

One of the early considerations and discussions for ME was related to accessibility and circulation, including circulation in and out of the building, horizontal circulation within the building, and vertical circulation within the building. The most recent Gatewood Addition provides accessible entrances in various locations, but the original ME areas of the overall building include entrances at stair towers that are at intermediate floor levels. Excluding any vertical transition at the exterior of the building, people must navigate up or down stairs to reach main floor levels upon entry at these stair tower locations, so accessibility is limited at these entrances.



ME Building, Interior Stair

ME Building, North Elevation

ME Building, East Elevation

Horizontal circulation within the overall building is also limited, and continuity and direct access between floors of the original ME areas of the building and the Gatewood Addition are only available at the ground floor and the second floor levels. The ground floor of the original ME building is lower than Gatewood, but existing ramping and a wheelchair lift are provided to accommodate better accessibility. The second floor also provides direct access between the two major portions of the building. On other public floors in ME (i.e., first and third), no access is provided between ME and the Gatewood Addition.

Vertical circulation is provided within ME and the Gatewood Addition through existing elevators. In ME, there is a single elevator that serves the ground, first, second, third and mechanical floors of ME. The newer Gatewood Addition has additional elevators that supplement the capacity and enhance the vertical circulation within the overall building, but as noted above, horizontal circulation between these areas of the building is limited to the ground and second floors.

In addition to circulation, building services and other support spaces within the building were considered as part of the study for proposed renovations. In the existing building, restrooms are located in consistent locations from floor to floor, but each floor typically only serves either men or women. As part of the proposed renovation, the proposed floor plans include considerations to provide facilities for men and women on each floor and in consistent locations from floor to floor.

The existing interior spaces of ME appear generally to be in good shape, but the finishes and layout are dated due to the age of the facilities. Materials in many areas of the building may contain asbestos or other related materials. Purdue University should provide consideration for investigation and abatement processes as these are not included as part of the study scope and documentation.

As identified in the plan diagrams (existing and proposed), the project scope for possible renovations within the ME building includes only part of the square footage within the overall building, and that creates unique challenges in terms of space utilization, efficiencies, circulation, and mechanical, electrical, plumbing and telecommunications (MEP/T) systems distribution and renovations. Refer to the MEP/T systems narratives, as well as the preliminary code study documentation, for additional information.

Also, the Mechanical Engineering department modified the program requirements from the original master plan quantities. In order to accommodate flexibility in programming of the existing building and space constraints, following is a list of revised program requirements provided by the Mechanical Engineering department, specifically for instructional labs.

Instructional Lab Summary

Computation + Build Space for 263, 444, and 497 FEA	1,200+300 =	1,500	SF
Instruction + Build Space for 297 How Stuff Works, 323, and 352	1,800+300=	2,100	SF
Instructional Mfg Labs Expansion (*additional space requirements*)		1,000	SF
General Prototyping Space		1,000	SF
Machine Shop Expansion (*additional space requirements*)		500	SF
E-Shop Expansion (*additional space requirements*)		500	SF
Computer Lab		1,200	SF
Tutorial Room Expansion (*additional space requirements*)		800	SF

Other notes:

- ME 463 would continue to use the PEARL rooms in Gatewood, with consideration of alternate uses when logistically possible.
- ME 557 would continue to use ME 1178 (also in Gatewood)

As the available existing space within the building cannot accommodate all of the necessary program functions and scope for the ME building, the focus of the study was on core academic functions, including classrooms, instruction / teaching labs, and offices and open-office workstations for faculty, staff, and graduate students. As indicated in earlier sections of the study, other functions that do not support the academic core mission will be relocated to other parts of campus. These include existing research lab spaces and support functions. At the conclusion of the study meetings and discussions, the Mechanical Engineering department was reviewing possible research lab functions that would remain within the building, based on the square footage available and identified on the proposed floor plans. No final conclusion was developed or proposed to incorporate into the study.

Various design solutions and considerations were discussed during team meetings, with an emphasis placed on efficiency, space adjacencies, and overall building organization and circulation. Proposed plan drawings are included on the following pages showing the proposed final layout. The basement was not included in the original project scope, but to accommodate electrical service recommendations and other system(s) modifications, renovation within the basement will be necessary to accommodate systems upgrades and distribution. These are also identified and outlined in the MEP/T systems narratives that follow.

GENERAL

As discussed in other sections of the study, a goal of the ME proposed plans is to provide better and consistent facilities for the building and occupants. As an example, upgraded restroom facilities are proposed to accommodate revised fixture layouts that will improve accessibility. Additional restroom facilities are also proposed on each floor and in consistent locations on each floor to serve both men and women. Other general layout considerations include circulation modifications of existing corridors to provide direct access and better flow, as well as flexibility for larger program spaces and functions. Other and more specific modifications are outlined below for each floor.

GROUND FLOOR

As discussed, the ground floor of ME has direct circulation connections to the Gatewood Addition. At the south part of the central wing is the existing shipping and receiving area, and these spaces will remain in their current location. North of this area is the existing ramp into Gatewood. Although there is an elevation change, the west end of the north wing has a direct circulation connection and is accessible via an existing wheelchair lift. This area of the north wing also includes the proposed SPIRA research lab relocated from the second floor to be closer to the existing SPIRA lab within the Gatewood Addition.

Renovations to the ground floor also include new classroom spaces that are being relocated from upper floors in order to accommodate improved space adjacencies and office suite configurations on those levels and to maximize efficiencies and overall space utilization. On this level, the small existing spaces are reorganized and reconfigured into larger blocks of space to accommodate larger classrooms, instructional / teaching labs and office suite arrangements. Although the research lab program and functions are yet to be determined by the department, larger blocks of space for research lab program elements are also created as part of the space reorganization. In the central wing, a larger teaching lab is identified for a later phase of the project. This space is flexible and may be sub-divided or opened up into a larger instructional space.

FIRST FLOOR

In the proposed plan, circulation to the first floor of the ME building will remain as it currently exists. The existing stair towers and elevator within ME provide access from the floors above and below.

The proposed first floor layout includes several existing SMAS classrooms that will remain. Additional classrooms relocated from other floors are also planned on this level to accommodate general student access and appropriate space adjacencies. A proposed student collaboration space is also included within the north wing to provide more student queuing space, which will help accommodate a higher volume of student traffic on this floor with the SMAS classrooms. This student collaboration and queuing space is incorporated with the creation of the new, adjacent restroom facilities.

Existing teaching labs in the central wing also remain on the first floor, and an additional larger teaching lab (with associated program build space) is planned for the east wing. See the revised instructional lab summary program spreadsheet included earlier in this section. In addition, on the first floor, a new office suite and proposed stair tower is shown in the east wing to accommodate additional departmental growth for faculty, staff and graduate assistants and exiting requirements. The office suite is at the end of the circulation and other student-centered program areas to provide better isolation and acoustical control from adjacent student spaces and public circulation elements.

SECOND FLOOR

The Mechanical Engineering department desires to maintain departmental core administrative and office functions in the north wing on the second floor due to the existing direct access to the Gatewood Addition. As shown on the proposed plans, this area will accommodate the departmental main offices, development office, business office, and graduate and undergraduate offices and advising spaces. In addition, the existing seminar room is relocated in the proposed plan layout. In the new proposed location, this seminar space may serve multiple functions for the department, including support for large group meetings.

In addition to administrative office suites, a proposed work café is centrally located and offers an opportunity for informal collaboration between faculty, staff and students. Tutoring rooms also support existing program functions within the adjacent Gatewood Addition, and these spaces provide additional growth for existing tutoring and student support functions.

In the central wing, the existing SPIRA research lab is relocated to the ground floor level, and the Electronics Shop is relocated to accommodate additional office and program requirements, as well as an opportunity for natural daylighting for office and open-office workstations. In addition, a new computer lab is proposed with an additional research lab (program to be determined by the department). These renovations are planned for a second or future phase of the project.

THIRD FLOOR

Similar to the first floor, the third floor does not have direct circulation to the Gatewood Addition, and access to the third floor will remain as existing. The existing stair towers and elevator provide access from the floors below.

In addition to the new and renovated restroom facilities, the third floor is planned to be a large, open-office suite, including spaces for amenities such as an informal work café and small, medium and large meeting and collaboration rooms, in addition to private offices and open-office workstations. Access to the floor may be flexible, and the department may consider open access during predetermined office hours for direct access by students and visitors and after-hours access-controlled security at the stairs and elevator.

FOURTH FLOOR

The fourth floor provides support and space for MEP/T systems. Refer to the MEP narratives for scope of renovation and other modifications required for this level.

PHASING

As described in earlier sections, proposed construction phases for ME follow the scope (Parts 1 and 2) of the study. Phase 1 includes spaces that directly impact the College of Engineering's growth plan, including spaces that support core academic functions within the building, including office and open-office workstations and suites, as well as instructional / teaching lab configurations that are impacted by the reorganization. Phase 2 includes additional considerations that impact overall building use and organization, and these areas also impact future flexibility in lab configuration, including existing instructional / teaching labs and research labs and related support spaces. Much of the phasing consideration for ME is based on increasing the quantities of private offices and open-office workstations and support functions within larger office suites throughout the building.

Looking at the proposed plans from the third floor down, the third floor will become a large office suite that encompasses the entire north wing of ME. The office suite concept provides flexibility for enclosed offices and adjacent workstations, while also incorporating similar qualitative considerations developed and implemented in other recent projects for the College of Engineering (i.e., Wang Hall, Grissom Hall). These spaces include small, medium, and large meeting spaces, as well as informal collaborative spaces and work cafes. However, with the implementation of the office suite on the third floor, core academic spaces that are currently located there will be displaced, and as we look at the overall building organization and where these spaces are relocated, we expand the Phase 1 construction scope of the project so that these core academic functions remain within the building.

Much of the north wing of the second floor is also incorporated into the Phase 1 project to accommodate the ME administrative offices. At the second floor level, core academic classrooms are impacted with the office suite configuration, and these spaces are to be relocated on the lower first and ground floor levels of the building. Other renovations on the second floor central wing do not have a direct impact on housing additional faculty, staff and graduate students, so renovations and reconfigurations of these areas are planned for a later, second phase project.

Many academic classrooms are planned to be relocated and incorporated into first floor spaces, adjacent to existing instructional / teaching labs and existing academic classrooms. The proposed plans also incorporate new instructional / teaching labs and office suites on the first floor in the east wing. All of these spaces are relevant to maintaining core academic spaces in the building, and these are planned as part of the first phase of the project.

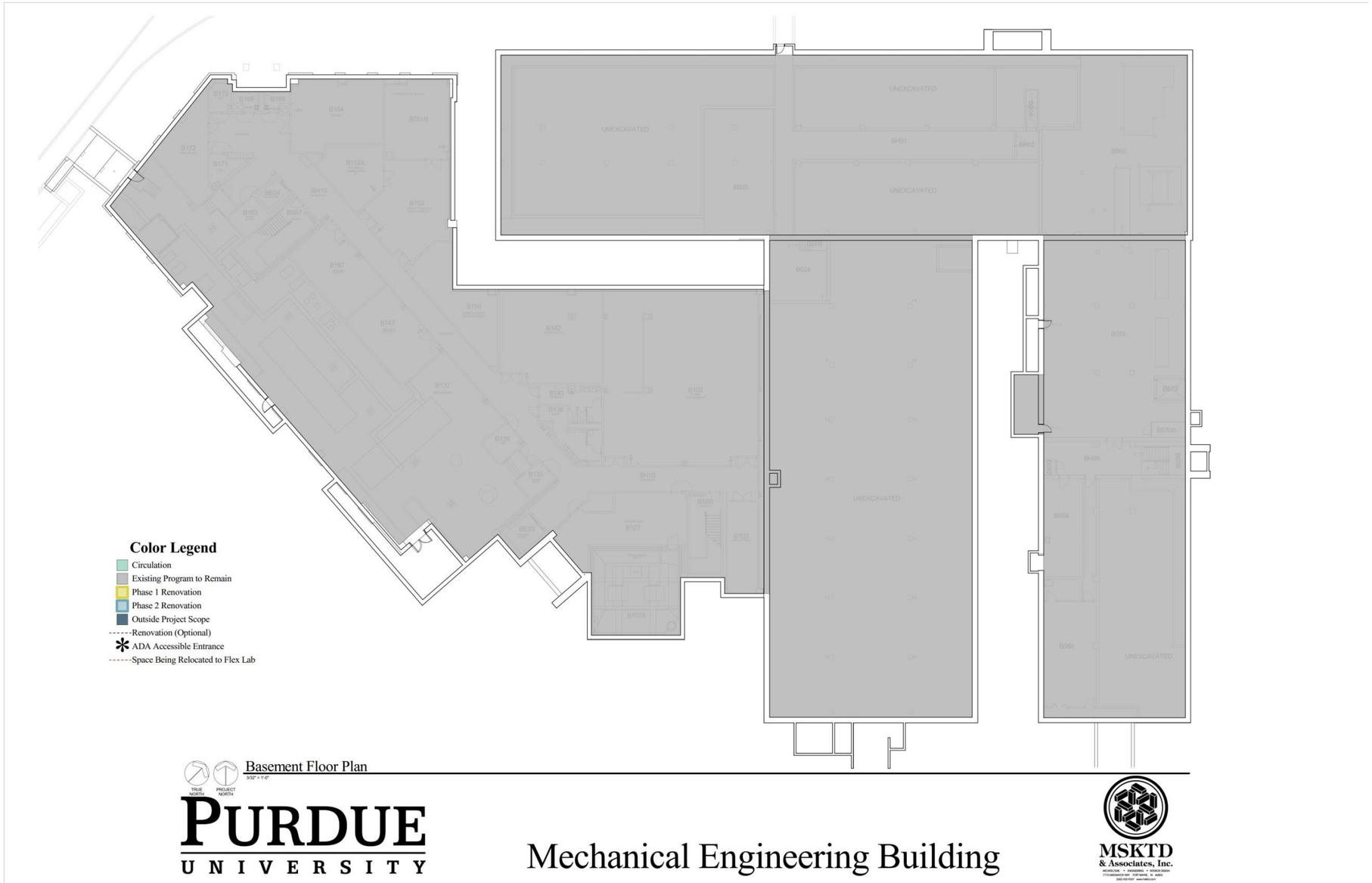
The ground floor level has a mixture of relocated classrooms and new instructional / teaching labs to address program requirements for the mechanical engineering department, and these are identified within Phase 1 of the project. Other relocated and reconfigured instructional / teaching labs and research lab spaces are incorporated into later phases.

In general, building improvements impacting circulation and core building services (e.g., restrooms, MEP/T systems) are planned as part of the Phase 1 scope of work.

OTHER CONSIDERATIONS

Towards the end of the study and related team meetings, there were discussions about considering an option to extend the east wing of ME up to the second and third floor for additional office suites and related functions (in lieu of renovating the existing building to accommodate these growth requirements). In this scenario, a fourth floor mechanical area would also be necessary to accommodate MEP/T system requirements. While this option does not address renovation and maintenance issues within the main ME building, the design team provided a high-level review for feasibility, and information is included in the study appendices and MEP narratives.

Architectural Narrative – Proposed Plans – ME Basement Floor



Architectural Narrative – Proposed Plans – ME Ground Floor

Ground Floor Plan - Proposed

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – ME First Floor

FIRST FLR. TOTALS

OVERALL TOTALS
(12) Offices
(25) Workstations
- (2 Staff, 23 GA)
(9) Classrooms
(6) Teaching Labs
(1) Restroom Renovation
(1) Restroom Addition

PHASE 1 TOTALS
(12) Offices
(25) Workstations
(3) Classrooms
(1) Teaching Lab
(1) Restroom Renovation
(1) Restroom Addition

Color Legend

- Circulation
- Existing Program to Remain
- Phase 1 Renovation
- Phase 2 Renovation
- Outside Project Scope
- Renovation (Optional)
- ADA Accessible Entrance
- Space Being Relocated to Flex Lab

First Floor Plan - Proposed

3/32" = 1'-0"

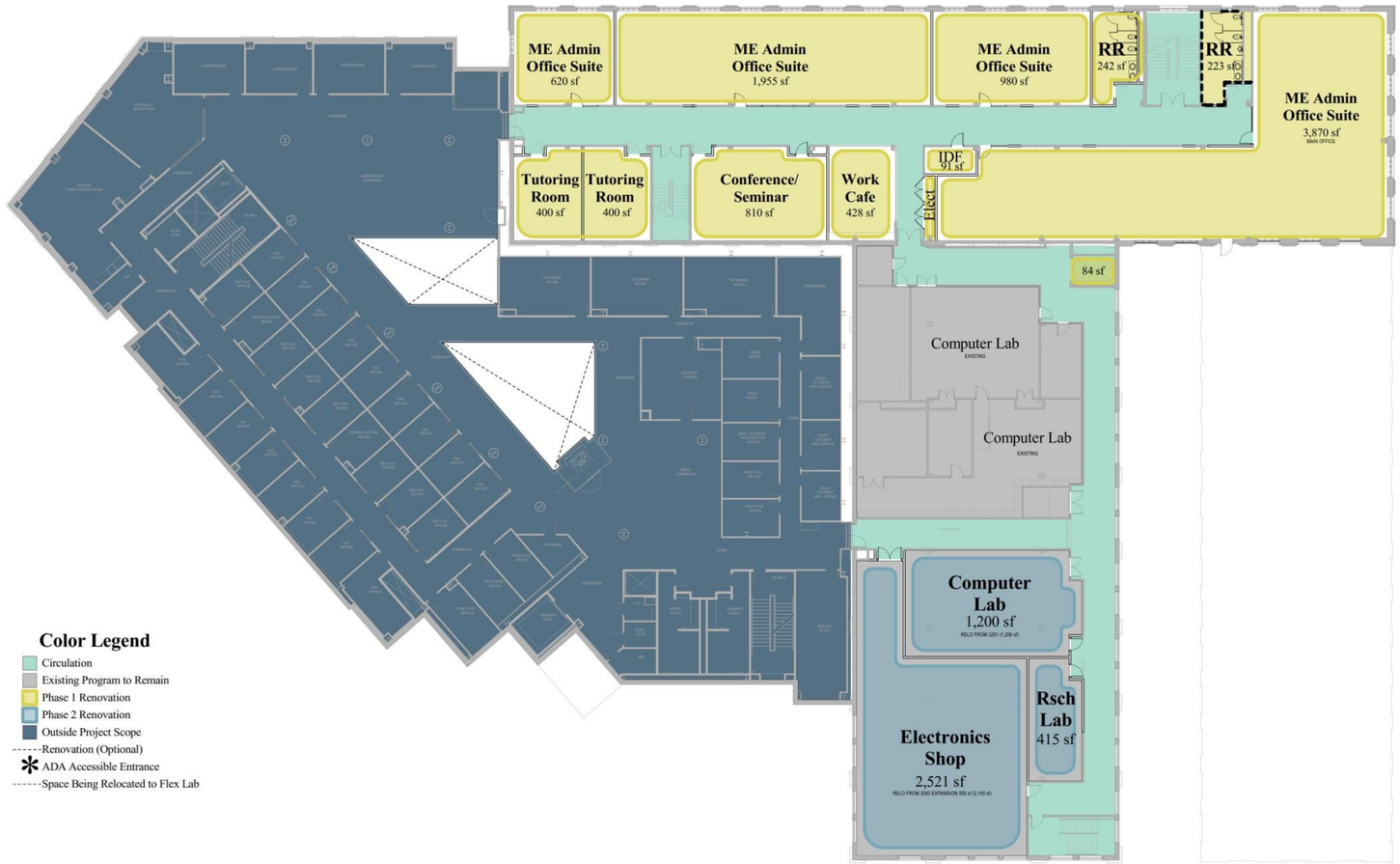
TRUE NORTH PROJECT NORTH

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – ME Second Floor



SECOND FLR. TOTALS

OVERALL TOTALS
(20) Offices
(25) Workstations
(4) Conference/ Meeting Rooms
(1) Work Cafe
(2) Tutoring Rooms
(3) Teaching Computer Lab
(1) Research Labs
(1) Restroom Renovation
(1) Restroom Addition

PHASE 1 TOTALS
(20) Offices
(25) Workstations
(4) Conference/ Meeting Rooms
(1) Work Cafe
(2) Tutoring Rooms
(1) Restroom Renovation
(1) Restroom Addition

PHASE 2 TOTALS
(1) Electronics Shop
(1) Teaching Lab
(1) Research Lab

Second Floor Plan - Proposed

3/32" = 1'-0"

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – ME Third Floor

THIRD FLR. TOTALS

OVERALL TOTALS
(36) Offices
(74) Workstations
- (24 Staff, 50 GA)
(1) Conference Rooms
(2) Work Rooms
(1) Work Cafe
(1) Restroom Renovation
(1) Restroom Addition

PHASE 1 TOTALS
(36) Offices
(74) Workstations
(1) Conference Rooms
(2) Work Rooms
(1) Work Cafe
(1) Restroom Renovation
(1) Restroom Addition

Third Floor Plan - Proposed

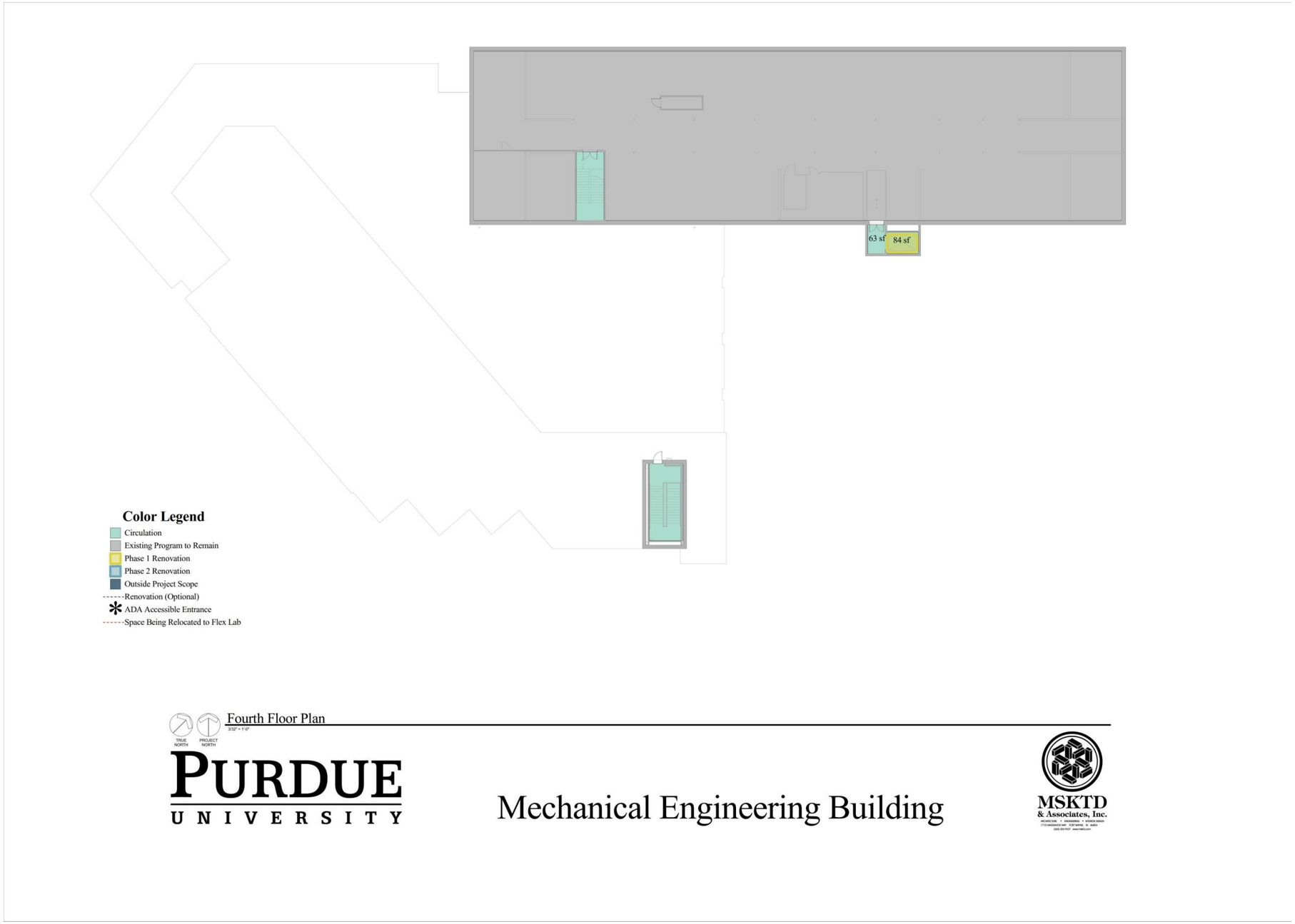
TRUE NORTH PROJECT NORTH

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – ME Fourth Floor



Additional Comments

Following the final presentation and review with faculty as part of the study, additional comments from the Mechanical Engineering department were provided, and these are included below for reference on future projects for the ME building:

Comments on ME/RAIL Renovation Plans Presented on June 2, 2015:

- *Concerns about the open office layout, in particular that it would not be conducive to productivity and creativity. Additional concerns about noise and privacy. Please, no "masking noise" added in. This is known to be disruptive and not conducive to creative, productive work and has potential for adverse health effects.*
- *It would be nice to have a more data-driven approach to the design of spaces. We should poll IE users of Wang to get their impressions, likes, dislikes, and suggestions for improvement. Perhaps also get feedback from existing users of ME/Gatewood to see what works and what doesn't.*
- *Faculty and grad students don't need to be collocated. It's better if grad students have office space near their labs. That being said, if much of the research labs move out of ME, then mostly TA grad students would remain in the building, thereby precluding the purpose for having faculty and students share the same open office area.*
- *Faculty having their own graduate students not too far away is fine, but right next to faculty is not so good. Faculty being able to see their students working at their desks, and grad students knowing that their advisors can see them will make for an intrusive environment that is counterproductive to working together collaboratively. Most people don't do well under continuous scrutiny. If faculty want to do that, they should not be encouraged to do so; it is not healthy.*
- *Schools will only work well when faculty talk to each other as a natural part of them going about their work. Faculty should not have to book a room to talk to each other. Faculty therefore need an environment where discussing problems or experiences can be done without disturbing graduate students who, in general, probably should not be part of such discussions, many of which are not of interest to them. Faculty should also be able to go into their offices and work without being disturbed. This is especially important when faculty teach courses that their grad students are taking, thus raising issues with FERPA compliance if graded materials are on their desks.*
- *The pod systems with the admin. staff person looking after the faculty members in that group, who have common teaching and research interests, works very well in creating good faculty communities within the school and allows them to do their job. Whether these communities have some shared offices or are all assigned doesn't necessarily matter, but they should be around 10-12 faculty communities. Too big doesn't work.*
- *When most people look back on their graduate experiences, the sense of community with their fellow graduate students is what they recall with warmth. The building layout should be there to encourage this sense of community among the graduate students. Ideally, grad students with similar research interests would share the same office space.*
- *Since each grad student will not be receiving a private office, where will TAs hold office hours?*
- *Admin office suites should be sized so as to encompass only the functions for that particular group of individuals, e.g., undergrad, grad, development, business, etc. Still need to decide who gets private offices and who gets workstations in a suite.*
- *Explain what all the 24 staff workstations in the Office Suite on the Third Floor would be for, since most staff would be working in the new Admin offices on the Second Floor. Would only need 2-3 secretaries in the Third Floor Office Suite.*
- *If possible, it would be advantageous to reduce some of the space allotted to classrooms, especially since some of the instructional lab spaces will replace classroom needs, e.g., using ME 2063 for ME 263. Is this even a remote possibility?*
- *Dedicate windowless spaces to such uses as computer lab, tutoring room, or conference rooms.*
- *Phase 1 versus Phase 2 sequencing on the Ground Floor under the main part of the building. Should the Research Labs be part of Phase 1 so that HVAC, Electrical, etc. can all be done at the same time?*
- *Can the entire First Floor be kept open with Phase 1 occurring at various locations?*
- *Explore the potential for adding two floors above the east wing.*

Some Specific Recommendations on Floor Plans:
Third Floor:

- *Leave existing office suites on both ends of the hallway as is. Expand remaining space for mix of grad student and faculty communities. This would not only preserve some existing office layouts that have been well-received by both faculty and staff, but would also save money in renovation costs.*

Second Floor:

- *Expand conference room to include the space currently slated for the Work Café.*
- *Move the Work Café across the hallway, where the Electrical room is currently drawn, thereby somewhat reducing the Admin Office Suite.*
- *Alternatively, rather than building a new seminar/conference room on the second floor, enlarge the existing Ford seminar room on both ends, still leaving room on the south end for a "kitchenette." The proposed conference room could then be a xeroxing/supply room.*
- *Break up the Admin Office Suite that is currently 3870 sq ft into two sections for better separation of functionality.*
- *Modify the Electronics Shop to have a wider L section (about 10-12 ft in total width), with a correspondingly shorter Computer Lab.*

First Floor:

- *Break up the Office Suite of 3670 sq ft into two sections, one for faculty, the other for grad students.*

Mechanical Narrative

This narrative describes the existing building mechanical systems, and the proposed changes to those systems. These changes are anticipated to be implemented as part of a multi-phase building renovation project.

The mechanical systems will be designed at a minimum in accordance with the following codes and standards as applicable:

1. Applicable Codes:
    a. 675 IAC 13 – Building Codes
        i. 2014 Indiana Building Code
        ii. International Building Code, 2012 Edition
    b. 675 IAC 16 – Plumbing Codes
        i. Indiana Plumbing Code, 2012 Edition
        ii. International Plumbing Code 2006 Edition
    c. 675 IAC 18 – Mechanical Codes
        i. Indiana Mechanical Code, 2014 Edition
        ii. International Mechanical Code, 2012 Edition
    d. 675 IAC 19 – Energy Conservation Codes
        i. Indiana Energy Conservation Code, 2010 Edition
        ii. ASHRAE Standard 90.1-2007 – Energy Standard for Buildings Except Low-Rise Residential Buildings.
    e. 675 IAC 22 – Fire Prevention Codes
        i. Indiana Fire Code, 2014 Edition
        ii. International Fire Code, 2012 Edition
2. Applicable Standards:
    a. NFPA 101 – Life Safety Code
    b. Purdue University Consultants Handbook

<u>Mechanical Engineering Building Existing Systems</u>

The existing building was constructed in phases. The west wing of the building was constructed in 1929. The north wing of the building was added in 1932. The east wing was built in 1941 with the final addition in 1948 that connected the north and east wings.

The mechanical infrastructure was added to the building in phases. Mechanical upgrades started in 1958 and continued with the latest updates being completed in 2013. A summary of existing system components is as follows:

1. Air side HVAC system – A dedicated 7,920 cfm air handling unit (ACG-1) was installed to provide cooling to the north wing Classroom 1061. This unit is the primary means of ventilation to Classroom 1061.
2. Air side HVAC system – In 1961 a constant volume cooling only unit was installed in the west wing to provide cooling to Lab 1030. This unit has been abandoned in place.
3. Air side HVAC system – In 1967 an 11,200cfm single zone constant volume air handling unit (AC1-4) was installed in the east wing to provide cooling to Lab 1093.
4. Air side HVAC system- In 1968 a 6,600 cfm dual duct air handling unit (ACG-2) was installed to serve the ground floor east wing. This unit has been renovated and utilizes the original hot deck ductwork as a duct for return air to the unit.
5. Air side HVAC system – In 1975 a number of fan coil units were installed in the 1030 suite of the west wing.
6. Air side HVAC system – In 1978 a 7,675 cfm constant volume air handling unit (AC1-3) was installed on the first floor of the east wing. This unit has been retrofit with a variable frequency drive.
7. Air side HVAC system- In 1981 two 22,000 cfm variable air volume air handling units (ACA-5 and ACA-6) were installed in the attic space of the north wing. These units provide cooling and ventilation to the ground, first, second, and third floors of the north wing.
8. Air side HVAC system- In 1984 a 6,000 cfm air handling unit (AC2-9) was installed to provide cooling for the computer lab and also provide additional cooling to adjacent areas served by fan coil units.
9. Air side HVAC system – In 2000 a 30,000 cfm air handling unit (ACG-10) was installed in the courtyard between the east and west wings. This unit serves the ground floor of the west wing. Renovations in 2013 have utilized this air handling unit to provide additional cooling to areas served by fan coils on the ground floor of the east wing.
10. Air side HVAC system- In 2009 a 6,000 cfm air handling unit (AC2-9) was installed on the second floor of the west wing. This unit replaced the 1984 AC2-9 air handling unit and now serves the computer labs and the electronics lab.
11. Hydronic Systems –Campus chilled water is delivered to the air handling units and fan coil units by a chilled water pump located in the crawl space. A heat exchanger and heating water pumps are also located within the crawl space.
12. Steam System – Low pressure (15 PSI) campus steam is delivered to the building through the steam tunnel below the building. Low pressure steam is piped to the hydronic heat exchanger, to the domestic hot water generator, air handling units, and to fan coil units. Steam condensate is returned to a steam condensate pump located in the building crawl space and pumped back to the campus return system.
13. Building HVAC Control System – Controls are believed to be a combination of DDC and pneumatic, varying with the age of equipment.
14. Domestic Water – The domestic water service is provided and enters the building in the crawl space. Domestic hot water is generated in the basement via a steam to hot water generator. A domestic hot water return system is also present in the building.
15. Sanitary Waste and Vent Systems are present for the main toilet group and miscellaneous fixtures through the building.
16. Fire Suppression – Areas renovated in any phase of the Mechanical Engineering Building will require that sprinkler heads be reworked as part of the scope of the renovation.

<u>Proposed Mechanical Scope</u>

This narrative proposes a multi-phase replacement of a significant portion of the existing airside HVAC systems within the building. A direct digital control system will be provided for HVAC system control and monitoring. The existing fire suppression system will remain, but will be modified as required to provide proper coverage of the renovated spaces.

The need for such a significant amount of system replacement is a result of the significant amount of general construction renovation work being proposed, as well as the current HVAC systems being over tasked and the units past their expected service life.

Due to the extent of the proposed renovations, project phasing may require that complete wings of the building be closed during construction.

Proposed Airside HVAC System

Re-programming and renovation of space is being planned for the ground, first, second, and third floors, presenting the need to modify the airside systems.
There is a significant amount of renovation in the north wing, Phase 1. This fact, combined with the fact that the existing north wing units are old and not of adequate capacity, makes replacement of these units a logical recommendation.

An additional consideration in the north wing is that the existing north wing terminal boxes close when there is no call for cooling which prevents ventilation air to the spaces. To meet current code and the ASHRAE 62.1 ventilation standards, the

renovated spaces will receive new terminal boxes with hot water reheat to allow them to be set to a minimum position that will provide the required ventilation to the spaces.

The design team proposes replacing the portions of the existing airside systems with new VAV systems with reheat. The components of the proposed system are:

Proposed Phase 1:

Proposed Air Side Equipment Systems

North Wing:

1. Replace the two existing air handling units in the attic with two new Variable Air Volume Air Handling Units located in the attic of the north wing, consisting of a steam pre-heat coil, chilled water cooling coil and fan controlled via a variable frequency drive.
2. Two return air fans, 26,000 CFM, fan controlled via a variable frequency drive, separate from the air handling units.
3. VAV Terminal Units throughout for zone control and airflow modulation.
4. Individual fan coil units to condition the IT rooms – chilled water, fractional horsepower.
5. Cabinet Heaters in the entry vestibules and stairs – fractional horsepower.
6. Unit heaters in the attic – fractional horsepower.
7. Toilet Exhaust Fan.

AIR SIDE EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
ACA-5	VAV Air Handler	26,000 CFM	460 V/50 HP
ACA-6	VAV Air Handler	26,000 CFM	460V/50 HP
RAA-5	VAV Return Fan	26,000 CFM	460V/20 HP
RAA-6	VAV Return Fan	26,000 CFM	460V/ 20HP
ATU-X (x 70)	Air Terminal Unit	8100 BTUH	24V control power
FCU-X (x 4)	Fan Coil Unit	12,000 BTUH	120V/Fractional HP
CUH-X (x 5)	Cabinet Unit Heater	15,000 BTUH	120V/Fractional
PUH-X (x6)	Propeller Unit Heater	30,000 BTUH	120V/ Fractional HP.
TEA-1	Toilet Exhaust Fan	3500 CFM	460 V/ 1.5 HP

East Wing:

1. Duct 12,000 cfm of supply air from existing air handling unit ACG-10 to the east wing.
2. VAV Terminal Units throughout for zone control and airflow modulation.
3. Individual fan coil units to condition the IT rooms – chilled water, fractional horsepower.
4. Cabinet Heaters in the entry vestibules and stairs – fractional horsepower.

AIR SIDE EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
ATU-X (x 30)	Air Terminal Unit	8100 BTUH	24V control power
FCU-X (x 1)	Fan Coil Unit	12,000 BTUH	120V/Fractional HP
CUH-X (x 2)	Cabinet Unit Heater	15,000 BTUH	120V/Fractional

Proposed Steam and Hydronic System

The existing perimeter finned tube radiators will be removed, and new radiators provided. This approach will also provide the opportunity to coordinate the finned tube locations with new room and furniture layouts.

An additional heat exchanger and heating water pumps will be required to supply heating water to new terminal units.

The existing chilled water pipe sizes are adequate for the proposed system changes.

A summary of the hydronic equipment is listed below.

STEAM AND HYDRONIC EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
HX-1	Shell and Tube Heat Exchanger	160 GPM/1600 lb/hr	-
HYPB-1	Hydronic Heating Water Pump	160 GPM/60 FT	460V/7.5 HP
HYPB-2	Hydronic Heating Water Pump	160 GPM/60 FT	460V/7.5 HP

Proposed Building Automation Modernization

The full extent of mechanical work being proposed will necessitate the need for a temperature control system to be provided. A new direct digital control building automation system will be provided. Actuators will be electrically actuated. The system will be either an Automated Logic or a Siemens system, and the front end shall be accessible at the campus control center.

Proposed Plumbing Upgrades

Portions of the sanitary and waste piping will need to be modified to accommodate the restroom upgrades. New sanitary and waste mains will need to be added as well as a new piping chase for the restrooms on the west side of the stairwell. New toilet fixtures to be ADA compliant.

Portions of the domestic water piping will need to be modified to accommodate the restroom upgrades. New domestic water mains will need to be added as well as a new piping chase for the restrooms on the west side of the stairwell.

Proposed Fire Suppression Upgrades

The design intent is to re-use the existing fire service, and sprinkler pipe zone valves and mains. Branch piping and heads will be replaced, and spaced to provide proper coverage.

Proposed Phase 2:

North Wing:

1. Provide a Laboratory Airflow Control System for the proposed research labs. Supply air from the new air handling unit provided as part of Phase 1. Exhaust valves will be connected to a new manifold exhaust fan located on the low roof of the Gatewood Wing.

AIR SIDE EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
SAV-X (x 3)	Laboratory Supply Air Valve w/ Reheat	8100 BTUH	24V control power
EAV-X(x4)	Laboratory Exhaust Air Valve	700 CFM Each	24 V control power
MHER-X	Manifolded Exhaust fan	2800 CFM	460V/3 HP

West Wing:

1. First Floor: Re-configure existing VAV Terminal Units for the Teaching Lab.
2. Second Floor: Duct supply air from existing air handling unit ACG-10 to the west wing second floor.
3. VAV Terminal Units throughout for zone control and airflow modulation.

AIR SIDE EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
ATU-X (x 7)	Air Terminal Unit	8100 BTUH	24V control power

Proposed Building Automation Modernization

The full extent of mechanical work being proposed will necessitate the need for a temperature control system to be provided. A new direct digital control building automation system will be provided. Actuators will be electrically actuated. The system will be either an Automated Logic or a Siemens system, and the front end shall be accessible at the campus control center.

Proposed Plumbing Upgrades

Provide services for the converted laboratory spaces. Services include cold water, hot water, lab gases and safety showers.

Proposed Fire Suppression Upgrades

The design intent is to re-use the existing fire service, and sprinkler pipe zone valves and mains. Branch piping and heads will be replaced, and spaced to provide proper coverage.

Future Work:

There are areas of the east and west wing that are currently served by fan coil units that are beyond their expected life. These units serve areas outside of the scope of the areas being proposed for space reprogramming, and therefore replacement of these unit is not addressed in this study. A recommended course of replacement would be to add an additional unit in the courtyard to serve these spaces, outside of the project that this study addresses.

Optional Addition

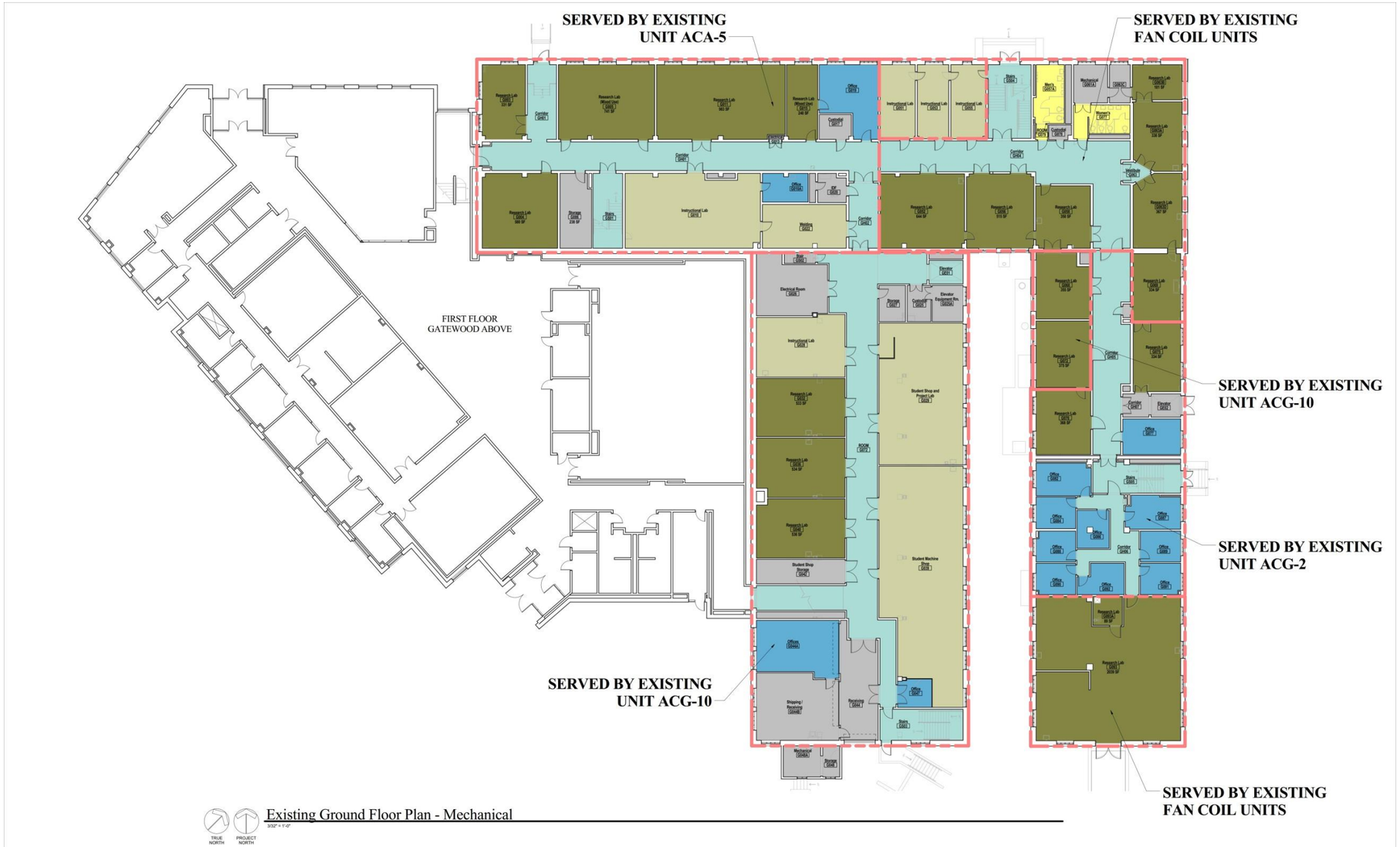
As an option to the Phase 1 and Phase 2 work proposed, the owner asked the team to consider an addition to the east wing. This study has looked at this option as square footage that would require dedicated air handling and hydronic equipment. A penthouse to house air handling units would be provided and space for the hydronic equipment would also be required. Under this option, the existing building systems would remain as they currently are, except where drains, exhaust, etc. need to be relocated in order to allow the additional two levels to be installed.

Attached Drawings

The following drawings are attached to help convey the written design intent.

1. Mechanical Engineering Building – Existing Ground Floor Plan
2. Mechanical Engineering Building – Existing First Floor Plan
3. Mechanical Engineering Building – Existing Second Floor Plan
4. Mechanical Engineering Building – Existing Third Floor Plan
5. Mechanical Engineering Building – Existing Fourth Floor Plan
6. Mechanical Engineering Building – Proposed Ground Floor Plan
7. Mechanical Engineering Building – Proposed First Floor Plan
8. Mechanical Engineering Building – Proposed Second Floor Plan
9. Mechanical Engineering Building – Proposed Third Floor Plan
10. Mechanical Engineering Building – Proposed Fourth Floor Plan

Mechanical Narrative – Existing Plans – ME Ground Floor



Existing Ground Floor Plan - Mechanical

PURDUE UNIVERSITY

Mechanical Engineering Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – ME First Floor

Existing First Floor Plan - Mechanical

PURDUE UNIVERSITY

Mechanical Engineering Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – ME Second Floor

Existing Second Floor Plan - Mechanical

PURDUE UNIVERSITY

Mechanical Engineering Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – ME Third Floor

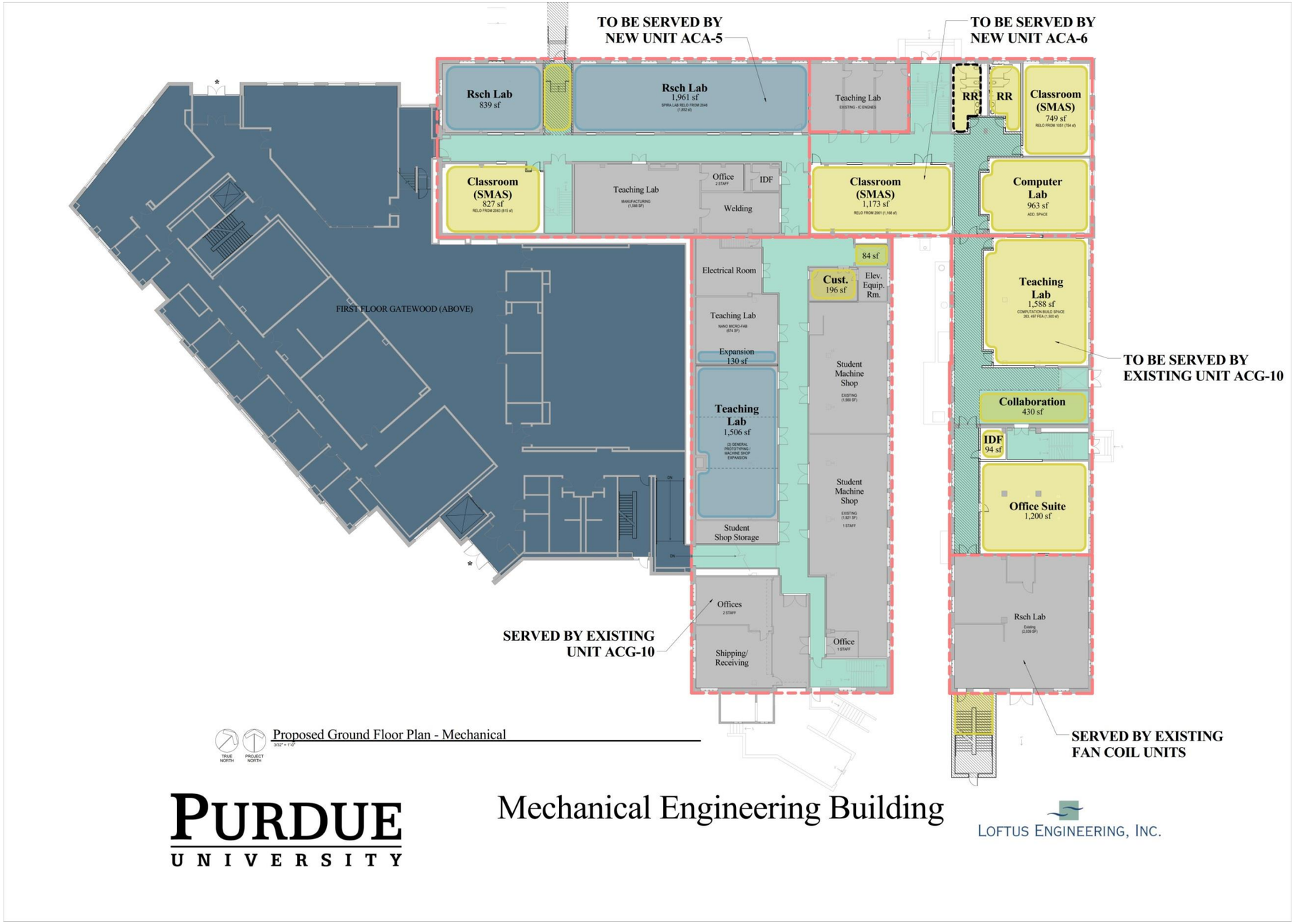
Existing Third Floor Plan - Mechanical

PURDUE UNIVERSITY

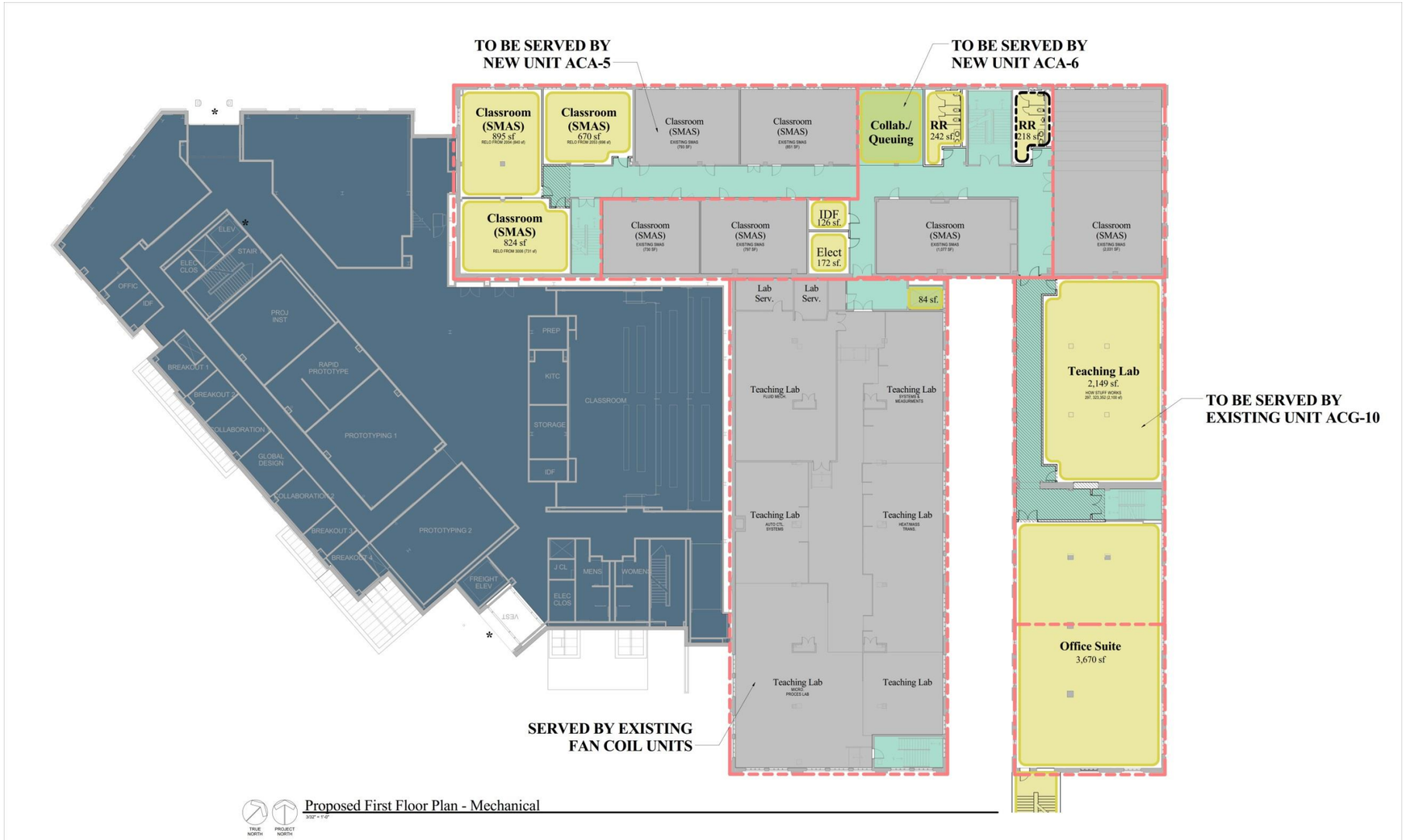
Mechanical Engineering Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – ME Ground Floor



Mechanical Narrative – Proposed Plans – ME First Floor

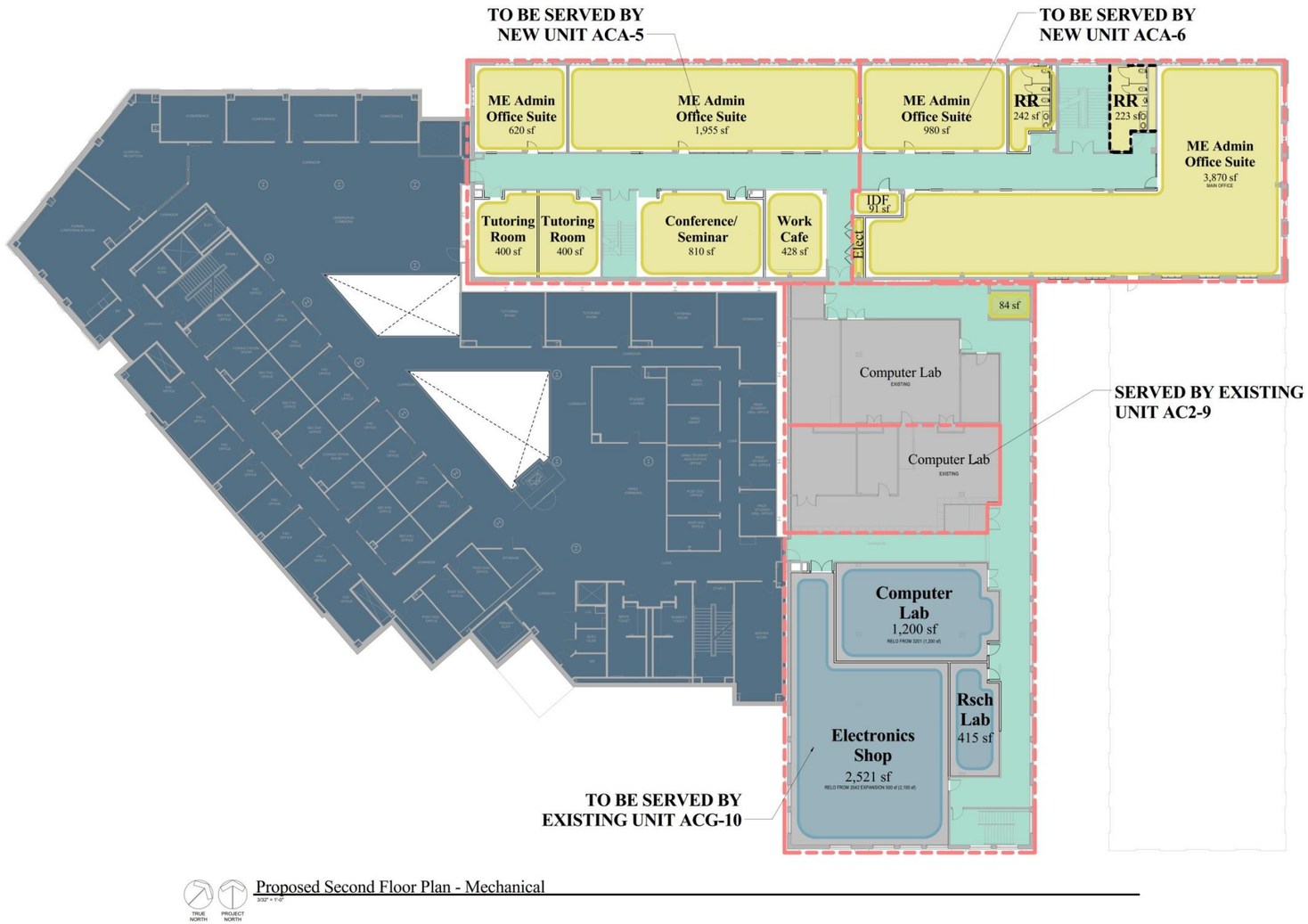


PURDUE UNIVERSITY

Mechanical Engineering Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – ME Second Floor



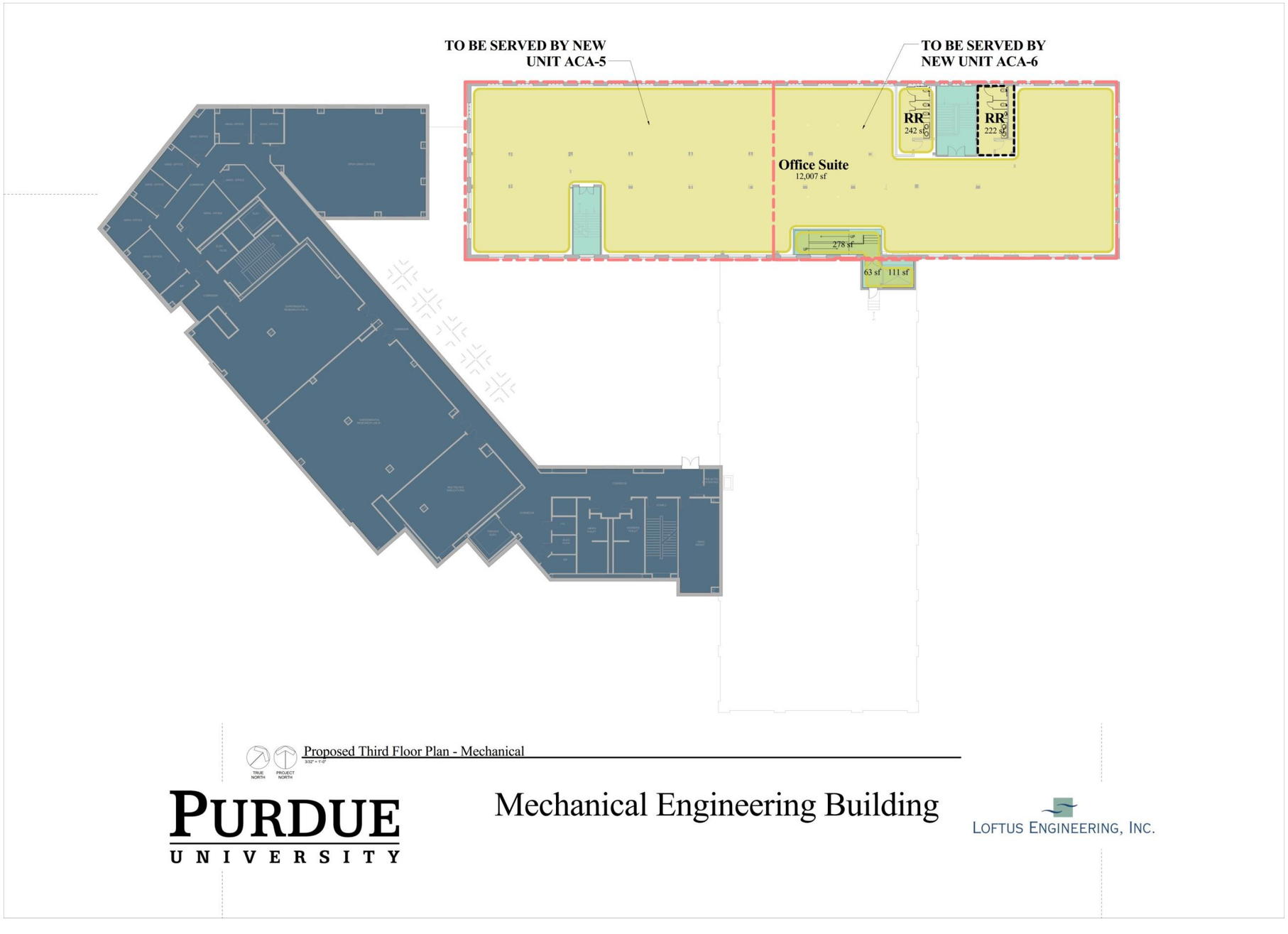
Proposed Second Floor Plan - Mechanical

PURDUE UNIVERSITY

Mechanical Engineering Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – ME Third Floor



Electrical Narrative

This narrative describes the existing electrical systems for the Mechanical Engineering Building, and the proposed changes to those respective systems. These changes are anticipated to potentially be implemented as part of a multi-phased building renovation project.

The electrical systems will be designed at a minimum in accordance with the following codes and standards as applicable:

1. Applicable Codes:
    a. 675 IAC 13 Building Codes.
        i. 2008 Indiana Building Code.
        ii. International Building Code, 2006 Edition.
    b. 675 IAC 17 – Electrical Codes.
        i. Indiana Electrical Code, 2009 Edition.
        ii. NFPA 70 – National Electrical Code, 2008 Edition.
    c. 675 IAC 19 – Energy Conservation Codes.
        i. 2010 Indiana Energy Conservation Code.
        ii. Energy Standard for Buildings Except Low-Rise Residential Buildings ANSI/ASHRAE 90.1, 2007 Edition.
    d. 675 IAC 22 – Fire Prevention Codes
        i. Indiana Fire Code, 2008 Edition
        ii. International Fire Code, 2006 Edition
    e. NPFA 72 – National Fire Alarm Code

2. Applicable Standards:
    a. NFPA 101 – Life Safety Code
    b. Purdue University Consultants Handbook

Power Distribution Systems – General Service

The complete Mechanical Engineering Building was constructed by multiple separate projects. The existing West Wing, North Wing, East Wing and Gatewood Addition comprise the complete building. The complete building is served by two separate electrical services. The Gatewood Addition was constructed around 2011 and was constructed with its own electrical service to provide electrical power to that addition as well as backfeed existing switchgear in the other wings of the facility. The West, North and Center Wings of the existing facility are served by another electrical service in addition to power that is provided out of Gatewood.

Power Distribution Systems – Existing Conditions – West, Center and North Wings

Electrical power to the existing facility is presently provided by from two separate sources. Prior to construction of the Gatewood Addition, power was provided solely from an existing pad mounted transformer located on grade between the West and East Wings. The existing pad mounted transformer is fed from an existing high voltage metal enclosed switchgear lineup "HVS-1" located on grade adjacent to the Transformer and cable connected. The existing 15kV feed for the high voltage metal enclosed switchgear lineup originates in a vault style SF6 switch located in existing manhole 19E-2. The existing transformer originally provided the designated service as follows:

1. **Pad Mounted Transformer:** This transformer provides 277/480 volt, 3-phase, 4-wire, wye electrical service to main switchboard "MDP".

    a. This transformer presently serves panelboards that provide power to lighting, receptacles, lab equipment, and non-HVAC equipment loads for the facility.
    b. This transformer serves motor control centers, etc. that provide power for motor loads associated with the HVAC equipment and building plumbing.
    c. This transformer is presently 1000 KVA in size.
    d. This transformer presently provides service to main switchboard "MDP" via (4) 350 MCM conductors in each of four parallel conduits.

Main distribution switchboards "MDP is located in a main electrical room in the basement in the central area of the east wing of the building. Existing Switchboard "MDP" is rated for 1200A, 480V, 3-phase, 4-wire service. The existing switchboard is manufactured by Square D, utilizes switches with Type L and Type R fuses, and appears to have been installed as part of an electrical upgrade project in 2008.

There is an existing sub-distribution switchboard "DP" located in the basement of the original building which receives 277/480V, 3-phase, 4-wire service from existing switchboard "MDP". All 480V loads for the facility appear to be fed directly out of switchboard "MDP" or "DP" directly.

There are two separate dry type 480 volt primary transformers, #T-P and T-S, located in a room in the north part of the basement of the West Wing of the building. Transformer #T-P is 500 KVA size and provides 120/240 volt, 3-phase, 4-wire electrical service to two existing switchboards, "MDP-3" located in Room 28 and an existing switchboard in Room 38. Transformer #T-S is 250 KVA size and provides 120/240 volt 1-phase, 3-wire electrical service to and existing switchboard "MDP-1" located in Room 28 and an existing "emergency" panel also located in Room 28. All panels for lighting and receptacle loads are fed electrical power from the distribution equipment located in Rooms 28 and 38 listed above in addition to existing bus ducts that feed existing laboratory spaces. Prior to construction of the Gatewood Addition, transformers #T-P and #T-S were fed from switches in switchboard "MDP" located in the basement of the East Wing. As part of the Gatewood Addition construction, these feeds were removed and the transformers were refed from a new main switchboard provided in that addition.

Existing branch circuit panelboards are located throughout the facility as necessary to serve respective branch circuit loads and are generally the switch/fuse type of panelboards that utilize plug fuses. Purdue University is presently undergoing systematic replacement projects across campus to replace existing switch/fuse panelboards with new circuit breaker type panelboards.

Power Distribution Systems – Arc Flash Study

An arc flash hazard study has been previously commissioned by Purdue University. According to this study, arc flash incident energy levels at existing switchboards are Category 4 or higher. Current Purdue University design standards require the design for new construction and major renovation projects to mitigate these levels to Category 1 or less at the line side of the main distribution switchgear. Presently no mitigation of arc flash hazard has been provided for this facility.

Power Distribution Systems – Renovations – Proposed Phase 1:

Any modifications to the existing electrical service to the Gatewood Addition are outside of the scope of this study, only those modifications that are associated with work in the existing West, North and East Wings will be provided. Building personnel

have identified that there is insufficient power capacity and distribution available in the existing West, North and East Wings of the building.

The existing pad mounted transformer that supplies the 277/480 volt, 3-phase, 4-wire electrical service will remain and be re-utilized. A second pad mounted transformer, to provide a 120/208 volt, 3-phase, 4-wire electrical service, will be provided adjacent to the existing pad mounted transformer and the existing high voltage switchgear lineup will be expanded to feed the new transformer. The existing main power distribution system for the West, North and East Wings will be completely replaced, to the extent possible, including main switchboards, sub-distribution switchboards, feeders and motor control centers. The existing dry type transformers, #T-P and #T-S, will be removed along with the associated feeders that originate in Gatewood.

Panelboards and bus ducts will be replaced in the areas to be renovated under the scope of the project however existing panelboards outside of the project scope will remain but will be re-fed from new switchboard sources. Existing bus ducts located within specific laboratory spaces that are outside of the project scope will remain and be re-fed from new sources. New main feeder bus ducts will be provided as part of the overall building distribution

Equipment to mitigate the arc flash hazard to levels compliant with Purdue University standards will be provided as part of the power distribution system replacement. It is assumed that the strategy for mitigating the arc flash hazard will follow strategies employed on past university projects. A main low voltage insulated case solid state circuit breaker with adjustable LSI trip will be provided in the system between each service transformer in the vault and main circuit breaker in the respective main distribution switchboards. These circuit breakers will be located in a separate lockable room and will be remotely operated from the main distribution equipment electrical room.

New sub-distribution switchboards will be provided on the each floor of the facility to provide distribution to local panelboards, bus ducts, etc. to be located on the respective floor. The exact quantity and voltages of sub-distribution switchboards will be determined during a subsequent design project as there is not enough information available at this time to determine loading requirements.

In the areas to be renovated, new branch circuit panelboards will be provided to replace existing switch/fuse panelboards. The new panelboards will be located on each floor throughout the facility, as required to provide 120/208 volt, 3-phase, 4-wire electrical power for lighting and receptacle loads. Every effort will be made to locate panelboards such that they will be central to their respective loads served. Each individual computer, research and teaching lab will be provided with a respective branch circuit panelboard to serve only loads located within that lab space. Bus Ducts will be provided as required during design in teaching and research labs to provide flexibility within that lab space. Panelboards will receive their power from the "L" service transformer via main and distribution switchboards as required.

The existing motor control centers and power panelboards will be replaced and be located within mechanical spaces to provide electrical power to mechanical, HVAC, and plumbing equipment motors. Each motor control center will be replaced with a new power panelboard. Each power panelboard will contain overcurrent protection devices as required for each respective item of equipment served. These power panelboards will receive their power from the "P" service transformer via main and distribution switchboards. Where required, combination overcurrent protective devices and motor controllers will be provided to serve mechanical loads, although most 3-phase motors will be controlled by variable frequency drives.

Each switchboard and panelboard will be sized to provide a minimum of 10% spare capacity and will be populated by a minimum of 10% spare devices.

Power Distribution Systems – Renovations – Proposed Phase 2:

New branch circuit panelboards and general power circuits will be provided as described in the Proposed Phase 1 work for areas to be renovated under Proposed Phase 2 as required.

Emergency Power Systems – Existing System

The West, North and East Wings are presently provided power for emergency lighting supplied by central battery inverter (EMAC) units with circuits distributed to selected lighting fixtures. The Fire Alarm System is provided with its own integral battery emergency power source. The Gatewood Addition is provided emergency and standby power by a 125 KW engine-generator installation. This generator was sized to provide only power for the Gatewood addition and does not have spare capacity to provide power for loads in the West, North and East Wings.

Emergency Power Systems – Renovations

A central battery inverter system (EMAC) system will be provided under the Proposed Phase 1 scope to feed lighting fixtures for all emergency lighting, egress lighting and exit signage. The Fire Alarm System will be provided with its own integral battery emergency power source.

Power Monitoring

Electronic metering will be provided at the main to each replacement main distribution switchboard. It is anticipated that the new electronic meters will be connected to Purdue's standard SCADA box. It is not anticipated that sub-metering will be provided at any other points on the power distribution system.

Interior Lighting

Lighting will be designed utilizing a combination of direct, indirect, and direct/indirect luminaires utilizing Light Emitting Diode (LED) technology as the prime illumination source. The exact type of luminaire to be utilized in each respective space will depend on room layout, function ceiling type, aesthetic requirements, and control requirements, among other factors.

In general, lighting will be designed per the requirements contained in the Purdue University Consultant's Handbook and supplemented by Illumination Engineering Society of North America (IESNA) standards with particular attention paid to the following publications:

1. ANSI/IESNA RP-1 – American National Standard Practice for Office Lighting.
2. ANSI/IESNA RP-3 – Lighting for Educational Facilities.
3. The IESNA Lighting Handbook – Ninth Edition.
4. ANSI/IESNA RP-31 – Recommended Practice for the Economic Analysis of Lighting.

Lighting luminaires will be selected based on:

1. Suitability for use in the respective area in which they will be installed.
2. Energy efficiencies required to obtain goals for lighting power densities.
3. Maintenance requirements.

Average lighting levels for each specific building space will be designed in accordance with the following requirements. All illumination level are horizontal values referenced to a work plane located 30 inches above finished floor unless noted otherwise.

1. Lecture Halls: 50 footcandles with full range dimming.
2. Classrooms: 50 footcandles at each desktop with full range dimming.
3. Computer Labs: 20-30 footcandles.
4. Conference Rooms: 40 footcandles with full range dimming.
5. Collaboration Rooms: 40 footcandles.
6. Open, Research, Simulation and Teaching Laboratories: 30-40 footcandles general illumination, 80-100 footcandles on work surface.
7. Offices: 25-30 footcandles general illumination with owner furnished supplemental task lighting.
8. Restrooms: 20-30 footcandles general illumination with 45 footcandles vertical illumination at the mirror.
9. Tutoring Rooms: 40 footcandles.
10. Electronics Shop: 50 footcandles.
11. Corridors: 10-15 footcandles at the floor level.
12. Mechanical and Electrical Rooms: 15-25 footcandles at floor level.
13. Stairwells: 10-15 footcandles maximum, 1-5 footcandles minimum
14. All Other Areas: Default to IESNA recommended standards.

Luminaires utilizing Light Emitting Diode (LED) technology will be provided wherever possible based on the following criteria:

1. Suitability for use in the respective area in which they will be installed.
2. Suitability to render color with proper color temperature.
3. Efficacies required to obtain goals for lighting power densities.
4. Life and maintenance requirements.
5. Lighting control requirements.

All light emitting diode (LED) lighting luminaires shall utilize electronic drivers.

Emergency and egress lighting will be provided in accordance with relevant codes by means of a combination of “Nightlight” circuits and switched circuits connected to the respective building’s emergency power source.

Illuminated exit signage will be provided in accordance with relevant codes and will be connected to the respective building’s emergency power source.

Exterior Lighting

Exterior lighting will be designed and limited to utilize building mounted luminaires. Exterior lighting will be designed to meet the exterior lighting power densities as prescribed by Indiana Energy Conservation Code, where applicable.

Sources for exterior lighting environments will primarily be Light Emitting Diode (LED) technology, as determined during design, and per campus standards. Luminaires will generally be constructed from cast aluminum material. All luminaires will meet IESNA “Full Cutoff” criteria.

Lighting Controls

Various degrees of lighting controls will be provided in the facility to provide functionality, optimize energy performance, and to comply with the requirements of the Indiana Energy Conservation Code. It is anticipated that this facility will utilize a combination of automatic occupancy sensor lighting controls, networkable relay based lighting control systems, daylight harvesting systems and dimming systems.

Unit type occupancy sensors with manual “on” switch will be utilized in all small offices, storage rooms, restrooms, and similar spaces. These controls will be designed such that the occupant of the room must turn the lights on via a switch located at the room entrance and the occupancy sensor will automatically turn off the lights in the respective space if it determines the space is unoccupied.

Relay based lighting control systems will be provided for control of lighting in common areas such as corridors, open office areas, large open workroom areas and similar spaces. The system will provide manual, time-of-day scheduling, and occupancy sensor control to the respective areas. The systems could be designed to be networked and/or stand alone, as determined during the design process.

Daylight harvesting systems will provide automatically adjustable full range dimming based on ambient lighting levels. Daylight harvesting will be provided in appropriate open office areas and open workroom areas along the exterior walls of the building where there will be the possibility of a large natural light contribution due to existing windows.

Architectural dimming systems will be provided for dimming control of luminaires in the meeting rooms, conference rooms, classrooms, lecture halls and multi-purpose rooms. The system will provide full range dimming capability of multiple sources and zones as required via local control stations located in the respective spaces. The exact features and requirements of each architectural dimming system will be determined during design.

Mechanical rooms, electrical rooms, and telecommunications rooms will be provided with standard manually operated switches for control of the lights in the respective spaces.

Audio/Visual Systems

Provisions for Audio/Visual systems will be provided for the following areas of the building, as determined through meetings with Purdue University Technology personnel:

1. Classrooms.
2. Conference Rooms.
3. Lecture Halls.
4. Collaboration Rooms.

The building will contain provisions to support overhead projectors and/or flat-screen televisions in the spaces mentioned above. The provisions will include rough-in and conduits, power and mounting provisions required to support overhead projectors or televisions. Exact requirements will be determined during design by consulting with Purdue University. It is assumed that the owner will furnish all overhead projectors. Other requirements and features will be incorporated into the design, as directed by Purdue University.

Fire Alarm System

The existing fire alarm system for the complete Mechanical Building and Gatewood Addition is a Simplex 4100 system and was installed approximately in 2011. The fire alarm system will be revised and extended as required by renovations of spaces within the building. The system will fully comply with the Americans with Disabilities Act (ADA), Indiana Fire Code, and all other state codes.

The existing system is an intelligent, addressable type system. The complete system will include all required manual and automatic initiating devices such as pull stations, smoke detectors, heat detectors, water flow switches, etc. The complete system will include all signaling devices such as audible/visible signals, visible only signals, sprinkler alarm bell, etc. The system will monitor sprinkler system actuation. The system will monitor and control smoke evacuation system functions, fan shutdown functions, door hold open release functions, and elevator recall functions.

Exact specifications for the complete system will be entirely coordinated with Purdue University's Safety and Security Department.

Card Access and Security Systems

It is anticipated that some degree of access control and security will be required for this project. The extent of this scope has not been identified at this phase of the project but it is anticipated that some degree of access control will be provided to limit access to the second and third floors. The requirements for these systems will be discussed with appropriate Purdue University staff, but at a minimum, rough-in and conduits will be provided to accommodate system equipment and cabling requirements.

Basic Materials

All wiring and cabling for power distribution, branch circuitry, low voltage systems, etc. will be routed throughout the facility utilizing conduits.

Separate closed conduit systems will be provided for each power distribution system and each low voltage system such as:

1. Normal Power.
2. Emergency Power.
3. Equipment Power.
4. Audio/Visual Systems.
5. Fire Alarm System.
6. Card Access and Security Systems.
7. Low Voltage and Digital Lighting Control Systems.

In general conduit types will be as follows:

1. Electrical Metallic Tubing (EMT): Permitted for branch circuit conduits and conduits smaller than 1-1/4" trade size.
2. Rigid Metal Conduit (RMC) - Steel: Required for all feeders, and conduits larger than 1-1/4" trade size.
3. Intermediate Metal Conduit (IMC): Permitted in lieu of Rigid Metal Conduit where located within the building above slab on grade.
4. Polyvinyl Chloride (PVC): Permitted below grade for site lighting circuitry only.
5. Flexible Metal Conduit (FMC): Required for final connections to motorized equipment, lighting fixtures and equipment subject to vibration.
6. Liquidtight Flexible Metal Conduit (LFMC): Required in lieu of Flexible Metal Conduit in damp or wet locations.

All 600V conductors will be copper material Type THHN/THWN insulation.

All switchboards and panelboards will be furnished with 1000A/in^2 plated copper bus bars.

Overcurrent protection for all switchboards will be provided by either solid state low voltage insulated case circuit breakers or solid state molded case circuit breakers.

Panelboards will be furnished complete with bolt-on circuit breakers and hinged fronts.

Dry-type transformers will be furnished complete with copper windings and electrostatic shield.

All wiring devices will be heavy duty grade. Switches will be rated for 20A, 120/277 VAC. Receptacles will have integral brass strap and be rated for 20A, 120VAC. Plates will be high impact nylon or stainless steel as required.

Seismic Restraint Requirements

Seismic restraints are not expected to be required for this project.

Arc Flash Study

It is understood that a comprehensive three part Arc Flash Study (as-drawn, construction and final) will be performed concurrently with any project design. An Arc Flash Consultant will be contracted to perform the Arc Flash Study and make recommendations of mitigation strategies.

Based on previous projects at the University, the project will initially be designed with main circuit breakers with adjustable LSI trip units located in switchboards. These switchboards will be located in their own separate electrical room adjacent to the transformer vault and will serve to disconnect all normal power entering the building from the vault. The circuit breakers will be remotely operable from the main electrical room and will be provided with remote annunciation of the breaker position at the remote operator location.

Each new main switchboard to replace existing switchboards will have a main circuit breaker with adjustable LSI trip unit as required to provide additional arc flash mitigation.

Additional strategies may be explored during design after an initial "as-drawn" Arc Flash Study has been performed.

Technology Narrative

I. TECHNOLOGY

A. Telecommunications Cabling Infrastructure:

1. Overview:

a. The systems referenced within this study pertain to all three facilities.

b. This infrastructure will provide support for Voice, Data and Cable TV distribution systems and services. In addition it will support interconnection of control panels for other ancillary building systems such as Fire Alarm, Electronic Security and Building Automation.

c. The Cabling Systems will be designed utilizing the Purdue ITaP guidelines and specifications as outlined in the Consultants Handbook.

d. It will be assumed that all offices, conference rooms, and other appropriate spaces shall be equipped with telephone circuits and one or more computer workstations. These will be connected via separate intra-building networks as well as separate inter-building networks. Currently, the telephone circuits, including dedicated 2-wire and 4-wire, security, fire, etc. operate on a copper backbone. The data network operates on copper from the workstation to the telecommunication room and then is connected via fiber optic cable to the Purdue fiber optic backbone.

e. CATV and Purdue Cable will continue to operate over a mixture of coaxial cable and fiber optic cables.

2. Demolition and Phasing:

a. Railway – see related RAIL section and narrative

b. ME

1) The existing BDF telecommunications room located within Gatewood supports the telecommunication requirements for ME. Due to this there is no BDF located within ME.

2) The existing telecommunications rooms that serve as IDF's shall be demolished as part of any renovation to ME. The existing spaces do not meet the current campus standards and must be replaced with new room's as part of any renovation to ME.

3) Phase 1 of ME currently identifies all work needing to be completed for the establishment of new IDF rooms on the Ground floor, First floor and Second floor. These new IDF locations will serve the new outlet locations throughout the facility.

4) The existing riser cabling serving all existing telecommunications rooms shall be demolished as part of any renovation to the ME building. The existing riser cabling does not meet the current campus standards for grade and type of fiber optic cabling. As well all riser cabling must be replaced as part of the creation of the new telecommunications rooms.

5) The existing data horizontal cabling serving the ME Building shall be demolished as a part of any renovation to the building. The existing Category 5e cabling infrastructure for data cabling does not meet the current campus standards.

6) The existing telephone horizontal cabling serving the ME Building shall be demolished as part of any renovation to the building. The existing Category 5e cabling meets the current campus standards but terminates in existing telecom rooms that must be replaced.

c. Hampton – see related HAMP section and narrative

3. Topology:

a. This infrastructure shall include workstation outlets (WO) at specific locations throughout the facility. The cabling system shall utilize a "Star" homerun topology. Each individual WO shall be run to a dedicated Telecommunication Room (TR) where it will be terminated to a patch panel within the 90 meter distance limitations dictated by industry standards.

b. All Telecommunication Rooms shall then connect to a central point known as the Main Cross Connect, aka Building Distribution Frame (BDF) located in the Main Equipment Room (ER). This connection shall include a combination of fiber optic cable as well as a multi-pair copper cable.

c. That BDF shall connect via backbone copper and fiber cabling to Campus Telecommunications Systems.

4. Communications Pathways

a. The building's new communications pathways will be designed as an easily accessible system that provides for flexibility and relative ease of modification throughout the lifecycle of the building.

b. Pathways will be sized for voice, data, video, and any other Owner approved low voltage cabling. Systems, may include fire alarm, security, building automation, etc. each will be provided with separate pathway for wire management. This may be a separate tier on the corridor cable tray system.

c. The cable tray system will be sized to handle required cable loading for building plus 50% expansion for future applications, above the ceilings in the corridors and elsewhere as required.

d. The distribution cable tray system within the corridors will allow a minimum 3" of clearance between the tray and the ceiling, a minimum 6" of clearance in front of the tray, and a minimum 12" of clearance above the tray.

e. Basket type cable tray will be provided and will be wall and/or ceiling hung and mounted above the accessible ceiling in the corridors as required. The cable tray will be a minimum of 12" W by 4"H.

5. Telecommunications Rooms:

a. Railway Building – see related RAIL section and narrative

b. ME Building

1) The project scope for ME includes the establishment of all new telecommunications rooms within the project area and it is anticipated that all work shall be completed in two phases. However, all work associated with the telecommunication rooms is currently identified for completion during phase 1. This will include the creation of a new IDF on the ground floor, a new IDF on the first floor and a new IDF on the second floor.

2) In Phase 1 of ME, a new IDF will be created on the ground floor in the project east wing next to the south stairwell. The existing first floor IDF will be fully renovated and refurbished to bring it up to ITaP standards. A new IDF will be created on the first floor in the project north wing adjacent to a new electrical room. A new IDF will be created on the second floor in the project north wing adjacent to a small electrical room. These telecommunications rooms shall be located so that every new or future PIC is by wire length within 90 meters of its distribution point in the telecommunications room. This includes areas where PICs are not currently designed to be installed but may be installed in the future (i.e. Far ends of the building, attic spaces, mechanical rooms, etc.).

3) The new telecommunications spaces have been designed to operate 24/7 at the required temperature and humidity range required for these types of spaces. As well no HVAC ducts, plumbing, gas lines, air lines, clean outs, etc. will be located in or routed through the shown telecommunications spaces.

4) The IDF telecommunication rooms for ME have been sized by area served (ASF).

a) The ground floor will include the addition of a single telecommunications room to serve the eastern portion and to support the existing IDF located on this floor. This ground floor IDF shall be approximately 99 SF and shall also provide service to the first floor office spaces located above.

b) The first floor will have 1 new telecommunications room that will be part of the Phase 1 renovation. This IDF will be a 101 SF area and be located directly above the refurbished ground floor IDF and be used to serve the northern portion of the floor. This should allow for ease of routing communications cable from floor to floor.

c) The second floor will have 1 new telecommunications room that will be located on the plan north wing. This IDF will be a 102 SF area and be located adjacent to the new small electrical room and will serve the intended renovated space on this level and the open office suite located above on the third floor.

c. Hampton Hall – see related HAMP section and narrative

d. All new or renovated telecommunications room walls will have an added layer of 3/4" plywood, B-B ext. grade 5. Plywood will be fire-retardant. Plywood will be installed horizontally on walls from 2'-0" to 6'-0"AFF from corner to corner.

e. All BDF and IDF telecommunication rooms shall have ladder type cable tray installed around the perimeter of the room at approximately 7'-2" AFF.

f. One dedicated, 110V, 20A double –duplex electrical outlets will be provided on each wall. One of those outlets in particular will be provided on the wall near the equipment frames.

g. Two dedicated, 208V, 30A, twist lock (NEMA L6-30R), single phase electrical outlet will be located 25" behind the front rail of the equipment frame on the wall in the telecommunications room BDF for the owner provided UPS.

h. Two dedicated, 208V, 20A, twist lock (NEMA L6-20R), single phase electrical outlet will be located 25" behind the front rail of the equipment frame on the wall in telecommunications room IDF for the owner provided UPS.

6. Workstation Outlets Requirements:

a. Workstation outlets will be placed in general office space, conference rooms, commons areas, laboratories, classrooms and other facility spaces as required. Special Circuit Requirements:

1) Wireless Access Point location will require one data cable each.

2) Every FACP (Fire Alarm Control Panel) requires two voice cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.

3) Every environmental control cabinet requires two data cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.

4) Every building electrical switch gear meter requires two data cables, terminated in an enclosure outside of the metering cabinet per telecom specifications, and to be installed to the nearest telecommunications room.

5) Every elevator requires two voice cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.

b. Each standard PIC Station will be one 4-11/16" square by 2-1/8" deep flush mounted box with double gang, square drawn extension or tile ring. Provide 1" EMT conduit from the box to cable tray in corridor.

7. Video Wiring System:

a. This system shall provide the signal to the television / display from the Cable Television (CATV) Distribution System.

b. Each outlet location shall be supplied with the required RG-6 or RG-11 horizontal cabling that is connected to a tap/splitter within the IDF serving that area. All new or renovated IDFs will be provided with new RG-11 Riser cabling to new BDF for distribution of CATV signals from the backbone cabling entering the building.

B. Wireless Network Systems for Data:

1. Overview:

a. Wireless network systems shall be engineered by Purdue's Information Technology Networks and Security (ITNS) department during the design phase of the project. ITNS will provide the A&E team with wireless access point locations based on a computer generated site survey for the existing buildings so they may be incorporated into the construction documents. The contractor will be responsible for installing a turn-key wireless system as part of the project. The overall wireless system shall be included in the building project budget.

C. Telephone System and Service:

1. Overview:

a. The telephone system equipment and handsets will be provided by the University.

b. Service to the system will be provided by connecting these facilities to the existing Campus Telecommunications Systems.

c. It is assumed that a majority of the new handset locations within each facility will be Voice over IP (VoIP) rather than traditional digital handsets. The riser and horizontal cabling quantity will reflect this type of handset being installed in all offices, conference rooms, and classrooms as required.

D. Emergency Communications:

1. Overview:

a. Emergency Communications in these facilities will be provided by "Voice Evacuation" capabilities of the Fire Alarm System.

b. It is yet to be determined whether any other emergency phones are required.

E. Paging System

1. No paging system will be provided.

F. Clock System:

1. No clock system will be provided.

2. Battery powered clocks may be installed by Owner in selected locations.

G. Local Area Network (LAN) Electronics:

1. Overview:

a. This includes Hubs, Routers, and Switches required to activate data ports. The University will provide all Network Electronics for these Projects. Space will be allocated in the BDF and IDF rack to accommodate these requirements.

H. Security Systems:

1. Overview:

a. This includes cabling, controllers, card readers and programming and activate the system in each building. The University will provide all cabling, controllers, and card readers for these Projects. Space will be allocated in the Electrical and mechanical rooms to accommodate these systems requirements.

2. Access Control:

a. A new/renovated electronic access control system will be provided. All readers, controllers, and wiring shall be providing, installed, and programmed by the University. The design of the facilities shall include all required rough-ins, conduit, junction boxes, and power to allow for the installation at each opening to be completed.

b. A single gang box with a single gang plaster ring and a 1" EMT conduit to above ceiling cable tray will be provided for each card reader location.

c. All keypads, card readers, cables, security cabinet, etc. shall be provided by the University.

d. During the design phase of the project key openings on the exterior and interior of these facilities will need to be identified that require access control and card readers.

3. CCTV:

a. A CCTV system will be provided to serve these facilities.

b. Interior camera locations will be recorded by the existing IP NVR located on campus to record all new camera locations. It may be determined during design that placing all cameras on the IP system may not be feasible as they may not accessible by staff and faculty, only campus police. In these cases a local DVR and cameras will be provided within the area. The cameras will be recorded by the DVR and will be accessible by the department only.

c. Currently cameras may be required to view and monitor any active lab spaces to ensure the security of ongoing experiments taking place in the spaces.

d. During the design phases other key areas of the facilities will need to be identified that require camera coverage.

e. All exterior camera locations will be provided with a two gang box with a single gang ring and a 1” EMT conduit to nearest above ceiling cable tray. All interior camera location will be provided with a two gang box with single gang plaster ring and 1" EMT conduit to the nearest above ceiling cable tray.

l. Audio Visual Systems:

1. Overview:

a. The Audio Visual (AV) system requirements for these facilities have not yet been fully determined and are not currently included in the design responsibility of the Architect or its consultants. It will be critical that these requirements be determined in the initial phase as they will impact the design of electrical pathways and power systems as well CATV and Network Connections.

b. We expect the AV requirements at a minimum to include technologies such as "way finding" and general information displays in common areas, projectors, and large LCD screens in conference rooms and dedicated audio systems in gathering and presentation areas.

II. SPECIFICATIONS

A. Main Bid Package:

1. Division 27:

a. 27 00 00 Communications (Common Work Results)

b. 27 05 26 Grounding and Bonding for Communication Systems

c. 27 05 28 Pathways for Communications Systems

d. 27 05 53 Identification for Communication Systems

e. 27 11 01 Telecommunications Room Finishes

f. 27 11 16 Cabinets and Frames

g. 27 11 19 Termination Blocks and Patch Panels

h. 27 14 13 Copper Riser Cabling

i. 27 14 23 Fiber Riser Cabling

j. 27 14 33 Coaxial Riser Cabling

k. 27 15 13 Copper Horizontal Cabling

l. 27 15 33 Coaxial Horizontal Cabling

m. 27 15 43 Faceplates and Connectors

The executive summary and calculated arc flash data follow. This is just a summary; calculated data and the full report are provided under separate cover.

PURDUE UNIVERSITY

Arc Flash Risk Assessment with Fault-Current and Protective Device Coordination Analysis

ME-EAST
As found Study Report

Applied#15-051
June 30, 2015

Applied Engineering Services

PURDUE UNIVERSITY

ARC FLASH REPORT

Table of Contents

1 Executive Summary .....
2 Calculated Arc Flash Data.....
3 Single Line Diagrams .....
4 TCC Plot and Device Settings.....

1 Executive Summary

Warning

The PPE requirements, when working on the line side of main disconnects, busswork and cabling between main transformers, main disconnects, and the connections within main transformers are not affected by the recommendations made in this report. That part of the system is protected by devices on the input side of the transformers that are normally owned and set by the utility.

- All of the panelboards and switchboards fall within NFPA 70E limits for PPE (personal protective equipment) categorized as Category 0, 1, or 2 unless noted below. Refer to **Section 2** of this report for a table describing the categories and results of the arc flash hazard calculations.
- The rated fault current for all busses is sufficient to clear a bolted fault unless noted below. Refer to **Section 4** of this report for a table describing the results.
- The rated fault current for all protective devices is sufficient to clear a bolted fault unless noted below. Refer to **Section 4** of this report for a table describing the results.

Findings

In the following table, the calculated incident energy is above 8.0 cal/cm^2 and PPE Category is 3 or above at buses. **Purdue University has provided available low and high fault current.**

ME-EAST

- Table – Shows Bus PPE Value 3 or higher

Bus Name	Protective Device Name	Bus kV	Bus Bolted Fault (kA)	Bus Arcing Fault (kA)	Arc Flash Boundary (in)	Incident Energy (cal/cm2)	PPE Level
MDP	Xfmr ME	0.480	16.81	8.69	90	17	Category 3
MDP-L	Xfmr L	0.240	6.85	3.48	104	21	Category 3

Recommendation

ME-EAST

- Provide new main breaker switchboard IM-MDP between transformer & MDP, new breaker with electronic trip unit (LSIG).
- Replace fuse switch MDP-DP in MDP with new circuit breaker with electronic trip unit (LSI) or reduce fuse size to 500 ampere.
- In existing fuse switchboard MDP-L, replace main switch (400a) with circuit breakers with electronic trip unit (LSI), MDP-L may be replace to accomplish this.

2 Calculated Arc Flash Data

The table included in this section is the screen displayed at the conclusion of the SKM arc-flash analysis. It was printed as a pdf file and merged into the report.

The arc-flash hazard analysis is done to conform to OSHA recommendations described in 29CFR 1910.333. Procedures for arc-flash hazard analysis are described in both NFPA Standard 70E-2009 and IEEE Standard 1584-2002. IEEE Standard 1584-2002 describes a method for calculating the potential arc-flash energy based on the anticipated “bolted” fault current. The results reported within have been calculated using the more conservative, interpretation of IEEE Standard 1584-2002.

IEEE Standard 1584-2002 states that equipment rated below 208V does not need be considered unless it involves a low-impedance transformer larger than 125 kVA for its immediate supply. The reason for this exclusion is the inability of such small capacity systems to initiate and support arcing conditions. NFPA Standard 70E-2009 applies the same reasoning to equipment rated 208V and less. NFPA Standard 70E is recognized as the guiding document.

The table beginning on the next page shows arc-flash data calculated. Included in the tables are the critical distances, the incident energies, the fault currents, and the PPE category level required for the indicated working distance. Arc-flash hazard warning labels for the equipment are created from this table.

If future changes are made in the electrical system or to the protective device settings, new calculations should be completed on the specific circuits and new warning labels printed to reflect any changes in the hazard.

- Arc Flash value as describe in table above and shown on one line drawing

3 Single-Line Drawings

The single-line drawings that follow show the system topology, the input, and calculated data for the facility. It shows circuit breaker and/or fuse data, cable data, transformer data, motor data, source data, calculated fault current at bus, and calculated arc flash data including incident energy, boundary distance and PPE category. Source fault current and X/R ratio are obtained from Purdue University.

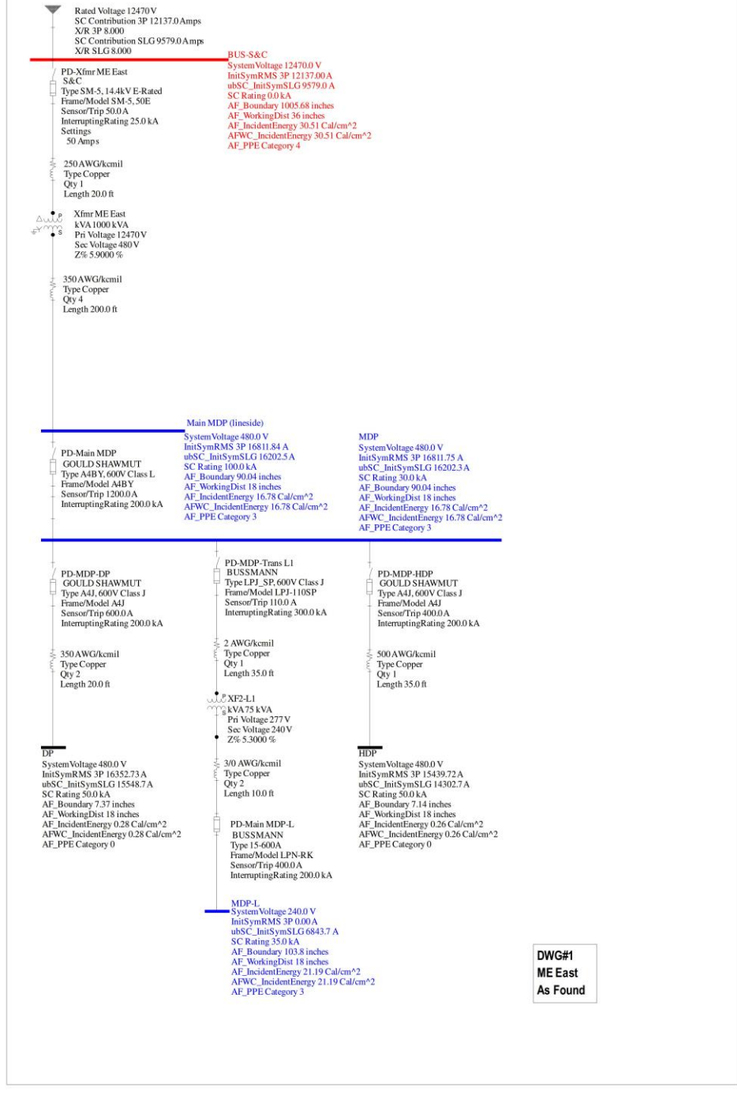
Drawing List

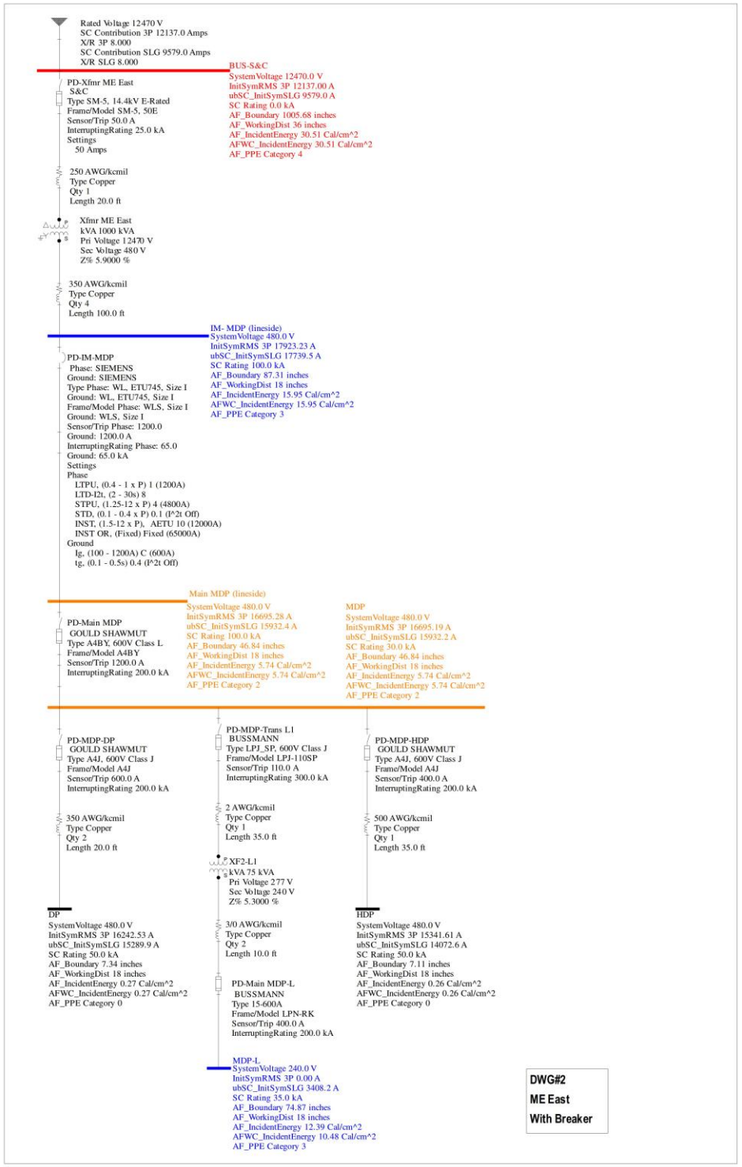
ME-EAST

DWG #1 Fuse	One line drawing w/fuse
DWG #2 BKR	One line drawing w/Breaker

Purdue University
ME-EAST
As found Study Report
Applied Project No. 15-051

June 30, 2015





4 TCC Plots and Device Settings

The following coordination curves graphically depict the time versus current characteristics for the overcurrent protective device shown, and the relationship of each device to the other devices that are in electrical series.

An ideally coordinated system would have the overcurrent protective device closest to a short circuit, or fault, condition detect the fault and open the circuit. In the event the closest device fails to operate, the next device between the fault and the power source would detect the fault and open the circuit. Realistically, the ideal is often not achievable, but an attempt is made to come as close as practical.

Each set of plotted curves shows at least two protective devices.

All recommendations presume the protective devices will be properly tested and maintained to operate in accordance with the manufacturer's specifications.

The page(s) following set of curves present(s) the pertinent data and settings for each protective device represented in the system. All recommendations presume the protective devices have been, and will continue to be, well and properly maintained to operate in accordance with the manufacturer's specifications.

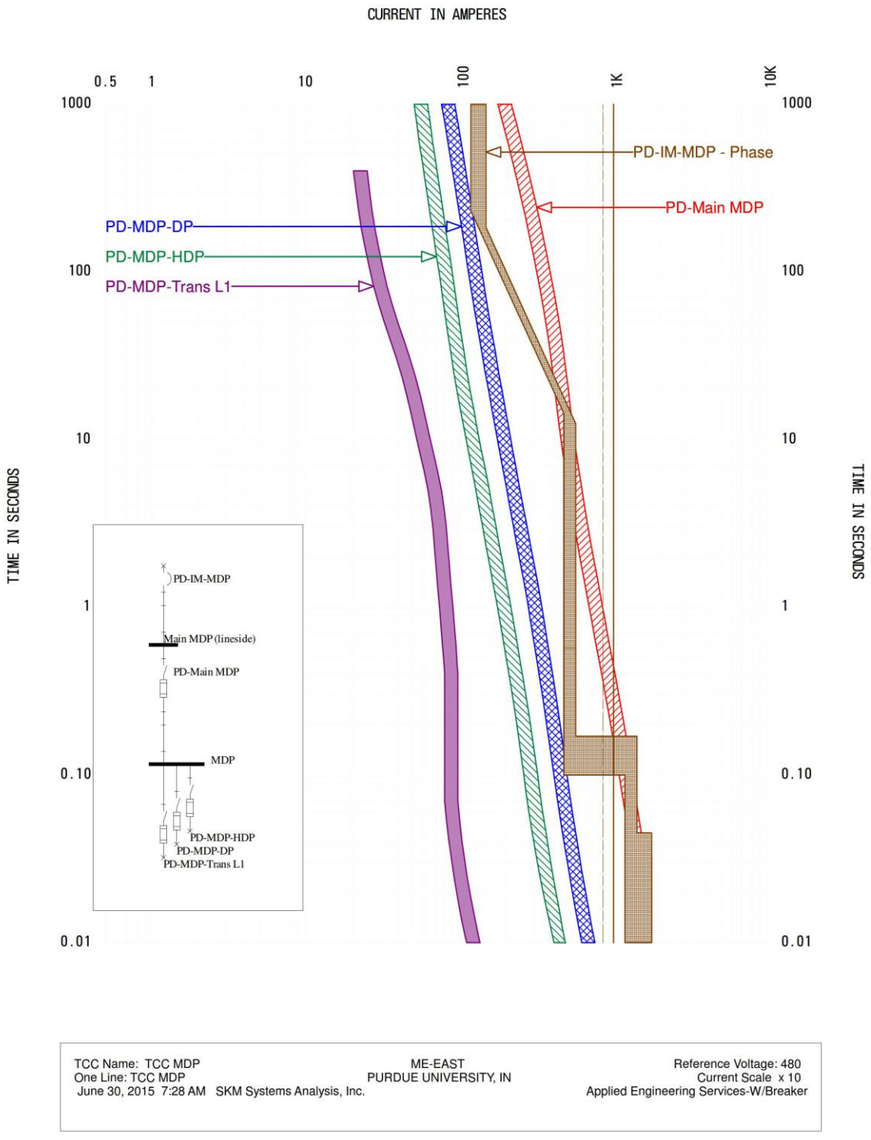
TCC Plot:

ME-EAST:

- TCC MDP

Purdue University
ME-EAST
As found Study Report
Applied Project No. 15-051

June 30, 2015



TCC Name: TCC MDP
One Line: TCC MDP
June 30, 2015 7:28 AM SKM Systems Analysis, Inc.

ME-EAST
PURDUE UNIVERSITY, IN

Reference Voltage: 480
Current Scale x 10
Applied Engineering Services-W/Breaker

Existing Plans

The study encompasses the complete American Railway (RAIL) Building.

The total gross square footage of the renovated space is approximately 2,500 GSF, with the following breakdown by floor.

First Floor .....1,250 GSF
Second Floor.....1,250 GSF

As a brief summary, we are showing approximately 2,072 NSF of proposed office space primarily housing graduate students. These NSF calculations do not include support, toilet rooms or circulation.

Existing Plans – RAIL First Floor

TRUE NORTH PROJECT NORTH

Existing First Floor Plan

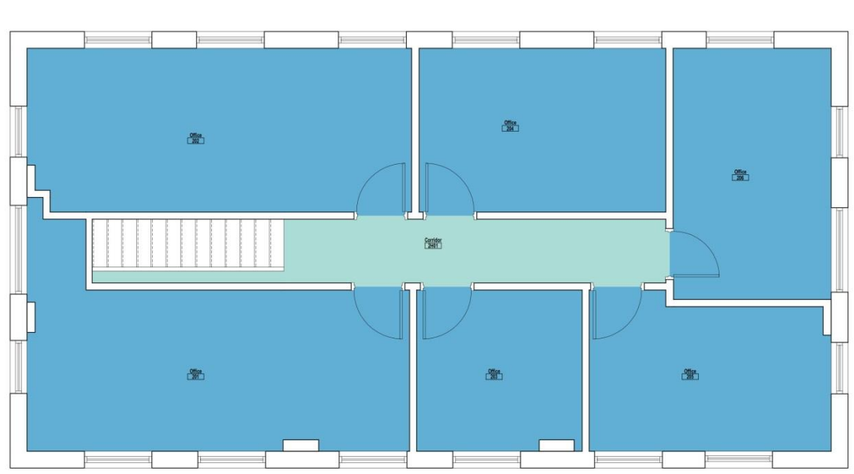
3/8" = 1'-0"

PURDUE UNIVERSITY

American Railway Building

MSKTD & Associates, Inc.

Existing Plans – RAIL Second Floor



Existing Second Floor Plan

PURDUE UNIVERSITY

American Railway Building

MSKTD & Associates, Inc.

EXISTING SPACE USE

Existing Space Use

The following spreadsheets are similar to standard forms used by architects and engineers to quantify and describe space use in a typical project. These sheets identify current/existing room names, room numbers, area or assignable square footage of rooms, departmental assignment, quantities of faculty, staff or graduate assistants accommodated, and other comments regarding space usage and adjacencies.

The room data information originated from the Space Management department (SMAS) at Purdue University, and the following sheets contain modified information that was confirmed with the Mechanical Engineering department to validate current space usage and existing quantities. The following information on the existing spaces provides a snapshot in time to understand graduate student offices and open-office workstations accommodated in RAIL.

As RAIL is approximately 2,500 square feet, the existing program summary page that follows is combined with the program summary information included with the Mechanical Engineering (ME) building as part of the overall study documentation. The Mechanical Engineering department currently uses the space within RAIL for graduate student workstations and related collaborative space.

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Purdue University ME/RAIL/HAMP Study

Mechanical Engineering (ME) and American Railway Building (RAIL)

EXISTING PROGRAM SUMMARY OF FACULTY, STAFF AND SPACE COUNTS											
	Basement	**Ground Floor**	**First Floor**	**Second Floor**	**Third Floor**	**Fourth Floor**		**Current Quantities**	*GATE*	*TOTAL*	
TEACHING FACULTY											
Mechanical Engineering (ME)	N/A	9	13	2	22	0		**46**	31	77	
American Railway (RAIL)	N/A	N/A	N/A	N/A	N/A	N/A		**N/A**			
COLLEGE/DEPARTMENT STAFF											
Mechanical Engineering (ME)	N/A	6	15	25	3	0		**49**	10	59	
American Railway (RAIL)	N/A	N/A	N/A	N/A	N/A	N/A		**N/A**			
GRADUATE ASSISTANTS											
Mechanical Engineering (ME)	N/A	20	26	0	61	0		**107**	60	167	
American Railway (RAIL)	N/A	N/A	7	19	N/A	N/A		**26**	---	---	

PROPOSED SUMMARY OF FACULTY, STAFF AND SPACE COUNTS (Based on CoE Master Plan Document) - Mechanical Engineering							
	Mechanical Engineering (ME)						
	Quantity	*Workstation Size (SF)*	*SF Info*				*Comments*
FACULTY	**74**		**9,000**				
Academic Department Head	1	180	180				
Academic Associate Head / Asst. Head	2	150	300				
Tenured/TT Faculty	66	120	7,920				
Instructor/Lecturer	4	120	480				
Emeritus/Emeritas	1	120	120				
OTHER *(Wkstns)*	**19**		**1,710**				
Adjuncts or Joint Appointments	10	60	600				
Visiting Scholars	1	630	630				
Research / Visiting Faculty	8	60	480				
STAFF *(Offices)*	**19**		**10,320**				
Professional Staff	12	120	1,440				
Business Manager	1	120	120				
ECN IT Staff	2	120	240				
Management	4	120	480				
STAFF *(Wkstns)*	**48**		**3,420**				
Post Doctoral Assistants	14	90	1,260				
Business Staff	7	90	630				
Admin/Operations Staff	6	90	540				
Clerical	11	90	990				
Service Staff	10	120	1,200				
GRADUATE ASSISTANTS	**169**		**6,760**				
Grad - Thesis	169	40	6,760				0.7 desk / grad
Subtotal	**329**		**31,210**				

ME/RAIL/HAMP Study
ME and RAIL

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other			
RAIL - First Floor													
Women's Restroom	101	56		Gen Building	Gen Building	REST ROOM					Women's Restroom - 1WC,1L-HCA		
Office Serv	102	150	Office Service - Duvall	Mech Engr	Mech Engr	OFF SERV					Conference Room		
Men's Restroom	103	75		Gen Building	Gen Building	REST ROOM					Men's Restroom - 1WC,1U,1L-HCA		
Mechanical	105	22		Gen Building	Gen Building	MECHANICAL					Water Heater + Service Sink		
Office	106	105		Mech Engr	Mech Engr	OFFICE			2		Graduate Assistants		
Office Serv	107	96		Mech Engr	Mech Engr	OFF SERV							
Office	108	148		Mech Engr	Mech Engr	OFFICE			3		Graduate Assistants		
BDF	109	97		Gen Building	Gen Building	MECHANICAL							
Office	110	126		Mech Engr	Mech Engr	OFFICE					Graduate Assistants		
Corridor	1H01	241		Gen Building	Gen Building	CIRCULATE			3				
Stair	1S01	49		Gen Building	Gen Building	CIRCULATE							
Subtotals		**1,165**	**SF**				0	0	8	0			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other			
RAIL - Second Floor													
Office	201	258		Mech Engr	Mech Engr	OFFICE			4		Graduate Assistants		
Office	202	241		Mech Engr	Mech Engr	OFFICE			4		Graduate Assistants		
Office	203	103		Mech Engr	Mech Engr	OFFICE			2		Graduate Assistants		
Office	204	157		Mech Engr	Mech Engr	OFFICE			3		Graduate Assistants		
Office	205	139		Mech Engr	Mech Engr	OFFICE			3		Graduate Assistants		
Office	206	152		Mech Engr	Mech Engr	OFFICE			3		Graduate Assistants		
Corridor	2H01	92		Gen Building	Gen Building	CIRCULATE					Main Corridor		
Stair	2S01	46		Gen Building	Gen Building	CIRCULATE					Stairway		
Subtotals		**1188**	**SF**				0	0	19	0			

SYSTEMS NARRATIVES
(Existing and Proposed)

Architectural Narrative

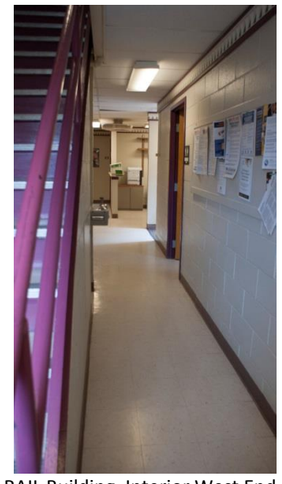
The American Railway (RAIL) building was originally built in 1926 and has received more recent renovations and updates to accommodate current needs for graduate student workspaces. RAIL is a two-story building, with offices and building support spaces such as restrooms, mechanical / electrical space, and telecommunications (BDF) room on the first floor. The second floor is currently subdivided into multiple, small offices shared by graduate students.

Renovations and proposed increases in efficiencies and workspace quantities for RAIL may be considered in conjunction with those in the adjacent Mechanical Engineering (ME) building, as these spaces are used by the Mechanical Engineering department

RAIL Building, West Elevation

One of the early considerations and discussions for RAIL relates to accessibility and circulation, including circulation in and out of the building, as well as vertical circulation within the building. Recent concrete paving and site improvements on the west side of the building appear to address accessibility considerations for the building, and a similar ramp could be considered at the entrance on the south side of the building to improve access. Within the building, vertical circulation is limited to the existing stair at the west-side entrance. Due to the limited building size, there is no elevator, so accessibility is limited.

In addition to circulation, building services and other support spaces within the building were considered as part of the proposed renovations. For example, space for HVAC equipment is limited in the current plan configuration, so consideration was given to a plan configuration to provide additional system cooling equipment to eliminate the need for window-mounted room air conditioning units. See the following MEP/T narratives for additional information and considerations.


RAIL Building, Interior West End

Existing materials and systems within the building may contain asbestos or other related materials. Purdue University should provide consideration for investigation and abatement processes as these are not included as part of the study documentation.

Various design solutions and considerations were discussed during team meetings, with an emphasis placed on efficiency, space adjacencies, and overall building organization and circulation. Proposed plan drawings are included on the following pages showing the final layout considered.

FIRST FLOOR

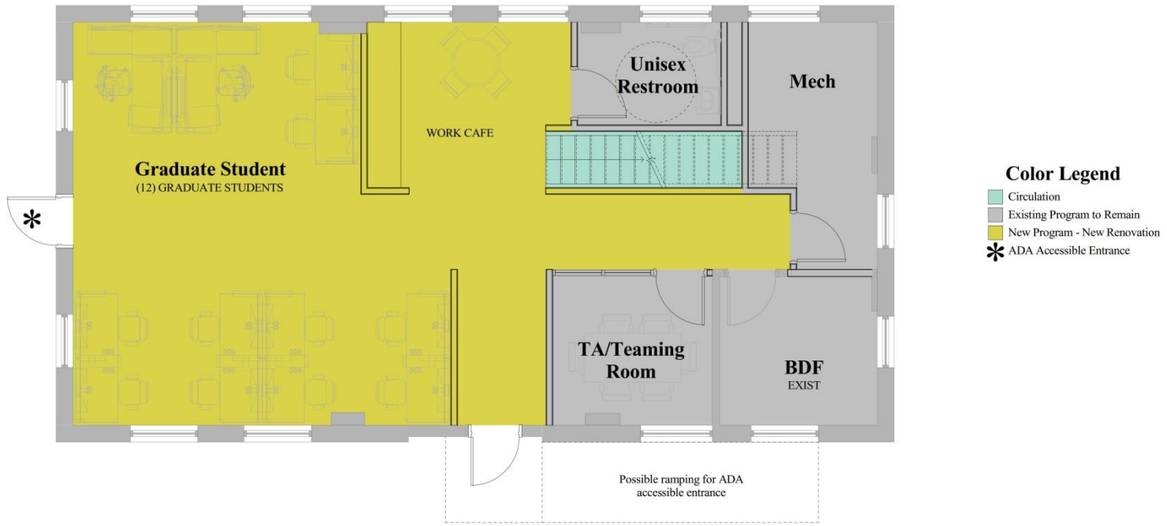
A primary consideration for the building layout was the vertical circulation and stair location. Based on input from reviews, the stair is proposed to be relocated to provide additional space at the west entrance, as well as consolidate building support functions at the east end of the building.

At the central part of the plan, collaboration spaces, including a work café and small meeting space, are planned between the support space functions and open office functions shown. Another alternative for this floor, at the west end, includes open collaboration and TA teaming spaces. A similar function was integrated into the west wing/annex at Grissom Hall.

SECOND FLOOR

Based on the stair reconfiguration and removal of the smaller offices and wall partitions, additional open-office workstations can be accommodated on the second floor of rail. Workstations can vary in size and configuration to accommodate resident graduate student workstations, as well as mobile workstations.

Architectural Narrative – Proposed Plans – RAIL First Floor



First Floor Plan - Proposed

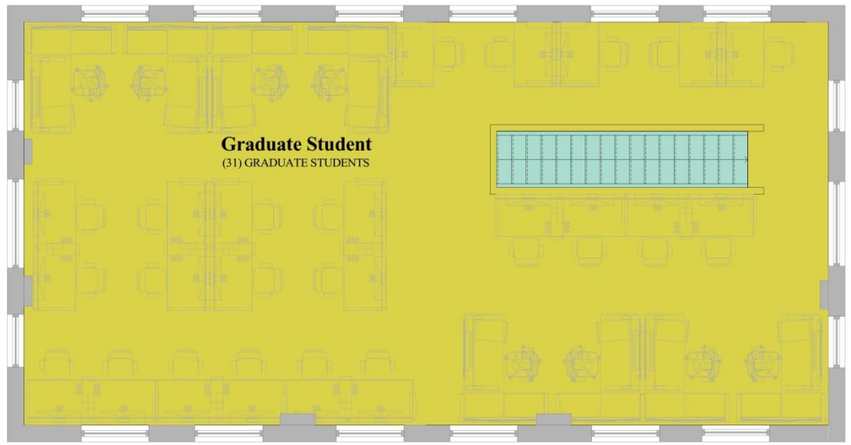
TRUE NORTH PROJECT NORTH

PURDUE UNIVERSITY

American Railway Building

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – RAIL Second Floor



TRUE NORTH PROJECT NORTH

Second Floor Plan - Proposed

3/8" = 1'-0"

PURDUE UNIVERSITY

American Railway Building

MSKTD & Associates, Inc.

Mechanical Narrative

This narrative describes the existing building mechanical systems, and the proposed changes to those systems. These changes are anticipated to be implemented as part of a multi-phase building renovation project.

The mechanical systems will be designed at a minimum in accordance with the following codes and standards as applicable:

1. Applicable Codes:
    a. 675 IAC 13 – Building Codes
        i. 2014 Indiana Building Code
        ii. International Building Code, 2012 Edition
    b. 675 IAC 16 – Plumbing Codes
        i. Indiana Plumbing Code, 2012 Edition
        ii. International Plumbing Code 2006 Edition
    c. 675 IAC 18 – Mechanical Codes
        i. Indiana Mechanical Code, 2014 Edition
        ii. International Mechanical Code, 2012 Edition
    d. 675 IAC 19 – Energy Conservation Codes
        i. Indiana Energy Conservation Code, 2010 Edition
        ii. ASHRAE Standard 90.1-2007 – Energy Standard for Buildings Except Low-Rise Residential Buildings.
    e. 675 IAC 22 – Fire Prevention Codes
        i. Indiana Fire Code, 2014 Edition
        ii. International Fire Code, 2012 Edition
2. Applicable Standards:
    a. NFPA 101 – Life Safety Code
    b. Purdue University Consultants Handbook

American Rail Building Existing Systems

The existing building was constructed in 1926.

The building has gone through two renovations. The first renovation was in 1950 and the second renovation was in 1979. A summary of existing system components is as follows:

1. In 1950 a direct expansion type air handling unit was installed on the second floor.
2. In 1979 the air handling unit was removed, and replaced with twelve window air conditioning units to provide cooling for the building.
3. Steam System – Low pressure (15 PSI) campus steam is delivered to fin tube radiators throughout the building.
4. Domestic Water – Domestic hot water is generated via an electric water heater located in a janitor's closet on the first floor.
5. Sanitary Waste and Vent Systems are present for the main toilet group.
6. Fire Suppression – The existing building is not currently sprinkled.

Proposed Mechanical Scope

This narrative proposes a replacement of the existing airside HVAC systems within the building. A direct digital control system will be provided for HVAC system control and monitoring. This narrative assumes a new wet pipe fire suppression system will be added.

Due to the extent of the proposed renovations the entire building will have to be vacated during construction.

Proposed HVAC System

The existing system of perimeter steam radiation and window air conditioning units will be replaced with newer more reliable equipment.

The design team proposes replacing the existing system units with two new packaged single zone direct expansion rooftop units. The system would also utilize a new steam to hot water heat exchanger to provide hot water for perimeter radiation. Hot water duct coils would be provided to temper the mixed air from the packaged rooftop unit before delivering it to the space. The components of the proposed system are:

Proposed Air Side Equipment Systems

1. Two packaged direct expansion cooling only rooftop units.
2. Split system air conditioning unit to serve the IT room.
3. Toilet Exhaust Fan.

AIR SIDE EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
ACR-1	Packaged RTU	5 Tons	460 V
ACR-2	Packaged RTU	5 Tons	460 V
FCU-X (x 1)	Split System AC	12,000 BTUH	120V/Fractional HP
TEA-1	Toilet Exhaust Fan	300 CFM	460 V/ 1/2 HP

Proposed Steam and Hydronic System

The existing steam perimeter finned tube radiators will be removed, and new hot water finned tube radiators provided.

A heat exchanger and heating water pumps will be required for the hydronic heating system.

A summary of the hydronic equipment is listed below.

STEAM AND HYDRONIC EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
HX-1	Shell and Tube Heat Exchanger	10 GPM/100 lb/hr	-
HYPB-1	Hydronic Heating Water Pump	10 GPM/40 FT	460V/1 HP
HYPB-2	Hydronic Heating Water Pump	10 GPM/40 FT	460V/1 HP

Proposed Building Automation Modernization

A new direct digital control building automation system will be provided. Actuators will be electrically actuated. The system will be either an Automated Logic or a Siemens system, and the front end shall be accessible at the campus control center.

Proposed Plumbing Upgrades

New sanitary and waste mains will need to be added as well as a new piping chase for the restroom. New toilet fixtures to be ADA compliant.

New domestic water mains will need to be added for the restroom.

Domestic hot water for the new restroom lavatory will be supplied via a new electric instantaneous water heater installed below the sink.

Proposed Fire Suppression Upgrades

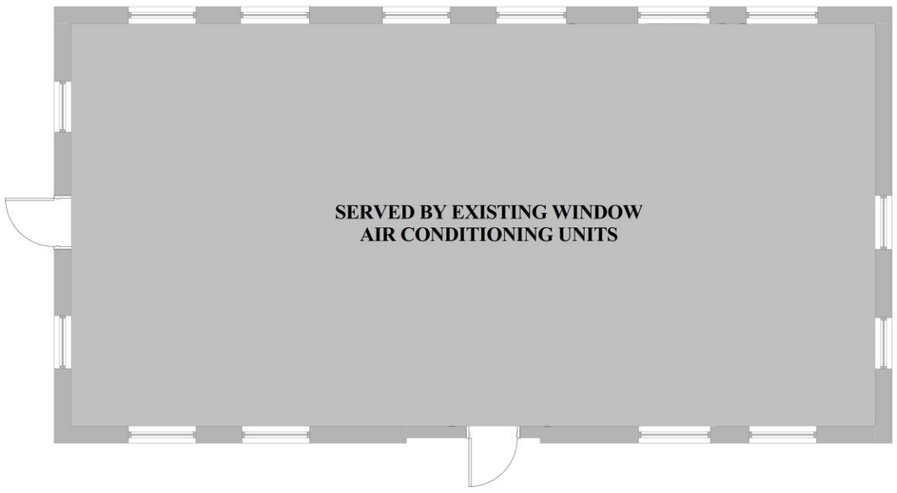
This study assumes the owner will elect to install an NFPA-13 fire suppression system in the building as part of an overall renovation. A new fire service to the building will be provided.

Attached Drawings

The following drawings are attached to help convey the written design intent.

1. American Railway Building - Existing First Floor Plan
2. American Railway Building – Existing Second Floor Plan
3. American Railway Building – Proposed First Floor Plan
4. American Railway Building – Proposed Second Floor Plan

Mechanical Narrative – Existing Plans – RAIL First Floor



Existing First Floor Plan - Mechanical

PURDUE UNIVERSITY

American Railway Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – RAIL Second Floor

TRUE NORTH PROJECT NORTH

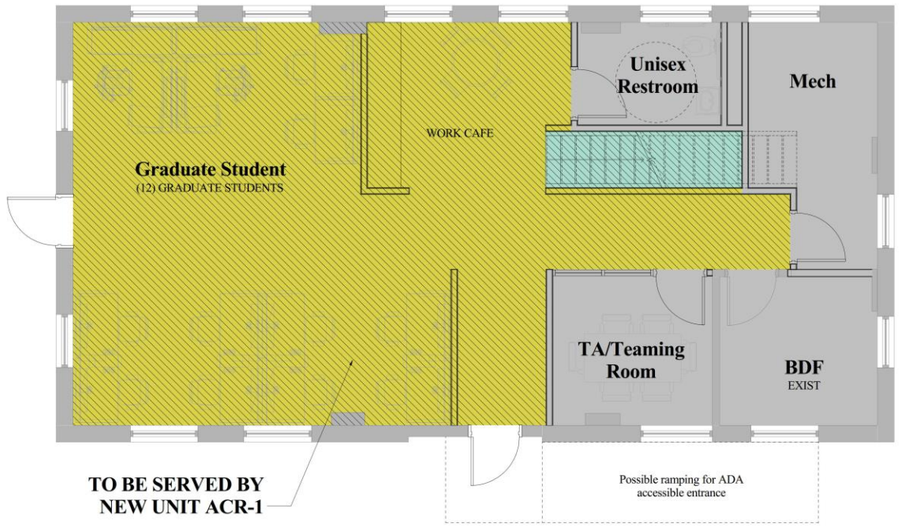
Existing Second Floor Plan - Mechanical

PURDUE UNIVERSITY

American Railway Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – RAIL First Floor



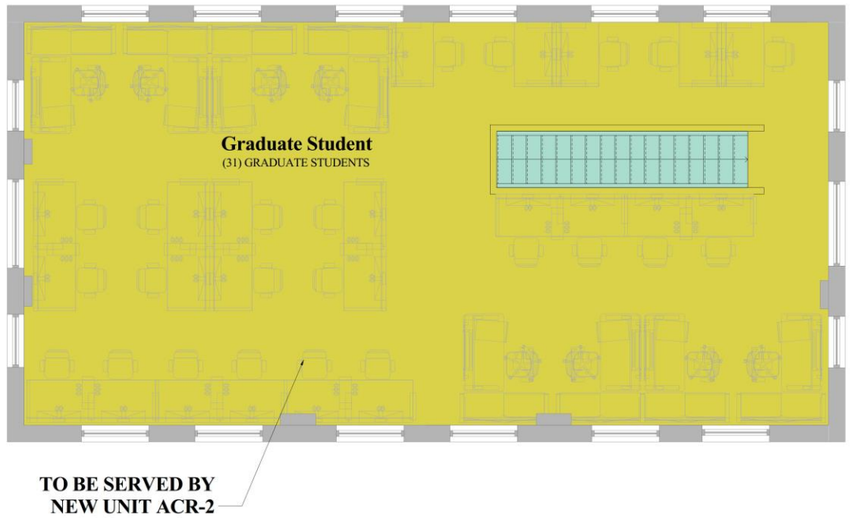
Proposed First Floor Plan - Mechanical

PURDUE UNIVERSITY

American Railway Building

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – RAIL Second Floor



Proposed Second Floor Plan - Mechanical

PURDUE UNIVERSITY

American Railway Building

LOFTUS ENGINEERING, INC.

Electrical Narrative

This narrative describes the existing electrical systems for the American Railway Building, and the proposed changes to those respective systems. These changes are anticipated to potentially be implemented as part of multi-phased building renovation projects.

The electrical systems will be designed at a minimum in accordance with the following codes and standards as applicable:

1. Applicable Codes:
    a. 675 IAC 13 Building Codes.
        i. 2008 Indiana Building Code.
        ii. International Building Code, 2006 Edition.
    b. 675 IAC 17 – Electrical Codes.
        i. Indiana Electrical Code, 2009 Edition.
        ii. NFPA 70 – National Electrical Code, 2008 Edition.
    c. 675 IAC 19 – Energy Conservation Codes.
        i. 2010 Indiana Energy Conservation Code.
        ii. Energy Standard for Buildings Except Low-Rise Residential Buildings ANSI/ASHRAE 90.1, 2007 Edition.
    d. 675 IAC 22 – Fire Prevention Codes
        i. Indiana Fire Code, 2008 Edition
        ii. International Fire Code, 2006 Edition
    e. NPFA 72 – National Fire Alarm Code

2. Applicable Standards:
    a. NFPA 101 – Life Safety Code
    b. Purdue University Consultants Handbook

Power Distribution Systems – Existing Conditions

Electrical power to the existing facility is presently provided with 120/240 volt, 1-phase 3-wire electrical power. Power originates at a switchboard in the existing Mechanical Engineering Building and feeds an existing panelboard located on the first floor. The existing panelboard supplies all electrical power for lighting, receptacles, and cooling for the facility.

Power Distribution Systems – Renovations

The existing power distribution system for the facility with be completely replaced. The existing main panelboard and associated 120/240 volt feeder that originates in the existing Mechanical Engineering Building will be removed. A new main branch circuit panelboard will be provided and located in the Mechanical/Electrical Room on the First Floor. This new panelboard will provide power for all lighting, receptacle, cooling and plumbing loads. The new feeder circuit for the new panelboard will receive is power from an existing spare fusible switch unit located in existing sub-distribution switchboard "SDP-LL" in the basement of the Gatewood Addition to the Mechanical Engineering Building. The new panelboard will provide 120/208 volt, 3-phase, 4-wire electrical service to the renovated building.

Emergency Power Systems – Existing System

Based on examination of the drawings for the existing facility, there does not appear to be any existing emergency lighting provided for the building.

Emergency Power Systems – Renovations

A small central battery inverter system (EMAC) system or individual battery units will be provided in lighting fixtures for all emergency lighting, egress lighting and exit signage. The Fire Alarm System will be provided with its own integral battery emergency power source.

Power Monitoring

Since power for this building will originate at the Gatewood Addition to the Mechanical Engineering Building, it is not anticipated that metering will be provided for the power distribution system. The existing electrical services to Gatewood are presently metered at the main distribution switchboards.

Interior Lighting

Lighting will be designed utilizing a combination of direct, indirect, and direct/indirect luminaires utilizing Light Emitting Diode (LED) technology as the prime illumination source. The exact type of luminaire to be utilized in each respective space will depend on room layout, function ceiling type, aesthetic requirements, and control requirements, among other factors.

In general, lighting will be designed per the requirements contained in the Purdue University Consultant's Handbook and supplemented by Illumination Engineering Society of North America (IESNA) standards with particular attention paid to the following publications:

1. ANSI/IESNA RP-1 – American National Standard Practice for Office Lighting.
2. ANSI/IESNA RP-3 – Lighting for Educational Facilities.
3. The IESNA Lighting Handbook – Ninth Edition.
4. ANSI/IESNA RP-31 – Recommended Practice for the Economic Analysis of Lighting.

Lighting luminaires will be selected based on:

1. Suitability for use in the respective area in which they will be installed.
2. Energy efficiencies required to obtain goals for lighting power densities.
3. Maintenance requirements.

Average lighting levels for each specific building space will be designed in accordance with the following requirements. All illumination level are horizontal values referenced to a work plane located 30 inches above finished floor unless noted otherwise.

1. TA/Teaming Rooms: 40 footcandles with full range dimming.
2. Offices: 25-30 footcandles general illumination with owner furnished supplemental task lighting.
3. Restrooms: 20-30 footcandles general illumination with 45 footcandles vertical illumination at the mirror.
4. Corridors: 10-15 footcandles at the floor level.
5. Mechanical and Electrical Rooms: 15-25 footcandles at floor level.
6. Stairwells: 10-15 footcandles maximum, 1-5 footcandles minimum
7. All Other Areas: Default to IESNA recommended standards.

Luminaires utilizing Light Emitting Diode (LED) technology will be provided wherever possible based on the following criteria:

1. Suitability for use in the respective area in which they will be installed.
2. Suitability to render color with proper color temperature.
3. Efficacies required to obtain goals for lighting power densities.
4. Life and maintenance requirements.
5. Lighting control requirements.

All light emitting diode (LED) lighting luminaires shall utilize electronic drivers.

Emergency and egress lighting will be provided in accordance with relevant codes by means of a combination of "Nightlight" circuits and switched circuits connected to the respective building's emergency power source.

Illuminated exit signage will be provided in accordance with relevant codes and will be connected to the respective building's emergency power source.

Exterior Lighting

Exterior lighting will be designed and limited to utilize building mounted luminaires. Exterior lighting will be designed to meet the exterior lighting power densities as prescribed by Indiana Energy Conservation Code, where applicable.

Sources for exterior lighting environments will primarily be Light Emitting Diode (LED) technology, as determined during design, and per campus standards. Luminaires will generally be constructed from cast aluminum material. All luminaires will meet IESNA "Full Cutoff" criteria.

Lighting Controls

Various degrees of lighting controls will be provided in the facility to provide functionality, optimize energy performance, and to comply with the requirements of the Indiana Energy Conservation Code. It is anticipated that this facility will utilize a combination of automatic occupancy sensor lighting controls, networkable relay based lighting control systems, daylight harvesting systems and dimming systems.

Unit type occupancy sensors with manual "on" switch will be utilized in all small offices, storage rooms, restrooms, and similar spaces. These controls will be designed such that the occupant of the room must turn the lights on via a switch located at the room entrance and the occupancy sensor will automatically turn off the lights in the respective space if it determines the space is unoccupied.

Relay based lighting control systems will be provided for control of lighting in common areas such as corridors, open office areas, large open workroom areas and similar spaces. The system will provide manual, time-of-day scheduling, and occupancy sensor control to the respective areas. The systems could be designed to be networked and/or stand alone, as determined during the design process.

Daylight harvesting systems will provide automatically adjustable full range dimming based on ambient lighting levels. Daylight harvesting will be provided in appropriate open office areas and open workroom areas along the exterior walls of the building where there will be the possibility of a large natural light contribution due to existing windows.

Architectural dimming systems will be provided for dimming control of luminaires in the meeting rooms, conference rooms, classrooms, lecture halls and multi-purpose rooms. The system will provide full range dimming capability of multiple sources and zones as required via local control stations located in the respective spaces. The exact features and requirements of each architectural dimming system will be determined during design.

Mechanical rooms, electrical rooms, and telecommunications rooms will be provided with standard manually operated switches for control of the lights in the respective spaces.

Audio/Visual Systems

Provisions for Audio/Visual systems will be provided for the following areas of the building, as determined through meetings with Purdue University Technology personnel:

1. TA/Teaming Rooms.

The building will contain provisions to support overhead projectors and/or flat-screen televisions in the spaces mentioned above. The provisions will include rough-in and conduits, power and mounting provisions required to support overhead projectors or televisions. Exact requirements will be determined during design by consulting with Purdue University. It is assumed that the owner will furnish all overhead projectors. Other requirements and features will be incorporated into the design, as directed by Purdue University.

Fire Alarm System

A fire alarm system will be provided as required by renovations of spaces within the building. The system will fully comply with the Americans with Disabilities Act (ADA), Indiana Fire Code, and all other state codes.

The system will be an intelligent, addressable type system. The complete system will include all required manual and automatic initiating devices such as pull stations, smoke detectors, heat detectors, water flow switches, etc. The complete system will include all signaling devices such as audible/visible signals, visible only signals, sprinkler alarm bell, etc. The system will monitor sprinkler system actuation. The system will monitor and control smoke evacuation system functions, fan shutdown functions, door hold open release functions, and elevator recall functions.

Exact specifications for the complete system will be entirely coordinated with Purdue University's Safety and Security Department.

Card Access and Security Systems

It is anticipated that some degree of access control and security will be required for this project. The extent of this scope has not been identified at this phase of the project but it is anticipated that some degree of access control will be provided to limit access to the second and third floors. The requirements for these systems will be discussed with appropriate Purdue University staff, but at a minimum, rough-in and conduits will be provided to accommodate system equipment and cabling requirements.

Basic Materials

All wiring and cabling for power distribution, branch circuitry, low voltage systems, etc. will be routed throughout the facility utilizing conduits.

Separate closed conduit systems will be provided for each power distribution system and each low voltage system such as:

1. Normal Power.
2. Emergency Power.
3. Equipment Power.
4. Audio/Visual Systems.
5. Fire Alarm System.
6. Card Access and Security Systems.
7. Low Voltage and Digital Lighting Control Systems.

In general conduit types will be as follows:

1. Electrical Metallic Tubing (EMT): Permitted for branch circuit conduits and conduits smaller than 1-1/4" trade size.
2. Rigid Metal Conduit (RMC) - Steel: Required for all feeders, and conduits larger than 1-1/4" trade size.
3. Intermediate Metal Conduit (IMC): Permitted in lieu of Rigid Metal Conduit where located within the building above slab on grade.
4. Polyvinyl Chloride (PVC): Permitted below grade for site lighting circuitry only.
5. Flexible Metal Conduit (FMC): Required for final connections to motorized equipment, lighting fixtures and equipment subject to vibration.
6. Liquidtight Flexible Metal Conduit (LFMC): Required in lieu of Flexible Metal Conduit in damp or wet locations.

All 600V conductors will be copper material Type THHN/THWN insulation.

All panelboards will be furnished with 1000A/in^2 plated copper bus bars.

Overcurrent protection for all switchboards will be provided by either solid state low voltage insulated case circuit breakers or solid state molded case circuit breakers.

Panelboards will be furnished complete with bolt-on circuit breakers and hinged fronts.

All wiring devices will be heavy duty grade. Switches will be rated for 20A, 120/277 VAC. Receptacles will have integral brass strap and be rated for 20A, 120VAC. Plates will be high impact nylon or stainless steel as required.

Seismic Restraint Requirements

Seismic restraints are not expected to be required for this project.

Arc Flash Study

Since this building will derive its electrical service from the existing Gatewood Addition to the Mechanical Engineering Building power distribution system, it is not anticipated that Arc Flash Mitigation will be provided as part of any subsequent construction project. Arc Flash warning labels will be provided on the new panelboard as determined by any Arc Flash Studies that Purdue University has performed in the past. All requirements for labelling will be closely coordination with Purdue University.

Technology Narrative

I. TECHNOLOGY

A. Telecommunications Cabling Infrastructure:

1. Overview:

a. The systems referenced within this study pertain to all three facilities.

b. This infrastructure will provide support for Voice, Data and Cable TV distribution systems and services. In addition it will support interconnection of control panels for other ancillary building systems such as Fire Alarm, Electronic Security and Building Automation.

c. The Cabling Systems will be designed utilizing the Purdue ITaP guidelines and specifications as outlined in the Consultants Handbook.

d. It will be assumed that all offices, conference rooms, and other appropriate spaces shall be equipped with telephone circuits and one or more computer workstations. These will be connected via separate intra-building networks as well as separate inter-building networks. Currently, the telephone circuits, including dedicated 2-wire and 4-wire, security, fire, etc. operate on a copper backbone. The data network operates on copper from the workstation to the telecommunication room and then is connected via fiber optic cable to the Purdue fiber optic backbone.

e. CATV and Purdue Cable will continue to operate over a mixture of coaxial cable and fiber optic cables.

2. Demolition and Phasing:

a. Railway

1) The existing BDF telecommunications room located within Railway on the first floor shall remain in its current location and is not included as part of any renovation to the building. This is the only telecommunications room located within Railway.

b. ME – see related ME section and narrative

c. Hampton – see related HAMP section and narrative

3. Topology:

a. This infrastructure shall include workstation outlets (WO) at specific locations throughout the facility. The cabling system shall utilize a "Star" homerun topology. Each individual WO shall be run to a dedicated Telecommunication Room (TR) where it will be terminated to a patch panel within the 90 meter distance limitations dictated by industry standards.

b. All Telecommunication Rooms shall then connect to a central point known as the Main Cross Connect, aka Building Distribution Frame (BDF) located in the Main Equipment Room (ER). This connection shall include a combination of fiber optic cable as well as a multi-pair copper cable.

c. That BDF shall connect via backbone copper and fiber cabling to Campus Telecommunications Systems.

4. Communications Pathways

a. The building's new communications pathways will be designed as an easily accessible system that provides for flexibility and relative ease of modification throughout the lifecycle of the building.

b. Pathways will be sized for voice, data, video, and any other Owner approved low voltage cabling. Systems, may include fire alarm, security, building automation, etc. each will be provided with separate pathway for wire management. This may be a separate tier on the corridor cable tray system.

c. The cable tray system will be sized to handle required cable loading for building plus 50% expansion for future applications, above the ceilings in the corridors and elsewhere as required.

d. The distribution cable tray system within the corridors will allow a minimum 3" of clearance between the tray and the ceiling, a minimum 6" of clearance in front of the tray, and a minimum 12" of clearance above the tray.

e. Basket type cable tray will be provided and will be wall and/or ceiling hung and mounted above the accessible ceiling in the corridors as required. The cable tray will be a minimum of 12" W by 4"H.

5. Telecommunications Rooms:

a. Railway Building

1) The only telecommunication room servicing Railway is the BDF located on the first floor. This space is not included in the study and shall stay intact.

b. ME Building – see related ME section and narrative

c. Hampton Hall – see related HAMP section and narrative

d. All new or renovated telecommunications room walls will have an added layer of 3/4" plywood, B-B ext. grade 5. Plywood will be fire-retardant. Plywood will be installed horizontally on walls from 2'-0" to 6'-0"AFF from corner to corner.

e. All BDF and IDF telecommunication rooms shall have ladder type cable tray installed around the perimeter of the room at approximately 7'-2" AFF.

f. One dedicated, 110V, 20A double –duplex electrical outlets will be provided on each wall. One of those outlets in particular will be provided on the wall near the equipment frames.

g. Two dedicated, 208V, 30A, twist lock (NEMA L6-30R), single phase electrical outlet will be located 25" behind the front rail of the equipment frame on the wall in the telecommunications room BDF for the owner provided UPS.

h. Two dedicated, 208V, 20A, twist lock (NEMA L6-20R), single phase electrical outlet will be located 25" behind the front rail of the equipment frame on the wall in telecommunications room IDF for the owner provided UPS.

6. Workstation Outlets Requirements:

a. Workstation outlets will be placed in general office space, conference rooms, commons areas, laboratories, classrooms and other facility spaces as required. Special Circuit Requirements:

1) Wireless Access Point location will require one data cable each.

2) Every FACP (Fire Alarm Control Panel) requires two voice cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.

3) Every environmental control cabinet requires two data cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.

4) Every building electrical switch gear meter requires two data cables, terminated in an enclosure outside of the metering cabinet per telecom specifications, and to be installed to the nearest telecommunications room.

5) Every elevator requires two voice cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.

b. Each standard PIC Station will be one 4-11/16" square by 2-1/8" deep flush mounted box with double gang, square drawn extension or tile ring. Provide 1" EMT conduit from the box to cable tray in corridor.

7. Video Wiring System:

a. This system shall provide the signal to the television / display from the Cable Television (CATV) Distribution System.

b. Each outlet location shall be supplied with the required RG-6 or RG-11 horizontal cabling that is connected to a tap/splitter within the IDF serving that area. All new or renovated IDFs will be

provided with new RG-11 Riser cabling to new BDF for distribution of CATV signals from the backbone cabling entering the building.

B. Wireless Network Systems for Data:
   1. Overview:
      a. Wireless network systems shall be engineered by Purdue's Information Technology Networks and Security (ITNS) department during the design phase of the project. ITNS will provide the A&E team with wireless access point locations based on a computer generated site survey for the existing buildings so they may be incorporated into the construction documents. The contractor will be responsible for installing a turn-key wireless system as part of the project. The overall wireless system shall be included in the building project budget.

C. Telephone System and Service:
   1. Overview:
      a. The telephone system equipment and handsets will be provided by the University.
      b. Service to the system will be provided by connecting these facilities to the existing Campus Telecommunications Systems.
      c. It is assumed that a majority of the new handset locations within each facility will be Voice over IP (VoIP) rather than traditional digital handsets. The riser and horizontal cabling quantity will reflect this type of handset being installed in all offices, conference rooms, and classrooms as required.

D. Emergency Communications:
   1. Overview:
      a. Emergency Communications in these facilities will be provided by "Voice Evacuation" capabilities of the Fire Alarm System.
      b. It is yet to be determined whether any other emergency phones are required.

E. Paging System
   1. No paging system will be provided.

F. Clock System:
   1. No clock system will be provided.
   2. Battery powered clocks may be installed by Owner in selected locations.

G. Local Area Network (LAN) Electronics:
   1. Overview:
      a. This includes Hubs, Routers, and Switches required to activate data ports. The University will provide all Network Electronics for these Projects. Space will be allocated in the BDF and IDF rack to accommodate these requirements.

H. Security Systems:
   1. Overview:
      a. This includes cabling, controllers, card readers and programming and activate the system in each building. The University will provide all cabling, controllers, and card readers for these Projects. Space will be allocated in the Electrical and mechanical rooms to accommodate these systems requirements.
   2. Access Control:
      a. A new/renovated electronic access control system will be provided. All readers, controllers, and wiring shall be providing, installed, and programmed by the University. The design of the facilities shall include all required rough-ins, conduit, junction boxes, and power to allow for the installation at each opening to be completed.
      b. A single gang box with a single gang plaster ring and a 1" EMT conduit to above ceiling cable tray will be provided for each card reader location.
      c. All keypads, card readers, cables, security cabinet, etc. shall be provided by the University.
      d. During the design phase of the project key openings on the exterior and interior of these facilities will need to be identified that require access control and card readers.
   3. CCTV:
      a. A CCTV system will be provided to serve these facilities.
      b. Interior camera locations will be recorded by the existing IP NVR located on campus to record all new camera locations. It may be determined during design that placing all cameras on the IP system may not be feasible as they may not accessible by staff and faculty, only campus police. In these cases a local DVR and cameras will be provided within the area. The cameras will be recorded by the DVR and will be accessible by the department only.
      c. Currently cameras may be required to view and monitor any active lab spaces to ensure the security of ongoing experiments taking place in the spaces.
      d. During the design phases other key areas of the facilities will need to be identified that require camera coverage.
      e. All exterior camera locations will be provided with a two gang box with a single gang ring and a 1" EMT conduit to nearest above ceiling cable tray. All interior camera location will be provided with a two gang box with single gang plaster ring and 1" EMT conduit to the nearest above ceiling cable tray.

I. Audio Visual Systems:
   1. Overview:
      a. The Audio Visual (AV) system requirements for these facilities have not yet been fully determined and are not currently included in the design responsibility of the Architect or its consultants. It will be critical that these requirements be determined in the initial phase as they will impact the design of electrical pathways and power systems as well CATV and Network Connections.
      b. We expect the AV requirements at a minimum to include technologies such as "way finding" and general information displays in common areas, projectors, and large LCD screens in conference rooms and dedicated audio systems in gathering and presentation areas.

II. SPECIFICATIONS
   A. Main Bid Package:
      1. Division 27:
         a. 27 00 00 Communications (Common Work Results)
         b. 27 05 26 Grounding and Bonding for Communication Systems
         c. 27 05 28 Pathways for Communications Systems
         d. 27 05 53 Identification for Communication Systems
         e. 27 11 01 Telecommunications Room Finishes
         f. 27 11 16 Cabinets and Frames
         g. 27 11 19 Termination Blocks and Patch Panels
         h. 27 14 13 Copper Riser Cabling
         i. 27 14 23 Fiber Riser Cabling
         j. 27 14 33 Coaxial Riser Cabling
         k. 27 15 13 Copper Horizontal Cabling
         l. 27 15 33 Coaxial Horizontal Cabling
         m. 27 15 43 Faceplates and Connectors

The executive summary and calculated arc flash data follow. The full report is provided under separate cover.

PURDUE
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Arc Flash Risk Assessment with Fault-Current and Protective Device Coordination Analysis

Gatewood Buildings
As found Study Report

Applied#15-051
June 17, 2015

Applied
Engineering Services

PURDUE
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ARC FLASH REPORT

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1 Executive Summary 1
2 Calculated Arc Flash Data 2
3 Single Line Diagrams 3
4 TCC Plot and Device Settings 6

1 Executive Summary

Warning

The PPE requirements, when working on the line side of main disconnects, busswork and cabling between main transformers, main disconnects, and the connections within main transformers are not affected by the recommendations made in this report. That part of the system is protected by devices on the input side of the transformers that are normally owned and set by the utility.

- All of the panelboards and switchboards fall within NFPA 70E limits for PPE (personal protective equipment) categorized as Category 0, 1, or 2 unless noted below. Refer to **Section 2** of this report for a table describing the categories and results of the arc flash hazard calculations.
- The rated fault current for all busses is sufficient to clear a bolted fault unless noted below. Refer to **Section 4** of this report for a table describing the results.
- The rated fault current for all protective devices is sufficient to clear a bolted fault unless noted below. Refer to **Section 4** of this report for a table describing the results.

Findings

In the following table, the calculated incident energy is above 8.0 cal/cm^2 and PPE Category is 3 or above at buses. **It is to be noted that existing model file is provided by Purdue University with available low and high fault current.**

Gatewood

- Table – Shows Bus PPE Value 3 or higher

Bus Name	Protective Device Name	Bus kV	Bus Bolted Fault (kA)	Bus Arcing Fault (kA)	Arc Flash Boundary (in)	Incident Energy (cal/cm2)	PPE Level
MDP-P	Xfmr P	0.480	20.40			29	Category 4
MDP-L	Xfmr L	0.208	43.02			82	Dangerous!

Recommendation

Gatewood

- Provide new main breaker switchboard MDP-IML between transformer & MDP-L, new breaker with electronic trip unit (LSIG).
- Replace fuse switch switchboard MDP-L with circuit breaker type switch board, new breaker with electronic trip unit (LSI).
- Provide new main breaker switchboard MDP-IMP between transformer & MDP-P, new breaker with electronic trip unit (LSIG).

2 Calculated Arc Flash Data

The table included in this section is the screen displayed at the conclusion of the SKM arc-flash analysis. It was printed as a pdf file and merged into the report.

The arc-flash hazard analysis is done to conform to OSHA recommendations described in 29CFR 1910.333. Procedures for arc-flash hazard analysis are described in both NFPA Standard 70E-2009 and IEEE Standard 1584-2002. IEEE Standard 1584-2002 describes a method for calculating the potential arc-flash energy based on the anticipated “bolted” fault current. The results reported within have been calculated using the more conservative, interpretation of IEEE Standard 1584-2002.

IEEE Standard 1584-2002 states that equipment rated below 208V does not need be considered unless it involves a low-impedance transformer larger than 125 kVA for its immediate supply. The reason for this exclusion is the inability of such small capacity systems to initiate and support arcing conditions. NFPA Standard 70E-2009 applies the same reasoning to equipment rated 208V and less. NFPA Standard 70E is recognized as the guiding document.

The table beginning on the next page shows arc-flash data calculated. Included in the tables are the critical distances, the incident energies, the fault currents, and the PPE category level required for the indicated working distance. Arc-flash hazard warning labels for the equipment are created from this table.

If future changes are made in the electrical system or to the protective device settings, new calculations should be completed on the specific circuits and new warning labels printed to reflect any changes in the hazard.

- Arc Flash value as describe in table above and shown on one line drawing

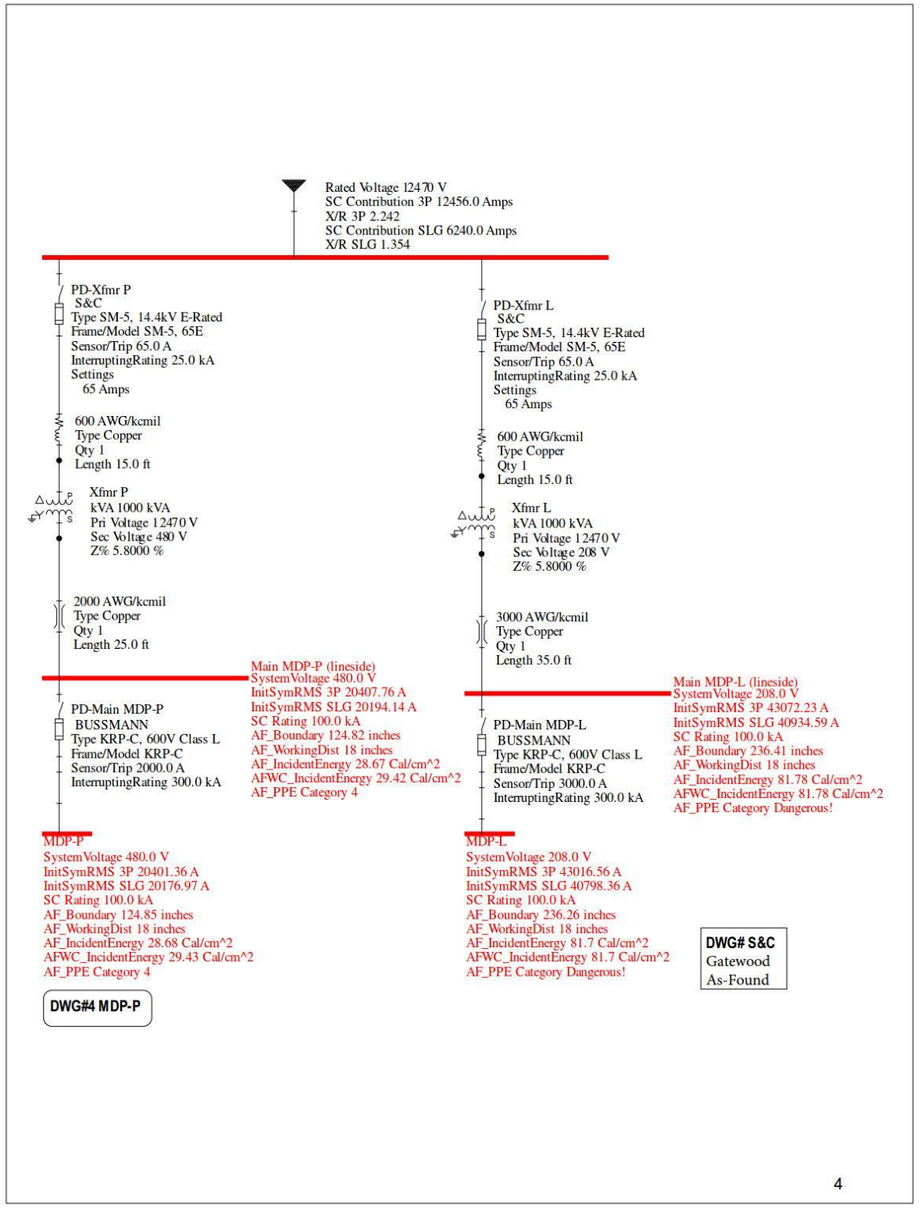
3 Single-Line Drawings

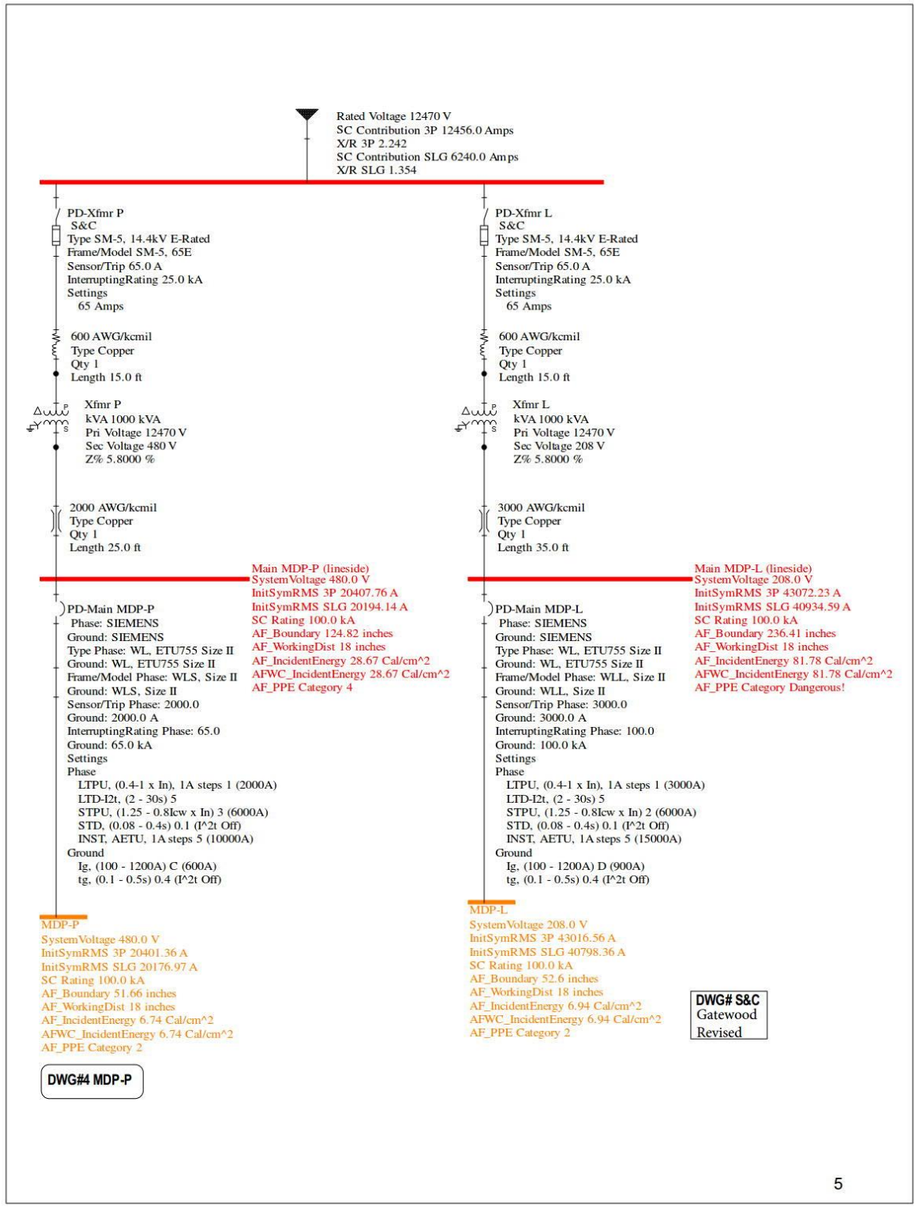
The single-line drawings that follow show the system topology, the input, and calculated data for the facility. It shows circuit breaker and/or fuse data, cable data, transformer data, motor data, source data, calculated fault current at bus, and calculated arc flash data including incident energy, boundary distance and PPE category. Source fault current and X/R ratio are obtained from Purdue University.

Drawing List

Gatewood

DWG #S&C Fuse	One line drawing w/fuse
DWG #S&C BKR	One line drawing w/Breaker





5

4 TCC Plots and Device Settings

The following coordination curves graphically depict the time versus current characteristics for the overcurrent protective device shown, and the relationship of each device to the other devices that are in electrical series.

An ideally coordinated system would have the overcurrent protective device closest to a short circuit, or fault, condition detect the fault and open the circuit. In the event the closest device fails to operate, the next device between the fault and the power source would detect the fault and open the circuit. Realistically, the ideal is often not achievable, but an attempt is made to come as close as practical.

Each set of plotted curves shows at least two protective devices.

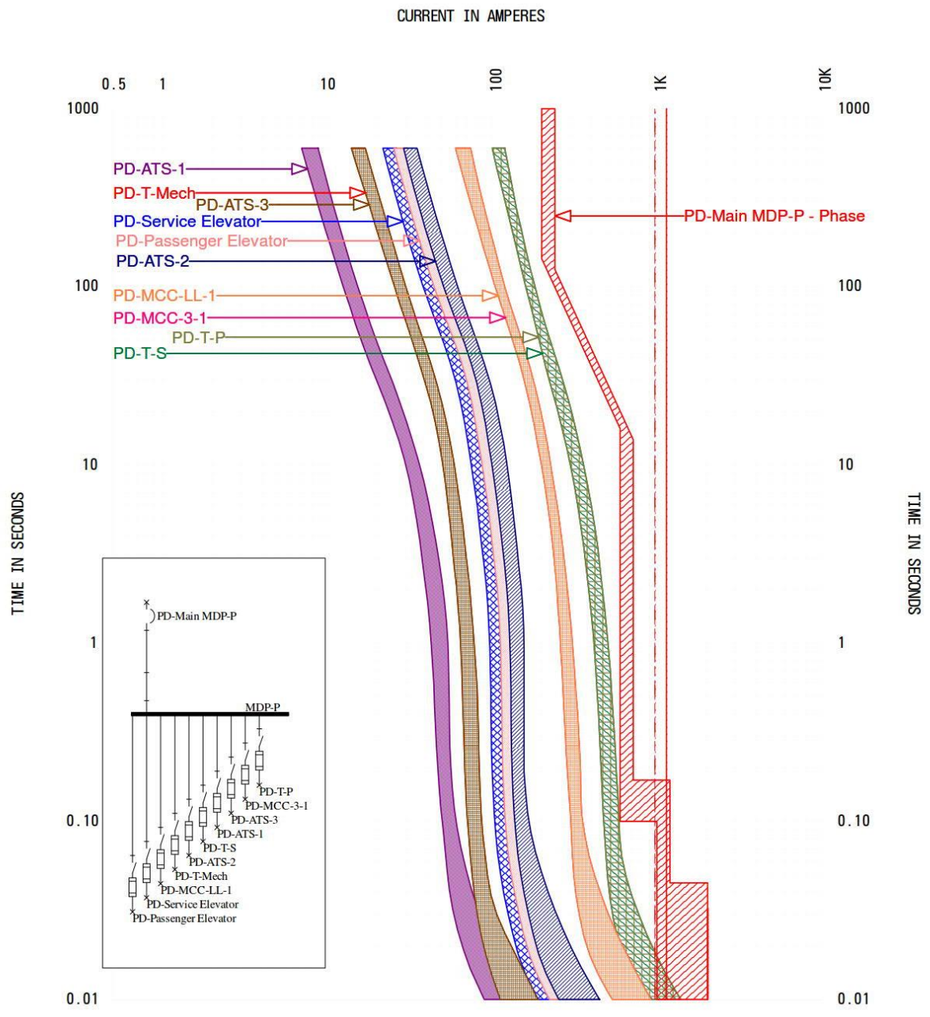
All recommendations presume the protective devices will be properly tested and maintained to operate in accordance with the manufacturer's specifications.

The page(s) following set of curves present(s) the pertinent data and settings for each protective device represented in the system. All recommendations presume the protective devices have been, and will continue to be, well and properly maintained to operate in accordance with the manufacturer's specifications.

TCC Plot:

Gatewood:

- TCC MDP-P
- TCC MDP-L

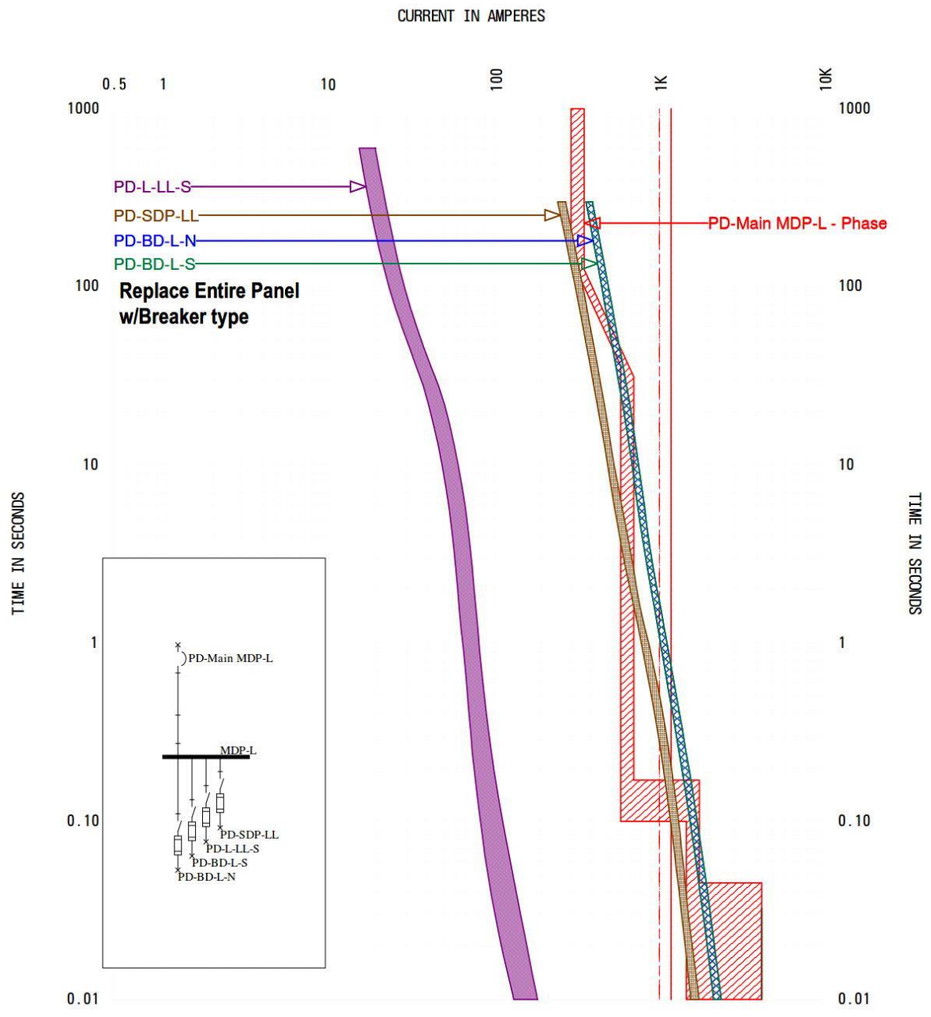


TCC Name: TCC MDP-P
One Line: TCC MDP-P
June 10, 2015 2:58 PM SKM Systems Analysis, Inc.

Gatewood Revised
PURDUE UNIVERSITY, IN

Reference Voltage: 480
Current Scale x 10
Applied Engineering Services

7



TCC Name: TCC MDP-L
One Line: TCC MDP-L
June 10, 2015 2:57 PM SKM Systems Analysis, Inc.

Gatewood Revised
PURDUE UNIVERSITY, IN

Reference Voltage: 208
Current Scale x 10
Applied Engineering Services

8

Existing Plans

The study encompasses only specific areas within Hampton Hall (HAMP), as indicated on the following color-coded floor plans of the existing building.

The total gross square footage of the renovated space is approximately 91,955 GSF, with the following breakdown by floor.

Sub-Basement .....0 GSF
Basement .....10,210 GSF
Ground Floor ..... 22,904 GSF
First Floor .....21,376 GSF
Second Floor.....6,027 GSF
Third Floor .....17,765 GSF
Fourth Floor.....13,673 GSF

As a brief summary, we are showing approximately 17,564 NSF of proposed instructional lab space, including existing labs to remain; 59,292NSF of proposed office space, including existing office spaces to remain; 26,389 NSF of proposed research lab space, including existing research labs to remain. These NSF calculations do not include support, restrooms or circulation (existing and new).

Existing Plans – HAMP Sub-Basement Floor

Sub-Basement Floor Plan - Existing

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Plans – HAMP Basement Floor

Basement Floor Plan - Existing

3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Plans – HAMP Ground Floor

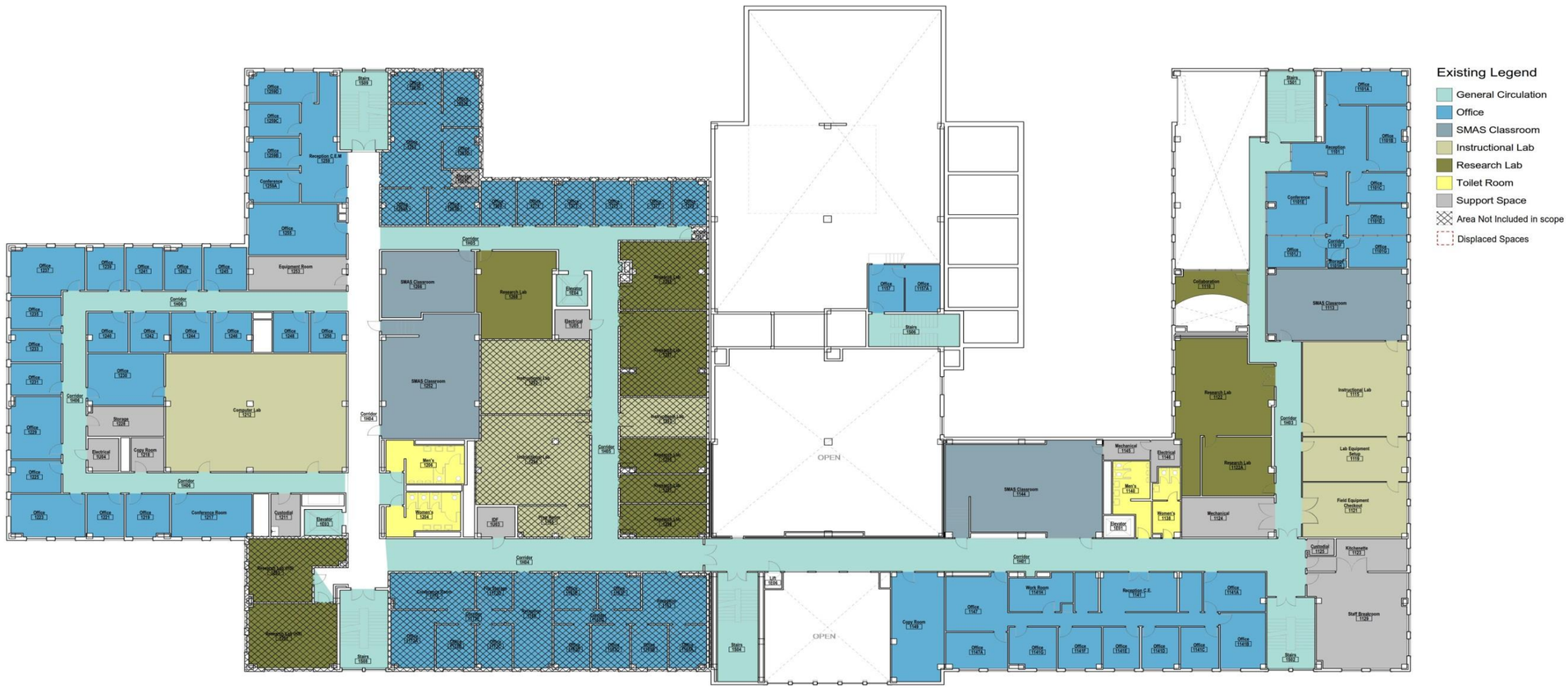
Ground Floor Plan - Existing

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Plans – HAMP First Floor



First Floor Plan - Existing

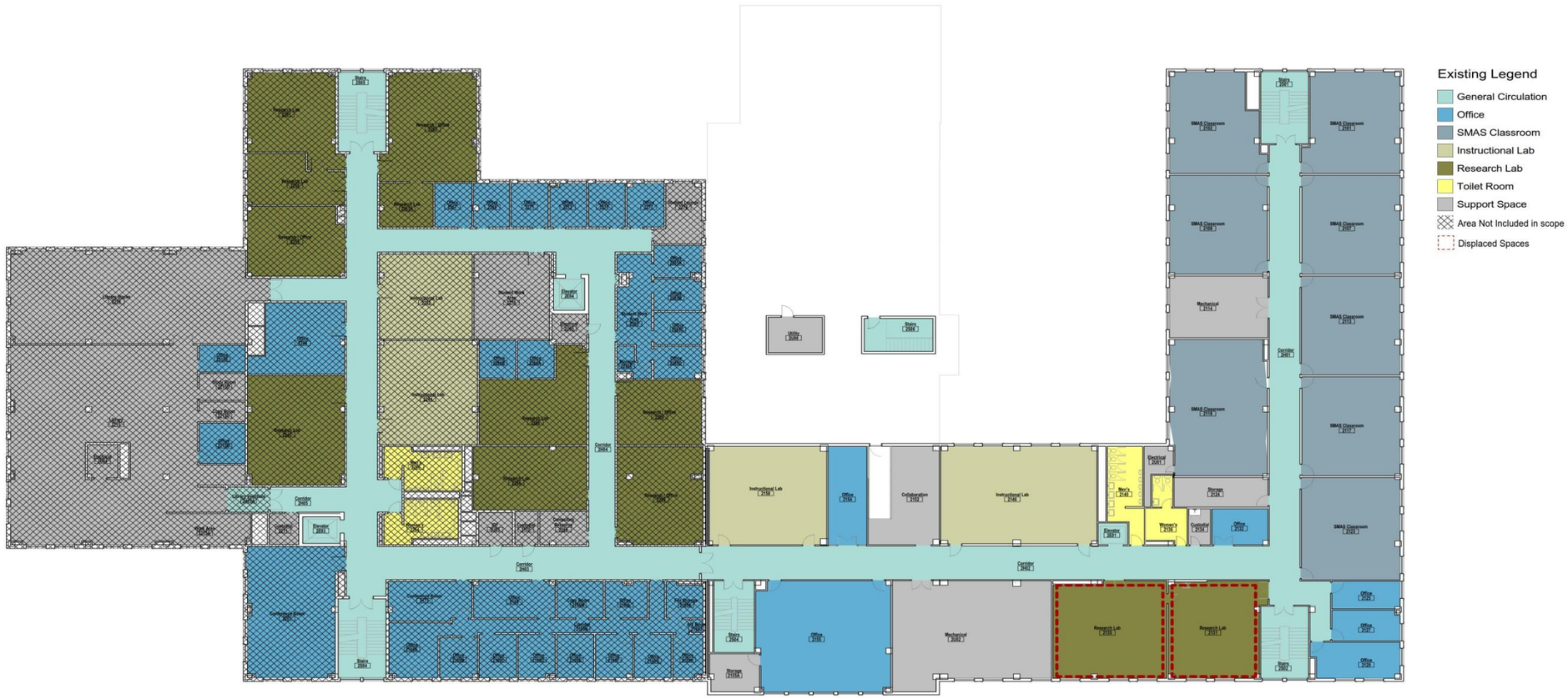
3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Plans – HAMP Second Floor



Second Floor Plan - Existing

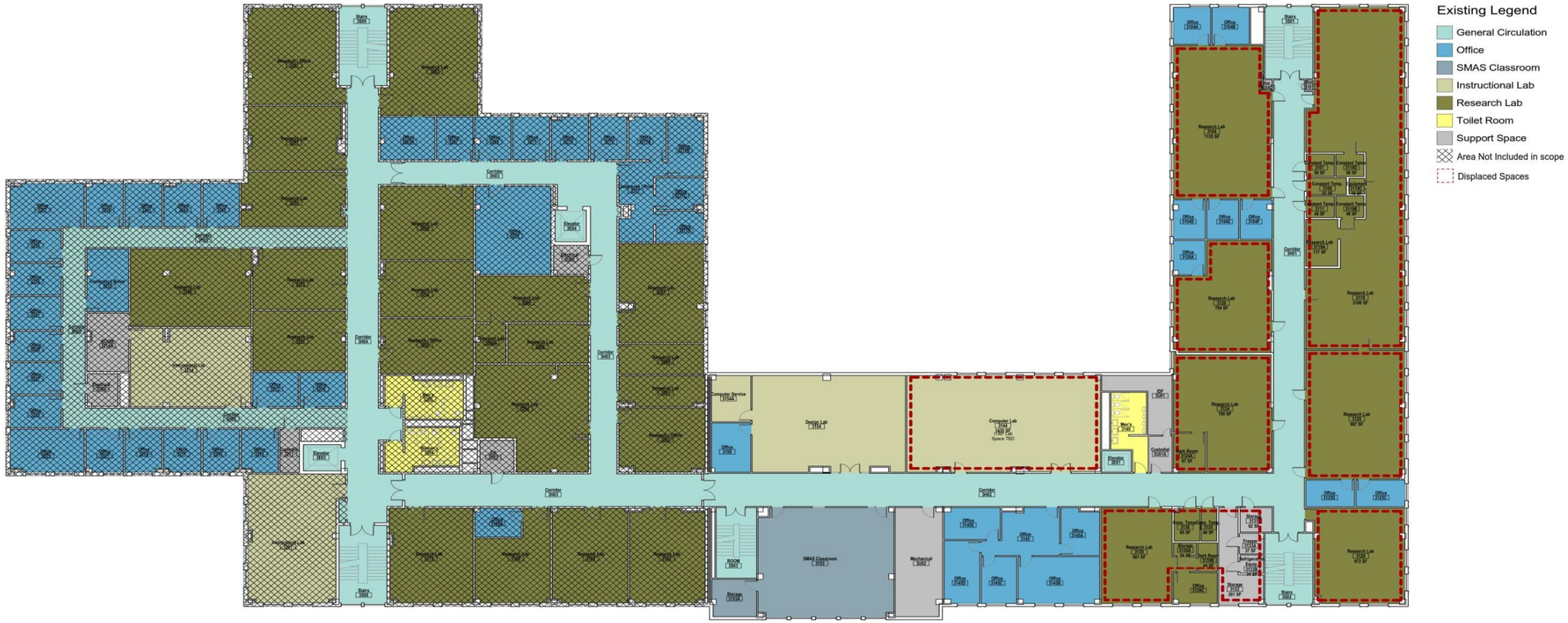
3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Plans – HAMP Third Floor



Third Floor Plan - Existing
3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Plans – HAMP Fourth Floor

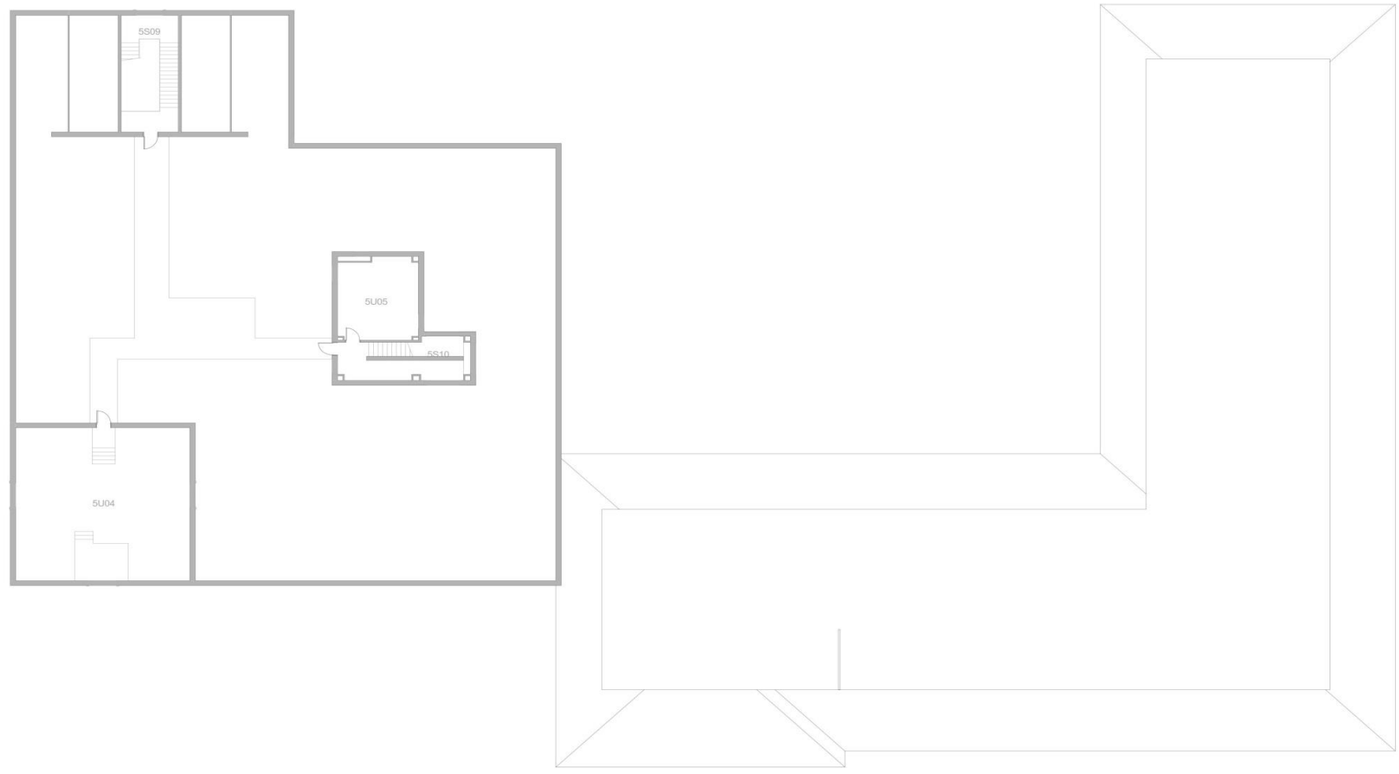
Fourth Floor Plan - Existing
3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Plans – HAMP Fifth Floor



Fifth Floor Plan

PURDUE UNIVERSITY

Hampton Hall

MSKTD & Associates, Inc.

Existing Space Use

The following spreadsheets are similar to standard forms used by architects and engineers to quantify and describe space use in a typical project. These sheets identify current/existing room names, room numbers, area or assignable square footage of rooms, departmental assignment, quantities of faculty, staff or graduate assistants accommodated, and other comments regarding space usage and adjacencies.

The room data information originated from the Space Management department (SMAS) at Purdue University, and the following sheets contain modified information that was confirmed with the Civil Engineering (CE) and Construction Engineering and Management (CEM) departments to validate space usage and existing quantities. The following information on the existing spaces provides a snapshot in time to understand current faculty, staff and graduate student offices and open-office workstations accommodated in specific areas of Hampton Hall. During our team meetings and discussions, it was observed that growth within departments had already begun (as compared to the master plan document, which identified projected future growth). In essence, the goals identified as part of the master planning process have already been impacted by the initial departmental growth that is summarized in the following pages.

In order to develop the final program of spaces for the design team, quantities, spatial relationships and important adjacencies will need to be confirmed. At the next phase of the project, more detailed, individual room data sheets will provide additional space requirements and other necessary information. These room data sheets will also include specific requirements for instructional / teaching labs and reconfigured / relocated research labs. Sample room data summary sheets are included, but relevant lab information data sheets were not available during the review process for the study.

Purdue University ME/RAIL/HAMP Study

Hampton Hall (HAMP)

EXISTING SQUARE FOOTAGE SUMMARY OF TEACHING AND RESEARCH LAB SPACES									
	Sub-Basement	**Basement**	**Ground Floor**	**First Floor**	**Second Floor**	**Third Floor**	**Fourth Floor**	**Existing Sqaure Footage**	
TEACHING LABS	0	2,218	3,477	1,598	2,023	1,309	0	**10,625**	
RESEARCH LABS	4,227	21,447	8,863	2,095	1,525	8,009	663	**46,829**	

PURDUE UNIVERSITY

Purdue University ME/RAIL/HAMP Study

Hampton Hall (HAMP)

EXISTING PROGRAM SUMMARY OF FACULTY, STAFF AND SPACE COUNTS

	Sub-Basement	Basement	Ground Floor	First Floor	Second Floor	Third Floor	Fourth Floor	Current Quantities	*CE*	*CEM*	
TEACHING FACULTY	0	0	28	16	3	5	22	**74**	66	8	
COLLEGE/DEPARTMENT STAFF	0	10	29	23	4	1	7	**74**	61	13	
GRADUATE ASSISTANTS	0	17	95	34	27	27	43	**243**	235	8	

PROPOSED SUMMARY OF FACULTY, STAFF AND SPACE COUNTS (Based on CoE Master Plan Document) - Hampton Hall

	Civil Engineering (CE)			Construction Engineering and Mgmt (CEM)			
	Quantity	*Workstation Size (SF)*	*SF Info*	*Quantity*	*Workstation Size (SF)*	*SF Info*	*Comments*
FACULTY	**61**		**7,380**	**12**		**1,500**	Proposed quantities change to 71 faculty. Quantities reflected above.
Academic Department Head	1	180	180	1	180	180	
Academic Associate Head / Asst. Head	0	150	0				Proposed quantity changes to 1 Academic Associate Head / Asst. Head
Tenured/TT Faculty	55	120	6,600	10	120	1,200	Existing quantities to be updated to reflect 64 current faculty. Reflected above.
Instructor/Lecturer	4	120	480	1	120	120	
Emeritus/Emeritas	1	120	120				
OTHER *(Wkstns)*	**24**		**1,440**	**1**		**97**	
Visiting Scholars	12	60	720	1	97	97	single room; 475 SF
Research / Visiting Faculty	12	60	720				
STAFF *(Offices)*	**33**		**3,960**	**5**		**600**	
Professional Staff	22	120	2,640	3	120	360	
Business Manager	1	120	120				
ECN IT Staff	2	120	240				
Management	2	120	240				
Service Staff	3	120	360	2	120	240	
JTRP Professional Staff	3	120	360				
STAFF *(Wkstns)*	**62**		**5,580**	**10**		**900**	
Post Doctoral Assistants	30	90	2,700	8	90	720	
Business Staff	5	90	450	1	90	90	
Admin/Operations Staff	9	90	810	1	90	90	
Clerical	11	90	990				
JTRP Admin	7	90	630				
GRADUATE ASSISTANTS	**163**		**6,520**	**22**		**880**	
Grad - Thesis	163	40	6,520	22	40	880	0.7 desk / grad
Subtotal	**343**		**24,880**	**50**		**3,977**	

ME/RAIL/HAMP Study
Hampton Hall

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities: Faculty	Quantities: Staff	Quantities: GAs	Quantities: Other	Exist Lab SF Calcs: Teaching Labs (SF)	Exist Lab SF Calcs: Research Labs (SF)	Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
SUB BASEMENT															
Hydraulics Towing Tank	S120	4227		Civil Engr	Civil Engr	NONCLAS LAB						4,227	Hydraulics Towing Tank		
Stair	SS07	49		Gen Building	Gen Building	CIRCULATE									
Electrical Vault	SU02	655		Gen Building	Gen Building	MECHANICAL							Transformer Vault		
Subtotals		4931	SF				0	0	0	0	0	4,227			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
BASEMENT															
Hydraulics Lab Control	B102	217		Civil Engr	Civil Engr	NONCLS SV						217			
Hydraulics Research Lab	B115	5573		Civil Engr	Civil Engr	NONCLAS LAB						5,573			
Custodial Closet	B115A	41		Gen Building	Gen Building	BLDG SERV									
Machine Shop	B115B	807		Civil Engr	Civil Engr	SHOP									
Machine Shop	B115C	85		Civil Engr	Civil Engr	SHOP SERVICE									
Instrument Storage and Workroom	B115D	338		Civil Engr	Civil Engr	NONCLS SV						338			
Hydraulics Research Lab	B115E	935		Civil Engr	Civil Engr	NONCLAS LAB						935			
Teaching Lab	B115F	847		Civil Engr	Civil Engr	CLASS LAB					847		Sensor Based Hydraulics Lab		
Teaching Lab	B115G	1371		Civil Engr	Civil Engr	CLASS LAB					1,371		Undergrad Hydromechanic Lab		
GPS Lab	B129	373		Civil Engr	Civil Engr	NONCLAS LAB						373			
GPS Lab Equipment	B129A	65		Civil Engr	Civil Engr	NONCLS SV						65			
Office	B133	222	J. Harley + S. Kelly + Students	Civil Engr	Civil Engr	OFFICE		2							
Equipment Prep/Storage	B133	223		Civil Engr	Civil Engr	OFF SERV									
Custodial Closet	B138	76		Gen Building	Gen Building	BLDG SERV									
Toilet Room	B140	227		Gen Building	Gen Building	REST ROOM							Unisex + Lockers: 2wc,2u,2l,2s		*Existing showers.*
Research Lab	B141	268	RESEARCH-DUNSTON+HASTAK	Civil Engr	Civil Engr	NONCLAS LAB						268			
Office	B141A	172		Civil Engr	Civil Engr	OFFICE			4				Construction GAs		
Office	B141B	108		Civil Engr	Civil Engr	OFFICE			3				Construction GAs		
Office	B141C	108		Civil Engr	Civil Engr	OFFICE			3				Construction GAs		
Research Lab	B141D	279		Civil Engr	Civil Engr	NONCLAS LAB						279			
Hall	B141E	88		Civil Engr	Civil Engr	NONCLS SV									
Server Room (ECN)	B146	630		Eng Cpt Ntwk	Civil Engr	ECN COMPUTER							ECN		*ECN is vacating this area. No HVAC. Existing Liebert and power mgr equipment removed.*
Research Lab	B147	630		Civil Engr	Civil Engr	NONCLAS LAB			4			630	Construction		
Research Lab	B149	120		Civil Engr	Civil Engr	NONCLAS LAB			3			120	Construction		
Research Lab	B150	2036		Civil Engr	Civil Engr	NONCLAS LAB						2,036	Bituminuous materials research		*Existing machine size may be issue to remove. Current research to be determined.*
Research Lab	B150A	475		Civil Engr	Civil Engr	NONCLAS LAB						475	Bituminuous materials research		
Research Lab	B150B	85		Civil Engr	Civil Engr	NONCLAS LAB						85	Bituminuous materials research		
Research Lab	B157	1587		Civil Engr	Civil Engr	NONCLAS LAB						1,587	Concrete/Cement Lab	*B157B, B157C, B157D, B157E, B157F*	
Pump Room	B157A	788		Civil Engr	Civil Engr	NONCLS SV						788		*Refrigeration for G157*	*Includes foundation for test machine. Additional refrigeration for environ. Chamber above.*
Curing Room	B157B	310		Civil Engr	Civil Engr	NONCLS SV						310	Concrete curing room		
Aggregate Storage	B157C	70		Civil Engr	Civil Engr	NONCLS SV							Aggregate storage (gravel)		*Includes chute for aggregate delivery.*
Aggregate Storage	B157D	70		Civil Engr	Civil Engr	NONCLS SV							Aggregate storage (sand)		*Includes chute for aggregate delivery.*
Cement Storage	B157E	110		Civil Engr	Civil Engr	NONCLS SV							Cement storage		*Includes chute for aggregate delivery.*
Aggregate Storage	B157F	72		Civil Engr	Civil Engr	NONCLS SV							CONC AGGR CR STN STOR		*Includes chute for aggregate delivery.*
Office	B170	335		Civil Engr	Civil Engr	OFFICE		8					Post Docs + Visitors		
Office	B173	496	P. Hill	REM	REM	OFFICE									
Office	B173A	153	J. Schweitzer	REM	REM	OFFICE									
Office	B173B	120	B. McDonald	REM	REM	OFFICE									
Office	B173C	120	A. Gerth	REM	REM	OFFICE									
Office	B173D	120	K. Evans	REM	REM	OFFICE									
Office	B173E	120	J. Young	REM	REM	OFFICE									
Office	B173F	120	S. Rainey + M. Nicholson	REM	REM	OFFICE									
Office	B173G	120	H. Bentley	REM	REM	OFFICE									
Office	B173H	108	Z. Tribbett	REM	REM	OFFICE									

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
BASEMENT															
Storage	B173J	9		REM	REM	OFF SERV									
Office	B173K	107	M. Tang	REM	REM	OFFICE									
Office	B173L	133	L. Swihart + J. Henseler	REM	REM	OFFICE									
Office	B173M	114	D. Petros	REM	REM	OFFICE									
Office	B173N	114	E. Butt	REM	REM	OFFICE									
Corridor	B173P	232		REM	REM	OFF SERV									
Copy Room	B201	58		REM	REM	HAZARD SRV									
Special Procedures Lab	B201A	459		REM	REM	HAZARD SRV									
Dispensing / Storage Room	B201B	223		REM	REM	HAZARD MTRL									
Office	B203	117	S. Rudolph	REM	REM	OFFICE									
Women's Restroom	B204	213		Gen Building	Gen Building	REST ROOM							Women's Restroom - 4WC,3L-HCA		
Men's Restroom	B206	223		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,3U,3L-HCA		
Custodial	B211	53		Gen Building	Gen Building	BLDG SERV									
Sample Preparation	B243	239		REM	REM	HAZARD SRV							Sample Preparation		
Special Counting	B245	387		REM	REM	HAZARD SRV							Special Counting		
Special Counting	B245	119		Health Sci	REM	CLASS LAB							Special Counting		
Office	B245	89	J. Gibbs	REM	REM	OFFICE									
Office	B251	100	R. Derudder + EH Student	REM	REM	OFFICE									
Industrial Hygiene Lab	B253	380		REM	REM	HAZARD SRV							Industrial Hygiene Lab		
Break Room / Storage	B255	287		REM	REM	OFF SERV									
Materials Lab	B257	1247		Civil Engr	Civil Engr	NONCLAS LAB						1,247			*old machine shop, includes OH crane and hood.*
Instrument Analysis	B263	493		Civil Engr	Civil Engr	NONCLAS LAB						493			
Optical Microscopy	B263A	247		Civil Engr	Civil Engr	NONCLAS LAB						247			
Chemistry Lab	B265	1140		Civil Engr	Civil Engr	NONCLAS LAB						1,140			
Research Lab	B268	791		Civil Engr	Civil Engr	NONCLAS LAB						791			
Research Lab	B277	857		Civil Engr	Civil Engr	NONCLAS LAB						857			
Research Lab	B286	1717		Civil Engr	Civil Engr	NONCLAS LAB						1,717	Geotechnical Research		
Temp/Humidity Control	B289	747		Civil Engr	Civil Engr	NONCLS SV						747	Temp and Humidity Room		*Room includes modular temp/humidity rooms.*
Research Lab	B290	129	research lab-santagata	Civil Engr	Civil Engr	NONCLAS LAB						129			
Office	B292	262	A. Pettigrew	REM	REM	OFFICE									
Research Lab	B293		Research Lab - Pinal	Ind&PhysPhrm	Ind&PhysPhrm	NONCLAS LAB									*Not included in study*
Elevator	BE01	52		Gen Building	Gen Building	CIRCULATE									
Elevator	BE03	77		Gen Building	Gen Building	CIRCULATE							West		
Elevator	BE04	77		Gen Building	Gen Building	CIRCULATE							East		
Lift	BE05	15		Gen Building	Gen Building	CIRCULATE							Lift		
Corridor	BH01	908		Gen Building	Gen Building	CIRCULATE									
Corridor	BH03	436		Gen Building	Gen Building	CIRCULATE									*Old freight elevator was located in this area but was removed and infilled.*
Corridor	BH04	1982		Gen Building	Gen Building	CIRCULATE									
Corridor	BH05	1011		Gen Building	Gen Building	CIRCULATE									
Corridor	BH06	105		Gen Building	Gen Building	CIRCULATE									
Corridor	BH07	116		Gen Building	Gen Building	CIRCULATE									
Stair	BS01	248		Gen Building	Gen Building	CIRCULATE									
Stair	BS02	203		Gen Building	Gen Building	CIRCULATE									
Stair	BS03	76		Gen Building	Gen Building	CIRCULATE									
Stair	BS04	337		Gen Building	Gen Building	CIRCULATE									
Stair	BS06	184		Gen Building	Gen Building	CIRCULATE									

ME/RAIL/HAMP Study
Hampton Hall

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
BASEMENT															
Stair	BS08	285		Gen Building	Gen Building	CIRCULATE									
Stair	BS09	285		Gen Building	Gen Building	CIRCULATE									
Elevator Equipment Room	BU01	73		Gen Building	Gen Building	MECHANICAL									
Electrical	BU02	644		Gen Building	Gen Building	MECHANICAL							Elect and Phone Equipment		*Vault?*
Mechanical	BU03	2117		Gen Building	Gen Building	MECHANICAL									
Mechanical	BU04	401		Gen Building	Gen Building	MECHANICAL									
Electrical	BU05	113		Gen Building	Gen Building	MECHANICAL							Electrical distribution panel		
BDF	BU06	85		Gen Building	Gen Building	MECHANICAL							Telecom BDF		
Electrical	BU07	81		Gen Building	Gen Building	MECHANICAL									
Mechanical	BU08	5384		Gen Building	Gen Building	MECHANICAL									
Electrical	BU09	718		Gen Building	Gen Building	MECHANICAL							Switchgear Room		
Electrical Vault	BU10	1027		Gen Building	Gen Building	MECHANICAL							Transformer Vault		
Subtotals		**49544**	**SF**				0	10	17	0	2,218	21,447			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
GROUND FLOOR															
Mechanical	G101	347		Gen Building	Gen Building	MECHANICAL									
Service Entrance	G102	455		Gen Building	Gen Building	CIRCULATE									*Existing OH crane; OH door for service access.*
Office	G103	113		Civil Engr	Civil Engr	OFFICE		1					Lab Coordinator		
Office	G107	451		Civil Engr	Civil Engr	OFFICE			16				Hydraulics Graduate Assistants		
Office	G109	177		Civil Engr	Civil Engr	OFFICE		4					Hydraulics Post Docs		
Collaboration Space	G110	314		Civil Engr	Civil Engr	NONCLAS LAB						314	Data Visualization Room		
Mechanical	G114	238		Gen Building	Gen Building	MECHANICAL									
Corridor	G129	48		Civil Engr	Civil Engr	OFF SERV									
Office	G129A	200	M. SOZEN	Civil Engr	Civil Engr	OFFICE	1								
Office	G129B	215		Civil Engr	Civil Engr	OFFICE		5					Post Docs + Visitors		
Office	G131	183		Civil Engr	Civil Engr	OFFICE		1					Business Office		
Office	G133	161	Business Office - C. Brawner +	Civil Engr	Civil Engr	OFFICE		1	1				Business Office		
Office	G135	452	Business Office - S. Nemeth +	Civil Engr	Civil Engr	OFFICE		2	1				Business Office		
Office	G137	134		Civil Engr	Civil Engr	OFFICE		1	1				Business Office		
Custodial	G138	79		Gen Building	Gen Building	BLDG SERV									
Electrical	G138A	143		Gen Building	Gen Building	MECHANICAL									
Office	G139	134	Business Manager - L. Ripley	Engineering	Civil Engr	OFFICE		1					Business Manager		
Unisex Restroom	G140	153		Gen Building	Gen Building	REST ROOM							Unisex Restroom-2wc,3u,2l		
Office	G144	642		Civil Engr	Civil Engr	OFFICE			17				Graduate Assistants		
Research Lab	G146	622		Civil Engr	Civil Engr	NONCLAS LAB						622	Laser Lab		
Commons	G149	997		Civil Engr	Civil Engr	BLDG COMMONS							Leonard & Margaret Wood Commons		
Instructional Lab (I2I)	G150A	756		Civil Engr	Civil Engr	CLASS LAB					756		Lyles Family I2I Lab - Northwest		
Instructional Lab (I2I)	G150B	826		Civil Engr	Civil Engr	CLASS LAB					826		Lyles Family I2I Lab - Northeast		
Instructional Lab (I2I)	G150C	844		Civil Engr	Civil Engr	CLASS LAB					844		Lyles Family I2I Lab - Southeast		
Instructional Lab (I2I)	G150D	805		Civil Engr	Civil Engr	CLASS LAB					805		Lyles Family I2I Lab - Southwest		
Electrical	G150E	51		Gen Building	Gen Building	MECHANICAL									
Research Lab	G157	2050		Civil Engr	Civil Engr	NONCLAS LAB						2,050	Testing Area	*B157A (pump); B157A refrigeration equipment*	*Existing test machine located here. G157D Enviro Chamber is modular. Strong Floor (Testing).*
Electrical	G157A	64		Gen Building	Gen Building	MECHANICAL									
Equipment Room	G157B	264		Civil Engr	Civil Engr	NONCLS SV						264			
Environmental Chamber	G157C	616		Civil Engr	Civil Engr	NONCLS SV						616			*Modular environmental chambers.*
Environmental Chamber	G157D	138		Civil Engr	Civil Engr	NONCLS SV						138			
Concrete Mixing	G157E	1658		Civil Engr	Civil Engr	NONCLAS LAB						1,658			*Existing OH crane*
Electrical	G157F	31		Gen Building	Gen Building	MECHANICAL									
Dust Collection	G159	216		Civil Engr	Civil Engr	NONCLS SV						216			
Aggregate and Cement Shop	G159A	252		Civil Engr	Civil Engr	NONCLS SV						252			
Rock Drill	G159B	216		Civil Engr	Civil Engr	NONCLS SV						216			
Storage (I2I)	G159C	246		Civil Engr	Civil Engr	CLS LB SV					246		I2I Storage		
Reception	G167	158	Reception - D. Miller	Constr Engr	COE PerfSpac	OFFICE		1					Reception		
Office	G167A	125		Constr Engr	COE PerfSpac	OFFICE	1								
Office	G167B	125	Faculty	Civil Engr	COE PerfSpac	OFFICE	1								
Office	G167C	125	j. fricker	Civil Engr	COE PerfSpac	OFFICE	1								
Office	G167D	113	s. ukkusuri	Civil Engr	COE PerfSpac	OFFICE	1								
Storage	G167E	8		Constr Engr	COE PerfSpac	OFF SERV									
Office	G167F	111		Civil Engr	COE PerfSpac	OFFICE		1							
Office	G167G	111		Civil Engr	COE PerfSpac	OFFICE		1							

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
GROUND FLOOR															
Reception	G175	279		Civil Engr	COE PerfSpac	OFF SERV		1							
Office	G175	120	C. Ralston	Civil Engr	COE PerfSpac	OFFICE	1								
Office	G175A	152	s. labi	Civil Engr	COE PerfSpac	OFFICE	1								
Office	G175B	125	a. tarko	Civil Engr	COE PerfSpac	OFFICE	1								
Office	G175C	125	s. peeta	Civil Engr	COE PerfSpac	OFFICE	1								
Office	G175D	125	f. mannering	Civil Engr	COE PerfSpac	OFFICE	1								
Storage	G175E	16		Civil Engr	COE PerfSpac	OFF SERV									
Office	G175F	119		Civil Engr	COE PerfSpac	OFFICE	2						Emeritus Faculty		
Office	G175G	108		Civil Engr	COE PerfSpac	OFFICE		1	0						
Commons	G201	455		Univ Commons	Univ Commons	BLDG COMMONS							Student Commons		
Women's Restroom	G204	213		Gen Building	Gen Building	REST ROOM							Women's Restroom - 4WC,3L-HCA		
Men's Restroom	G206	223		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,3U,3L-HCA		
Custodial	G211	99		Gen Building	Gen Building	BLDG SERV									
Conference Room	G212	0		Civil Engr	Engineering	CONFERENCE									*Not included in study*
Equipment Storage (Conf Rm)	G212A	23		Civil Engr	Engineering	CONF SERV									
Office	G215	142	J. WEISS	Civil Engr	Civil Engr	OFFICE	1								
Student Organization Office	G216	318		Engineering	Engineering	OFFICE			2				Purdue Engr Student Council		
Student Organization Office	G216A	150		Engineering	Engineering	OFFICE			1				Purdue Engr Student Council		
Student Organization Office	G216B	132		Engineering	Engineering	OFFICE			1				Purdue Student Engr Foundation		
Student Organization Office	G216C	117		Engineering	Engineering	OFFICE			1				Purdue Student Engr Foundation		
Student Organization Office	G216D	131		Engineering	Engineering	OFFICE			1				Purdue FIRST		
Office	G217	124	P. Zavattieri	Civil Engr	Civil Engr	OFFICE	1								
Office	G219	124	T. Horton	Civil Engr	Civil Engr	OFFICE	1								
Office	G221	124	J. Olek	Civil Engr	Civil Engr	OFFICE	1								
Office	G223	272		Civil Engr	Civil Engr	OFFICE			8				Graduate Assistants		
Office	G225	127	T. Tzempelikos	Civil Engr	Civil Engr	OFFICE	1								
Office	G227	124	J. Haddock	Civil Engr	Civil Engr	OFFICE	1								
Office	G229	124	M. Santagata	Civil Engr	Civil Engr	OFFICE	1								
Office	G230	410		Civil Engr	Civil Engr	OFFICE			12				Graduate Assistants		
Office	G231	124	J. Sinfield	Civil Engr	Civil Engr	OFFICE	1								
Office	G232	371		Civil Engr	Civil Engr	OFFICE			9				Graduate Assistants		
Office	G233	124	P. Bourdeau	Civil Engr	Civil Engr	OFFICE	1								
Office	G235	124	R. Salgado	Civil Engr	Civil Engr	OFFICE	1								
Office	G237	258	k. sinha	Civil Engr	Civil Engr	OFFICE	1								
Office	G239	124	A. Bobet	Civil Engr	Civil Engr	OFFICE	1								
Office	G241	124	d. bullock	Civil Engr	Civil Engr	OFFICE	1								
Office	G242	365		Civil Engr	Civil Engr	OFFICE			9				Graduate Assistants / TA Room		
Office	G243	124	M. Qu	Civil Engr	Civil Engr	OFFICE	1								
Office	G245	124	M. Prezzi	Civil Engr	Civil Engr	OFFICE	1								
Storage	G252	0		EarthAtm&Pln	EarthAtm&Pln	NONCLS SV							Rock Storage		*Not included in study*
Research Lab	G252A	0	Research Project Room-Grang	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Shower	G252B	0		EarthAtm&Pln	EarthAtm&Pln	NONCLS SV									*Not included in study*
Teaching Lab (EAPS)	G257	0		EarthAtm&Pln	EarthAtm&Pln	CLASS LAB									*Not included in study*
Research Lab	G259	0	Research Lab - Cannon	Health Sci	Health Sci	NONCLAS LAB									*Not included in study*
Research Lab	G263	802		Civil Engr	Civil Engr	NONCLAS LAB						802	Testing Laboratory		
Computer Room	G263A	136		Civil Engr	Civil Engr	NONCLS SV						136			
Research Lab	G266	0	Research Lab - Haq	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Building Receiving	G270	220		Gen Building	Gen Building	RECEIVING							Building Receiving		

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							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
GROUND FLOOR															
Office	G271	91	shipping+receiving-a. holtman	Matrl Distr	Civil Engr	OFFICE		1					Shipping and Receiving		
Research Lab	G277	865		Civil Engr	Civil Engr	NONCLAS LAB						865	Testing Laboratory		
Research Lab	G286	714		Civil Engr	Civil Engr	NONCLAS LAB						714	Soil Mechanics Lab		
Research Lab	G288	0	Research Lab - Granger	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	G288A	0	Research Project Room-Grang	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Machine/Carpenter Shop	G289	0		EarthAtm&Pln	EarthAtm&Pln	SHOP									*Not included in study*
Office	G292	269		Civil Engr	Civil Engr	OFFICE			5				Graduate Assistants		
Office	G292A	141		Civil Engr	Civil Engr	OFFICE		4					Post Docs + Visiting Scholars		
Office	G292B	135		Civil Engr	Civil Engr	OFFICE			4				Graduate Assistants		
Office	G292C	174		Civil Engr	Civil Engr	OFFICE			6				Graduate Assistants		
Reception	G293	156		Engineering	COE PerfSpac	OFF SERV							Engineering Honors		
Office	G293	127	Engineering Honors - M. Hale	Engineering	COE PerfSpac	OFFICE		1					Engineering Honors Reception		
Office	G293A	116	Engineering Honors - T. Whale	Engineering	COE PerfSpac	OFFICE		1					Engineering Honors		
Office	G293B	115	Engineering Honors - J. Alter	Engineering	COE PerfSpac	OFFICE		1					Engineering Honors		
Office	G293C	116	Engineering Honors - E. Naum	Engineering	COE PerfSpac	OFFICE	1						Engineering Honors		
Elevator	GE01	52		Gen Building	Gen Building	CIRCULATE									
Elevator	GE03	77		Gen Building	Gen Building	CIRCULATE									
Elevator	GE04	77		Gen Building	Gen Building	CIRCULATE									
Lift	GE05	15		Gen Building	Gen Building	CIRCULATE									
Lift	GE06	21		Gen Building	Gen Building	CIRCULATE							Lift Inside G149		
Corridor	GH01	378		Gen Building	Gen Building	CIRCULATE									
Corridor	GH02	193		Gen Building	Gen Building	CIRCULATE									
Corridor	GH03	703		Gen Building	Gen Building	CIRCULATE									
Corridor	GH05	159		Gen Building	Gen Building	CIRCULATE									
Corridor	GH06	121		Gen Building	Gen Building	CIRCULATE									
Corridor	GH07	2196		Gen Building	Gen Building	CIRCULATE									
Corridor	GH08	701		Gen Building	Gen Building	CIRCULATE									
Corridor	GH09	802		Gen Building	Gen Building	CIRCULATE									
Corridor	GH10	547		Gen Building	Gen Building	CIRCULATE									*Old freight elevator was located in this area but was removed and infilled.*
Corridor	GH11	1155		Gen Building	Gen Building	CIRCULATE									
Vestibule	GH12	126		Gen Building	Gen Building	CIRCULATE									*Existing ADA accessible entrance.*
Vestibule	GH13	463		Gen Building	Gen Building	CIRCULATE									*Existing ADA accessible entrance. Behind exterior fence. Locked?*
Corridor	GH14	249		Gen Building	Gen Building	CIRCULATE									
Stair	GS01	248		Gen Building	Gen Building	CIRCULATE									
Stair	GS02	203		Gen Building	Gen Building	CIRCULATE									
Stair	GS03	76		Gen Building	Gen Building	CIRCULATE									
Stair	GS04	337		Gen Building	Gen Building	CIRCULATE									
Stair	GS06	224		Gen Building	Gen Building	CIRCULATE									
Stair	GS08	285		Gen Building	Gen Building	CIRCULATE									
Stair	GS09	285		Gen Building	Gen Building	CIRCULATE									
Mechanical	GU01	748		Gen Building	Gen Building	MECHANICAL									
Building Engineer's Shop	GU01	402		Oper & Maint	Oper & Maint	SHOP							*This is currently used as mechanical space with (3) AHUs.*		*Main OSP voice/data/telecommunications lines located here.*
IDF	GU01A	91		Gen Building	Gen Building	MECHANICAL									
IDF	GU03	85		Gen Building	Gen Building	MECHANICAL									
Electrical	GU04	79		Gen Building	Gen Building	MECHANICAL									

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							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
GROUND FLOOR															
Electrical	GU05	79		Gen Building	Gen Building	MECHANICAL									
Subtotals		**37984**	**SF**				28	29	95	0	3,477	8,863			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
FIRST FLOOR															
Reception	1101	210	Reception + Secretary - C. Bo	Civil Engr	Civil Engr	OFFICE		1							
Corridor	1101	210		Civil Engr	Civil Engr	OFF SERV									
Office	1101A	209	R.S. Govindaraju	Civil Engr	Civil Engr	OFFICE	1								
Office	1101B	200	R. JACKO	Civil Engr	Civil Engr	OFFICE	1								
Office	1101C	135	D. Lyn	Civil Engr	Civil Engr	OFFICE	1								
Office	1101D	141	C. Troy	Civil Engr	Civil Engr	OFFICE	1								
Conference Room	1101E	282		Civil Engr	Civil Engr	CONFERENCE				14			Conference Room		
Office	1101G	147	V. Merwade	Civil Engr	Civil Engr	OFFICE	1								
Storage	1101H	23		Civil Engr	Civil Engr	OFF SERV									
Office	1101J	140		Civil Engr	Civil Engr	OFFICE	2						Emeritus Faculty		
Collaboration Space	1110	145		Civil Engr	Civil Engr	NONCLS SV						145	Data Visualization Room-upper		
SMAS Classroom	1113	726		Gen Acad Cls	Gen Acad Cls	CLASSROOM				32			Tables + Chairs		
Teaching Lab	1115	790		Civil Engr	Civil Engr	CLASS LAB					790		Geomatics Design Lab		
Equipment Setup	1119	336		Civil Engr	Civil Engr	CLS LB SV					336		Geomatics Lab Equipment Setup		
Equipment Checkout	1121	472		Civil Engr	Civil Engr	CLS LB SV					472		Geomatics Field Equip Checkout		
Research Lab	1122	828		Civil Engr	Civil Engr	NONCLAS LAB						828	Traffic Signal Lab		
Research Lab	1122A	334		Civil Engr	Civil Engr	NONCLAS LAB						334	Transportation Lab		
Mechanical	1124	256		Gen Building	Gen Building	MECHANICAL									
Custodial	1125	35		Gen Building	Gen Building	BLDG SERV									
Kitchenette	1127	196		Civil Engr	Civil Engr	OFF SERV									
Faculty Lounge	1129	681		Civil Engr	Civil Engr	DEPT COMMONS				25					
Women's Restroom	1138	182		Gen Building	Gen Building	REST ROOM							Women's Restroom - 2WC, 1L		
Men's Restroom	1140	146		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,3U,3L		
Reception	1141	149		Civil Engr	Civil Engr	OFFICE				2			Main Office Reception		
Waiting	1141	224		Civil Engr	Civil Engr	OFF SERV				6					
Office	1141A	181	C. Crozier	Civil Engr	Civil Engr	OFFICE		1							
Office	1141B	195	R. Govindaraju	Civil Engr	Civil Engr	OFFICE	1								
Conference Room	1141C	121		Civil Engr	Civil Engr	CONFERENCE				6			Conference Room		
Office	1141D	120	Advancement Director - D. Fry	PRF Devlpmnt	Civil Engr	OFFICE		1							
Office	1141E	116	program coordinator-k. heath	Civil Engr	Civil Engr	OFFICE		1							
Office	1141F	132	undergrad prgrm admin-m. lah	Civil Engr	Civil Engr	OFFICE		1							
Office	1141G	155	J. Ricksy	Civil Engr	Civil Engr	OFFICE	0	1							
Office	1141H	162		Civil Engr	Civil Engr	OFFICE		1	0						
SMAS Classroom	1144	1094		Gen Acad Cls	Gen Acad Cls	CLASSROOM				100			Tablet Arm Chairs		
Electrical	1144A	72		Gen Building	Gen Building	MECHANICAL									
Office	1147	261	J. Jesnik + Student Worker	Civil Engr	Civil Engr	OFFICE		2	0						
Office	1147A	173	c. lawley	Civil Engr	Civil Engr	OFFICE		1							
Copy Room / Files	1149	477		Civil Engr	Civil Engr	OFF SERV									
Office (Research Lab)	1157	135	research lab-zavattieri	Civil Engr	Civil Engr	NONCLAS LAB			3			135	Graduate Assistants	*G157*	
Office (Research Lab)	1157A	132	research lab-zavattieri	Civil Engr	Civil Engr	NONCLAS LAB			4			132	Graduate Assistants	*G157*	
Office	1163	0	Clerical - H. Terrell	Health Sci	Health Sci	OFFICE									*Not included in study*
Reception	1163	0		Health Sci	Health Sci	OFF SERV									*Not included in study*
Office	1163A	0	T. Strange	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1163B	0	R. Ricci	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1163C	0	L. Hilliard	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1163D	0	S. Liu	Health Sci	Health Sci	OFFICE									*Not included in study*
Conference Room	1163E	0		Health Sci	Health Sci	CONFERENCE							OSS Meeting Room		*Not included in study*
Student Resource Room	1163F	0		Health Sci	Health Sci	STUDY									*Not included in study*
Corridor	1163G	0		Health Sci	Health Sci	OFF SERV									*Not included in study*

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							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
FIRST FLOOR															
Prep Room	1168	0		Health Sci	Health Sci	CLS LB SV									Not included in study
Reception	1169	0	Reception - D. Tate	Health Sci	Health Sci	OFFICE									Not included in study
Conference Room	1173	0		Health Sci	Health Sci	CONFERENCE									Not included in study
Office	1173A	0	W. Zheng	Health Sci	Health Sci	OFFICE									Not included in study
Office	1173B	0	K. Walker	Health Sci	Health Sci	OFFICE									Not included in study
Office	1173C	0	K. Stantz	Health Sci	Health Sci	OFFICE									Not included in study
File Storage	1173D	0		Health Sci	Health Sci	OFF SERV									Not included in study
Corridor	1173E	0		Health Sci	Health Sci	OFF SERV									Not included in study
Research Lab	1201	0	Research Lab - Zheng	Health Sci	Health Sci	NONCLAS LAB									Not included in study
Research Lab	1203	0	Research Lab - Freeman	Health Sci	Health Sci	NONCLAS LAB									Not included in study
Women's Restroom	1204	213		Gen Building	Gen Building	REST ROOM							Women's Restroom - 4WC,3L-HCA		
Men's Restroom	1206	212		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,3U,3L-HCA		
Custodial	1211	101		Gen Building	Gen Building	BLDG SERV									
Computer Lab	1212	1661		Civil Engr	COE PerfSpac	OPEN LAB				37			D-CL Computer Lab (COE)		
Conference Room	1217	265		Constr Engr	Constr Engr	CONFERENCE				16			Conference Room		
Copy Room	1218	83		Constr Engr	Constr Engr	OFF SERV									
Office	1219	148	dir of internships - fulk	Constr Engr	Constr Engr	OFFICE		1							
Office	1221	126	B. Sondgeroth	Constr Engr	Constr Engr	OFFICE		1							
Office	1223	252	Dept Head - M. Hastak	Constr Engr	Constr Engr	OFFICE	1								
Office	1225	129	Clerical - M. Geiger	Constr Engr	Constr Engr	OFFICE		1							
Office	1227	257	N. Haab	Constr Engr	Constr Engr	OFFICE		1							
Storage	1228	176		Constr Engr	Constr Engr	OFF SERV									
Office	1230	153		Constr Engr	Civil Engr	OFFICE			5				Graduate Assistants		
Office	1230	153		Civil Engr	Civil Engr	OFFICE			4				Graduate Assistants		
Office	1231	126	A. Kandil	Constr Engr	Constr Engr	OFFICE	1								
Office	1233	126	Lecturer - V. Gervais	Constr Engr	Constr Engr	OFFICE		1					Lecturer		
Office	1235	126	H. Cai	Constr Engr	Constr Engr	OFFICE	1								
Office	1237	264		Constr Engr	Constr Engr	OFFICE	0	4					Visiting Scholars		
Office	1239	126	P. Karava	Constr Engr	Constr Engr	OFFICE	1								
Office	1240	125		Constr Engr	COE PerfSpac	OFFICE			4				Graduate Assistants		
Office	1241	126	D. Abraham	Constr Engr	Constr Engr	OFFICE	1								
Student Organization Office	1242	122		Engineering	Engineering	OFFICE			1				Tau Beta Pi		
Office	1243	121	P. Dunston	Constr Engr	Constr Engr	OFFICE	1								
Student Organization Office	1244	123		Engineering	Engineering	OFFICE			2				SWE		
Office	1245	148	T. Weidner	Constr Engr	Constr Engr	OFFICE	1								
Student Organization Office	1246	123		Engineering	Engineering	OFFICE			1				NSBE		
Student Organization Office	1248	110		Engineering	Engineering	OFFICE			1				HPE		
Student Organization Office	1250	111		Engineering	Engineering	OFFICE			1				PSPE		
SMAS Classroom	1252	932		Gen Acad Cls	Gen Acad Cls	CLASSROOM				64			Tablet Arm Chairs		
Equipment Room	1253	103		Health Sci	REM	NONCLS SV									
Equipment Room	1253	155		**REM/Health Sc**	**REM/Health Sc**	OFF SERV									
Office	1255	386		Constr Engr	Constr Engr	OFFICE			6				Graduate Assistants		
Office	1259	375	Area was "COE PerfSpac" for	Engineering	**CEM**	OFFICE		1	2				CEM		
Conference Room	1259A	125	Area was "COE PerfSpac" for	Engineering	**CEM**	CONFERENCE				6			Conference Room		
Office	1259B	125	Area was "COE PerfSpac" for	Engineering	**CEM**	OFFICE		2					CEM		
Office	1259C	125	Area was "COE PerfSpac" for	Engineering	**CEM**	OFFICE		1					CEM		

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
FIRST FLOOR															
Office	1259D	164	Area was "COE PerfSpac" for	Engineering	**CEM**	OFFICE	1						CEM		
Office	1263	0	Y. Nash	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1263A	0		Health Sci	Health Sci	OFFICE							Graduate Assistants		*Not included in study*
Office	1263B	0		Health Sci	Health Sci	OFFICE							Graduate Assistants		*Not included in study*
Storage	1263C	0		Health Sci	Health Sci	OFF SERV									*Not included in study*
Office	1263D	0	J. Freeman	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1263E	0	J. McGlothlin	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1263F	0	U. Dydak	Health Sci	Health Sci	OFFICE									*Not included in study*
SMAS Classroom	1266	467		Gen Acad Cls	Gen Acad Cls	CLASSROOM				30			Tablet Arm Chairs		
Research Lab	1268	521		Civil Engr	Civil Engr	NONCLAS LAB						521	Trans. Comp. Lab		
Office	1269	0	E. Wells	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1271	0	J. Cannon	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1273	0	F. Rosenthal	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1275	0	C. Tsai	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1277	0	L. Nie	Health Sci	Health Sci	OFFICE									*Not included in study*
Office	1279	0	W. Jiang + Post Doc	Health Sci	Health Sci	OFFICE									*Not included in study*
Storage	1281	0		Health Sci	Health Sci	OFF SERV									*Not included in study*
Research Lab	1285	0	Ergonomics Lab - McGlothlin	Health Sci	Health Sci	NONCLAS LAB							Ergonomics Lab		*Not included in study*
Research Lab	1287	0	Research Lab - Tsai	Health Sci	Health Sci	NONCLAS LAB									*Not included in study*
Teaching Lab	1292	0		Health Sci	Health Sci	CLASS LAB							Counting Lab		*Not included in study*
Teaching Lab	1292	0		Pharm Pract	Pharm Pract	CLASS LAB							Counting Lab - NUPH		*Not included in study*
Teaching Lab	1293	0		Health Sci	Health Sci	CLS LB SV							Diagnostic Imaging Lab		*Not included in study*
Teaching Lab	1293	0		Pharm Pract	Pharm Pract	CLS LB SV							Diagnostic Imaging Lab - NUPH		*Not included in study*
Teaching Lab	1294	0		Health Sci	Health Sci	CLASS LAB							Teaching Lab		*Not included in study*
Teaching Lab	1294	0		Pharm Pract	Pharm Pract	CLASS LAB							Teaching Lab - NUPH		*Not included in study*
Research Lab	1295	0		Health Sci	Health Sci	NONCLAS LAB							Surgery Room		*Not included in study*
Research Lab	1297	0		Health Sci	Health Sci	NONCLAS LAB							Cell Culture Room		*Not included in study*
Research Lab	1299	0	Research Lab - Freeman	Health Sci	Health Sci	NONCLAS LAB									*Not included in study*
Elevator	1E01	52		Gen Building	Gen Building	CIRCULATE									
Elevator	1E03	77		Gen Building	Gen Building	CIRCULATE									
Elevator	1E04	77		Gen Building	Gen Building	CIRCULATE									
Lift	1E06	21		Gen Building	Gen Building	CIRCULATE							Lift Inside G149		
Corridor	1H01	1490		Gen Building	Gen Building	CIRCULATE									
Corridor	1H03	697		Gen Building	Gen Building	CIRCULATE									
Corridor	1H04	2015		Gen Building	Gen Building	CIRCULATE									
Corridor	1H05	1159		Gen Building	Gen Building	CIRCULATE									
Corridor	1H06	1155		Gen Building	Gen Building	CIRCULATE									
Stair	1S01	248		Gen Building	Gen Building	CIRCULATE									
Stair	1S02	248		Gen Building	Gen Building	CIRCULATE									
Stair	1S04	337		Gen Building	Gen Building	CIRCULATE									
Stair	1S06	227		Gen Building	Gen Building	CIRCULATE									
Stair	1S08	295		Gen Building	Gen Building	CIRCULATE									
Stair	1S09	295		Gen Building	Gen Building	CIRCULATE									
Electrical	1U01	70		Gen Building	Gen Building	MECHANICAL									
IDF	1U03	85		Gen Building	Gen Building	MECHANICAL									
Electrical	1U04	79		Gen Building	Gen Building	MECHANICAL									
Electrical	1U05	81		Gen Building	Gen Building	MECHANICAL									
Subtotals		**28715**	**SF**				16	23	34	338	1,598	2,095			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
SECOND FLOOR															
SMAS Classroom	2101	714		Gen Acad Cls	Gen Acad Cls	CLASSROOM				45			Tablet Arm Chairs		
SMAS Classroom	2102	660		Gen Acad Cls	Gen Acad Cls	CLASSROOM				40			Tablet Arm Chairs		
SMAS Classroom	2107	744		Gen Acad Cls	Gen Acad Cls	CLASSROOM				32			Tables + Chairs		
SMAS Classroom	2108	973		Gen Acad Cls	Gen Acad Cls	CLASSROOM				44			Tables + Chairs		
SMAS Classroom	2113	741		Gen Acad Cls	Gen Acad Cls	CLASSROOM				32			Tables + Chairs		
Mechanical	2114	305		Gen Building	Gen Building	MECHANICAL							HVAC for 2101-2123		
SMAS Classroom	2117	741		Gen Acad Cls	Gen Acad Cls	CLASSROOM				45			Tablet Arm Chairs		
SMAS Classroom	2118	888		Gen Acad Cls	Gen Acad Cls	CLASSROOM				40			Tables + Chairs		
SMAS Classroom	2123	765		Gen Acad Cls	Gen Acad Cls	CLASSROOM				40			Tables + Chairs		
Equipment Storage	2124	210		Civil Engr	Civil Engr	NONCLS SV								*Main CE Office Storage*	
Office	2125	155		Civil Engr	Civil Engr	OFFICE	1								*changed from Visiting Prof + Post Doc*
Office	2127	155	I. Hua	Civil Engr	Civil Engr	OFFICE	1								
Office	2129	231	E. Blatchley	Civil Engr	Civil Engr	OFFICE	1								
Research Lab	2131	676		Civil Engr	Civil Engr	NONCLAS LAB						676			
Office	2132	162		Civil Engr	Civil Engr	OFFICE		4					Post Docs		
Custodial	2134	67		Gen Building	Gen Building	BLDG SERV									
Research Lab	2135	849		Civil Engr	Civil Engr	NONCLAS LAB						849			
Women's Restroom	2136	188		Gen Building	Gen Building	REST ROOM							Women's Restroom - 2WC, 1L		
Men's Restroom	2140	251		Gen Building	Gen Building	REST ROOM							Men's Restroom - 4WC,5U,4L		
Teaching Lab	2146	577		Civil Engr	Civil Engr	CLASS LAB					577				
Teaching Lab	2146	576		Env&EcolEngr	Env&EcolEngr	CLASS LAB					576				
Student Lounge	2152	554		Civil Engr	Civil Engr	BLDG COMMONS				31			Pankow Student Lounge		*Access to green roof area above G157*
Student Organization Office	2154	290		Civil Engr	Civil Engr	OFFICE			3				asce + ceeas + chi epsilon		
Office	2155	1105		Civil Engr	Civil Engr	OFFICE			24				Transportation Grad Assistants		
Storage	2155A	164		Civil Engr	Civil Engr	NONCLS SV							Equipment Storage		
Teaching Lab	2158	870		Civil Engr	Civil Engr	CLASS LAB				12	870		Self-Paced Struct Mechanic		
Office	2169	0	Main Office Receptionist - T. M	EarthAtm&Pln	EarthAtm&Pln	OFFICE							Main Office Reception		*Not included in study*
Office	2169A	0	Department Head - I. Chaubey	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	2169B	0	Asst Department Head - B. Gil	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	2169C	0	Communications Coordinator -	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	2169D	0	Graduate Office - K. Kincade	EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Office		*Not included in study*
Office	2169E	0	Undergrad Counseling-N.Stew	EarthAtm&Pln	EarthAtm&Pln	OFFICE							Undergrad Counseling		*Not included in study*
Office	2169F	0	Business Office - K. Levitt	EarthAtm&Pln	EarthAtm&Pln	OFFICE							Business Office		*Not included in study*
Office	2169G	0	business office - s. cordell	EarthAtm&Pln	EarthAtm&Pln	OFFICE							Business Office		*Not included in study*
Office	2169H	0	Business Office - M. Guinn	Science	EarthAtm&Pln	OFFICE							Business Office		*Not included in study*
A/V Room	2169J	0		EarthAtm&Pln	EarthAtm&Pln	OFF SERV									*Not included in study*
Files Storage	2169K	0		EarthAtm&Pln	EarthAtm&Pln	OFF SERV									*Not included in study*
Office	2169L	0	Business Office - D. Vanhorn	EarthAtm&Pln	EarthAtm&Pln	OFFICE							Business Office		*Not included in study*
Copy Room	2169M	0		EarthAtm&Pln	EarthAtm&Pln	OFF SERV							Duplicating/Fax/Printer/Mail		*Not included in study*
Corridor	2169N	0		EarthAtm&Pln	EarthAtm&Pln	OFF SERV									*Not included in study*
Custodial	2170	0		Gen Building	Gen Building	BLDG SERV									*Not included in study*
Conference Room	2173	0		EarthAtm&Pln	EarthAtm&Pln	CONFERENCE									*Not included in study*
Conference Room	2201	0		EarthAtm&Pln	EarthAtm&Pln	CONFERENCE									*Not included in study*
Women's Restroom	2204	213		Gen Building	Gen Building	REST ROOM							Women's Restroom - 4WC,3L-HCA		
Men's Restroom	2206	223		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,3U,3L-HCA		
Custodial	2211	101		Gen Building	Gen Building	BLDG SERV									
Library	2215	0		Libraries	Libraries	STUDY							EAS Library		*Not included in study*

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
SECOND FLOOR															
Library	2215	0		Libraries	Libraries	STACK							EAS Library		Not included in study
Work Area	2215A	0	work area - t. wade	Libraries	Libraries	LIBR PROC									Not included in study
Office	2215B	0	M. Nelson	Libraries	Libraries	OFFICE									Not included in study
Copy Room	2215C	0		Libraries	Libraries	STUDY SRV									Not included in study
Study Room	2215D	0		Libraries	Libraries	STUDY									Not included in study
Office	2215E	0		Libraries	Libraries	OFFICE									Not included in study
Teaching Lab	2244	0		EarthAtm&Pln	EarthAtm&Pln	CLASS LAB									Not included in study
Research Lab	2245	0	Research Lab - Ogg	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2249	0	Computer Support - F. Bakhit	EarthAtm&Pln	EarthAtm&Pln	OFFICE							Computer Support		Not included in study
Computer Machine Room	2249	0		EarthAtm&Pln	EarthAtm&Pln	CEN COMPUTER									Not included in study
Teaching Lab	2252	0		EarthAtm&Pln	EarthAtm&Pln	CLASS LAB									Not included in study
Library Stacks	2253	0		Libraries	Libraries	OPEN STACK							Map Collection		Not included in study
Research Lab	2255	0	Research Lab - Cushman	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2255	0		EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assts + Post Doc		Not included in study
Research Lab	2259	0	Research Lab - Melosh	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Research Lab	2261	0	Research Lab - Melosh	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2261	0		EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assts + Visiting Scholars		Not included in study
Research Lab	2263	0	Research Lab - Horgan + Mint	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2263	0		EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		Not included in study
Research Lab	2263A	0	Research Project Room - Horg	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2267	0		EarthAtm&Pln	EarthAtm&Pln	OFFICE							Post Doc		Not included in study
Office	2269	0	Faculty - Q. Zhuang	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Research Lab	2270	0	Outreach Lab - Smith	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2270	0	Staff - S. Smith	EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE									Not included in study
Student Work Area	2270	0		EarthAtm&Pln	EarthAtm&Pln	NONCLS SV									Not included in study
Office	2271	0	Faculty - L. Braile	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	2273	0	Faculty - J. Ogg	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	2275	0	Faculty - R. Nowack	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	2277	0	Faculty - S. Haq	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Graduate Student Lounge	2279	0		EarthAtm&Pln	EarthAtm&Pln	DEPT COMMONS									Not included in study
Student Work Area	2285	0		EarthAtm&Pln	EarthAtm&Pln	OFF SERV									Not included in study
Office	2285A	0	J. Bell	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	2285B	0	Faculty - D. Shepardson	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	2285C	0	Faucity - G. Krockover	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	2285D	0	Faculty - D. Ballotti	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Storage	2285E	0		EarthAtm&Pln	EarthAtm&Pln	OFF SERV									Not included in study
Research Lab	2286	0	Research Lab - Harbor + Lifto	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2286	0		EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assts + Visiting Scholars		Not included in study
Office	2286A	0	Post Doc - Heyman	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	2286B	0	Post Doc - Goehring	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Research Lab	2289	0	Research Lab - Harbor + Lifto	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2289	0		EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistant		Not included in study
Research Lab	2294	0	Research Lab - Harbor + Lifto	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2294	0		EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assts + Visiting Scholars		Not included in study
Research Lab	2295	0	research lab - nowack	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	2295	0		EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistant + Post Doc		Not included in study
Computing Resource Room	2296	0		EarthAtm&Pln	EarthAtm&Pln	OFF SERV									Not included in study

ME/RAIL/HAMP Study
Hampton Hall

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
SECOND FLOOR															
Elevator	2E01	52		Gen Building	Gen Building	CIRCULATE									
Elevator	2E03	77		Gen Building	Gen Building	CIRCULATE									
Elevator	2E04	77		Gen Building	Gen Building	CIRCULATE									
Corridor	2H01	1327		Gen Building	Gen Building	CIRCULATE									
Corridor	2H02	1232		Gen Building	Gen Building	CIRCULATE									
Corridor	2H03	2222		Gen Building	Gen Building	CIRCULATE									
Corridor	2H04	1152		Gen Building	Gen Building	CIRCULATE									
Corridor	2H05	78		Gen Building	Gen Building	CIRCULATE									
Stair	2S01	248		Gen Building	Gen Building	CIRCULATE									
Stair	2S02	248		Gen Building	Gen Building	CIRCULATE									
Stair	2S04	248		Gen Building	Gen Building	CIRCULATE									
Stair	2S06	223		Gen Building	Gen Building	CIRCULATE									
Stair	2S08	274		Gen Building	Gen Building	CIRCULATE									
Stair	2S09	274		Gen Building	Gen Building	CIRCULATE									
Electrical	2U01	65		Gen Building	Gen Building	MECHANICAL									
Mechanical	2U02	1237		Gen Building	Gen Building	MECHANICAL									*Autoclave (x2) through into Research Lab 2135.*
IDF	2U03	85		Gen Building	Gen Building	MECHANICAL									
Electrical	2U04	78		Gen Building	Gen Building	MECHANICAL									
Electrical	2U05	81		Gen Building	Gen Building	MECHANICAL									
Subtotals		**23426**	**SF**				3	4	27	361	2,023	1,525			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
THIRD FLOOR															
Storage	3101	7		Civil Engr	Civil Engr	NONCLS SV									
Research Lab	3104	1138		Civil Engr	Civil Engr	NONCLAS LAB						1,138	Environmental Lab; Wet lab		*Emergency shower w/o drain. Possible relo. To Flex Lab.*
Office (Research Lab)	3104A	105		Civil Engr	Civil Engr	OFFICE			2				Post Doc	*3104*	
Office (Research Lab)	3104B	105		Civil Engr	Civil Engr	OFFICE			2				Visiting Scholar + Post Doc	*3104*	
Storage	3104C	7		Civil Engr	Civil Engr	NONCLS SV						7			
Office (Research Lab)	3104D	99		Civil Engr	Civil Engr	OFFICE			3				ENV/HYD Graduate Assistants	*3104*	
Office (Research Lab)	3104E	99		Civil Engr	Civil Engr	OFFICE			3				ENV/HYD Graduate Assistants	*3104*	
Office (Research Lab)	3104F	99		Civil Engr	Civil Engr	OFFICE			3				ENV/HYD Graduate Assistants	*3104*	
Constant Temperature Room	3107	48		Civil Engr	Civil Engr	NONCLS SV						48			*Reported to be in poor condition*
Constant Temperature Room	3109	42		Civil Engr	Civil Engr	NONCLS SV						42			*Reported to be in poor condition*
Constant Temperature Room	3111	49		Civil Engr	Civil Engr	NONCLS SV						49			*Reported to be in poor condition*
Research Lab	3119	2089		Civil Engr	Civil Engr	NONCLAS LAB						2,089	Analytical Instrument Lab; Wet lab		*Emergency shower w/o drain. Possible relo. To Flex Lab.*
Research Lab	3119A	126		Civil Engr	Civil Engr	NONCLS SV						126	Computer Room		
Constant Temperature Room	3119B	49		Civil Engr	Civil Engr	NONCLS SV						49			*Reported to be in poor condition*
Equipment Room	3119C	16		Civil Engr	Civil Engr	NONCLS SV						16	Refrig Equip + Contr		
Constant Temperature Room	3119D	48		Civil Engr	Civil Engr	NONCLS SV						48			*Reported to be in poor condition*
Research Lab	3120	770		Civil Engr	Civil Engr	NONCLAS LAB						770	Wet lab		*Emergency shower w/o drain. Possible relo. To Flex Lab.*
Office (Research Lab)	3120A	99		Civil Engr	Civil Engr	OFFICE			2				Graduate Assistants	*3120*	
Research Lab	3124	789		Civil Engr	Civil Engr	NONCLAS LAB						789	Wet lab		*Emergency shower w/o drain. Possible relo. To Flex Lab.*
Dark Room	3124A	99		Civil Engr	Civil Engr	NONCLS SV						99		*3124*	
Research Lab	3125	1033		Civil Engr	Civil Engr	NONCLAS LAB						1,033	Environmental Research Lab; Wet lab		*Emergency shower w/o drain. Possible relo. To Flex Lab.*
Office (Research Lab)	3125C	108		Civil Engr	Civil Engr	OFFICE			3				Environmental Graduate Assts	*3125*	
Office (Research Lab)	3125D	104		Civil Engr	Civil Engr	OFFICE			3				Environmental Graduate Assts	*3125*	
Research Lab	3129	675		Civil Engr	Civil Engr	NONCLAS LAB						675			*Existing refrigerators and freezers. Existing biohazard hood. No sink.*
Storage	3131	59		Civil Engr	Civil Engr	NONCLS SV						59			
Storage	3133	152		Civil Engr	Civil Engr	NONCLS SV						152	Tank Storage		*Has an area open to below.*
Freezer	3133A	39		Civil Engr	Civil Engr	NONCLS SV						39			
Refrigeration Equipment	3133B	35		Civil Engr	Civil Engr	NONCLS SV						35			*Refrigeration eqiupment.*
Constant Temperature Room	3135	40		Civil Engr	Civil Engr	NONCLS SV						40			
Constant Temperature Room	3137	70		Civil Engr	Civil Engr	NONCLS SV						70			
Research Lab	3139	568		Civil Engr	Civil Engr	NONCLAS LAB						568	Wet lab		*Emergency shower w/o drain. Possible relo. To Flex Lab.*
Storage	3139A	24		Civil Engr	Civil Engr	NONCLS SV						24			
Dark Room	3139B	44		Civil Engr	Civil Engr	NONCLS SV						44			
Office	3139C	100		Civil Engr	Civil Engr	OFFICE			3				Environmental Graduate Assts		
Men's Restroom	3140	181		Gen Building	Gen Building	REST ROOM							Men's Restroom - 3WC,4U,3L		
Computer Lab	3144	1463		Gen Acad Lab	Gen Acad Lab	STUDY							ITAP - 60 stations		
Office	3145	237	C. Boone	Civil Engr	Civil Engr	OFFICE		1							
Office	3145A	120		Civil Engr	Civil Engr	OFFICE	1								*Changed from "(2) Visiting Scholars"*
Office	3145B	274	S. RAO	Civil Engr	Civil Engr	OFFICE	1								
Office	3145C	129	L. Nies	Civil Engr	Civil Engr	OFFICE	1								
Office	3145D	174	C. Jafvert	Civil Engr	Civil Engr	OFFICE	1								

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
THIRD FLOOR															
Office	3145E	139		Civil Engr	Civil Engr	OFFICE	1								*Changed from "(2) Visiting Scholars"*
SMAS Classroom	3153	1087		Gen Acad Cls	Gen Acad Cls	CLASSROOM				48			Tables + Chairs		
SMAS Classroom Storage	3153A	149		Gen Acad Cls	Gen Acad Cls	CLSRM SRV									
Senior Design Lab	3154	1165		Civil Engr	Civil Engr	OPEN LAB				16	1,165		D-CL Senior Design Project Lab		
Computer Service	3154A	144		Civil Engr	Civil Engr	OPEN SERV					144		Computer Service	*3154*	
Office	3158	148		Civil Engr	Civil Engr	OFFICE			3				Graduate Assistants	*3154*	
Research Lab	3163			EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3165		Research Lab - Granger	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3169		Research Lab - Flesch	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Office	3169			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		*Not included in study*
Research Lab	3169A		Research Project Room - Flesc	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3175		Research lab - Filley	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Teaching Lab	3201			EarthAtm&Pln	EarthAtm&Pln	CLASS LAB									*Not included in study*
Women's Restroom	3204	213		Gen Building	Gen Building	REST ROOM							WOMEN 4WC,3L - HCA		
Men's Restroom	3206	223		Gen Building	Gen Building	REST ROOM							MEN 2WC,3U,3L - HCA		
Office	3210			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Visiting Scholars + Post Docs		*Not included in study*
Custodial	3211	67		Gen Building	Gen Building	BLDG SERV									
Office	3212			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Visiting Scholars + Post Docs		*Not included in study*
Office	3213			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Visiting Scholars + Post Docs		*Not included in study*
Teaching Lab	3214			EarthAtm&Pln	EarthAtm&Pln	CLASS LAB									*Not included in study*
Prep Room	3214A			EarthAtm&Pln	EarthAtm&Pln	CLS LB SV							3214 Prep Room		*Not included in study*
Office	3215		Faculty - P. Shepson	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3217		Faculty - Y. Wu	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3219		Faculty - W. Tung	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3221		Faculty - R. Trapp	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3223		Faculty - J. Cushman	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3225		Faculty - T. Filley	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3227		Faculty - W. Zinsmeister	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3229		Faculty - G. Michalski	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3231		Faculty - D. Minton	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Conference Room	3232			EarthAtm&Pln	EarthAtm&Pln	CONFERENCE									*Not included in study*
Office	3233		Faculty - B. Horgan	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3235		Faculty - L. Flesch	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3237		Faculty - J. Melosh	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3239		Faculty - J. Richardson	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3241		Faculty - H. Gilbert	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3243		Faculty - Vacant	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3245		Faculty - T. West	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Research Lab	3246		Research Lab - West + Braile	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3250		research lab - michalski	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Office	3250			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assts + Post Docs		*Not included in study*
Research Lab	3251		Research Lab - Filley	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3253		Research Lab - Chaubey	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3254		Research Lab - Filley	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3255		Research Lab - Elliott	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3257		Research Lab - Open	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3261		Research Lab - Freed+Gilbert	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Office	3261			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		*Not included in study*
Research Lab	3263		Research Lab - Andronicos + F	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
THIRD FLOOR															
Office	3263			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		*Not included in study*
Office	3263A			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Post Doc		*Not included in study*
Research Lab	3266		Research Lab - Ridgway	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Office	3267		Faculty - J. Elliott	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3269		Faculty - A. Freed	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3270			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		*Not included in study*
Research Lab	3270		Research Lab - Minton	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Office	3271		Faculty - Y.N. Shieh	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3273		Faculty - J. Harbor	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3275		Faculty - N. Lifton	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Printer	3277			EarthAtm&Pln	EarthAtm&Pln	OFF SERV									*Not included in study*
Conference Room	3277			EarthAtm&Pln	EarthAtm&Pln	CONFERENCE									*Not included in study*
Office	3277A		Faculty - D. Granger	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3277B		Faculty - K. Ridgway	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3277C		Faculty - J. Elliott	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Office	3277D		Faculty - C. Andronicos	EarthAtm&Pln	EarthAtm&Pln	OFFICE									*Not included in study*
Research Lab	3286		Research Lab - Michalski	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3286A		Research Projct Room-Michals	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3287		Research Lab - Shieh + Michal	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Research Lab	3288			EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB							Sample Prep Lab - Stable Isotope		*Not included in study*
Research Lab	3289			EarthAtm&Pln	EarthAtm&Pln	NONCLS SV							Research Sample Collection		*Not included in study*
Research Lab	3291			EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB							Wet Chemical Lab - Stable Isotope		*Not included in study*
Research Lab	3293		Research Lab - Ridgway	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Office	3293			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		*Not included in study*
Research Lab	3294			EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB							Research Lab - Stable Isotope		*Not included in study*
Elevator	3E01	52		Gen Building	Gen Building	CIRCULATE									
Elevator	3E03	77		Gen Building	Gen Building	CIRCULATE									
Elevator	3E04	77		Gen Building	Gen Building	CIRCULATE									
Corridor	3H01	1188		Gen Building	Gen Building	CIRCULATE									
Corridor	3H02	1232		Gen Building	Gen Building	CIRCULATE									
Corridor	3H03	1792		Gen Building	Gen Building	CIRCULATE									
Corridor	3H04	1225		Gen Building	Gen Building	CIRCULATE									
Corridor	3H05	1155		Gen Building	Gen Building	CIRCULATE									
Stair	3S01	248		Gen Building	Gen Building	CIRCULATE									
Stair	3S02	248		Gen Building	Gen Building	CIRCULATE									
Stair	3S04	248		Gen Building	Gen Building	CIRCULATE									
Stair	3S08	274		Gen Building	Gen Building	CIRCULATE									
Stair	3S09	274		Gen Building	Gen Building	CIRCULATE									
IDF	3U01	172		Gen Building	Gen Building	MECHANICAL									
Custodial	3U01A	48		Gen Building	Gen Building	BLDG SERV									
Mechanical	3U02	406		Gen Building	Gen Building	MECHANICAL									
IDF	3U03	85		Gen Building	Gen Building	MECHANICAL									
Electrical	3U04	78		Gen Building	Gen Building	MECHANICAL									
Electrical	3U05	81		Gen Building	Gen Building	MECHANICAL									
Subtotals		**23807**	**SF**				5	1	27	64	1,309	8,009			

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
FOURTH FLOOR															
Head Tank and Storage	4102	533		Civil Engr	Civil Engr	NONCLS SV							Constant Head Tank + Storage		*Head tank is no longer used/req'd; Unused piping to be removed; Roof access.*
Office	4103	181	Pre-Award - A. Gutterman	Spns Pgm Srv	Civil Engr	OFFICE		2					JTRP		*Changed from "Post Doc"*
Office	4103	181	Pre-Award - Service Area	Spns Pgm Srv	Civil Engr	OFF SERV									
Office	4105	174	Pre-Award - K. Gascho	Spns Pgm Srv	Civil Engr	OFFICE		1					JTRP		*Changed from "Post Doc"*
Office	4106	158	D. Perissin	Civil Engr	Civil Engr	OFFICE	1								
Office	4107	174	Pre-Award - R. Bryant + B. Scott	Spns Pgm Srv	Civil Engr	OFFICE	1								*Changed from "Post Doc"*
Office	4108	168	Orig: "Emeritus + Visiting Scholar"	Civil Engr	Civil Engr	OFFICE	1								
Office	4109	174	Pre-Award - T. Harper + B. Johnson	Spns Pgm Srv	Civil Engr	OFFICE		1					JTRP		*Changed from (2) staff*
Office	4110	168	J. Shan	Civil Engr	Civil Engr	OFFICE	1								
Office	4111	174	J. Bethel	Civil Engr	Civil Engr	OFFICE	1								
Office	4112	169	S. Johnson	Civil Engr	Civil Engr	OFFICE	1								
Office	4113	174	Orig: "Geomatics Graduate Assistan	Civil Engr	Civil Engr	OFFICE	1		0						*Low ceiling.*
Office	4114	169	Orig: "Visiting Faculty + Scholars"	Civil Engr	Civil Engr	OFFICE	1								
Office	4115	174	R. Connor	Civil Engr	Civil Engr	OFFICE	1								
Office	4116	163	Oriig: "Visiting Scholars"	Civil Engr	Civil Engr	OFFICE	1								
Office	4117	174	A. Irfanoglu	Civil Engr	Civil Engr	OFFICE	1								
Office	4118	168	S. Pujol	Civil Engr	Civil Engr	OFFICE	1								
Office	4119	174	A. Prakash	Civil Engr	Civil Engr	OFFICE	1								
Office	4120	114	B. Caffery	Civil Engr	Civil Engr	OFFICE		1							
Office	4121	174	J. Liu	Civil Engr	Civil Engr	OFFICE	1								
Office	4122	120		Civil Engr	Civil Engr	OFFICE	1						Visiting Faculty		*Low ceiling. HVAC fed from third floor.*
Office	4123	170	A. Varma	Civil Engr	Civil Engr	OFFICE	1								
Copy Room	4124	120		Civil Engr	Civil Engr	OFF SERV							Struct.		*HVAC fed from third floor.*
Office	4125	170	J. Ramirez	Civil Engr	Civil Engr	OFFICE	1								
Office	4127	170	M. Kreger	Civil Engr	Civil Engr	OFFICE	1								
Office	4129	344		Civil Engr	Civil Engr	OFFICE			9				Structures Graduate Assistants		*Low ceiling; ducts down to third floor from roof*
Office	4131	216	G. Jeong	Civil Engr	Civil Engr	OFFICE	1								
Office	4136	200	R. Frosch	Civil Engr	Civil Engr	OFFICE	1								
Custodial	4138	48		Gen Building	Gen Building	BLDG SERV									
Women's Restroom	4140	199		Gen Building	Gen Building	REST ROOM							Women's Restroom - 2WC, 1L		
Office	4141	435	Reception - M. Stetler	Civil Engr	Civil Engr	OFFICE		2					Reception		
Office	4141A	130	G. Haikal	Civil Engr	Civil Engr	OFFICE	1								
Storage	4141B	130		Civil Engr	Civil Engr	OFF SERV									
Men's Restroom	4142	175		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,2U,2L		
Office	4143	328	M. Bowman	Civil Engr	Civil Engr	OFFICE	1								
Conference Room	4144	342		Civil Engr	Civil Engr	CONFERENCE									
Office	4147	265	Orig: "Structures Graduate Assistant	Civil Engr	Civil Engr	OFFICE			5				Graduate Assistants		
Research Lab	4148	155	Orig: Geometics Lab	Civil Engr	Civil Engr	NONCLAS LAB						155	Geomatics		
Office	4149	265	Orig: "Structures Graduate Assistant	Civil Engr	Civil Engr	OFFICE			6				Graduate Assistants		
Research Lab	4150	232	Orig: Geometics Lab	Civil Engr	Civil Engr	NONCLAS LAB						232	Geomatics		
Research Lab	4152	276	Orig: Geometics Lab	Civil Engr	Civil Engr	NONCLAS LAB						276	Geomatics		
Office	4154	785		Civil Engr	Civil Engr	OFFICE			23				Graduate Assistants		
Storage	4155	712		Civil Engr	Civil Engr	CENT STORAGE									*Roof access*
Conference Room	4158	276		Civil Engr	Civil Engr	CONFERENCE				12			Conference Room		
Office	4161			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		*Not included in study*
Media	4162			EarthAtm&Pln	EarthAtm&Pln	MEDIA PROD							Video Express Site		*Not included in study*
Office	4163			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		*Not included in study*
Office	4165			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		*Not included in study*
Research Lab	4166		Research Lab - Francisco+Zhuang	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									*Not included in study*
Office	4166			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		*Not included in study*
Office	4167			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		*Not included in study*

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
FOURTH FLOOR															
Office	4169			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		Not included in study
Office	4171			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		Not included in study
Teaching Lab	4173			EarthAtm&Pln	EarthAtm&Pln	CLASS LAB							Purdue Weather Lab		Not included in study
Teaching Lab	4201			EarthAtm&Pln	EarthAtm&Pln	CLASS LAB							Instrumentation Lab		Not included in study
Women's Restroom	4204	203		Gen Building	Gen Building	REST ROOM							Women's Restroom - 4WC,3L-HCA		
Men's Restroom	4206	223		Gen Building	Gen Building	REST ROOM							Men's Restroom - 2WC,3U,3L-HCA		
Office	4210			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Visiting Faculty		Not included in study
Custodial	4211	94		Gen Building	Gen Building	BLDG SERV							Custodial Closet		
Office	4212			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Visiting Faculty		Not included in study
Office	4244			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		Not included in study
Conference Room	4244			EarthAtm&Pln	EarthAtm&Pln	CONFERENCE							Conference Area		Not included in study
Teaching Lab	4247			EarthAtm&Pln	EarthAtm&Pln	CLASS LAB							Instructional Computing Lab		Not included in study
Teaching Lab	4251			EarthAtm&Pln	EarthAtm&Pln	CLASS LAB									Not included in study
Copy Room	4257			EarthAtm&Pln	EarthAtm&Pln	OFF SERV									Not included in study
Research Lab	4259		Research Lab - Lasher-Trapp	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4259			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		Not included in study
Office	4259A		Faculty - S. Lasher-Trapp	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	4259B			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Graduate Assistants		Not included in study
Research Lab	4263		Research Lab - Gluhovsky	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4263		Faculty - A. Gluhovsky	EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE									Not included in study
Research Lab	4263A		Research Lab - E. Agee	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4263B		Faculty - E. Agee	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Research Lab	4267		Research Lab - Baldwin + Agee + G	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4267			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		Not included in study
Office	4267A		Faculty - M. Baldwin	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	4271			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Student Assistants		Not included in study
Research Lab	4277		Research Lab - Trapp	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4277			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		Not included in study
Office	4277A		Faculty - R. Trapp	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	4280		Faculty - W.Y. Sun	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Research Lab	4281		Research Lab - Sun+Tung	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4281			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		Not included in study
Research Lab	4281A		Research Project Room-Sun+Tung	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4282			EarthAtm&Pln	EarthAtm&Pln	OFFICE							Visiting Instructors		Not included in study
Office	4284		Faculty - D. Niyogi	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	4286		Faculty - Harshvardhan	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Research Lab	4287		Research Lab - Zhuang	EarthAtm&Pln	EarthAtm&Pln	NONCLAS LAB									Not included in study
Office	4287			EarthAtm&Pln	EarthAtm&Pln	R LAB OFFICE							Graduate Assistants		Not included in study
Office	4288		Faculty - P. Smith	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Office	4290		Faculty - J. Francisco	EarthAtm&Pln	EarthAtm&Pln	OFFICE									Not included in study
Elevator	4E01	52		Gen Building	Gen Building	CIRCULATE									
Elevator	4E03	77		Gen Building	Gen Building	CIRCULATE									
Elevator	4E04	77		Gen Building	Gen Building	CIRCULATE									
Corridor	4H01	1188		Gen Building	Gen Building	CIRCULATE									
Corridor	4H02	1232		Gen Building	Gen Building	CIRCULATE									
Corridor	4H03	1916		Gen Building	Gen Building	CIRCULATE									
Corridor	4H04	1002		Gen Building	Gen Building	CIRCULATE									
Stair	4S01	248		Gen Building	Gen Building	CIRCULATE									
Stair	4S02	248		Gen Building	Gen Building	CIRCULATE									
Stair	4S04	248		Gen Building	Gen Building	CIRCULATE									
Stair	4S08	196		Gen Building	Gen Building	CIRCULATE									

Room Description / Name	Room Number	Area (SF)	Faculty / Staff Assignment	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
							Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
FOURTH FLOOR															
Stair	4S09	274		Gen Building	Gen Building	CIRCULATE									
Stair	4S10	146		Gen Building	Gen Building	CIRCULATE									
Mechanical	4U01	348		Gen Building	Gen Building	MECHANICAL									*Access to attic*
IDF	4U03	89		Gen Building	Gen Building	MECHANICAL									
Electrical	4U05	76		Gen Building	Gen Building	MECHANICAL									
Mechanical	4U06	5216		Gen Building	Gen Building	MECHANICAL									
Electrical	4U07	21		Gen Building	Gen Building	MECHANICAL									
Subtotals		**23175**	**SF**				22	7	43	12	0	663			

Sample Room Data Sheets

Room data sheets were provided to each department to summarize important space requirements and adjacencies. At the conclusion of the study, the room data sheets were not complete, but a sample sheet is included for reference.

PURDUE UNIVERSITY

(TL) Room Data Sheets

PURDUE UNIVERSITY - ME/RAIL/HAMP Study - HAMP		1,200		
Project Name	Date Last Revised	Assignable SF	Room Number	MSKTD Proj
Civil Engineering (CE)	**Structures Lab**			
Department/Group Name	Room Name	Contact, Phone		

Room Function and Type (Laboratory/Lab Support (Dedicated)/Biosafety Level/Clean Room Classification): Existing room (2158) is approximately 870 ASF.

Room Adjacencies - *What other rooms/functions should this room be directly adjacent to, if any? Why? (Includes exterior adjacencies or direct access to exterior.)*

Number of Occupants: | Special Ceiling Height (if any): | Fire Rating (if known) | Special/Hazardous Room Conditions (if any):

Finishes and Materials

Floor/Base:
- ◯ Vinyl Composition Tile/4" Vinyl Base
- ◯ Seamless Sheet Vinyl/Integral Coved Vinyl Base
- ◯ Epoxy/Integral Coved Epoxy Base
- ◯ Carpet/4" Vinyl Base
- ◯ Sealed Concrete/4" Vinyl Base
- ◯ Other :

Ceiling:
- ◯ Standard Lay-In Tile ◯ Mylar Faced Tile
- ◯ Gypsum Board/Painted ◯ Exposed/Non-Painted
- ◯ Exposed/Painted ◯ Other :

Doors
- ◯ Hollow Metal ◯ Wood ◯ Vision Panel ◯ Light Seals ◯ Acoustical Seals ◯ Other:

Security/Access Control:
- ◯ Key Lock ◯ Alarmed Door ◯ Card Access ◯ Motion Sensors ◯ Closed Circuit TV ◯ Other:

Walls:
- ◯ Gypsum Wall Board
- ◯ Cementitous Wall Board
- ◯ Concrete Masonry Units
- ◯ Other :

Wall Finishes:
- ◯ Latex Enamel Paint
- ◯ Epoxy Paint
- ◯ Vinyl Wallcovering
- ◯ Seamless Sheet Vinyl
- ◯ Other :

Lab Casework:
- ◯ Fixed ◯ Portable
- ◯ Metal ◯ Wood
- ◯ Other :

Lab Benchtop:
- ◯ Chemical-resistant Plastic Laminate
- ◯ Epoxy Resin ◯ Stainless Steel
- ◯ Other :

Wall Cabinets/Shelving:
- ◯ Open ◯ Swing Doors
- ◯ Sliding Doors ◯ Adjustable

Structural Considerations:
- ◯ Special Floor Loading Requirements: _____ PSF ◯ Recessed Floor Area Requirements ◯ Ceiling Mounted Equipment

Environmental Considerations:
- ◯ Significant Noise/Vibrational Generator _____ ◯ Visual Privacy Required ◯ Natural Light Desirable ◯ Odor Producing Operations
- ◯ Significant Noise/Vibrational Sensitivity _____ ◯ Views Desirable ◯ Avoid Natural Light ◯ Waste Production : ◯ Low ◯ High

Mechanical, Plumbing and Electrical Considerations:

HVAC Considerations (Brief description of any special HVAC considerations for the space, if any, including temperature/relative humidity, air flow/changes, air quality/exhaust.)

Plumbing Considerations (Brief description of any special plumbing considerations for the space, if any, including fixtures/features, services, waste.)

Electrical Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)

Communications Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)
- ◯ Telephone Connections ◯ Data Connections ◯ Network Service Type: _____ ◯ Clocks ◯ Intercom System ◯ Public Address System

Signatures

User Representative | Date

PU Representative | Date

MSKTD Associates, Inc. TEACH-6 Lab Structures Page 6

PURDUE UNIVERSITY

(RSCH) Room Data Sheets

PURDUE UNIVERSITY - ME/RAIL/HAMP Study - HAMP				
Project Name	Date Last Revised	Assignable SF	Room Number	MSKTD Proj
Civil Engineering (CE)	**RSCH Dry Comp Collaboration**			
Department/Group Name	Room Name	Contact, Phone		

Room Function and Type (Laboratory/Lab Support (Dedicated)/Biosafety Level/Clean Room Classification):

Room Adjacencies - *What other rooms/functions should this room be directly adjacent to, if any? Why? (Includes exterior adjacencies or direct access to exterior.)*

Number of Occupants: | Special Ceiling Height (if any): | Fire Rating (if known) | Special/Hazardous Room Conditions (if any):

Finishes and Materials

Floor/Base:
- ◯ Vinyl Composition Tile/4" Vinyl Base
- ◯ Seamless Sheet Vinyl/Integral Coved Vinyl Base
- ◯ Epoxy/Integral Coved Epoxy Base
- ◯ Carpet/4" Vinyl Base
- ◯ Sealed Concrete/4" Vinyl Base
- ◯ Other :

Ceiling:
- ◯ Standard Lay-In Tile ◯ Mylar Faced Tile
- ◯ Gypsum Board/Painted ◯ Exposed/Non-Painted
- ◯ Exposed/Painted ◯ Other :

Doors
- ◯ Hollow Metal ◯ Wood ◯ Vision Panel ◯ Light Seals ◯ Acoustical Seals ◯ Other:

Security/Access Control:
- ◯ Key Lock ◯ Alarmed Door ◯ Card Access ◯ Motion Sensors ◯ Closed Circuit TV ◯ Other:

Walls:
- ◯ Gypsum Wall Board
- ◯ Cementitous Wall Board
- ◯ Concrete Masonry Units
- ◯ Other :

Wall Finishes:
- ◯ Latex Enamel Paint
- ◯ Epoxy Paint
- ◯ Vinyl Wallcovering
- ◯ Seamless Sheet Vinyl
- ◯ Other :

Lab Casework:
- ◯ Fixed ◯ Portable
- ◯ Metal ◯ Wood
- ◯ Other :

Lab Benchtop:
- ◯ Chemical-resistant Plastic Laminate
- ◯ Epoxy Resin ◯ Stainless Steel
- ◯ Other :

Wall Cabinets/Shelving:
- ◯ Open ◯ Swing Doors
- ◯ Sliding Doors ◯ Adjustable

Structural Considerations:
- ◯ Special Floor Loading Requirements: _____ PSF ◯ Recessed Floor Area Requirements ◯ Ceiling Mounted Equipment

Environmental Considerations:
- ◯ Significant Noise/Vibrational Generator _____ ◯ Visual Privacy Required ◯ Natural Light Desirable ◯ Odor Producing Operations
- ◯ Significant Noise/Vibrational Sensitivity _____ ◯ Views Desirable ◯ Avoid Natural Light ◯ Waste Production : ◯ Low ◯ High

Mechanical, Plumbing and Electrical Considerations:

HVAC Considerations (Brief description of any special HVAC considerations for the space, if any, including temperature/relative humidity, air flow/changes, air quality/exhaust.)

Plumbing Considerations (Brief description of any special plumbing considerations for the space, if any, including fixtures/features, services, waste.)

Electrical Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)

Communications Considerations (Brief description of any special electrical considerations for the space, if any, including power service, power supply features, illumination.)
- ◯ Telephone Connections ◯ Data Connections ◯ Network Service Type: _____ ◯ Clocks ◯ Intercom System ◯ Public Address System

Signatures

User Representative | Date

PU Representative | Date

MSKTD Associates, Inc. RSCH-1 Dry Comp Collaboration Page 14

Architectural Narrative

Hampton Hall was built in multiple phases. The original building was built in 1951 (central wing), with separate projects/additions in 1962 (south and west wing) and 1986 (north wing). The early portions of the building are in need of renovation to satisfy educational and organizational needs. These proposed renovations will also accommodate the core academic functions initially identified in the College of Engineering master plan, as well as recommended upgrades to the mechanical, electrical, plumbing and telecommunications systems, as identified within this study.

One of the initial study considerations relates to accessibility and circulation, including circulation in and out of the building and vertical circulation within the building. Due to the age of the building and relationship of main floor levels to the ground level, accessible entrances are limited to the northwest corner of the 1986 wing and the south side of the central wing (within the courtyard). At other locations, entrances at stairs are at intermediate floor levels. Excluding any vertical transition at the exterior of the building, people must navigate up or down stairs to reach main floor levels upon entry at these stair tower locations, so accessibility is limited.

Hampton Hall, West elevation

Hampton Hall, Southwest corner

Vertical circulation is provided within Hampton Hall through existing elevators, including one in the 1962 wing and one in the 1986 addition. These existing elevators serve the major floors but not necessarily all of the intermediate levels (e.g., Wood Commons).

In addition to circulation, building services and other support spaces within the building were considered as part of the study for proposed renovations. In the existing 1962 wing of the building, restrooms are located in consistent locations from floor to floor, but each floor typically only serves either men or women (excluding the 1986 wing). The proposed renovation plans include considerations to provide facilities for men and women on each floor and in consistent locations in the 1962 wing.

The existing interior spaces of HAMP appear generally to be in good shape, but the finishes and layout is dated due to the age of the facilities. Materials in many areas of the building may contain asbestos or other related materials. Purdue University should provide consideration for investigation and abatement processes as these are not included as part of the study documentation.

SYSTEMS NARRATIVES

(Existing and Proposed)

As identified in the plan diagrams (existing and proposed), the project scope for possible renovations within the HAMP building includes only part of the square footage for the overall existing building, and that creates unique challenges in terms of space utilization, efficiencies, circulation, and mechanical, electrical, plumbing and telecommunications (MEP/T) systems distribution and renovations. Refer to the MEP/T systems narratives, as well as the preliminary code study documentation, for additional information.

During review and meetings for the study, HAMP program requirements were modified from the original departmental master plan quantities. In order to accommodate flexibility in programming of the existing building and space constraints, instructional / teaching lab space requirements were modified to be more flexible and offer opportunity for multiple classes to utilize the spaces. For example, in the existing I2I labs on the ground floor, renovation of existing adjacent office space is proposed to provide teaching lab prep areas and lockable storage to accommodate multiple teaching functions within the existing, sub-dividable labs. This helps the space be more flexible in function and set up to meet the program needs for multiple class labs. For similar reasons, prep and storage areas are incorporated in adjacent spaces for other instructional / teaching labs.

As the available existing space within the building cannot accommodate all of the necessary program functions and scope for the HAMP building, the focus of the study was on core academic functions, including classrooms, instruction / teaching labs, and offices and open-office workstations for faculty, staff, and graduate students. As indicated in earlier sections of the study, other functions that do not support the academic core mission will be relocated to other parts of campus. These include existing research lab spaces and function. At the conclusion of the study meetings and discussions, the Civil Engineering and Construction Engineering and Management departments were reviewing possible research lab functions that would remain within the building, based on the square footage available and identified on the proposed floor plans. No final conclusion was developed or proposed to incorporate into the study.

Various design solutions and considerations were discussed during team meetings, with an emphasis placed on efficiency, space adjacencies, and overall building organization and circulation. Proposed plan drawings are included on the following pages showing the final layout considered. Limited parts of the building were not included in the original project scope, but to accommodate electrical service recommendations and other system modifications, renovation within these additional areas may be necessary to accommodate systems upgrades and distribution. These are also identified and outlined in the MEP/T systems narratives that follow.

GENERAL

As discussed in other sections of the study, a goal of the HAMP proposed plans is to provide better and consistent facilities for the building and occupants. As an example, upgraded restroom facilities are proposed to accommodate revised fixture layouts that will improve accessibility. Additional restroom facilities are also proposed on each floor and in consistent locations on each floor to serve both men and women. Other general layout considerations include circulation modifications of existing corridors to provide direct access and better flow, as well as flexibility for larger program spaces and functions. Other and more specific modifications are outlined below for each floor.

The existing elevator in the 1962 wing was reviewed, and due to location and size, a new elevator is proposed in a different location. The new elevator location will serve additional floor levels in the existing building (e.g., Wood Commons) and provide additional space for restroom renovations in the current elevator location. As the new elevator location will impact existing mechanical space and systems on the basement floor, please refer to the MEP/T narratives for additional information.

During the study, there were ongoing discussions regarding the various computer labs within Hampton Hall, and following the final study meetings, additional information was provided from ITaP and ECN regarding existing computer lab use. The existing CE-owned computer lab located in HAMP 1212 has 49PCs, plus 11 additional open stations for laptop users. There is also an ITaP lab in 3144, and it has about 45 seats and was converted to a teaching lab a year or two ago. HAMP 3154 is the CE senior design class and is a 17-seat lab.

In regard to current computer lab utilization, ECN and ITaP identified that the HAMP 1212 lab is heavily used during the semester, and usage runs over 50-percent from about 10:00 a.m. through 1:00 p.m., 100-percent most of the afternoon and over 50-percent until 9:00 p.m. on most week days. In addition, based on information regarding usage history, HAMP 3144 was identified as one of the most utilized ITaP labs on campus in the past.

Considering the high utilization, ECN and ITaP recommend keeping 50 seats, but these seats do not need to be in one large room. Two or more smaller labs could be useful, and the large graduate office suite space proposed on the fourth floor could also accommodate some “lab” PCs for general use. These recommendations should be reviewed and considered for future phases of the HAMP project.

As part of the original study discussions, as well as the College of Engineering master plan, there were two additional areas within HAMP that were to be considered for the College of Engineering growth plan, including proposed renovations to meet program requirements for CE and CEM. These included the REM offices and associated labs on the basement floor and the EAPS library on the second floor. As the study and other related discussions progressed, these areas were removed from consideration and thus are not included within this study.

Finally, there are multiple rooms and labs within Hampton Hall that are named spaces based on generous donations to the school of Civil Engineering. The school is reviewing and considering the impact on these spaces based on the proposed renovations reviewed and discussed as part of the study, including options to relocate the spaces within the building.

BASEMENT FLOOR

As shown on the proposed plans, much of the basement floor renovation includes areas within the 1951 and 1962 wings of HAMP. As a result of reorganization and consolidation of office and open-office workstations into suites on upper floors, space on the basement floor is available for larger research labs. In addition, a server room and associated equipment was removed from HAMP, and the space is available for other program functions. Due to space allocation on the upper floors, two existing research labs are proposed for relocation to the basement floor.

In addition to the research labs described above, the HAMP project includes proposed renovations to two large CE research labs in the central part of the building, including the concrete lab and the bituminous lab (and associated, adjacent support spaces). Recent renovations and program changes have impacted these spaces, and function and finish updates would be beneficial. In addition, recent renovations to the existing hydraulics lab on the south end of the building removed much needed power (e.g., power bus duct) to accommodate projects and flexibility. Refer to the MEP/T narrative for additional information.

GROUND FLOOR

Similar to the basement floor, much of the proposed renovation on the ground floor is planned for the 1951 and 1962 wings of HAMP. In order to accommodate flexibility in the centrally located I2I instructional / teaching lab, renovation and reallocation of existing, adjacent office space is proposed to provide teaching lab prep areas and lockable storage to accommodate multiple teaching functions within the existing, sub-dividable I2I labs. These support spaces help the lab to be more flexible in function, while offering opportunities for multiple classes to utilize the space. For similar reasons, prep and storage areas are proposed in adjacent spaces for other instructional / teaching labs in HAMP.

The instructional / teaching labs in the 1951 wing (east side of building) are planned to incorporate flexibility in space and function. The adjacent labs may be designed as flexible spaces to accommodate various classes and small or larger groups of students. This flexibility may be accomplished by utilizing large wall openings with overhead folding glass partitions / doors to permit the adjacent spaces to be opened or closed to accommodate flexible schedules and space needs. These teaching labs, including lab and support spaces for Geomatics, are located to provide exterior doors and easy access to exterior spaces to accommodate program requirements and adjacencies, as well as potential for large deliveries requiring overhead doors.

In addition to the other labs proposed in the 1951 wing, a structures instructional / teaching lab is proposed to take advantage of the large, existing test machine and strong floor system in this area. The test machine is not currently used and was covered by recent construction, but the machine could be reconditioned to provide valuable testing and teaching facilities for the school of Civil Engineering that does not currently exist on campus. Costs associated with reconditioning this equipment is available from the College of Engineering, based on separate review and considerations outside the scope of this study.

The CE computer lab (HAMP 1212) is also proposed to be relocated to the ground floor to accommodate computer lab facilities for the department (as described above), and the senior capstone instructional / teaching lab is proposed in an adjacent space. Other renovations are proposed for office suites, collaboration spaces and student organization spaces on the floor.

Finally, a new corridor and ramp is proposed on the south side of the existing I2I lab. This ramp will provide more direct circulation and access to the proposed new teaching labs, as well as an opportunity to take advantage of an existing accessible entrance from the courtyard. During discussions at the final study meeting, a building addition was proposed within the courtyard to accommodate additional student commons and collaboration spaces. In this location, this space can serve as a major, accessible entrance on the east side of the building. Additional information is needed to review and assess this option.

FIRST FLOOR

The Civil Engineering department desires to maintain core administrative and office functions on the first floor of the 1962 wing due to history and prominence of this location, as well as ease of access. As shown on the proposed plans, this area will accommodate the departmental main offices, development office, business office, and graduate and undergraduate offices and advising spaces. In addition, the existing classroom that is located on this floor is proposed to be relocated to accommodate an adjacent large conference room to serve the department and administrative offices to support large group meetings.

The Civil Engineering main office suite is planned to incorporate private and open-office workstations, including amenities such as an adjacent informal work café and small, medium and large meeting and collaboration spaces. Access to the office suite may be flexible, and the department may consider open access during predetermined office hours for direct access by students and visitors and after-hours access-controlled security as necessary.

The work café is planned on the east side of the office suite to provide direct access to the stair towers and public circulation without the need to travel through the office suite. Faculty and staff mail boxes are being considered in this area for central distribution. Secure access should be reviewed.

In the 1986 wing, additional office suites and instructional / teaching labs are planned. In recent moves, the CEM administration offices were located in this area of the building, and in the proposed plans, these are shown to be in the same location with renovations to match more current office configurations being incorporated into recent College of Engineering projects.

SECOND FLOOR

The proposed second floor layout includes several existing SMAS classrooms that will remain. Additional classrooms relocated from other floors are also planned on this level to accommodate general student access and appropriate space adjacencies. A proposed student collaboration space is also included within this wing to provide more student queuing space, which will help accommodate a higher volume of student traffic on this floor with the SMAS classrooms. This student collaboration and queuing space is incorporated with the creation of the new, adjacent restroom facilities.

As ITaP computer labs are discussed to determine whether they will remain in the building due to the department's space needs as identified in the master plan program, students on this floor for SMAS classroom spaces would benefit from an ITaP lab being located on this floor. However, as space is limited, consideration would need to be given to determine if the lab or additional SMAS classrooms would be located on this floor.

THIRD FLOOR

In addition to the new and renovated restroom facilities and elevator location, the third floor of the 1962 wing is planned to be a large, open-office suite, including spaces for amenities such as an informal work café and small, medium and large meeting and collaboration rooms, in addition to private offices and open-office workstations. Access to the floor may be flexible, and the department may consider open access during predetermined office hours for direct access by students and visitors and after-hours access-controlled security at the stairs and elevator.

FOURTH FLOOR

Similar to the third floor, on the fourth floor, in addition to the new and renovated restroom facilities and elevator location, the 1962 wing is planned to be a large, open-office suite, including spaces for amenities such as an informal work café and small, medium and large meeting and collaboration rooms, in addition to graduate student open-office workstations. Access to the floor may be flexible, and the department may consider open access during predetermined office hours for direct access by students and visitors and after-hours access-controlled security at the stairs and elevator.

The fourth floor is located within the roof lines of the existing building, so natural daylighting through windows does not exist. As part of the proposed renovation, MEP system will be reconfigured for the fourth floor. See the MEP narratives for additional information. Further review and discussion will be needed with Purdue University, but with the removal and reconfiguration of the ductwork that is currently installed on the roof, there may be options to incorporate skylights in the roof at regular intervals to bring natural daylighting into the renovated fourth floor space.

PHASING

As described in earlier sections, proposed construction phases for HAMP follow the scope (Parts 1 and 2) of the study. Phase 1 includes spaces that directly impact the College of Engineering's growth plan, including spaces that support core academic functions within the building, including office and open-office workstations and suites, as well as instructional / teaching lab configurations that are impacted by the reorganization. Phase 2 includes additional considerations that impact overall building use and organization, and these areas also impact future flexibility in lab configuration, including existing instructional / teaching labs and research labs and related support spaces. Much of the phasing consideration for HAMP is based on increasing the quantities of private offices and open-office workstations and support functions within larger office suites throughout the building.

Another consideration for HAMP includes the areas of proposed renovation and the impact that construction activities will have on use of the existing facilities in the rest of the building. During the study meetings, the team discussed the option to focus the Phase 1 construction and renovation activities on the 1951 and 1962 wings of the building, with future renovations in the more recent 1986 wing. These comments were incorporated into the proposed plan layout for phasing included within the study. In general, building improvements impacting circulation and core building services (e.g., restrooms, MEP/T systems) are also planned as part of the Phase 1 scope of work.

Looking at the proposed plans from the fourth floor down, the fourth floor and third floor will become large office suites that encompass the entire 1962 wing of HAMP. The office suite concept provides flexibility for enclosed offices and adjacent workstations, while also incorporating similar qualitative considerations developed and implemented in other recent projects for the College of Engineering (i.e., Wang Hall, Grissom Hall). These spaces include small, medium, and large meeting spaces, as well as informal collaborative spaces and work cafes. However, with the implementation of the office suite on these floors, core academic spaces that are currently located there will be displaced, and as we look at the overall building organization and where these spaces are relocated, we expand the Phase 1 construction scope of the project so that these core academic functions remain within the building.

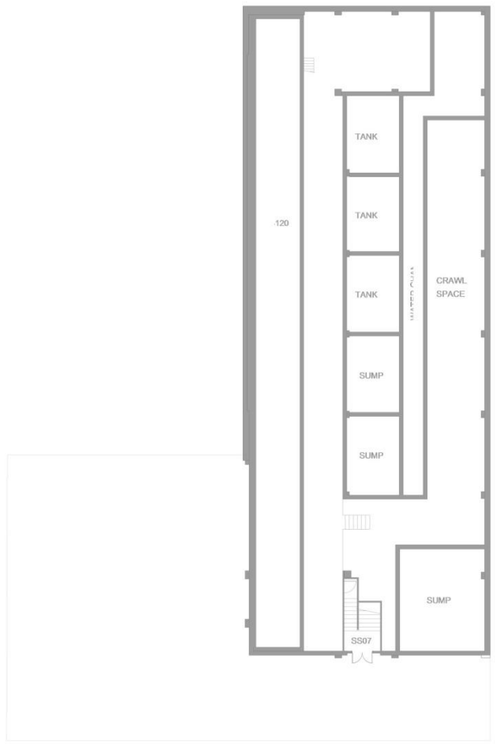
Many academic classrooms are planned to be relocated and incorporated into second floor spaces, adjacent to existing academic classrooms. The proposed plans also incorporate options for the relocated ITaP computer lab and collaboration / queuing space for students. All of these spaces are relevant to maintaining core academic spaces in the building, and these are planned as part of the first and second phases of the project, depending on impact in other areas of the building that will require their relocation.

Much of the 1962 wing of the first floor is also incorporated into the Phase 1 project to accommodate the ME main office and other related administrative offices. A core academic classroom and research lab that is required to stay in the building are impacted with the office suite configuration, and in the proposed plans, these spaces are relocated to the basement and second floor levels of the building. Other renovations on the second floor 1986 wing are planned for a later, second phase project.

The ground floor level has a mixture of relocated and new instructional / teaching labs to address program requirements for the department, and these are identified within Phase 1 of the project. Other relocated and reconfigured offices and open-office workstation suite are incorporated into later phases.

Architectural Narrative – Proposed Plans – HAMP Sub-Basement Floor

OVERALL PROGRAM COUNTS	Program	Proposed
Offices	(111)	(121)
Workstations	(86)	(122)
Teaching Labs	(10)	(10)
Classrooms (SMAS)	(13)	(13)
Research Labs		(16)
Mediascapes	(4)	(4)
Conference Rooms	(5)	(5)
Civil Engineering Administration	(1)	(1)
CEM Administration	(1)	(1)
CEM Simulation Labs	(2)	(2)
Dry Comp Collaboration	(2)	(2)
Open Lab Project	(1)	(1)
Fieldwork Prep Room	(1)	(1)
Computer Labs	(2)	(2)
Student Collaboration Spaces	(4)	(4)
Student Organizations	(1)	(1)
Graduate Student Desks	(185)	(291)



PURDUE UNIVERSITY

Sub-Basement Floor Plan

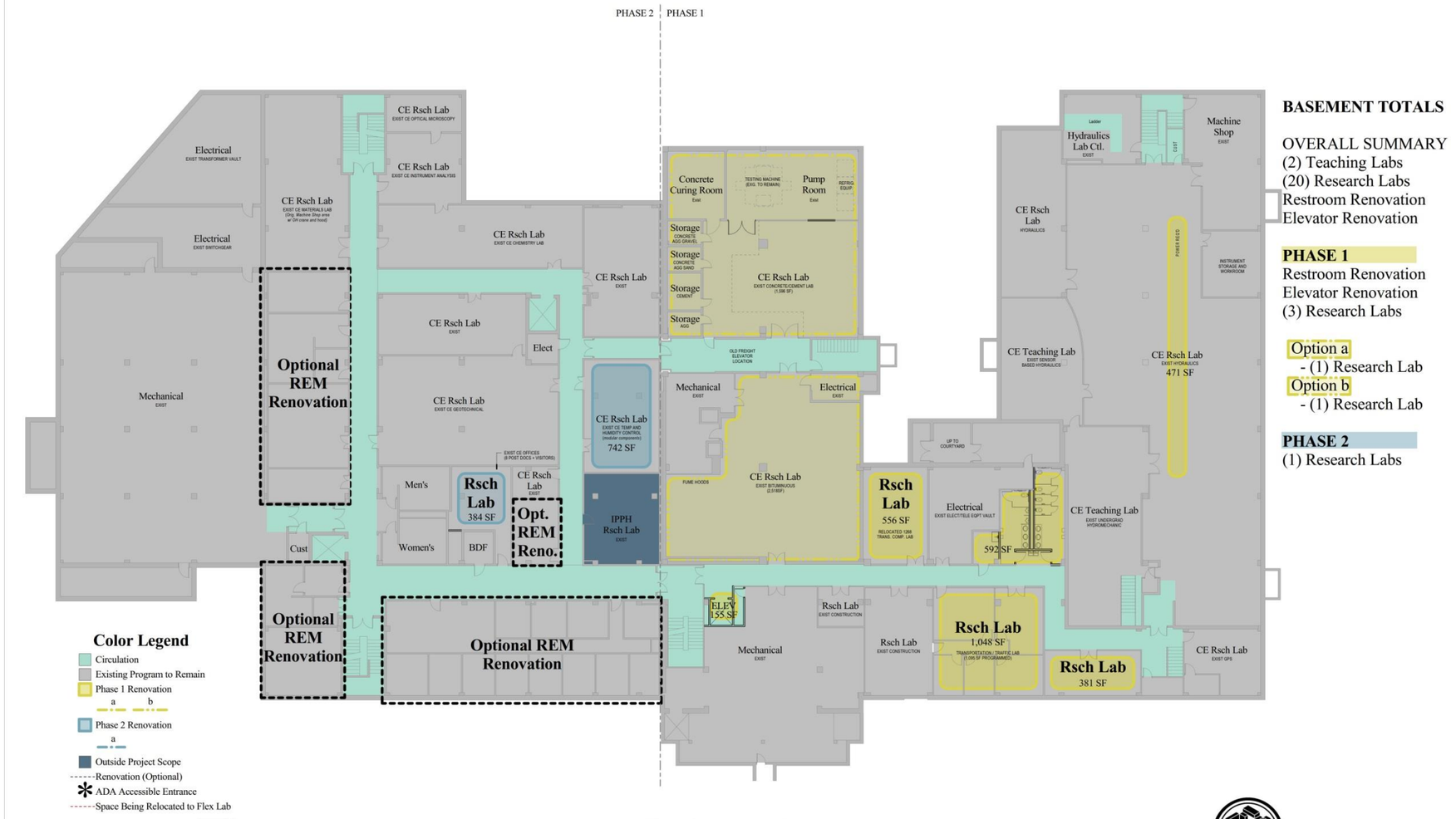
3/32" = 1'-0"

Hampton Hall

Preliminary Conceptual Plan

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – HAMP Basement Floor



Architectural Narrative – Proposed Plans – HAMP Ground Floor

Architectural Narrative – Proposed Plans – HAMP First Floor

FIRST FLOOR TOTALS

OVERALL SUMMARY
(51) Offices
(65) Workstations
(2) SMAS Classrooms
(2) Simulation Labs
(3) Conference Rooms
Elevator Renovation
Restroom Renovation

PHASE 1
(3) Office Suites
-(23) Offices
-(28) Workstations
(2) Conference Rooms
Work Cafe
Elevator Renovation
Restroom Renovation

PHASE 2
(2) Simulation Labs

Option a
- (2) Office Suites
  - (27) Offices
  - (30) Workstations
  - (1) Conference Rm.

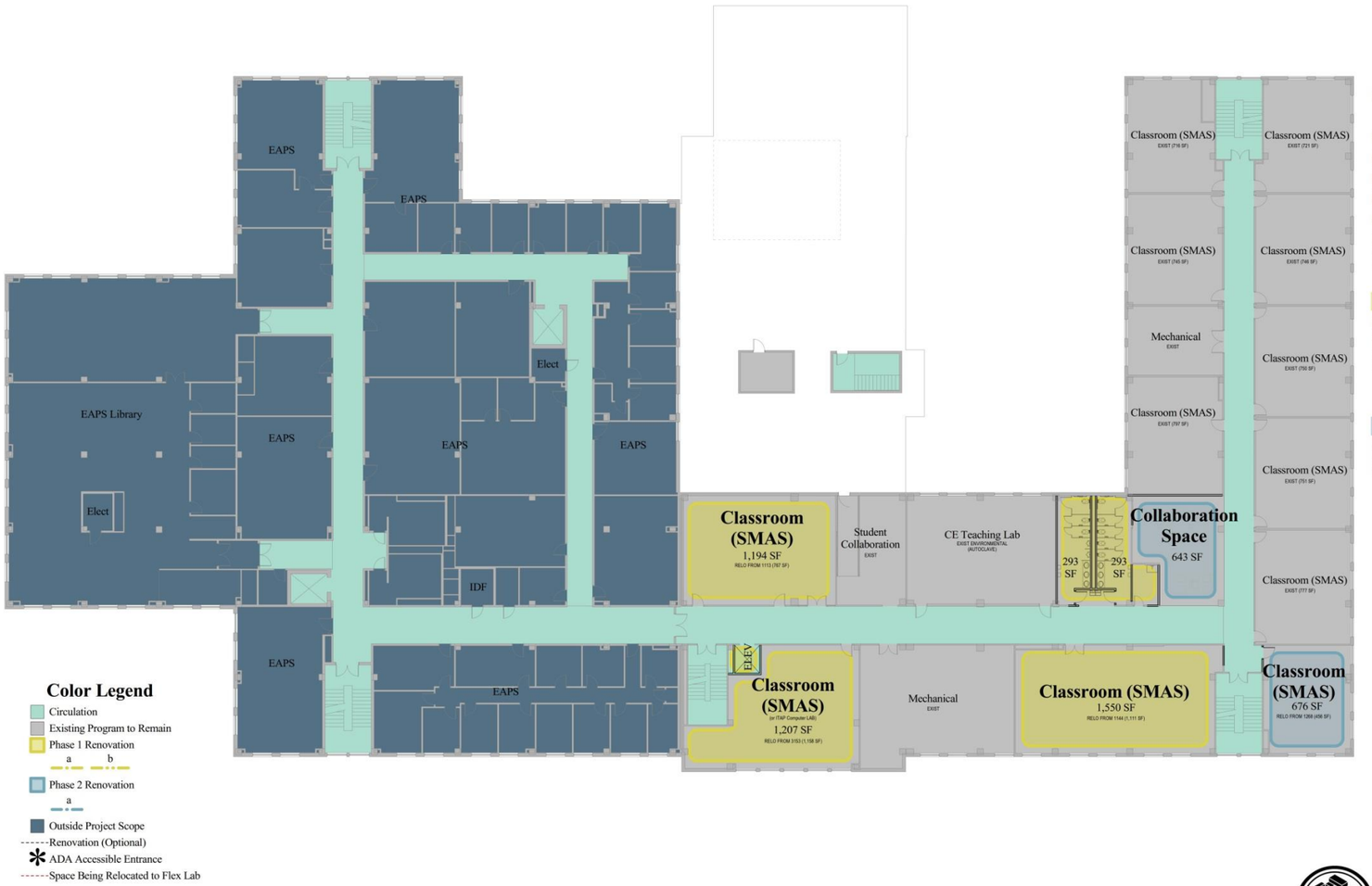
PURDUE UNIVERSITY

First Floor Plan - Proposed
3/32" = 1'-0"

Hampton Hall
Preliminary Conceptual Plan

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – HAMP Second Floor



Architectural Narrative – Proposed Plans – HAMP Third Floor

THIRD FLOOR TOTALS

OVERALL SUMMARY
(43) Offices
(109) Workstations
(3) Conference Rooms
(1) Teaching Lab
Elevator Renovation
Restroom Renovation

PHASE 1
(43) Offices
(109) Workstations
(3) Conference Rooms
Elevator Renovation
Restroom Renovation

PHASE 2
N/A

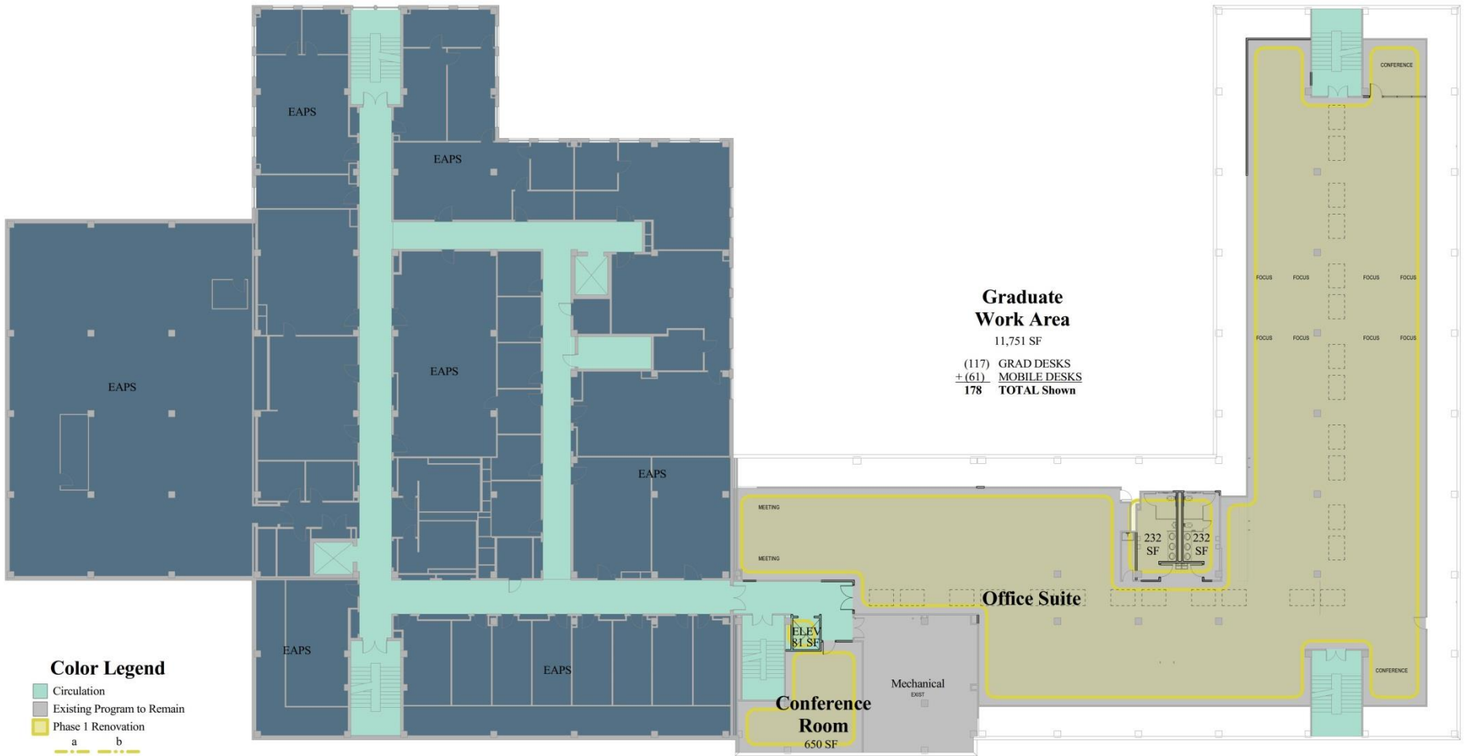
PURDUE UNIVERSITY

Third Floor Plan - Proposed
3/32" = 1'-0"

Hampton Hall
Preliminary Conceptual Plan

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – HAMP Fourth Floor



FOURTH FLOOR TOTALS

OVERALL SUMMARY
(178) GA Workstations
(3) Conference Rooms
Elevator Renovation
Restroom Renovation

PHASE 1
(178) GA Workstations
(3) Conference Rooms
Elevator Renovation
Restroom Renovation

PHASE 2
N/A

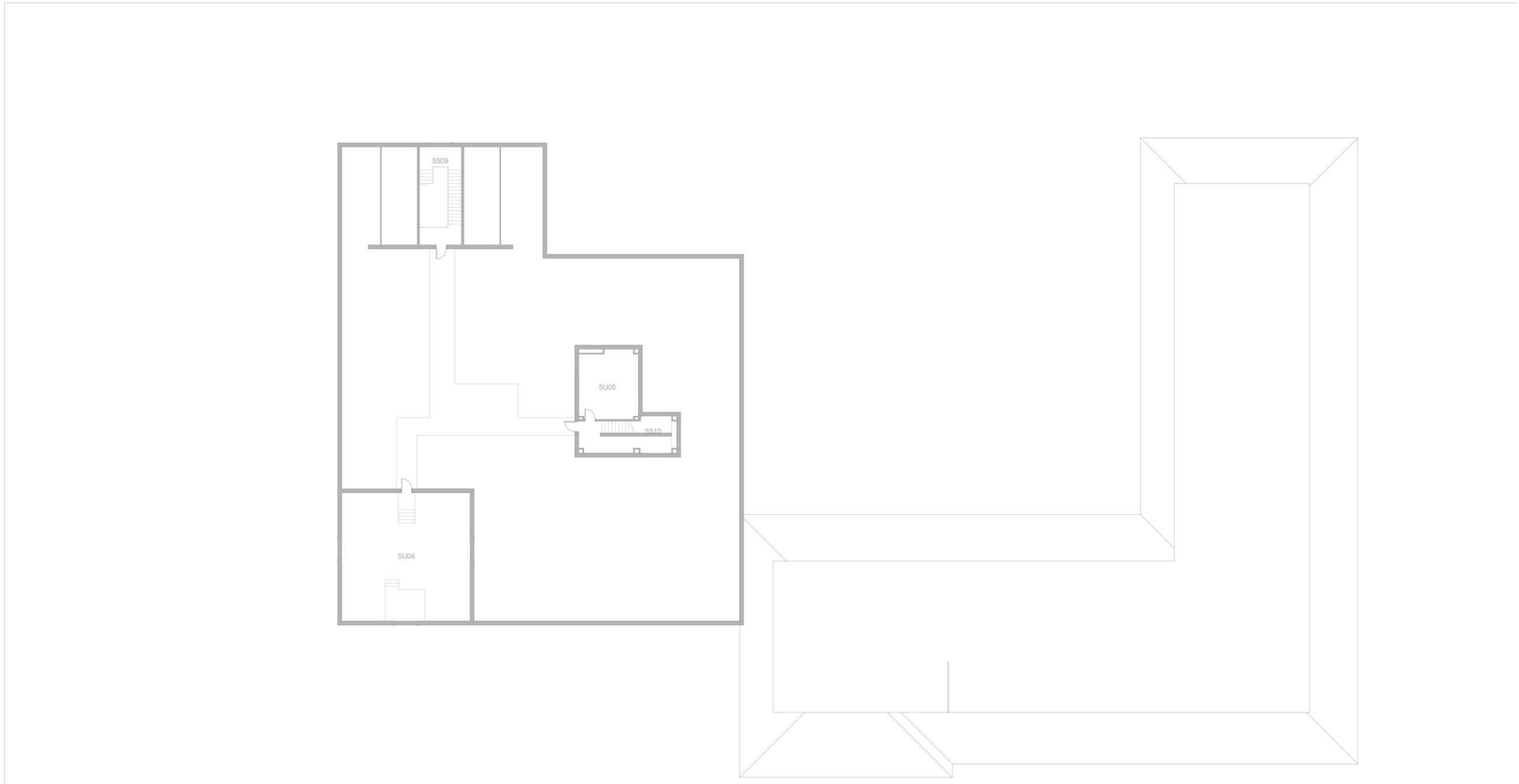
PURDUE UNIVERSITY

Fourth Floor Plan - Proposed
3/32" = 1'-0"

Hampton Hall
Preliminary Conceptual Plan

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – HAMP Fifth Floor



PURDUE UNIVERSITY

Fifth Floor Plan
3/32" = 1'-0"

Hampton Hall
Preliminary Conceptual Plan

MSKTD & Associates, Inc.

Preliminary Conceptual Plan

Mechanical Narrative

This narrative describes the existing building mechanical systems, and the proposed changes to those systems. These changes are anticipated to be implemented as part of a multi-phase building renovation project.

The mechanical systems will be designed at a minimum in accordance with the following codes and standards as applicable:

1. Applicable Codes:
    a. 675 IAC 13 – Building Codes
        i. 2014 Indiana Building Code
        ii. International Building Code, 2012 Edition
    b. 675 IAC 16 – Plumbing Codes
        i. Indiana Plumbing Code, 2012 Edition
        ii. International Plumbing Code 2006 Edition
    c. 675 IAC 18 – Mechanical Codes
        i. Indiana Mechanical Code, 2014 Edition
        ii. International Mechanical Code, 2012 Edition
    d. 675 IAC 19 – Energy Conservation Codes
        i. Indiana Energy Conservation Code, 2010 Edition
        ii. ASHRAE Standard 90.1-2007 – Energy Standard for Buildings Except Low-Rise Residential Buildings.
    e. 675 IAC 22 – Fire Prevention Codes
        i. Indiana Fire Code, 2014 Edition
        ii. International Fire Code, 2012 Edition
2. Applicable Standards:
    a. NFPA 101 – Life Safety Code
    b. Purdue University Consultants Handbook

Hampton Hall Existing Systems

The existing building was constructed in phases. The original building was constructed in 1950. The first addition to the building was completed in 1962. The last addition to the building was completed in 1986.

The 1950 and 1962 additions function as a separate building from the 1986 addition from a service perspective. Chilled water, steam, domestic water, and fire suppression systems are separate services for each wing.

The mechanical infrastructure was added to the building in phases. A summary of existing system components is as follows:

1. Air side HVAC system – The basement floor of the original building is served by ACB-7 a 7,900 cfm air handling unit.
2. Air side HVAC system – The ground and first floors of the original building are served by two air handling units: ACG-4 and ACG-10. ACG-4 is a 19,270 cfm air handling unit. ACG-10 is a 9,500 cfm air handling unit.
3. Air side HVAC system – The basement floor of the 1962 addition is served by three air handling units: ACG-4, ACB-9, and ACG-30. ACB-9 is a 2,840 cfm air handling unit. ACG-30 serves rooms B115E and B115F.
4. Air side HVAC system – The ground floor of the 1962 addition is served by three air handling units: ACG-5, ACG-6, and ACG-29. ACG-5 is a 5,820 cfm air handling unit. ACG-6 is a 6,330 cfm air handling unit. ACG-29 serves rooms B115B and B115D.
5. Air side HVAC system – The first floor of the 1962 addition is served by three air handling units: ACG-4, ACG-5, and AC1-25. AC1-25 is a constant volume recirculation unit.
6. Air side HVAC system – The second floor of the 1962 addition is served by two air handling units: AC2-3 and AC2-26. AC2-3 is an 8,800 cfm air handling unit. AC2-26 is a 7,850 cfm constant volume air handling unit.
7. Air side HVAC system – The third floor of the 1962 addition is served by fan coil units and two air handling units AC4-2 and AC4-24.
8. Air side HVAC system – The fourth floor of the 1962 addition is served by air handling unit AC4-24.
9. Air side HVAC system – The basement and ground floors of the 1986 addition are served by two air handling units: ACB-13 and ACB-14. ACB-13 is a 22,000 cfm air handling unit. ACB-14 is a 39,720 cfm air handling unit.
10. Air side HVAC system – The first floor of the 1986 addition is served by two air handling units: ACB-13 and ACB-14. Two fan coil units have been installed in Room 1212 to provide additional cooling.
11. Air side HVAC system – There is no proposed work on the second, third, and fourth floors of the 1986 addition. These spaces are served by air handling units serving only these floors.
12. Chilled Water Systems –Campus chilled water is delivered to the air handling units and fan coil units by chilled water pumps located throughout the building. There are two separate chilled water services, one to the 1962 wing, and one to the 1986 wing.
13. Hydronic Heating Systems: There are multiple steam to hot water heat exchangers and heating water pumps located throughout the building. There is a hydronic reheat loop and a perimeter heating loop in the 1986 building, with the equipment located in the basement mechanical room. The 1962 building side has a couple of smaller hydronic systems for isolated areas.
14. Steam System – There are two Low pressure (15 PSI) campus steam services delivered, one to each building side. This steam is delivered to the building through the steam tunnels below the buildings. Low pressure steam is piped to air handling units. Steam condensate is returned to a steam condensate pumps and pumped back to the campus return system.
15. Building HVAC Control System – Controls are believed to be a combination of pneumatic and DDC.
16. Domestic Water – There are two domestic water services, one for each side. Domestic hot water is generated for each side by a steam to hot water generator.
17. Sanitary Waste and Vent Systems are present for the main toilet group and miscellaneous fixtures through the building.
18. Fire Suppression – Areas renovated in either phase of the Hampton Hall projects will require that sprinkler heads be reworked as part of the scope of the renovation.

Proposed Mechanical Scope

This narrative proposes a multi-phase replacement of the existing airside HVAC systems within the building. A direct digital control system will be provided for HVAC system control and monitoring. The existing fire suppression system will remain, but will be modified as required to provide proper coverage of the renovated spaces.

Proposed Airside HVAC System

Re-programming and renovation of space is being planned for the basement, ground, first, second, third, and fourth floors, presenting the need to modify the airside systems.

The design team proposes replacing portions of the existing airside systems with new VAV systems with reheat. The components of the proposed system are:

Proposed Phase 1:

Proposed Air Side Equipment Systems

1. Install a new 100% outside air, 6,000 cfm VAV air handling unit to replace existing unit ACB-7. The unit would consist of a steam pre-heat coil, chilled water cooling coil and fan controlled via a variable frequency drive. The new unit would serve the basement floor of the 1950 building. Air handling unit ACB-7 is original to the building.

2. Install a new 3,000 cfm VAV air handling unit to replace existing unit ACB-9. The unit would consist of a steam pre-heat coil, chilled water cooling coil and fan controlled via a variable frequency drive. The new unit would serve the new research labs on the basement floor of the 1962 addition. Air handling unit ACB-9 is a multi-zone unit that is original to the building.
3. The existing air handling unit ACR-10 will continue to provide conditioning to the CE Teaching Labs and Teaching Lab Prep on the ground floor of the 1950 building. New zone ductwork will need to be installed to coordinate with the new program floor plan. Air handling unit ACG-10 is a multi-zone unit that was installed in 2008.
4. New VAV Terminal Units to be installed in the CE Teaching Lab Prep/Class, CE Research Lab, and Teaching Lab Prep (I2I) on the ground floor of the 1986 addition for zone control and airflow modulation
5. Install a new 6,500 cfm VAV air handling unit to replace the existing ACG-6. The unit would consist of a steam pre-heat coil, chilled water cooling coil and fan controlled via a variable frequency drive. The new unit would serve the ground floor of the 1962 addition. Air handling unit ACG-6 is a multi-zone unit that is original to the building.
6. Remove existing air handling unit AC1-25 and replace with a new VAV unit that will serve the first floor 1962 addition spaces. Air handling unit AC1-25 is a VAV unit installed in 1995. Replacement of this unit is necessary due to the mechanical room it is housed in being relocated.
7. Install a new 6,000 cfm VAV air handling unit to replace the existing AC2-3. The unit would consist of a steam pre-heat coil, chilled water cooling coil and fan controlled via a variable frequency drive. This unit will serve the three SMAS Classrooms on the second floor of the 1962 addition. Air handling unit AC2-3 is a multi-zone unit that is original to the building.
8. The third floor of the 1962 addition will continue to be served by air handling units AC4-2 and AC4-24. New VAV Terminal Units to be installed for zone control and airflow modulation.
9. The fourth floor of the 1962 addition will continue to be served by air handling unit AC4-24. New VAV Terminal Units to be installed for zone control and airflow modulation.
10. VAV Terminal Units to be installed throughout renovated areas for zone control and airflow modulation.
11. Individual fan coil units to condition the IT rooms – chilled water, fractional horsepower.
12. A manifolded Laboratory Exhaust Fan to serve the basement labs will be installed and will utilize the existing exhaust duct path.

13. Toilet Exhaust Fan.

AIR SIDE EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
ACB-7	VAV Air Handler	6,000 CFM	460 V/5 HP
ACB-9	VAV Air Handler	3,000 CFM	460V/3 HP
ACG-6	VAV Air Handler	6,500 CFM	460V/5 HP
AC2-3	VAV Air Handler	6,000 CFM	460V/5 HP
AC1-25	VAV Air Handler	13,000 CFM	460V/15 HP
ATU-X (x 85)	Air Terminal Unit	8100 BTUH	24V control power
FCU-X (x 6)	Fan Coil Unit	12,000 BTUH	120V/Fractional HP
PUH-X (x8)	Propeller Unit Heater	30,000 BTUH	120V/ Fractional HP.
TEA-1	Toilet Exhaust Fan	3,500 CFM	460 V/ 1.5 HP
MHER-X	Lab Exhaust	7500 CFM	460V/ 7.5 HP

Proposed Steam and Hydronic System

In areas where it is practical, existing perimeter finned tube radiators will be removed, and new radiators provided. This approach will also provide the opportunity to coordinate the finned tube locations with new room and furniture layouts.

A new heat exchanger and related pumps will be provided in the basement mechanical room to provide heating water for new VAV reheat valves in the basement and third floor. A heat exchanger and heating water pumps will be required to supply heating water to new terminal units.

The existing chilled water pipe sizes are adequate for the proposed system changes.

A summary of the hydronic equipment is listed below.

STEAM AND HYDRONIC EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
HX-1	Shell and Tube Heat Exchanger	80 GPM/800 lb/hr	-
HYPB-1	Hydronic Heating Water Pump	80GPM/60 FT	460V/5 HP
HYPB-2	Hydronic Heating Water Pump	80 GPM/60 FT	460V/5 HP

Proposed Building Automation Modernization

The full extent of mechanical work being proposed will necessitate the need for a temperature control system to be provided. New direct digital building automation controls will be provided. Actuators will be electrically actuated. The system will be either an Automated Logic or a Siemens system, and the front end shall be accessible at the campus control center.

Proposed Plumbing Upgrades

Portions of the sanitary and waste piping will need to be modified to accommodate the restroom upgrades. New sanitary and waste mains will need to be added as well as a new piping chase for the restroom group in the 1962 addition. New toilet fixtures to be ADA compliant.

Portions of the domestic water piping will need to be modified to accommodate the restroom upgrades. Domestic water mains will need to be added as well as a branch piping to restroom fixtures.

Proposed Fire Suppression Upgrades

The design intent is to re-use the existing fire service, and sprinkler pipe zone valves and mains. Branch piping and heads will be replaced, and spaced to provide proper coverage.

Proposed Phase 2:

1. The Research Lab on the basement floor of the 1986 addition will continue to be served by air handling unit ACB-14. The CE Research Lab will require a VAV terminal and branch ductwork to be installed from unit ACB-14. Air handling unit ACB-14 is a VAV unit that is original to the building.
2. The lab exhaust from the CE research lab will be connected to the manifold exhaust system furnished in phase I.
3. VAV Terminal Units to be replaced throughout the new Office Suites and Collaboration Space on the 1986 ground floor to accommodate zone control and airflow modulation for the new program floor plan.
4. VAV Terminal Units to be replaced throughout the new Office Suite, CEM Admin, and CEM Simulation Labs on the first floor of the 1986 addition to accommodate zone control and airflow modulation for the new program floor plan.
5. VAV Terminal Units to be replaced throughout the new Collaboration Space and Classroom (SMAS) on the second floor of the 1968 addition to accommodate zone control and airflow modulation for the new program floor plan.

AIR SIDE EQUIPMENT SUMMARY			
Equipment Tag	Description	Capacity	Electrical
ATU-X (x 50)	Air Terminal Unit	8100 BTUH	24V control power
SAV-X (x1)	Laboratory Supply Air Valve	500 CFM	24V control power
EAV-X (x1)	Laboratory Exhaust Air Valve	550 CFM	24 V control power

Proposed Building Automation Modernization

The full extent of mechanical work being proposed will necessitate the need for a temperature control system to be provided. New direct digital building automation controls will be provided. Actuators will be electrically actuated. The system will be either an Automated Logic or a Siemens system, and the front end shall be accessible at the campus control center.

Proposed Plumbing Upgrades

Provide services for the converted laboratory spaces. Services include cold water, hot water, lab gases and safety showers.

Proposed Fire Suppression Upgrades

The design intent is to re-use the existing fire service, and sprinkler pipe zone valves and mains. Branch piping and heads will be replaced, and spaced to provide proper coverage.

Attached Drawings

The following drawings are attached to help convey the written design intent.

1. Hampton Hall – Existing Basement Floor Plan
2. Hampton Hall – Existing Ground Floor Plan
3. Hampton Hall – Existing First Floor Plan
4. Hampton Hall – Existing Second Floor Plan
5. Hampton Hall – Existing Third Floor Plan
6. Hampton Hall – Existing Fourth Floor Plan
7. Hampton Hall – Proposed Basement Floor Plan
8. Hampton Hall – Proposed Ground Floor Plan
9. Hampton Hall – Proposed First Floor Plan
10. Hampton Hall – Proposed Second Floor Plan
11. Hampton Hall – Proposed Third Floor Plan
12. Hampton Hall – Proposed Fourth Floor Plan

Mechanical Narrative – Existing Plans – HAMP Basement Floor

Existing Basement Floor Plan - Mechanical

PURDUE UNIVERSITY

Hampton Hall

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – HAMP Ground Floor

Existing Ground Floor Plan - Mechanical
3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – HAMP First Floor

Existing First Floor Plan - Mechanical
3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – HAMP Second Floor

Existing Second Floor Plan - Mechanical

PURDUE UNIVERSITY

Hampton Hall

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – HAMP Third Floor

Existing Third Floor Plan - Mechanical

3/32" = 1'-0"

PURDUE UNIVERSITY

Hampton Hall

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Existing Plans – HAMP Fourth Floor

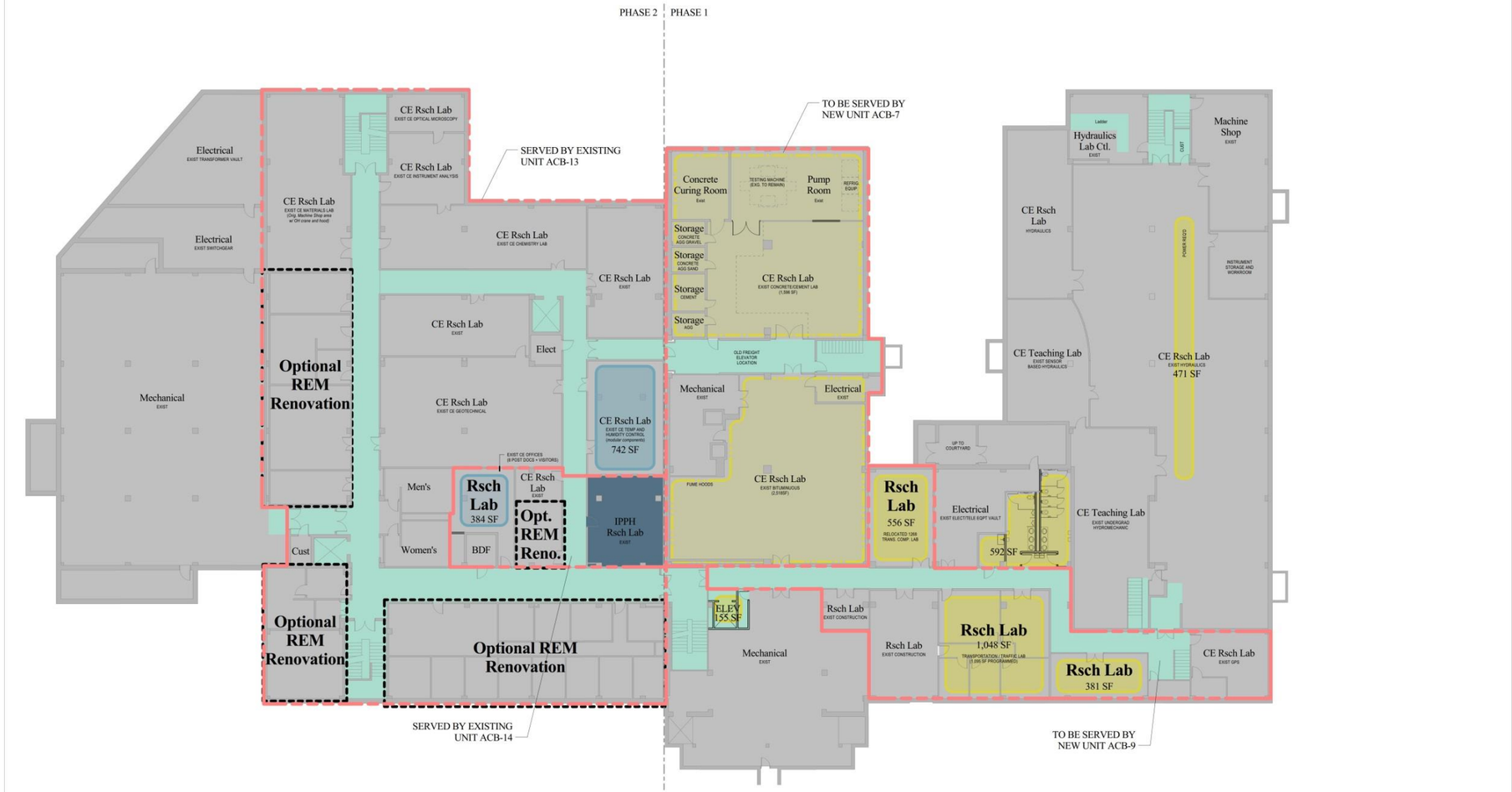
Existing Fourth Floor Plan - Mechanical

PURDUE UNIVERSITY

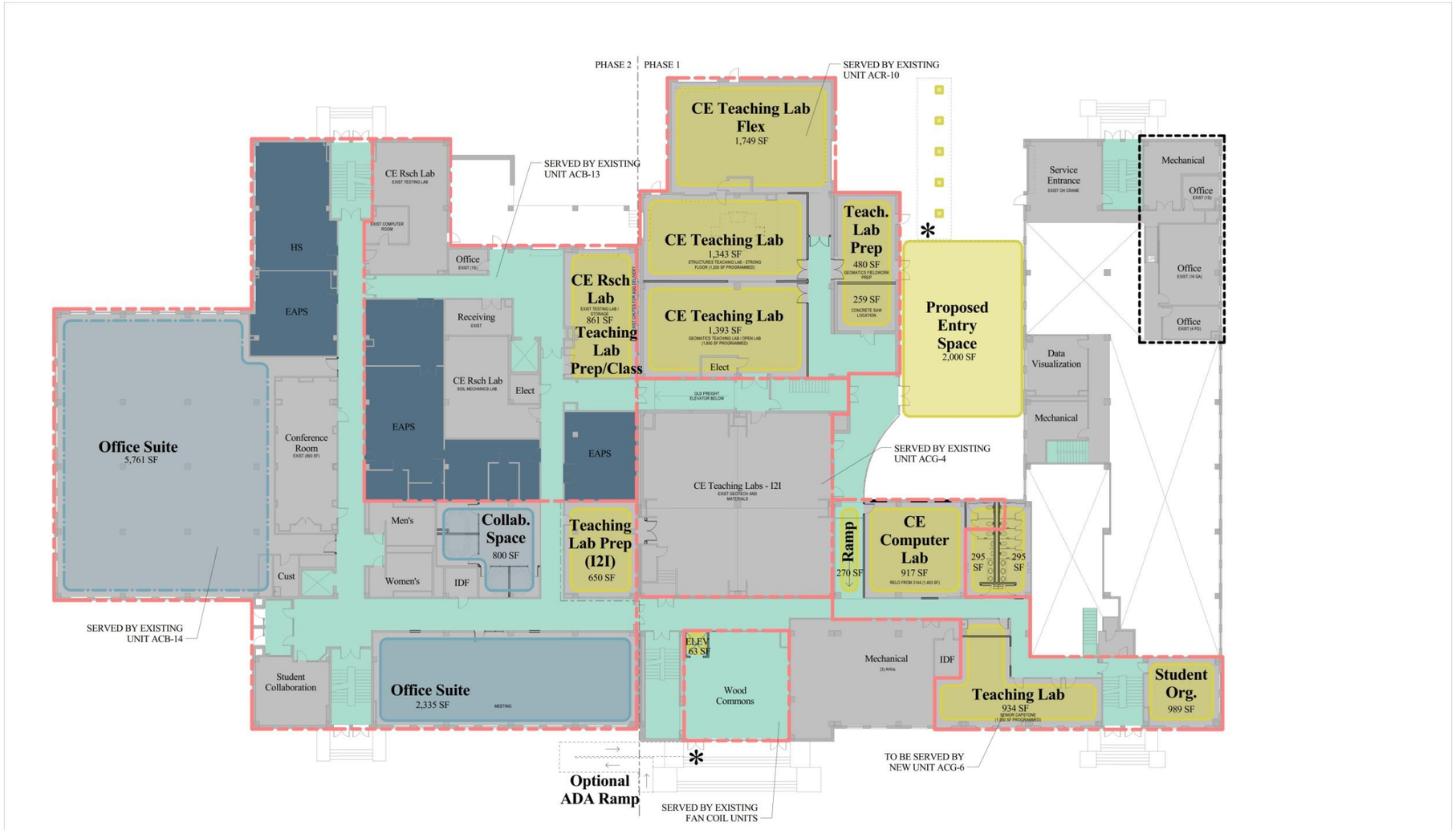
Hampton Hall

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – HAMP Basement Floor



Mechanical Narrative – Proposed Plans – HAMP Ground Floor



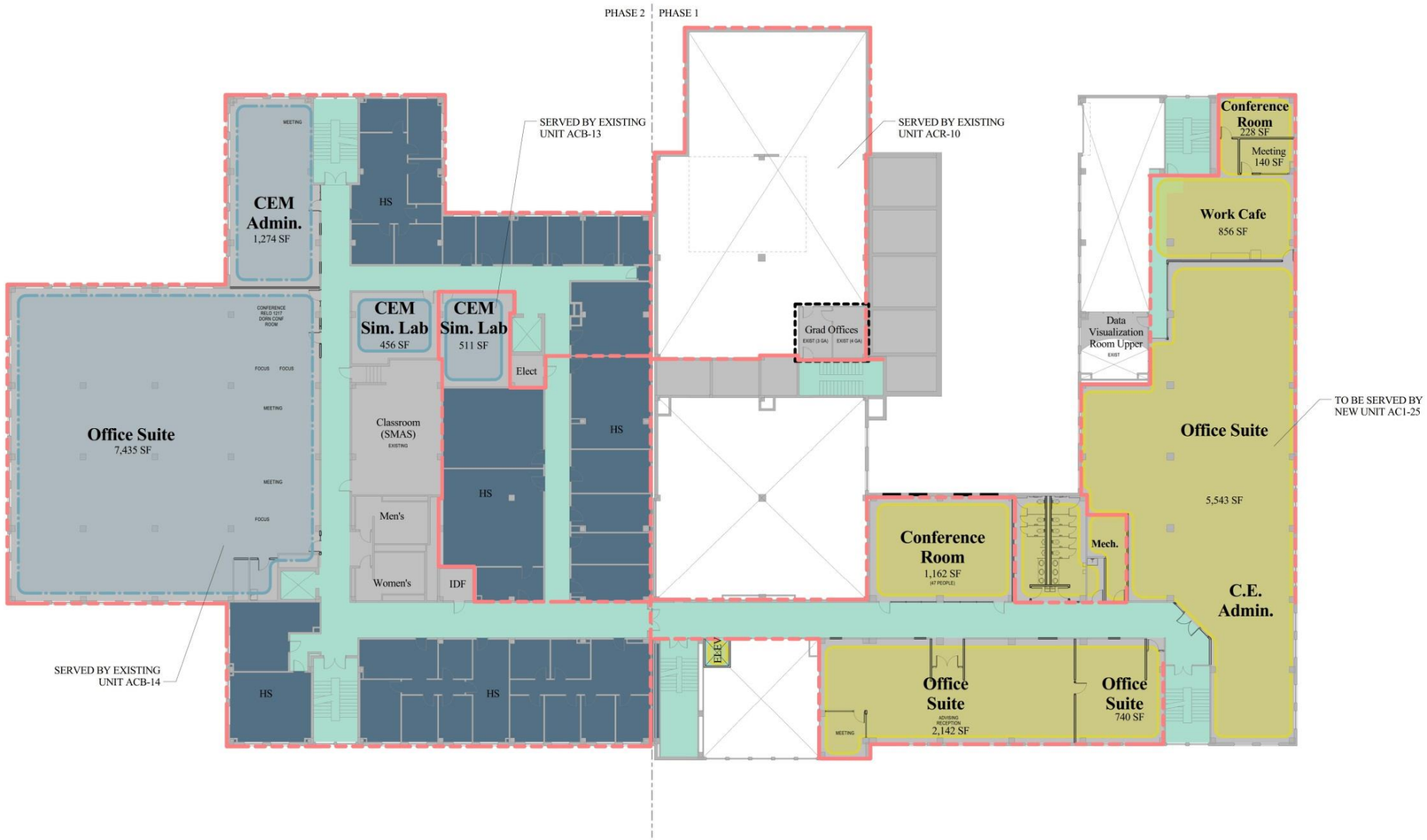
PURDUE UNIVERSITY

Proposed Ground Floor Plan - Mechanical
3/32" = 1'-0"

Hampton Hall
Preliminary Conceptual Plan

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – HAMP First Floor



PURDUE UNIVERSITY

Proposed First Floor Plan - Mechanical

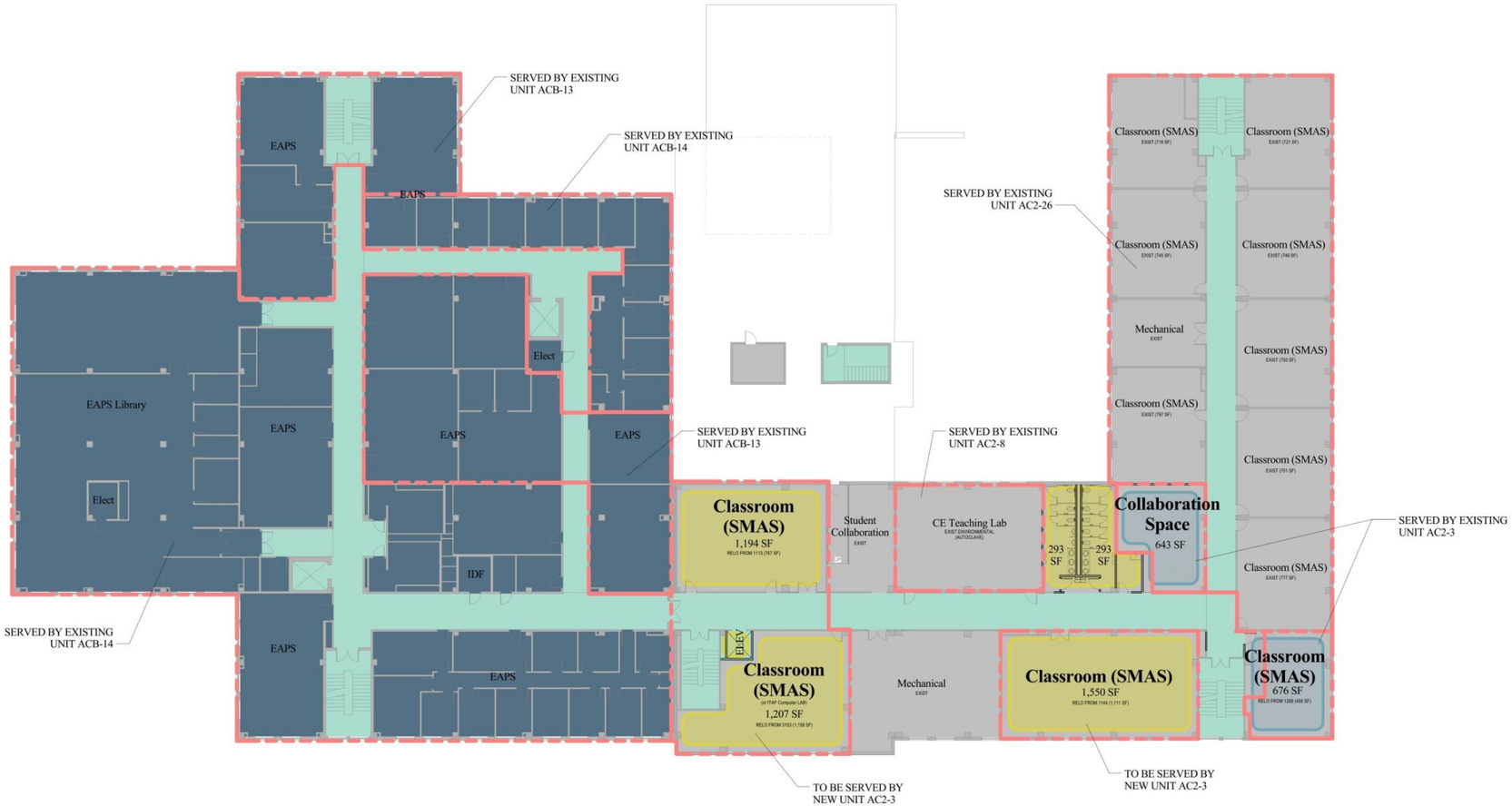
3/32" = 1'-0"

Hampton Hall

Preliminary Conceptual Plan

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – HAMP Second Floor



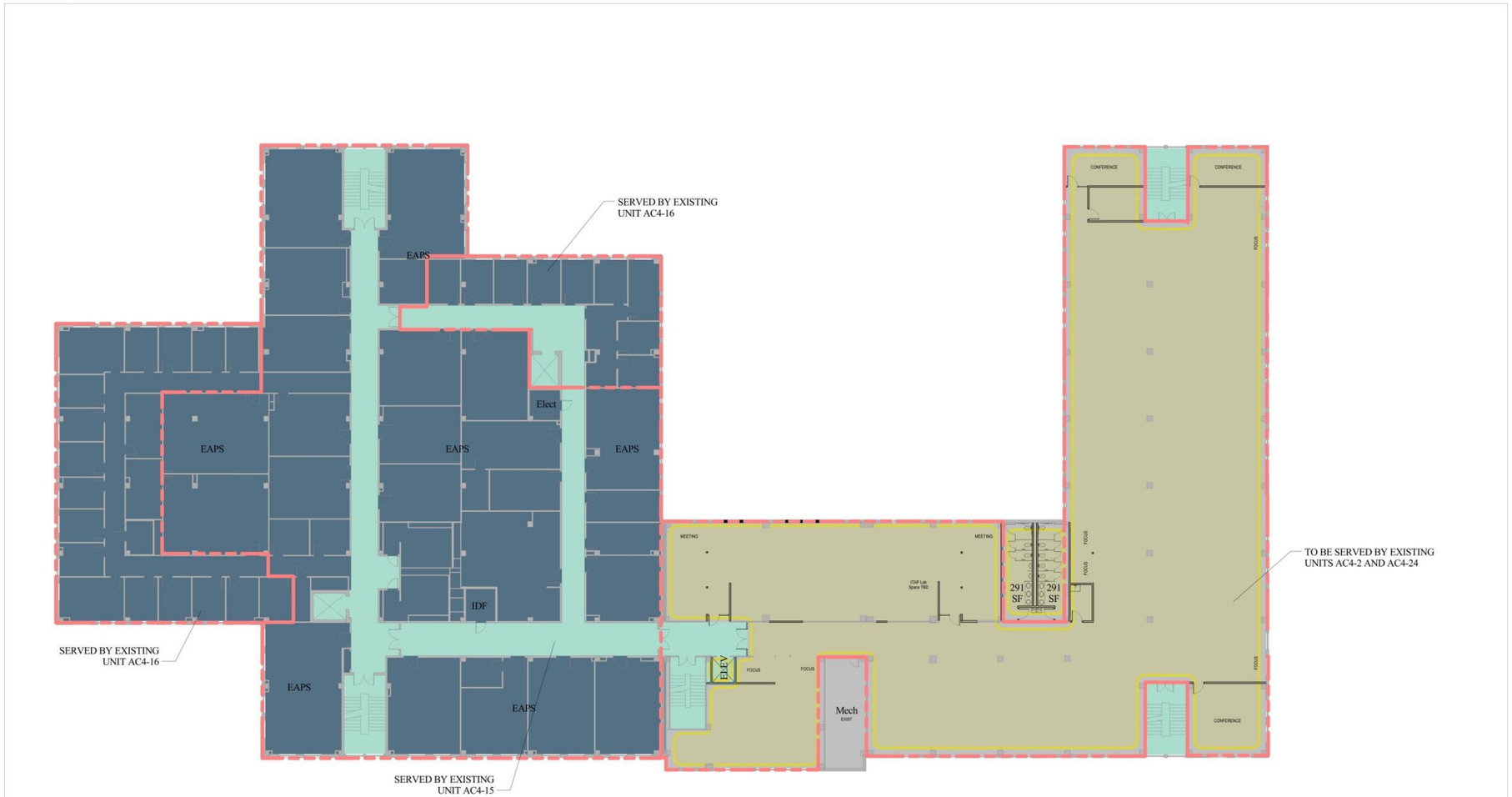
PURDUE UNIVERSITY

Proposed Second Floor Plan - Mechanical
3/32" = 1'-0"

Hampton Hall
Preliminary Conceptual Plan

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – HAMP Third Floor



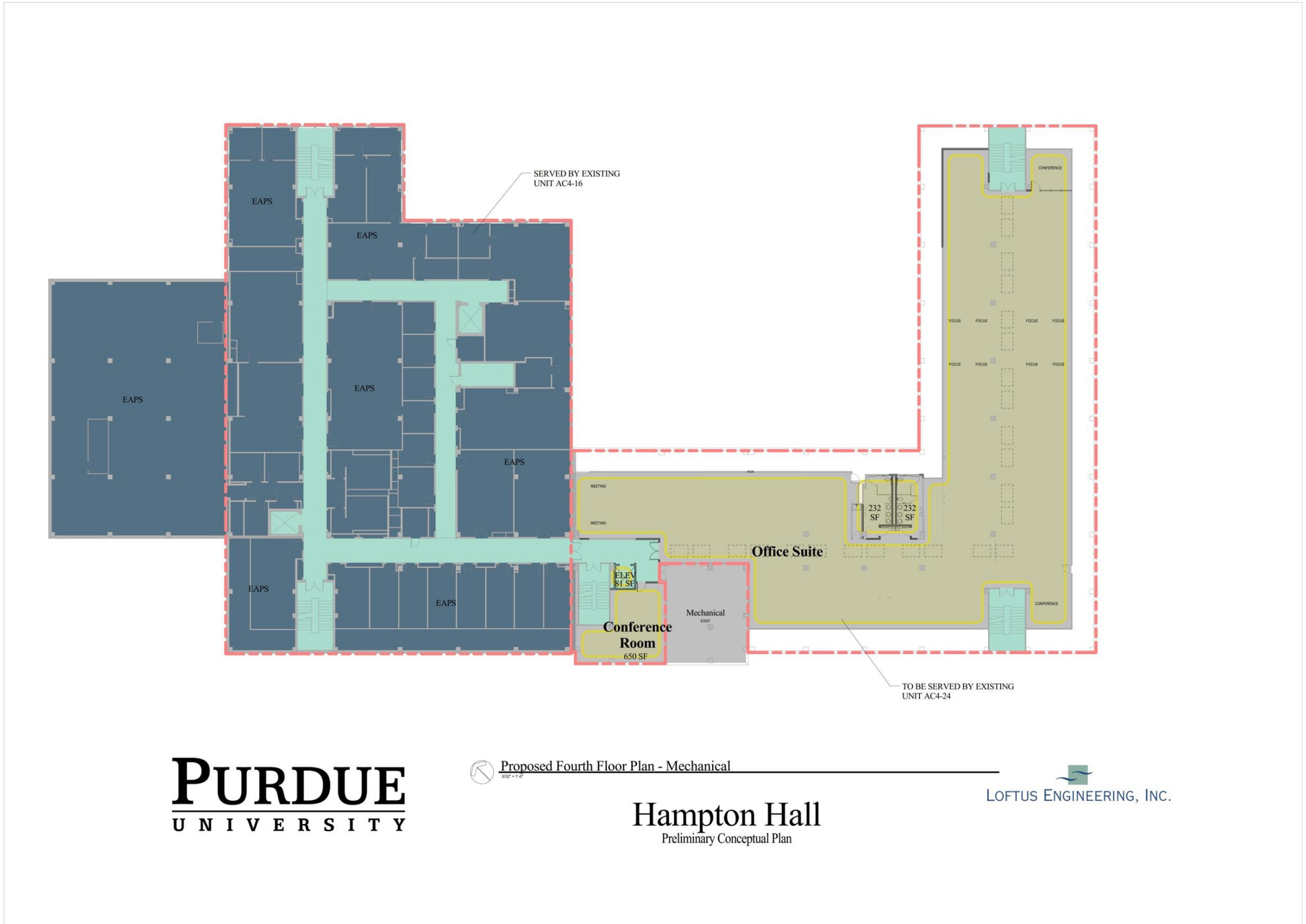
PURDUE UNIVERSITY

Proposed Third Floor Plan - Mechanical
3/32" = 1'-0"

Hampton Hall
Preliminary Conceptual Plan

LOFTUS ENGINEERING, INC.

Mechanical Narrative – Proposed Plans – HAMP Fourth Floor



PURDUE UNIVERSITY

Proposed Fourth Floor Plan - Mechanical

3/32" = 1'-0"

Hampton Hall

Preliminary Conceptual Plan

LOFTUS ENGINEERING, INC.

Electrical Narrative

This narrative describes the existing electrical systems for Hampton Hall, and the proposed changes to those respective systems. These changes are anticipated to potentially be implemented as part of a multi-phased building renovation project.

The electrical systems will be designed at a minimum in accordance with the following codes and standards as applicable:

1. Applicable Codes:
    a. 675 IAC 13 Building Codes.
        i. 2008 Indiana Building Code.
        ii. International Building Code, 2006 Edition.
    b. 675 IAC 17 – Electrical Codes.
        i. Indiana Electrical Code, 2009 Edition.
        ii. NFPA 70 – National Electrical Code, 2008 Edition.
    c. 675 IAC 19 – Energy Conservation Codes.
        i. 2010 Indiana Energy Conservation Code.
        ii. Energy Standard for Buildings Except Low-Rise Residential Buildings ANSI/ASHRAE 90.1, 2007 Edition.
    d. 675 IAC 22 – Fire Prevention Codes
        i. Indiana Fire Code, 2008 Edition
        ii. International Fire Code, 2006 Edition
    e. NPFA 72 – National Fire Alarm Code

2. Applicable Standards:
    a. NFPA 101 – Life Safety Code
    b. Purdue University Consultants Handbook

Power Distribution Systems – General Service

The complete Hampton Hall Building was constructed in three separate projects. The original portion of Hampton Hall was constructed around 1950 and was provided with two separate electrical services. One service provided 120/08 volt, 3-phase, 4-wire electrical power to the building and the second service provided 240 volt, 3-phase, 3-wire electrical power to the building. Power was provided from transformers located in a vault adjacent to the main electrical room.

An addition to Hampton Hall was constructed around 1962 to the south and west around the original structure. A new electrical service was provided as part of this project which consisted of four separate electrical services. One service provides 120/240 volt, 1-phase, 3-wire electrical power to the building, the second service provides 240v, 3-phase, 3-wire electrical power to the building, the third service provides 120/208v, 3-phase, 4-wire electrical service to the building and the fourth service provides 480v, 3-phase, 3-wire electrical service to the building. Power is provided from transformers located in a vault adjacent to the main electrical room. As part of this project, the main switchboards in the original building were backfed power from new main switchboards in the new addition electrical room and the original transformers were removed.

A second addition to Hampton Hall was constructed around 1986 to the north of the existing building. A new separate electrical service was provided to serve this new addition which consisted of three separate electrical services. One service provides 120/240 volt, 3-phase, 4-wire electrical power to the building, the second service provides 120/208 volt, 3-phase, 4-wire electrical power to the building and the third service provided 277/480 volt, 3-phase, 4-wire electrical power to the building.

Power Distribution Systems – Existing Conditions – Original Building/1962 Addition

Electrical power to the existing facility is presently provided by from four existing services via vault style transformer banks cable to an existing high voltage metal enclosed switchgear lineup “HVS-1” located in an existing transformer vault. Each existing transformer bank provides the designated service as follows:

1. **Transformer Bank “T-EQ-1”:** This transformer provides 120/240 volt, 1-phase, 3-wire electrical service to existing main switchboard “M-EQ-1”.
    a. This transformer presently serves panelboards that provide power typically to lab equipment, receptacles, and non-HVAC equipment loads for the facility.
    b. This transformer is presently 167 KVA in size.
    c. This transformer presently provides service to main switchboard “M-EQ-1” via an existing 800A bus duct.

2. **Transformer Bank “T-P-1”:** This transformer provides 240 volt, 3-phase, 3-wire, delta electrical service to main switchboard “M-P-1”.
    a. This transformer serves power panelboards, motor control centers, etc. that provide power typically for motor loads associated with the HVAC equipment and building plumbing.
    b. This transformer provides power for the building elevator.
    c. This transformer is presently 500 KVA in size.
    d. This transformer presently provides service to main switchboard “M-P-1” via an existing 1350A bus duct.

3. **Transformer Bank “T-L-1”:** This transformer provides 120/208 volt, 3-phase, 4-wire, wye electrical service to existing main switchboard “M-L-1”.
    a. This transformer presently serves panelboards that provide power typically to lighting, receptacles, and non-HVAC equipment loads for the facility.
    b. This transformer is presently 500 KVA in size.
    c. This transformer presently provides service to main switchboard “M-L-1” via an existing 1600A bus duct.

4. **Transformer Bank “T-P-2”:** This transformer provides 480 volt, 3-phase, 3-wire, delta electrical service to main switchboard “M-P-2”.
    a. This transformer serves motor control centers, etc. that provide power typically for motor loads associated with the HVAC equipment and building plumbing.
    b. This transformer is presently 500 KVA in size.
    c. This transformer presently provides service to main switchboard “M-P-2” via an existing 800A bus duct.

Main distribution switchboards “M-EQ-1”, “M-L-1”, “M-P-1” and “M-P-2” are located in a main electrical room in the basement in the central area of the building adjacent to the existing transformer vault. Existing switchboard “M-EQ-1” is rated for 800A, 120/240V, 1-phase, 3-wire service. Existing switchboard “M-L-1” is rated for 1600A, 120/208V, 3-phase, 4-wire service. Existing Switchboard “M-P-1” is rated for 1600A, 240V, 3-phase, 3-wire service. Existing Switchboard “M-P-2” is rated for 1200A, 480V, 3-phase, 3-wire service. Each existing switchboard is manufactured by Square D, utilizes switches with Type L and Type R fuses, and appears to have been installed as part of the mid 1962 addition project.

There is an existing sub-distribution switchboard “SD-L-1” located in the basement of the original building which receives 120/208V, 3-phase, 4-wire service from existing switchboard “M-L-1”. There also is an existing sub-distribution panel “SD-P-1” in the basement of the original building that receives 240V, 3-phase, 3-wire service from existing switchboard “M-P-1”. Each sub-distribution panel provides power for either existing loads directly, or existing branch circuit panelboards located in the original portion of the building.

Existing branch circuit panelboards are located throughout the facility as necessary to serve respective branch circuit loads and are generally the switch/fuse type of panelboards that utilize plug fuses. Purdue University is presently undergoing systematic replacement projects across campus to replace existing switch/fuse panelboards with new circuit breaker type panelboards.

Power Distribution Systems – Existing Conditions – 1986 Addition

Electrical power to the existing facility is presently provided by from three existing services via unit sub-station type transformers, throat-connected to an existing high voltage metal enclosed switchgear lineup “HVS-1” located in an existing transformer vault. Each existing transformer provides the designated service as follows:

1. **Transformer “T-L”:** This transformer provides 120/208 volt, 3-phase, 4-wire, wye electrical service to existing main switchboard “MDP-L”.
    a. This transformer presently serves panelboards that provide power typically to lighting, receptacles, and non-HVAC equipment loads for the facility.
    b. This transformer is presently 500 KVA in size.
    c. This transformer presently provides service to main switchboard “MDP-L” via an existing 2500A bus duct.
2. **Transformer “T-P”:** This transformer provides 277/480 volt, 3-phase, 4-wire, wye electrical service to main switchboard “MDP-P”.
    a. This transformer serves panelboards, motor control centers, etc. that provide power typically for motor loads associated with the HVAC equipment and building plumbing.
    b. This transformer provides power for the building elevator.
    c. This transformer is presently 750 KVA in size.
    d. This transformer presently provides service to main switchboard “MDP-P” via an existing 1200A bus duct.
3. **Transformer “T-LABS”:** This transformer provides 120/240 volt, 3-phase, 4-wire, grounded delta electrical service to main switchboard “MDP-LABS”.
    a. This transformer serves panelboards, etc. that provide power typically for laboratory receptacles and equipment.
    b. This transformer is presently 500 KVA in size.
    c. This transformer presently provides service to main switchboard “MDP-P” via an existing 2500A bus duct.

Main distribution switchboards “MDP-L”, “MDP-P” and “MDP-LABS” are located in a main electrical room in the basement in the northeast area of the building adjacent to the existing transformer vault. Existing switchboard “MDP-L” is rated for 2500A, 120/208V, 3-phase, 4-wire service. Existing Switchboard “MDP-P” is rated for 1200A, 480V, 3-phase, 4-wire service. Existing switchboard “MDP-LABS” is rated for 2500A, 120/208V, 3-phase, 4-wire service. Each existing switchboard is manufactured by Westinghouse, utilizes switches with Type L and Type R fuses, and appears to have been installed as part of the 1986 addition project.

Existing bus duct risers originate in both “MDP-L” and “MDP-LABS” that are routed vertically through the building. There are two bus duct risers that originate for each switchboard and are utilized to provide sub-distribution to branch circuit panelboards on each of the floors.

Existing panelboards are located throughout the facility as necessary to serve respective branch circuit loads and are generally the circuit breaker type of panelboards.

Power Distribution Systems – Arc Flash Study

An arc flash hazard study has been previously commissioned by Purdue University. According to this study, arc flash incident energy levels at existing switchboards are Category 3 or higher. Current Purdue University design standards require the design for new construction and major renovation projects to mitigate these levels to Category 1 or less at the line side of the main distribution switchgear. Presently no mitigation of arc flash hazard has been provided for this facility.

Power Distribution Systems – Renovations – Proposed Phase 1:

Original Building/1962 Addition:

The existing main power distribution system for the facility will be completely replaced, to the extent possible, including main switchboards, sub-distribution switchboards, feeders and motor control centers. Panelboards will be replaced in the areas to be renovated under the scope of the project however existing panelboards outside of the project scope will remain but will be re-fed from new switchboard sources. Equipment to mitigate the arc flash hazard to levels compliant with Purdue University standards will be provided as part of the power distribution system replacement. It is assumed that the strategy for mitigating the arc flash hazard will follow strategies employed on past university projects. A main low voltage insulated case solid state circuit breaker with adjustable LSI trip will be provided in the system between each service transformer in the vault and main circuit breaker in the respective main distribution switchboards. These circuit breakers will be located in a separate lockable room and will be remotely operated from the main distribution equipment electrical room.

New sub-distribution switchboards will be provided on the each floor of the facility to provide distribution to local panelboards, bus ducts, etc. to be located on the respective floor. The exact quantity and voltages of sub-distribution switchboards will be determined during a subsequent design project as there is not enough information available at this time to determine loading requirements.

In the areas to be renovated, new branch circuit panelboards will be provided to replace existing switch/fuse panelboards. The new panelboards will be located on each floor throughout the facility, as required to provide 120/208 volt, 3-phase, 4-wire electrical power for lighting and receptacle loads. Every effort will be made to locate panelboards such that they will be central to their respective loads served. Each individual computer, research and teaching lab will be provided with a respective branch circuit panelboard to serve only loads located within that lab space. Bus Ducts will be provided as required during design in teaching and research labs to provide flexibility within that lab space. A new bus duct and additional general power receptacles will be provided within the CE Research Lab to provide additional power and flexibility as requested. Panelboards will receive their power from the “L” or “EQ” service transformer via main and distribution switchboards as required.

The existing motor control centers and power panelboards will be replaced and be located within mechanical spaces to provide electrical power to mechanical, HVAC, and plumbing equipment motors. Each motor control center will be replaced with a new power panelboard. Each power panelboard will contain overcurrent protection devices as required for each respective item of equipment served. These power panelboards will receive their power from the "P" service transformer via main and distribution switchboards. Where required, combination overcurrent protective devices and motor controllers will be provided to serve mechanical loads, although most 3-phase motors will be controlled by variable frequency drives.

Each switchboard and panelboard will be sized to provide a minimum of 10% spare capacity and will be populated by a minimum of 10% spare devices.

1986 Addition:

The existing main power distribution system for the facility will be remain in place without significant renovation. Electrical service to new panelboards will be extended from the respective existing main switchboards or bus duct risers as required. In the areas to be renovated, new branch circuit panelboards will be provided to replace existing panelboards. The new panelboards will be located on the basement, ground and first floors in the facility, as required to provide 120/208 volt, 3-phase, 4-wire electrical power for lighting and receptacle loads. Every effort will be made to locate panelboards such that they will be central to their respective loads served. Each individual computer, research and teaching lab will be provided with a respective branch circuit panelboard to serve only loads located within that lab space. Bus Ducts will be provided as required during design in teaching and research labs to provide flexibility within that lab space. Panelboards will receive their power from the "L" or "LABS" service transformer via existing main switchboards and busduct risers as required.

Power Distribution Systems – Renovations – Proposed Phase 2:

New branch circuit panelboards and general power circuits will be provided as described in the Proposed Phase 1 work for areas to be renovated under Proposed Phase 2 as required.

Emergency Power Systems – Existing System

The Original Building/1962 Addition presently has a panelboard that is dedicated to feed emergency lighting fixtures however there is presently no emergency power source feeding this panel. The 1986 Addition has emergency lighting supplied by central battery inverter (EMAC) units with circuits distributed to selected lighting fixtures. The Fire Alarm System is provided with its own integral battery emergency power source.

Emergency Power Systems – Renovations

A central battery inverter system (EMAC) system will be provided to feed lighting fixtures for all emergency lighting, egress lighting and exit signage for the Original Building/1962 Addition under the Proposed Phase 1 scope. New circuits will be derived from the existing EMAC system as needed for work under in the 1986 Addition. This system will be expanded in the event that there is not sufficient capacity to handle the new load although it is not anticipated at this point. The Fire Alarm System will be provided with its own integral battery emergency power source.

Power Monitoring

Electronic metering will be provided at the main to each replacement main distribution switchboard in the Original Building/1962 Addition. The 1986 Addition presently has active electronic metering that will be maintained and not modified. It is anticipated that the new electronic meters will be connected to Purdue's standard SCADA box. It is not anticipated that sub-metering will be provided at any other points on the power distribution system.

Interior Lighting

Lighting will be designed utilizing a combination of direct, indirect, and direct/indirect luminaires utilizing Light Emitting Diode (LED) technology as the prime illumination source. The exact type of luminaire to be utilized in each respective space will depend on room layout, function ceiling type, aesthetic requirements, and control requirements, among other factors.

In general, lighting will be designed per the requirements contained in the Purdue University Consultant's Handbook and supplemented by Illumination Engineering Society of North America (IESNA) standards with particular attention paid to the following publications:

1. ANSI/IESNA RP-1 – American National Standard Practice for Office Lighting.
2. ANSI/IESNA RP-3 – Lighting for Educational Facilities.
3. The IESNA Lighting Handbook – Ninth Edition.
4. ANSI/IESNA RP-31 – Recommended Practice for the Economic Analysis of Lighting.

Lighting luminaires will be selected based on:

1. Suitability for use in the respective area in which they will be installed.
2. Energy efficiencies required to obtain goals for lighting power densities.
3. Maintenance requirements.

Average lighting levels for each specific building space will be designed in accordance with the following requirements. All illumination level are horizontal values referenced to a work plane located 30 inches above finished floor unless noted otherwise.

1. Lecture Halls: 50 footcandles with full range dimming.
2. Classrooms: 50 footcandles at each desktop with full range dimming.
3. Computer Labs: 20-30 footcandles.
4. Conference Rooms: 40 footcandles with full range dimming.
5. Collaboration Rooms: 40 footcandles.
6. Open, Research, Simulation and Teaching Laboratories: 30-40 footcandles general illumination, 80-100 footcandles on work surface.
7. Offices: 25-30 footcandles general illumination with owner furnished supplemental task lighting.
8. Restrooms: 20-30 footcandles general illumination with 45 footcandles vertical illumination at the mirror.
9. Tutoring Rooms: 40 footcandles.
10. Electronics Shop: 50 footcandles.
11. Corridors: 10-15 footcandles at the floor level.
12. Mechanical and Electrical Rooms: 15-25 footcandles at floor level.
13. Stairwells: 10-15 footcandles maximum, 1-5 footcandles minimum
14. All Other Areas: Default to IESNA recommended standards.

Luminaires utilizing Light Emitting Diode (LED) technology will be provided wherever possible based on the following criteria:

1. Suitability for use in the respective area in which they will be installed.
2. Suitability to render color with proper color temperature.
3. Efficacies required to obtain goals for lighting power densities.
4. Life and maintenance requirements.
5. Lighting control requirements.

All light emitting diode (LED) lighting luminaires shall utilize electronic drivers.

Emergency and egress lighting will be provided in accordance with relevant codes by means of a combination of "Nightlight" circuits and switched circuits connected to the respective building's emergency power source.

Illuminated exit signage will be provided in accordance with relevant codes and will be connected to the respective building's emergency power source.

Exterior Lighting

Exterior lighting will be designed and limited to utilize building mounted luminaires. Exterior lighting will be designed to meet the exterior lighting power densities as prescribed by Indiana Energy Conservation Code, where applicable.

Sources for exterior lighting environments will primarily be Light Emitting Diode (LED) technology, as determined during design, and per campus standards. Luminaires will generally be constructed from cast aluminum material. All luminaires will meet IESNA "Full Cutoff" criteria.

Lighting Controls

Various degrees of lighting controls will be provided in the facility to provide functionality, optimize energy performance, and to comply with the requirements of the Indiana Energy Conservation Code. It is anticipated that this facility will utilize a combination of automatic occupancy sensor lighting controls, networkable relay based lighting control systems, daylight harvesting systems and dimming systems.

Unit type occupancy sensors with manual "on" switch will be utilized in all small offices, storage rooms, restrooms, and similar spaces. These controls will be designed such that the occupant of the room must turn the lights on via a switch located at the room entrance and the occupancy sensor will automatically turn off the lights in the respective space if it determines the space is unoccupied.

Relay based lighting control systems will be provided for control of lighting in common areas such as corridors, open office areas, large open workroom areas and similar spaces. The system will provide manual, time-of-day scheduling, and occupancy sensor control to the respective areas. The systems could be designed to be networked and/or stand alone, as determined during the design process.

Daylight harvesting systems will provide automatically adjustable full range dimming based on ambient lighting levels. Daylight harvesting will be provided in appropriate open office areas and open workroom areas along the exterior walls of the building where there will be the possibility of a large natural light contribution due to existing windows.

Architectural dimming systems will be provided for dimming control of luminaires in the meeting rooms, conference rooms, classrooms, lecture halls and multi-purpose rooms. The system will provide full range dimming capability of multiple sources and zones as required via local control stations located in the respective spaces. The exact features and requirements of each architectural dimming system will be determined during design.

Mechanical rooms, electrical rooms, and telecommunications rooms will be provided with standard manually operated switches for control of the lights in the respective spaces.

Audio/Visual Systems

Provisions for Audio/Visual systems will be provided for the following areas of the building, as determined through meetings with Purdue University Technology personnel:

1. Classrooms.
2. Conference Rooms.
3. Lecture Halls.
4. Collaboration Rooms.

The building will contain provisions to support overhead projectors and/or flat-screen televisions in the spaces mentioned above. The provisions will include rough-in and conduits, power and mounting provisions required to support overhead projectors or televisions. Exact requirements will be determined during design by consulting with Purdue University. It is assumed that the owner will furnish all overhead projectors. Other requirements and features will be incorporated into the design, as directed by Purdue University.

Fire Alarm System

The existing fire alarm system for Hampton Hall is a Simplex 4100 system and appears to have been installed recently. Each existing fire alarm system will be revised and extended as required by renovations of spaces within the building. The system will fully comply with the Americans with Disabilities Act (ADA), Indiana Fire Code, and all other state codes.

The existing system is an intelligent, addressable type system. The complete system will include all required manual and automatic initiating devices such as pull stations, smoke detectors, heat detectors, water flow switches, etc. The complete system will include all signaling devices such as audible/visible signals, visible only signals, sprinkler alarm bell, etc. The system will monitor sprinkler system actuation. The system will monitor and control smoke evacuation system functions, fan shutdown functions, door hold open release functions, and elevator recall functions.

Exact specifications for the complete system will be entirely coordinated with Purdue University's Safety and Security Department.

Card Access and Security Systems

It is anticipated that some degree of access control and security will be required for this project. The extent of this scope has not been identified at this phase of the project but it is anticipated that some degree of access control will be provided to limit access to the second and third floors. The requirements for these systems will be discussed with appropriate Purdue University staff, but at a minimum, rough-in and conduits will be provided to accommodate system equipment and cabling requirements.

Basic Materials

All wiring and cabling for power distribution, branch circuitry, low voltage systems, etc. will be routed throughout the facility utilizing conduits.

Separate closed conduit systems will be provided for each power distribution system and each low voltage system such as:

1. Normal Power.
2. Emergency Power.
3. Equipment Power.
4. Audio/Visual Systems.
5. Fire Alarm System.
6. Card Access and Security Systems.
7. Low Voltage and Digital Lighting Control Systems.

In general conduit types will be as follows:

1. Electrical Metallic Tubing (EMT): Permitted for branch circuit conduits and conduits smaller than 1-1/4" trade size.
2. Rigid Metal Conduit (RMC) - Steel: Required for all feeders, and conduits larger than 1-1/4" trade size.
3. Intermediate Metal Conduit (IMC): Permitted in lieu of Rigid Metal Conduit where located within the building above slab on grade.
4. Polyvinyl Chloride (PVC): Permitted below grade for site lighting circuitry only.
5. Flexible Metal Conduit (FMC): Required for final connections to motorized equipment, lighting fixtures and equipment subject to vibration.
6. Liquidtight Flexible Metal Conduit (LFMC): Required in lieu of Flexible Metal Conduit in damp or wet locations.

All 600V conductors will be copper material Type THHN/THWN insulation.

All switchboards and panelboards will be furnished with 1000A/in^2 plated copper bus bars.

Overcurrent protection for all switchboards will be provided by either solid state low voltage insulated case circuit breakers or solid state molded case circuit breakers.

Panelboards will be furnished complete with bolt-on circuit breakers and hinged fronts.

Dry-type transformers will be furnished complete with copper windings and electrostatic shield.

All wiring devices will be heavy duty grade. Switches will be rated for 20A, 120/277 VAC. Receptacles will have integral brass strap and be rated for 20A, 120VAC. Plates will be high impact nylon or stainless steel as required.

Seismic Restraint Requirements

Seismic restraints are not expected to be required for this project.

Arc Flash Study

It is understood that a comprehensive three part Arc Flash Study (as-drawn, construction and final) will be performed concurrently with any project design. An Arc Flash Consultant will be contracted to perform the Arc Flash Study and make recommendations of mitigation strategies.

Based on previous projects at the University, the project will initially be designed with main circuit breakers with adjustable LSI trip units located in switchboards. These switchboards will be located in their own separate electrical room adjacent to the transformer vault and will serve to disconnect all normal power entering the building from the vault. The circuit breakers will be remotely operable from the main electrical room and will be provided with remote annunciation of the breaker position at the remote operator location.

Each new main switchboard to replace the existing switchboards will have a main circuit breaker with adjustable LSI trip unit as required to provide additional arc flash mitigation.

Additional strategies may be explored during design after an initial "as-drawn" Arc Flash Study has been performed.

Technology Narrative

I. TECHNOLOGY

A. Telecommunications Cabling Infrastructure:

1. Overview:

a. The systems referenced within this study pertain to all three facilities.

b. This infrastructure will provide support for Voice, Data and Cable TV distribution systems and services. In addition it will support interconnection of control panels for other ancillary building systems such as Fire Alarm, Electronic Security and Building Automation.

c. The Cabling Systems will be designed utilizing the Purdue ITaP guidelines and specifications as outlined in the Consultants Handbook.

d. It will be assumed that all offices, conference rooms, and other appropriate spaces shall be equipped with telephone circuits and one or more computer workstations. These will be connected via separate intra-building networks as well as separate inter-building networks. Currently, the telephone circuits, including dedicated 2-wire and 4-wire, security, fire, etc. operate on a copper backbone. The data network operates on copper from the workstation to the telecommunication room and then is connected via fiber optic cable to the Purdue fiber optic backbone.

e. CATV and Purdue Cable will continue to operate over a mixture of coaxial cable and fiber optic cables.

2. Demolition and Phasing:

a. Railway – see related RAIL section and narrative

b. ME – see related ME section and narrative

c. Hampton

1) The existing BDF shall remain in its current location and shall not be demolished as part of the renovation to the building.

2) The existing telecommunications rooms that are currently identified in the project scope shall be modified as part of any renovation to the building. Many of the existing spaces do not meet the current campus standards and must be replaced with new room's as part of any renovation to the building.

3) The telecommunication rooms located in the north end of the facility shall remain. However, all internal layout elements shall be replaced and brought up to the current ITaP Standards.

4) The facility requires new telecommunication rooms located in the south end that shall be stacked to provide better services to this portion of the facility.

5) As part of Phase 1, a new IDF telecommunications room will be created on the Basement level. Additionally, a new IDF telecommunications room will be created on the Ground floor along with one on the 1st Floor, 2nd Floor, 3rd Floor and 4th Floor.

6) As part of Phase 2, the BDF will remain in the current location however it will be updated to current ITaP Standards. Additionally, the IDF telecommunication rooms on the Ground Floor and 1st Floor will remain in the respective locations however they will be updated to the current ITaP Standards.

3. Topology:

a. This infrastructure shall include workstation outlets (WO) at specific locations throughout the facility. The cabling system shall utilize a "Star" homerun topology. Each individual WO shall be run to a dedicated Telecommunication Room (TR) where it will be terminated to a patch panel within the 90 meter distance limitations dictated by industry standards.

b. All Telecommunication Rooms shall then connect to a central point known as the Main Cross Connect, aka Building Distribution Frame (BDF) located in the Main Equipment Room (ER). This connection shall include a combination of fiber optic cable as well as a multi-pair copper cable.

c. That BDF shall connect via backbone copper and fiber cabling to Campus Telecommunications Systems.

4. Communications Pathways

a. The building's new communications pathways will be designed as an easily accessible system that provides for flexibility and relative ease of modification throughout the lifecycle of the building.

b. Pathways will be sized for voice, data, video, and any other Owner approved low voltage cabling. Systems, may include fire alarm, security, building automation, etc. each will be provided with separate pathway for wire management. This may be a separate tier on the corridor cable tray system.

c. The cable tray system will be sized to handle required cable loading for building plus 50% expansion for future applications, above the ceilings in the corridors and elsewhere as required.

d. The distribution cable tray system within the corridors will allow a minimum 3" of clearance between the tray and the ceiling, a minimum 6" of clearance in front of the tray, and a minimum 12" of clearance above the tray.

e. Basket type cable tray will be provided and will be wall and/or ceiling hung and mounted above the accessible ceiling in the corridors as required. The cable tray will be a minimum of 12" W by 4"H.

5. Telecommunications Rooms:

a. Railway Building – see related RAIL section and narrative

b. ME – see related ME section and narrative

c. Hampton Hall

1) The project scope for Hampton includes the establishment of all new telecommunications rooms within the scope areas identified as part of this study and it is anticipated that all work shall be completed in two phases.

2) In Phase 1 of Hampton, a new IDF will be created in the Basement level along with the 1st Floor, 2nd Floor, 3rd Floor and 4th Floor. These telecommunications rooms shall be located so that every new or future PIC is by wire length within 90 meters of its distribution point in the telecommunications room. This includes areas where PICs are not currently designed to be installed but may be installed in the future (i.e. Far ends of the building, attic spaces, mechanical rooms, etc.).

3) In Phase 2 of Hampton, the BDF will be included in the renovation. Additionally, the IDF's located on the Ground Floor and 1st Floor will be included in the renovation. All interior telecom elements of each room shall be fully replaced and updated to the current ITaP standards.

4) The new telecommunications spaces have been designed to operate 24/7 at the required temperature and humidity range required for these types of spaces. As well no HVAC ducts, plumbing, gas lines, air lines, clean outs, etc. will be located in or routed through the new or renovated telecommunications spaces.

5) The IDF telecommunication rooms for Hampton have been sized by area served (ASF).
   a) The ground floor will include the addition/renovation of the single telecommunications room to serve the eastern portion and to support the existing IDF located on this floor. This ground floor IDF shall be a minimum of 100 SF and shall also provide service to the first floor space located above.
   b) The first floor will have 1 new telecommunications room that will be part of the Phase 1 renovation. This IDF will be a 101 SF area and be located directly above the ground floor IDF and be used to serve the southern portion of the floor. This should allow for ease of routing communications cable from floor to floor.
   c) The second floor will have 1 new telecommunications room that will be located on the plan south wing. This IDF will be a 102 SF area and be located adjacent to the new restrooms and will serve the intended renovated space on this level.
   d) The third floor will have one new telecommunications room that will be located in the renovated Office Suite. This IDF will be a minimum of 100 SF and will serve the intended renovated space on this level.
   e) The fourth floor will have one new telecommunications room that will be located in the renovated Office Suite. This IDF will be a minimum of 100 SF and will serve the intended renovated space on this level.

d. All new or renovated telecommunications room walls will have an added layer of 3/4" plywood, B-B ext. grade 5. Plywood will be fire-retardant. Plywood will be installed horizontally on walls from 2'-0" to 6'-0"AFF from corner to corner.

e. All BDF and IDF telecommunication rooms shall have ladder type cable tray installed around the perimeter of the room at approximately 7'-2" AFF.

f. One dedicated, 110V, 20A double –duplex electrical outlets will be provided on each wall. One of those outlets in particular will be provided on the wall near the equipment frames.

g. Two dedicated, 208V, 30A, twist lock (NEMA L6-30R), single phase electrical outlet will be located 25" behind the front rail of the equipment frame on the wall in the telecommunications room BDF for the owner provided UPS.

h. Two dedicated, 208V, 20A, twist lock (NEMA L6-20R), single phase electrical outlet will be located 25" behind the front rail of the equipment frame on the wall in telecommunications room IDF for the owner provided UPS.

6. Workstation Outlets Requirements:
   a. Workstation outlets will be placed in general office space, conference rooms, commons areas, laboratories, classrooms and other facility spaces as required. Special Circuit Requirements:
      1) Wireless Access Point location will require one data cable each.
      2) Every FACP (Fire Alarm Control Panel) requires two voice cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.
      3) Every environmental control cabinet requires two data cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.
      4) Every building electrical switch gear meter requires two data cables, terminated in an enclosure outside of the metering cabinet per telecom specifications, and to be installed to the nearest telecommunications room.
      5) Every elevator requires two voice cables, terminated per telecom specifications, to be installed to the nearest telecommunications room.
   b. Each standard PIC Station will be one 4-11/16" square by 2-1/8" deep flush mounted box with double gang, square drawn extension or tile ring. Provide 1" EMT conduit from the box to cable tray in corridor.
7. Video Wiring System:
   a. This system shall provide the signal to the television / display from the Cable Television (CATV) Distribution System.
   b. Each outlet location shall be supplied with the required RG-6 or RG-11 horizontal cabling that is connected to a tap/splitter within the IDF serving that area. All new or renovated IDFs will be provided with new RG-11 Riser cabling to new BDF for distribution of CATV signals from the backbone cabling entering the building.

B. Wireless Network Systems for Data:
   1. Overview:
      a. Wireless network systems shall be engineered by Purdue's Information Technology Networks and Security (ITNS) department during the design phase of the project. ITNS will provide the A&E team with wireless access point locations based on a computer generated site survey for the existing buildings so they may be incorporated into the construction documents. The contractor will be responsible for installing a turn-key wireless system as part of the project. The overall wireless system shall be included in the building project budget.

C. Telephone System and Service:
   1. Overview:
      a. The telephone system equipment and handsets will be provided by the University.
      b. Service to the system will be provided by connecting these facilities to the existing Campus Telecommunications Systems.
      c. It is assumed that a majority of the new handset locations within each facility will be Voice over IP (VoIP) rather than traditional digital handsets. The riser and horizontal cabling quantity will reflect this type of handset being installed in all offices, conference rooms, and classrooms as required.

D. Emergency Communications:
   1. Overview:
      a. Emergency Communications in these facilities will be provided by "Voice Evacuation" capabilities of the Fire Alarm System.
      b. It is yet to be determined whether any other emergency phones are required.

E. Paging System
   1. No paging system will be provided.

F. Clock System:
   1. No clock system will be provided.
   2. Battery powered clocks may be installed by Owner in selected locations.

G. Local Area Network (LAN) Electronics:
   1. Overview:
      a. This includes Hubs, Routers, and Switches required to activate data ports. The University will provide all Network Electronics for these Projects. Space will be allocated in the BDF and IDF rack to accommodate these requirements.

H. Security Systems:
   1. Overview:
      a. This includes cabling, controllers, card readers and programming and activate the system in each building. The University will provide all cabling, controllers, and card readers for these Projects.

Space will be allocated in the Electrical and mechanical rooms to accommodate these systems requirements.

2. Access Control:
   a. A new/renovated electronic access control system will be provided. All readers, controllers, and wiring shall be providing, installed, and programmed by the University. The design of the facilities shall include all required rough-ins, conduit, junction boxes, and power to allow for the installation at each opening to be completed.
   b. A single gang box with a single gang plaster ring and a 1" EMT conduit to above ceiling cable tray will be provided for each card reader location.
   c. All keypads, card readers, cables, security cabinet, etc. shall be provided by the University.
   d. During the design phase of the project key openings on the exterior and interior of these facilities will need to be identified that require access control and card readers.
3. CCTV:
   a. A CCTV system will be provided to serve these facilities.
   b. Interior camera locations will be recorded by the existing IP NVR located on campus to record all new camera locations. It may be determined during design that placing all cameras on the IP system may not be feasible as they may not accessible by staff and faculty, only campus police. In these cases a local DVR and cameras will be provided within the area. The cameras will be recorded by the DVR and will be accessible by the department only.
   c. Currently cameras may be required to view and monitor any active lab spaces to ensure the security of ongoing experiments taking place in the spaces.
   d. During the design phases other key areas of the facilities will need to be identified that require camera coverage.
   e. All exterior camera locations will be provided with a two gang box with a single gang ring and a 1" EMT conduit to nearest above ceiling cable tray. All interior camera location will be provided with a two gang box with single gang plaster ring and 1" EMT conduit to the nearest above ceiling cable tray.

I. Audio Visual Systems:
   1. Overview:
      a. The Audio Visual (AV) system requirements for these facilities have not yet been fully determined and are not currently included in the design responsibility of the Architect or its consultants. It will be critical that these requirements be determined in the initial phase as they will impact the design of electrical pathways and power systems as well CATV and Network Connections.
      b. We expect the AV requirements at a minimum to include technologies such as "way finding" and general information displays in common areas, projectors, and large LCD screens in conference rooms and dedicated audio systems in gathering and presentation areas.

II. SPECIFICATIONS
   A. Main Bid Package:
      1. Division 27:
         a. 27 00 00 Communications (Common Work Results)
         b. 27 05 26 Grounding and Bonding for Communication Systems
         c. 27 05 28 Pathways for Communications Systems
         d. 27 05 53 Identification for Communication Systems
         e. 27 11 01 Telecommunications Room Finishes
         f. 27 11 16 Cabinets and Frames
         g. 27 11 19 Termination Blocks and Patch Panels
         h. 27 14 13 Copper Riser Cabling
         i. 27 14 23 Fiber Riser Cabling
         j. 27 14 33 Coaxial Riser Cabling
         k. 27 15 13 Copper Horizontal Cabling
         l. 27 15 33 Coaxial Horizontal Cabling
         m. 27 15 43 Faceplates and Connectors

The executive summary and calculated arc flash data follow. This is just a summary; calculated data and the full report are provided under separate cover.

PURDUE UNIVERSITY

Arc Flash Risk Assessment with Fault-Current and Protective Device Coordination Analysis

Hampton Buildings (North & South)

As found Study Report

Applied#15-051

Applied Engineering Services

PURDUE UNIVERSITY

ARC FLASH REPORT

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Purdue University
Hampton North & South
As found Study Report
Applied Project No. 15-051

June 17, 2015

1 Executive Summary

Warning

The PPE requirements, when working on the line side of main disconnects, busswork and cabling between main transformers, main disconnects, and the connections within main transformers are not affected by the recommendations made in this report. That part of the system is protected by devices on the input side of the transformers that are normally owned and set by the utility.

- All of the panelboards and switchboards fall within NFPA 70E limits for PPE (personal protective equipment) categorized as Category 0, 1, or 2 unless noted below. Refer to **Section 2** of this report for a table describing the categories and results of the arc flash hazard calculations.
- The rated fault current for all busses is sufficient to clear a bolted fault unless noted below. Refer to **Section 4** of this report for a table describing the results.
- The rated fault current for all protective devices is sufficient to clear a bolted fault unless noted below. Refer to **Section 4** of this report for a table describing the results.

Findings

In the following table, the calculated incident energy is above 8.0 cal/cm^2 and PPE Category is 3 or above at buses. **It is to be noted that existing model file is provided by Purdue University with available low and high fault current.**

Hampton Building-North

- Table – Shows Bus PPE Value 3 or higher

Bus Name	Protective Device Name	Bus kV	Bus Bolted Fault (kA)	Bus Arcing Fault (kA)	Arc Flash Boundary (in)	Incident Energy (cal/cm2)	PPE Level
MDP-P	Xfmr P	0.480	14.42			19	Category 3
MDP-L	Xfmr L	0.208	26.02			47	Dangerous!
MDP-LABS	Xfmr EQ	0.240	25.07			33	Category 4

Hampton Building-South

- Table – Shows Bus PPE Value 3 or higher

Bus Name	Protective Device Name	Bus kV	Bus Bolted Fault (kA)	Bus Arcing Fault (kA)	Arc Flash Boundary (in)	Incident Energy (cal/cm2)	PPE Level
M-P-1	Xfmr P2	0.240	35.08			27	Category 4
M-P-2	XfmrP1	0.480	14.97			9	Category 3
M-L-1	Xfmr L	0.208	25.80			47	Dangerous!

Recommendation

Hampton Building-North

- Provide new main breaker switchboard MDP-IML between transformer & MDP-L, new breaker with electronic trip unit (LSIG).
- Replace PD-N BD Riser & PD-S BD Riser fuse in switchboard MDP-L w/Siemens NMG-586-LSI Breaker.
- Provide new main breaker switchboard MDP-IMLABS between transformer & MDP-LABS, new breaker with electronic trip unit (LSIG).
- Replace fuse switch switchboard MDP-LABS with circuit breaker type switch board, new breaker with electronic trip unit (LSI).
- Provide new main breaker switchboard MDP-IMP between transformer & MDP-P, new breaker with electronic trip unit (LSIG).
- Replace PD-N, PD-S, PD-S MCCB & PD-S MCC4 fuse with 325A fuse in switchboard MDP-P.

Hampton Building-South

- Provide new main breaker switchboard M-L-1-IML between transformer & M-L-1, new breaker with electronic trip unit (LSIG).
- Replace PD-SDL-1, PD-L3-3, PD-L-3-2, & PD-L-3-1 fuse with LPN-RK-SP fuse in switchboard M-L-1.
- Provide new main breaker switchboard M-P-1-IMP1 between transformer & M-P-1, new breaker with electronic trip unit (LSIG).
- Replace PD-SD-P1, PD-P-H-1 fuse with LPN-RK-SP-500A fuse in switchboard M-P-1.
- Provide new main breaker switchboard M-P-2-IMP2 between transformer & M-P-2, new breaker with electronic trip unit (LSIG).

2 Calculated Arc Flash Data

The table included in this section is the screen displayed at the conclusion of the SKM arc-flash analysis. It was printed as a pdf file and merged into the report.

The arc-flash hazard analysis is done to conform to OSHA recommendations described in 29CFR 1910.333. Procedures for arc-flash hazard analysis are described in both NFPA Standard 70E-2009 and IEEE Standard 1584-2002. IEEE Standard 1584-2002 describes a method for calculating the potential arc-flash energy based on the anticipated “bolted” fault current. The results reported within have been calculated using the more conservative, interpretation of IEEE Standard 1584-2002.

IEEE Standard 1584-2002 states that equipment rated below 208V does not need be considered unless it involves a low-impedance transformer larger than 125 kVA for its immediate supply. The reason for this exclusion is the inability of such small capacity systems to initiate and support arcing conditions. NFPA Standard 70E-2009 applies the same reasoning to equipment rated 208V and less. NFPA Standard 70E is recognized as the guiding document.

The table beginning on the next page shows arc-flash data calculated. Included in the tables are the critical distances, the incident energies, the fault currents, and the PPE category level required for the indicated working distance. Arc-flash hazard warning labels for the equipment are created from this table.

If future changes are made in the electrical system or to the protective device settings, new calculations should be completed on the specific circuits and new warning labels printed to reflect any changes in the hazard.

- Arc Flash value as describe in table above and shown on one line drawing

3 Single-Line Drawings

The single-line drawings that follow show the system topology, the input, and calculated data for the facility. It shows circuit breaker and/or fuse data, cable data, transformer data, motor data, source data, calculated fault current at bus, and calculated arc flash data including incident energy, boundary distance and PPE category. Source fault current and X/R ratio are obtained from Purdue University.

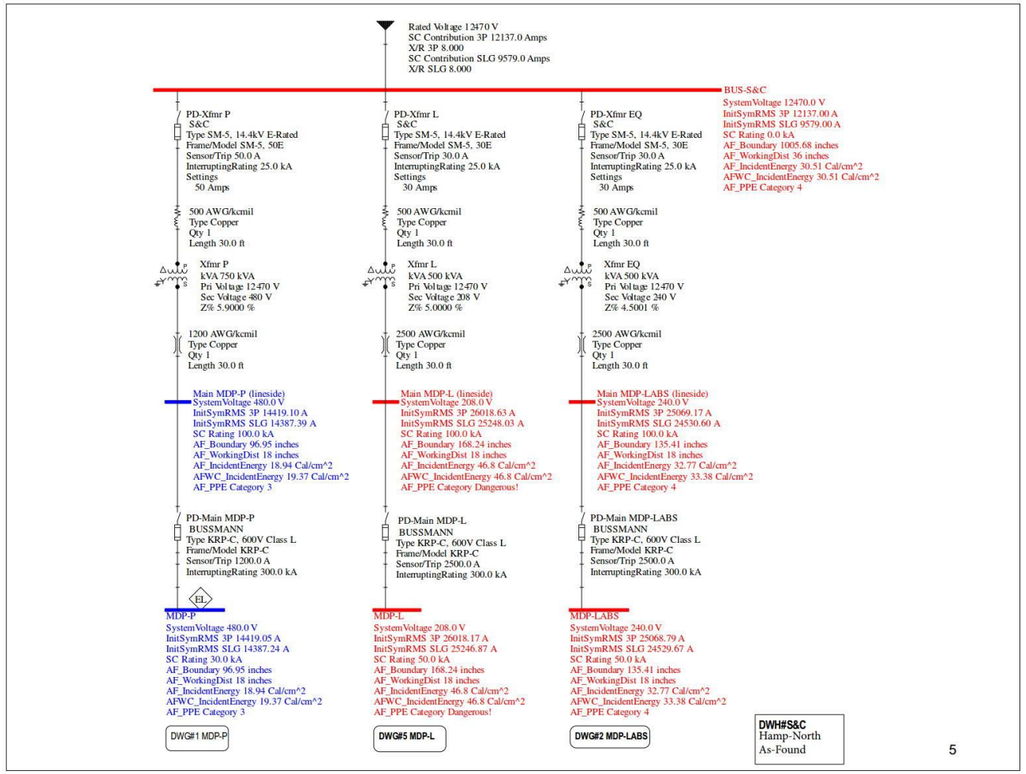
Drawing List

Hampton Building-North

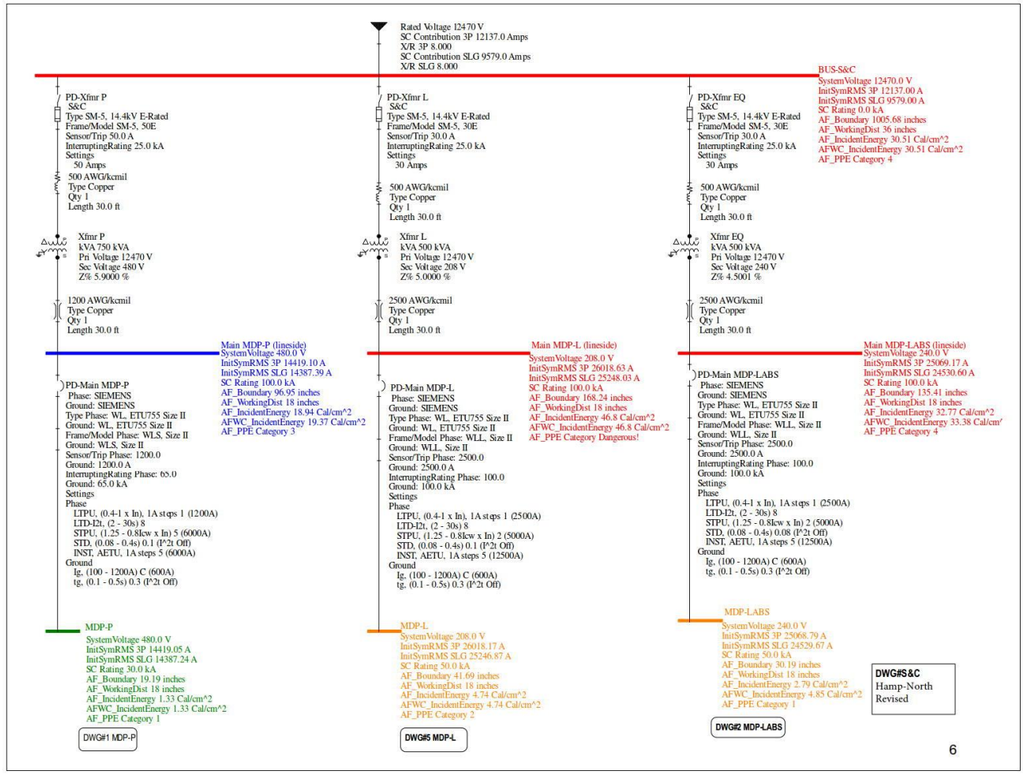
DWG #S&C Fuse	One line drawing w/fuse
DWG #S&C BKR	One line drawing w/Breaker

Hampton Building-South

DWG #S&C Fuse	One line drawing w/fuse
DWG #S&C BKR	One line drawing w/Breaker


Rated Voltage 12470 V
SC Contribution 3P 12137.0 Amps
X/R 3P 8.000
SC Contribution SLG 9579.0 Amps
X/R SLG 8.000
BUS-S&C
SystemVoltage 12470.0 V
InitSymRMS 3P 12137.00 A
InitSymRMS SLG 9579.00 A
SC Rating 0.0 kA
AF_Boundary 1005.68 inches
AF_WorkingDist 36 inches
AF_IncidentEnergy 30.51 Cal/cm^2
AFWC_IncidentEnergy 30.51 Cal/cm^2
AF_PPE Category 4
PD-Xfmr P
S&C
Type SM-5, 14.4kV E-Rated
Frame/Model SM-5, 50E
Sensor/Trip 50.0 A
InterruptingRating 25.0 kA
Settings
50 Amps
500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Xfmr P
kVA 750 kVA
Pri Voltage 12470 V
Sec Voltage 480 V
Z% 5.9000 %
1200 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Main MDP-P (lineside)
SystemVoltage 480.0 V
InitSymRMS 3P 14419.10 A
InitSymRMS SLG 14387.39 A
SC Rating 100.0 kA
AF_Boundary 96.95 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 18.94 Cal/cm^2
AFWC_IncidentEnergy 19.37 Cal/cm^2
AF_PPE Category 3
PD-Main MDP-P
BUSSMANN
Type KRP-C, 600V Class L
Frame/Model KRP-C
Sensor/Trip 1200.0 A
InterruptingRating 300.0 kA
EL
MDP-P
SystemVoltage 480.0 V
InitSymRMS 3P 14419.05 A
InitSymRMS SLG 14387.24 A
SC Rating 30.0 kA
AF_Boundary 96.95 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 18.94 Cal/cm^2
AFWC_IncidentEnergy 19.37 Cal/cm^2
AF_PPE Category 3
DWG#1 MDP-P
PD-Xfmr L
S&C
Type SM-5, 14.4kV E-Rated
Frame/Model SM-5, 30E
Sensor/Trip 30.0 A
InterruptingRating 25.0 kA
Settings
30 Amps
500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Xfmr L
kVA 500 kVA
Pri Voltage 12470 V
Sec Voltage 208 V
Z% 5.0000 %
2500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Main MDP-L (lineside)
SystemVoltage 208.0 V
InitSymRMS 3P 26018.63 A
InitSymRMS SLG 25248.03 A
SC Rating 100.0 kA
AF_Boundary 168.24 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 46.8 Cal/cm^2
AFWC_IncidentEnergy 46.8 Cal/cm^2
AF_PPE Category Dangerous!
PD-Main MDP-L
BUSSMANN
Type KRP-C, 600V Class L
Frame/Model KRP-C
Sensor/Trip 2500.0 A
InterruptingRating 300.0 kA
MDP-L
SystemVoltage 208.0 V
InitSymRMS 3P 26018.17 A
InitSymRMS SLG 25246.87 A
SC Rating 50.0 kA
AF_Boundary 168.24 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 46.8 Cal/cm^2
AFWC_IncidentEnergy 46.8 Cal/cm^2
AF_PPE Category Dangerous!
DWG#5 MDP-L
PD-Xfmr EQ
S&C
Type SM-5, 14.4kV E-Rated
Frame/Model SM-5, 30E
Sensor/Trip 30.0 A
InterruptingRating 25.0 kA
Settings
30 Amps
500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Xfmr EQ
kVA 500 kVA
Pri Voltage 12470 V
Sec Voltage 240 V
Z% 4.5001 %
2500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Main MDP-LABS (lineside)
SystemVoltage 240.0 V
InitSymRMS 3P 25069.17 A
InitSymRMS SLG 24530.60 A
SC Rating 100.0 kA
AF_Boundary 135.41 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 32.77 Cal/cm^2
AFWC_IncidentEnergy 33.38 Cal/cm^2
AF_PPE Category 4
PD-Main MDP-LABS
BUSSMANN
Type KRP-C, 600V Class L
Frame/Model KRP-C
Sensor/Trip 2500.0 A
InterruptingRating 300.0 kA
MDP-LABS
SystemVoltage 240.0 V
InitSymRMS 3P 25068.79 A
InitSymRMS SLG 24529.67 A
SC Rating 50.0 kA
AF_Boundary 135.41 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 32.77 Cal/cm^2
AFWC_IncidentEnergy 33.38 Cal/cm^2
AF_PPE Category 4
DWG#2 MDP-LABS
DWH#S&C
Hamp-North
As-Found

5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X/R SLG 8.000
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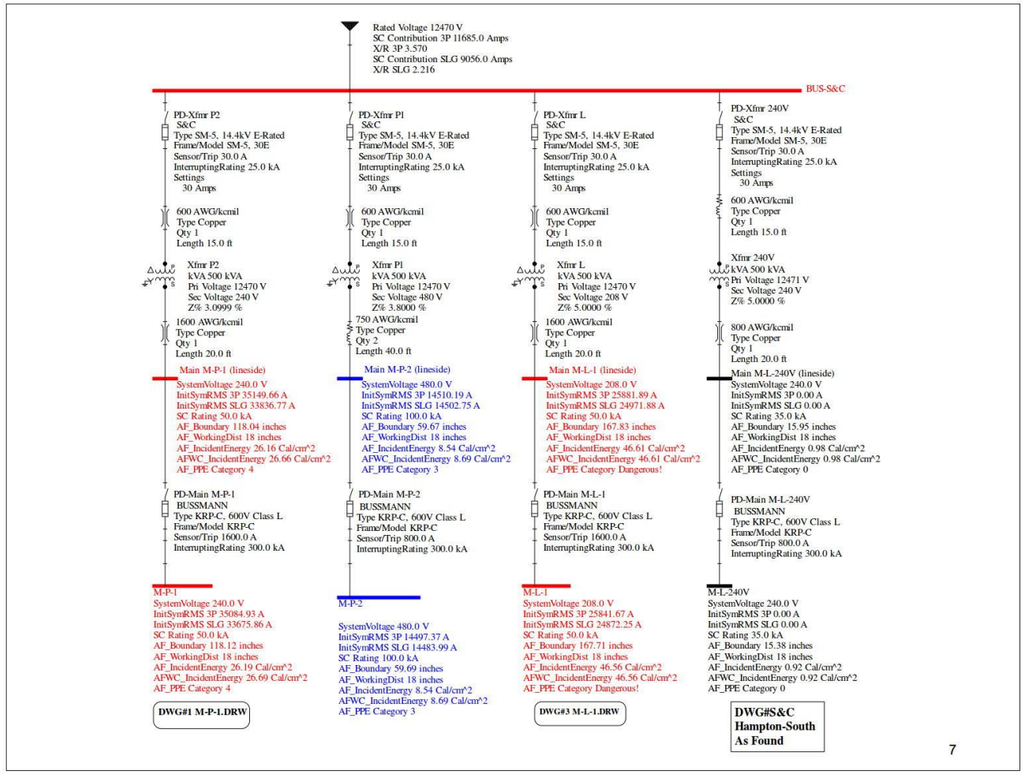
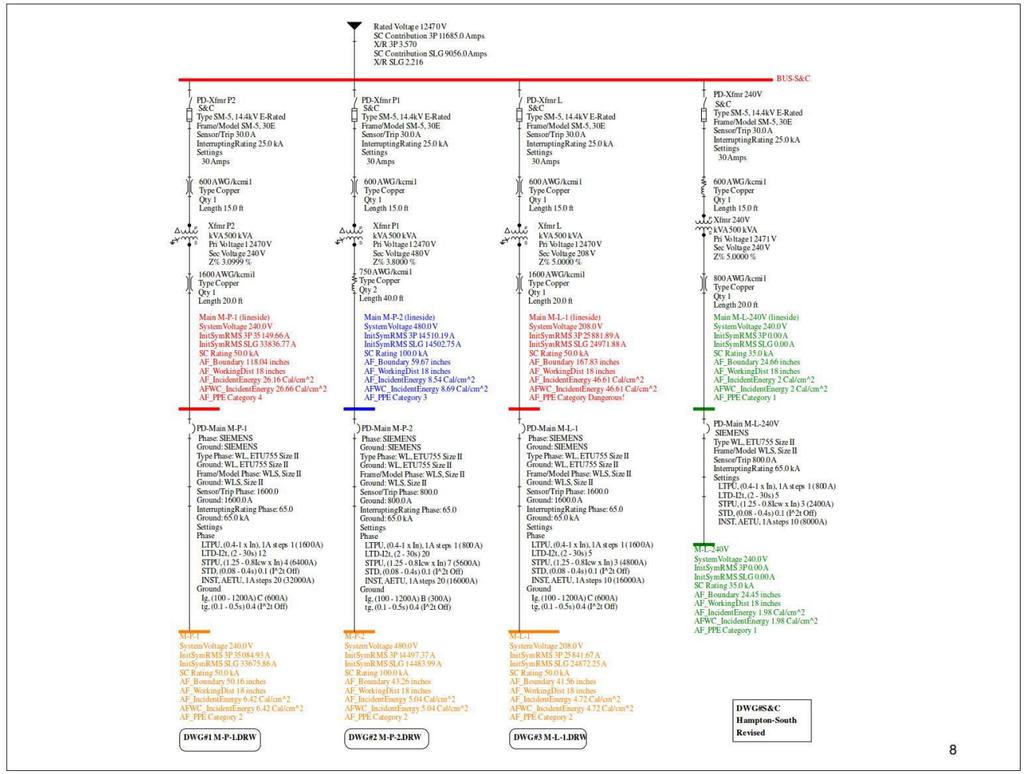
SystemVoltage 12470.0 V
InitSymRMS 3P 12137.00 A
InitSymRMS SLG 9579.00 A
SC Rating 0.0 kA
AF_Boundary 1005.68 inches
AF_WorkingDist 36 inches
AF_IncidentEnergy 30.51 Cal/cm^2
AFWC_IncidentEnergy 30.51 Cal/cm^2
AF_PPE Category 4
PD-Xfmr P
S&C
Type SM-5, 14.4kV E-Rated
Frame/Model SM-5, 50E
Sensor/Trip 50.0 A
InterruptingRating 25.0 kA
Settings
50 Amps
500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Xfmr P
kVA 750 kVA
Pri Voltage 12470 V
Sec Voltage 480 V
Z% 5.9000 %
1200 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Main MDP-P (lineside)
SystemVoltage 480.0 V
InitSymRMS 3P 14419.10 A
InitSymRMS SLG 14387.39 A
SC Rating 100.0 kA
AF_Boundary 96.95 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 18.94 Cal/cm^2
AFWC_IncidentEnergy 19.37 Cal/cm^2
AF_PPE Category 3
PD-Main MDP-P
Phase: SIEMENS
Ground: SIEMENS
Type Phase: WL, ETU755 Size II
Ground: WL, ETU755 Size II
Frame/Model Phase: WLS, Size II
Ground: WLS, Size II
Sensor/Trip Phase: 1200.0
Ground: 1200.0 A
InterruptingRating Phase: 65.0
Ground: 65.0 kA
Settings
Phase
LTPU, (0.4-1 x In), 1A steps 1 (1200A)
LTD-I2t, (2 - 30s) 8
STPU, (1.25 - 0.8Isc x In) 5 (6000A)
STD, (0.08 - 0.4s) 0.1 (I^2t Off)
INST, AETU, 1A steps 5 (6000A)
Ground
Ig, (100 - 1200A) C (600A)
tg, (0.1 - 0.5s) 0.3 (I^2t Off)
MDP-P
SystemVoltage 480.0 V
InitSymRMS 3P 14419.05 A
InitSymRMS SLG 14387.24 A
SC Rating 30.0 kA
AF_Boundary 96.95 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 18.94 Cal/cm^2
AFWC_IncidentEnergy 19.37 Cal/cm^2
AF_PPE Category 3
DWG#1 MDP-P
PD-Xfmr L
S&C
Type SM-5, 14.4kV E-Rated
Frame/Model SM-5, 30E
Sensor/Trip 30.0 A
InterruptingRating 25.0 kA
Settings
30 Amps
500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Xfmr L
kVA 500 kVA
Pri Voltage 12470 V
Sec Voltage 208 V
Z% 5.0000 %
2500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Main MDP-L (lineside)
SystemVoltage 208.0 V
InitSymRMS 3P 26018.63 A
InitSymRMS SLG 25248.03 A
SC Rating 100.0 kA
AF_Boundary 168.24 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 46.8 Cal/cm^2
AFWC_IncidentEnergy 46.8 Cal/cm^2
AF_PPE Category Dangerous!
PD-Main MDP-L
Phase: SIEMENS
Ground: SIEMENS
Type Phase: WL, ETU755 Size II
Ground: WL, ETU755 Size II
Frame/Model Phase: WLL, Size II
Ground: WLL, Size II
Sensor/Trip Phase: 2500.0
Ground: 2500.0 A
InterruptingRating Phase: 100.0
Ground: 100.0 kA
Settings
Phase
LTPU, (0.4-1 x In), 1A steps 1 (2500A)
LTD-I2t, (2 - 30s) 8
STPU, (1.25 - 0.8Isc x In) 2 (5000A)
STD, (0.08 - 0.4s) 0.1 (I^2t Off)
INST, AETU, 1A steps 5 (12500A)
Ground
Ig, (100 - 1200A) C (600A)
tg, (0.1 - 0.5s) 0.3 (I^2t Off)
MDP-L
SystemVoltage 208.0 V
InitSymRMS 3P 26018.17 A
InitSymRMS SLG 25246.87 A
SC Rating 50.0 kA
AF_Boundary 41.69 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 4.74 Cal/cm^2
AFWC_IncidentEnergy 4.74 Cal/cm^2
AF_PPE Category 2
DWG#5 MDP-L
PD-Xfmr EQ
S&C
Type SM-5, 14.4kV E-Rated
Frame/Model SM-5, 30E
Sensor/Trip 30.0 A
InterruptingRating 25.0 kA
Settings
30 Amps
500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Xfmr EQ
kVA 500 kVA
Pri Voltage 12470 V
Sec Voltage 240 V
Z% 4.5001 %
2500 AWG/kcmil
Type Copper
Qty 1
Length 30.0 ft
Main MDP-LABS (lineside)
SystemVoltage 240.0 V
InitSymRMS 3P 25069.17 A
InitSymRMS SLG 24530.60 A
SC Rating 100.0 kA
AF_Boundary 135.41 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 32.77 Cal/cm^2
AFWC_IncidentEnergy 33.38 Cal/cm^2
AF_PPE Category 4
PD-Main MDP-LABS
Phase: SIEMENS
Ground: SIEMENS
Type Phase: WL, ETU755 Size II
Ground: WL, ETU755 Size II
Frame/Model Phase: WLL, Size II
Ground: WLL, Size II
Sensor/Trip Phase: 2500.0
Ground: 2500.0 A
InterruptingRating Phase: 100.0
Ground: 100.0 kA
Settings
Phase
LTPU, (0.4-1 x In), 1A steps 1 (2500A)
LTD-I2t, (2 - 30s) 8
STPU, (1.25 - 0.8Isc x In) 2 (5000A)
STD, (0.08 - 0.4s) 0.08 (I^2t Off)
INST, AETU, 1A steps 5 (12500A)
Ground
Ig, (100 - 1200A) C (600A)
tg, (0.1 - 0.5s) 0.3 (I^2t Off)
MDP-LABS
SystemVoltage 240.0 V
InitSymRMS 3P 25068.79 A
InitSymRMS SLG 24529.67 A
SC Rating 50.0 kA
AF_Boundary 30.19 inches
AF_WorkingDist 18 inches
AF_IncidentEnergy 2.79 Cal/cm^2
AFWC_IncidentEnergy 4.85 Cal/cm^2
AF_PPE Category 1
DWG#2 MDP-LABS
DWG#S&C
Hamp-North
Revised

6

SPACE OVERVIEW – HAMP BUILDING





4 TCC Plots and Device Settings

The following coordination curves graphically depict the time versus current characteristics for the overcurrent protective device shown, and the relationship of each device to the other devices that are in electrical series.

An ideally coordinated system would have the overcurrent protective device closest to a short circuit, or fault, condition detect the fault and open the circuit. In the event the closest device fails to operate, the next device between the fault and the power source would detect the fault and open the circuit. Realistically, the ideal is often not achievable, but an attempt is made to come as close as practical.

Each set of plotted curves shows at least two protective devices.

All recommendations presume the protective devices will be properly tested and maintained to operate in accordance with the manufacturer's specifications.

The page(s) following set of curves present(s) the pertinent data and settings for each protective device represented in the system. All recommendations presume the protective devices have been, and will continue to be, well and properly maintained to operate in accordance with the manufacturer's specifications.

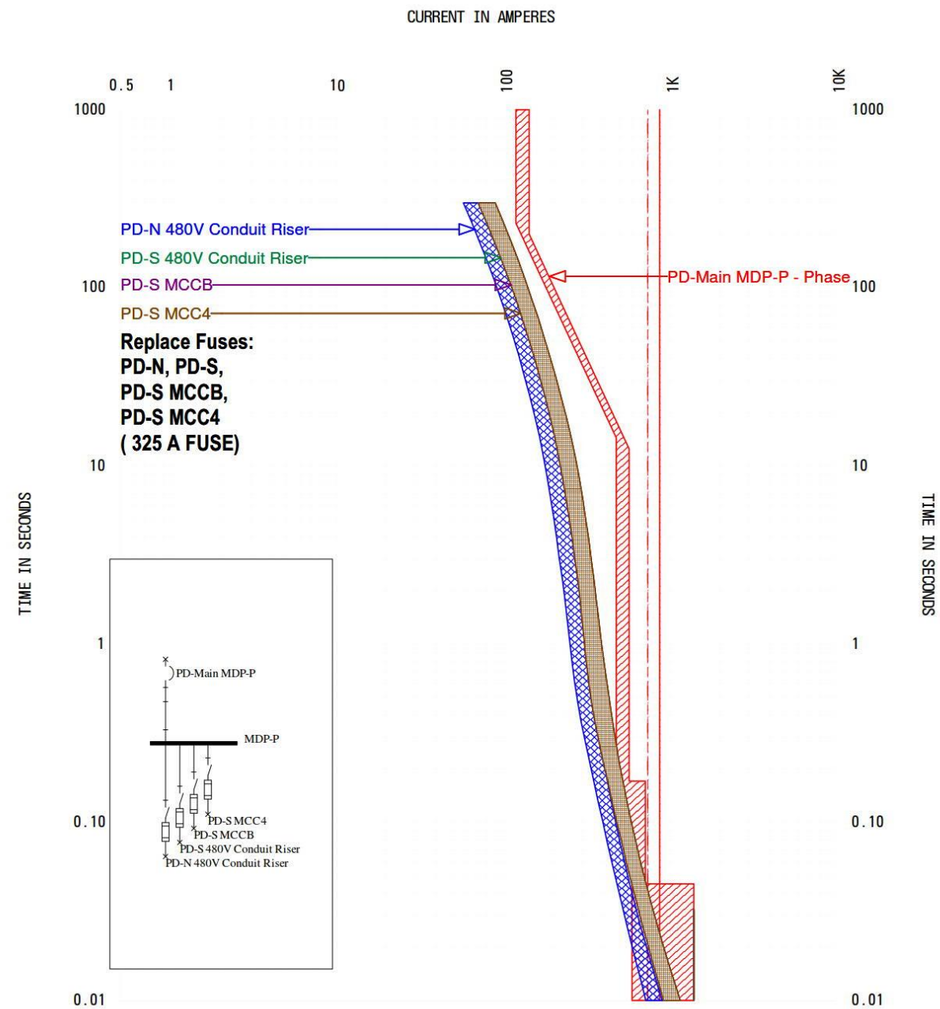
TCC Plot:

Hampton-North:

- TCC MDP-P
- TCC MDP-L
- TCC MDP-LABS

Hampton-South:

- TCC M-P-1
- TCC M-P-2
- TCC M-L-1



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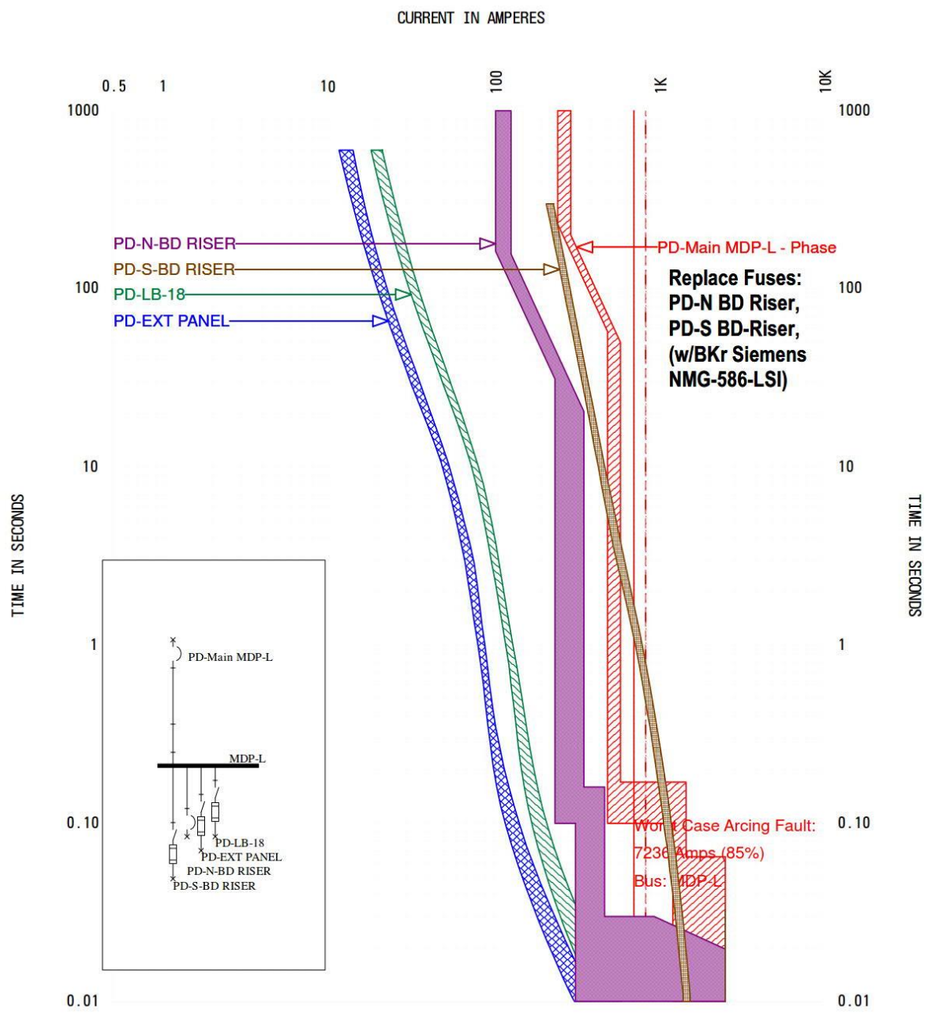
HAMP-North Revised
PURDUE UNIVERSITY, IN

Reference Voltage: 480
Current Scale x 10
Applied Engineering Services

10



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HAMP-North Revised
PURDUE UNIVERSITY, IN

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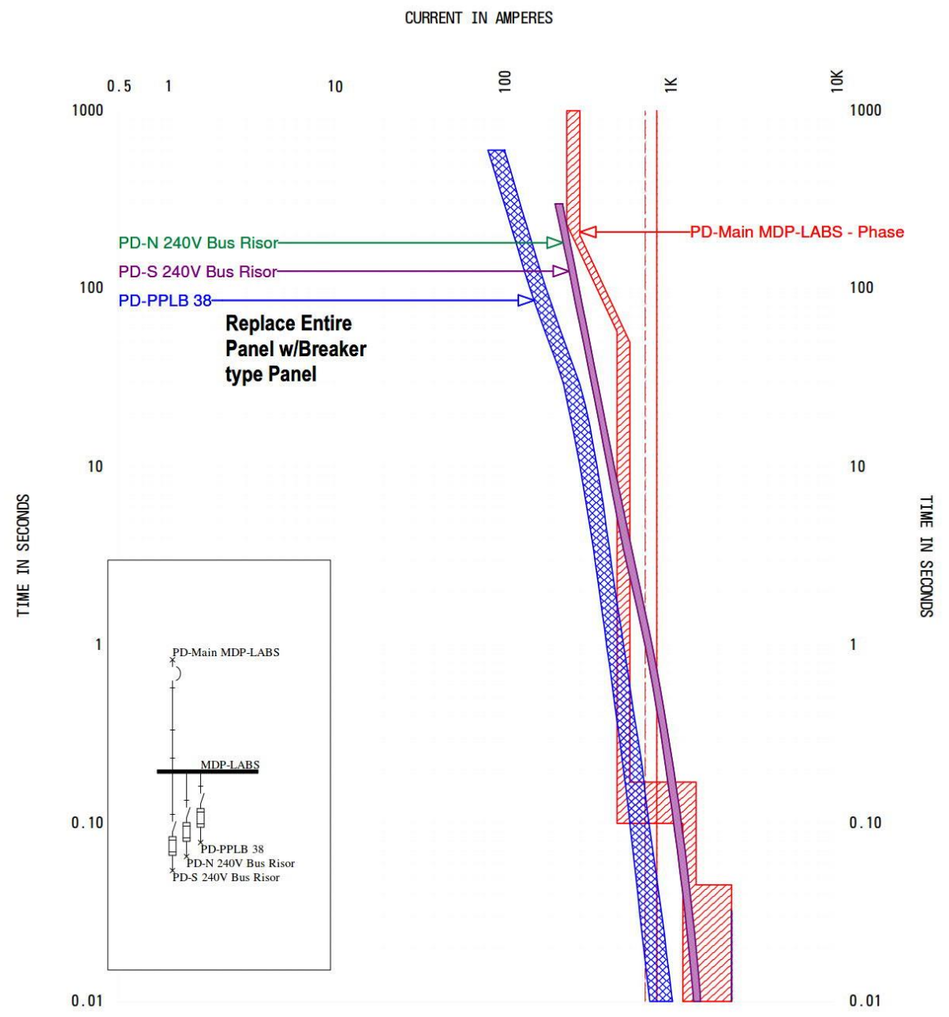
Current Scale x 10
Applied Engineering Services

11



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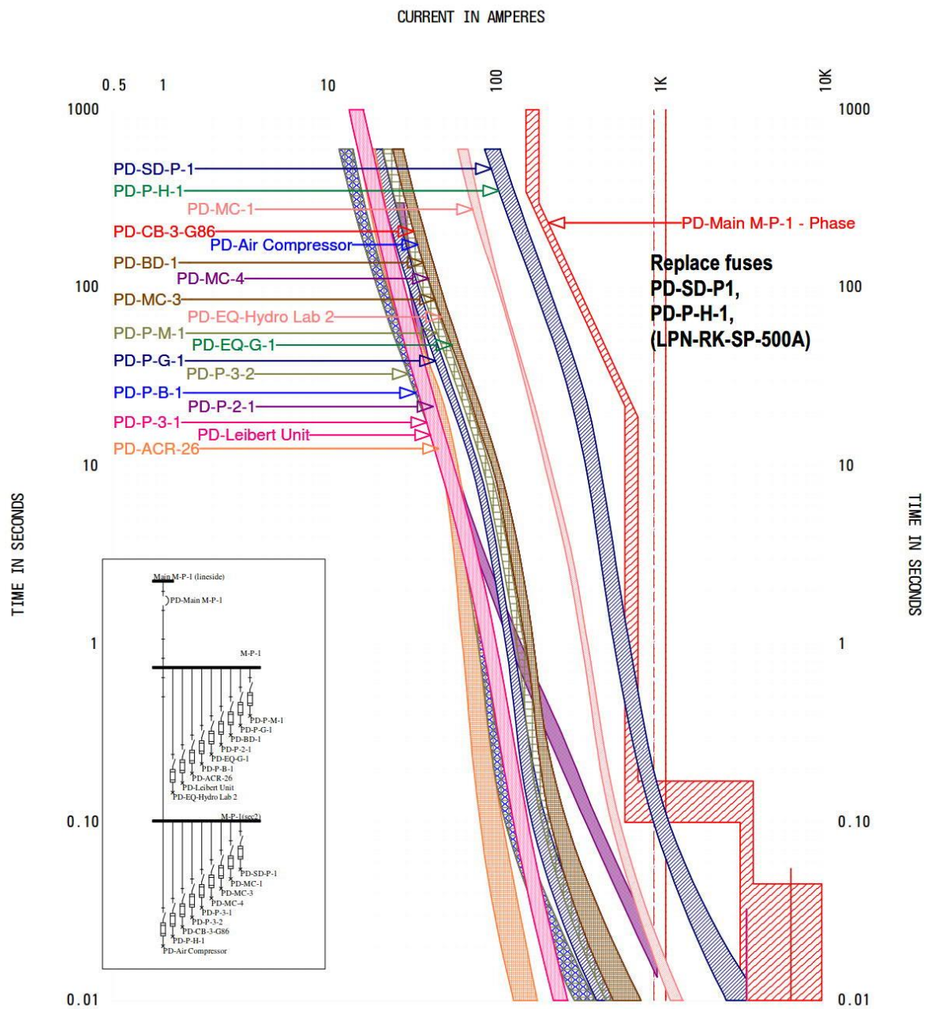
HAMP-North Revised
PURDUE UNIVERSITY, IN

Reference Voltage: 240
Current Scale x 10
Applied Engineering Services

12



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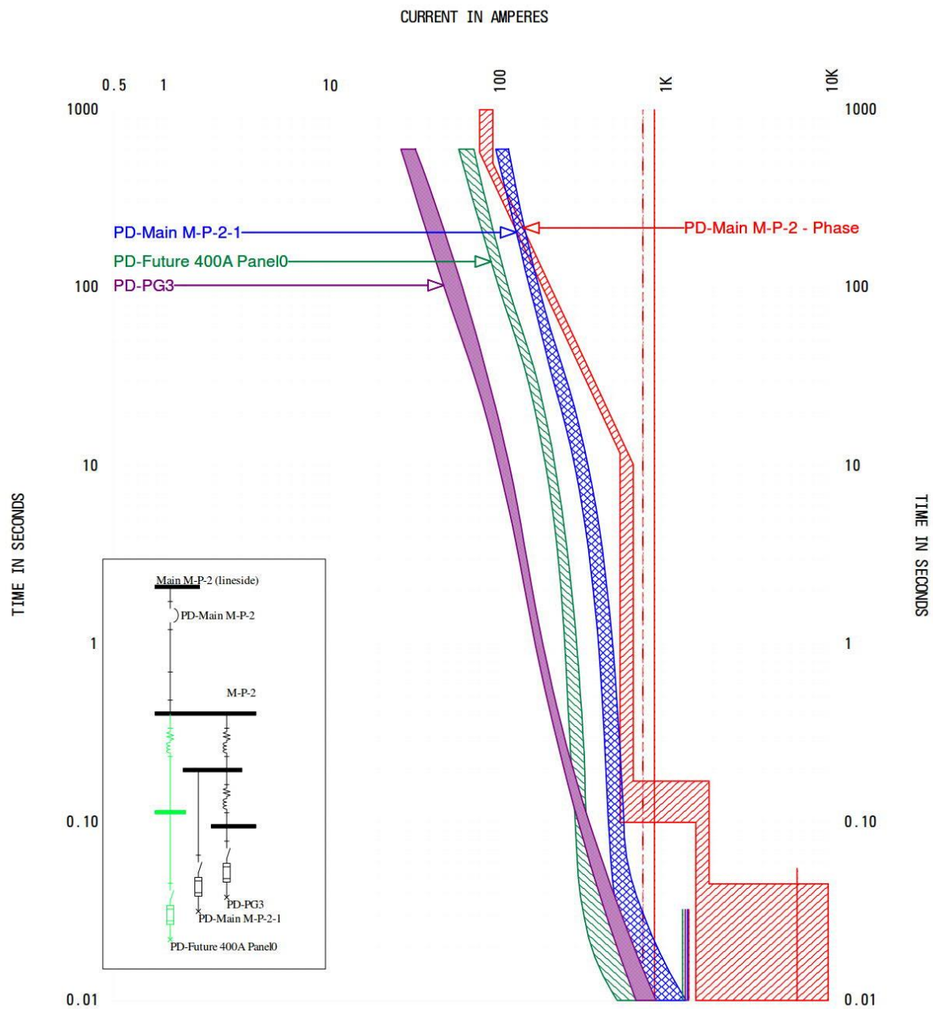
HAMP-South Revised
PURDUE UNIVERSITY, IN

Reference Voltage: 240
Current Scale x 10
Applied Engineering Services

13



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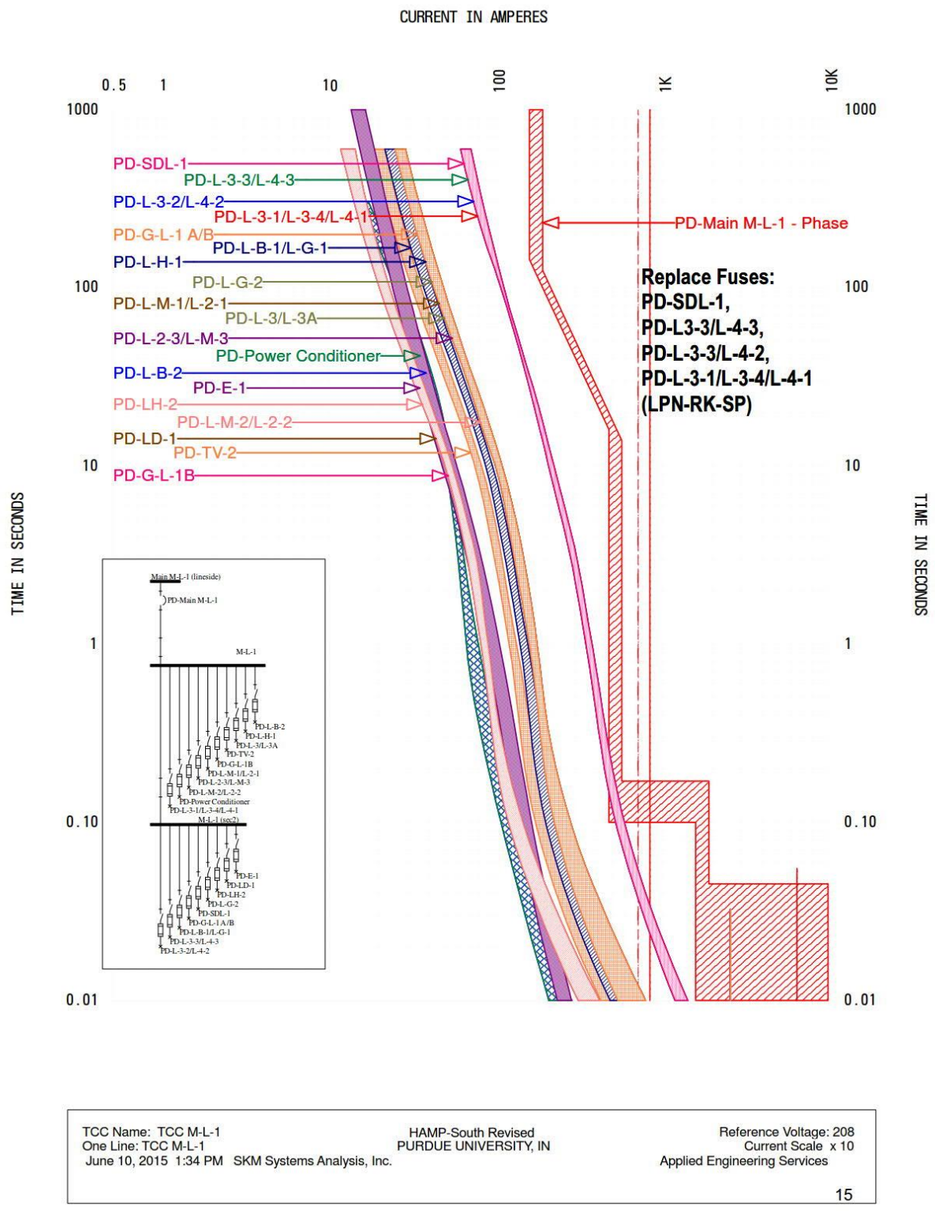
HAMP-South Revised
PURDUE UNIVERSITY, IN

Reference Voltage: 480
Current Scale x 10
Applied Engineering Services

14


CURRENT IN AMPERES
0.5
1
10
100
1K
10K
1000
100
10
1
0.10
0.01
TIME IN SECONDS
PD-SDL-1
PD-L-3-3/L-4-3
PD-L-3-2/L-4-2
PD-L-3-1/L-3-4/L-4-1
PD-G-L-1 A/B
PD-L-B-1/L-G-1
PD-L-H-1
PD-L-G-2
PD-L-M-1/L-2-1
PD-L-3/L-3A
PD-L-2-3/L-M-3
PD-Power Conditioner
PD-L-B-2
PD-E-1
PD-LH-2
PD-L-M-2/L-2-2
PD-LD-1
PD-TV-2
PD-G-L-1B
PD-Main M-L-1 - Phase
Replace Fuses:
PD-SDL-1,
PD-L3-3/L-4-3,
PD-L-3-3/L-4-2,
PD-L-3-1/L-3-4/L-4-1
(LPN-RK-SP)
Main M-L-1 (lineside)
PD-Main M-L-1
M-L-1
M-L-1 (480V)
TCC Name: TCC M-L-1
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June 10, 2015 1:34 PM SKM Systems Analysis, Inc.
HAMP-South Revised
PURDUE UNIVERSITY, IN
Reference Voltage: 208
Current Scale x 10
Applied Engineering Services
15

CONCEPTUAL COST OF CONSTRUCTION

As indicated in the following pages, the opinions of probable costs for the ME-RAIL-HAMP study include renovations to each building, as shown on the conceptual design plans included in other parts of this study. Our team, including CCS International, has reviewed the estimated construction costs based on parametric estimates for space type and extent of renovation at the conceptual level and provided the following estimates and information for each building. Escalation factors are identified in the estimates, as indicated on each sheet. Costs associated with general requirements and standard overhead and profit are based upon a conventional unified design bid build construction method of delivery.

In addition to the following, please see the "Clarifications" page of the estimate for each building.

DESIGN AND CONSTRUCTION CONTINGENCY

Careful review of the conceptual estimate reveals that we are showing a design contingency. It is normal to hold a relatively high level of design contingency at this stage of design due to the potential changes that can occur over the next few major design milestones. As indicated on the attached, the design contingency is shown at 15-percent. As we move toward a more solid solution and refined level of design detail, this contingency will be reduced. As indicated, the construction contingency is currently calculated at 10-percent.

PHASING

As indicated in previous sections of the study, as well as the following breakdown information, this estimate is based upon a multi-phased renovation of each building (except RAIL), along with a portion of the building remaining operational during construction. These factors add cost and complexity to the project.

FURNITURE, FURNISHINGS AND EQUIPMENT

The estimate does not include furniture, furnishings and equipment. Benches, computers, monitors, projectors, A/V systems, artwork, display cases, testing and lab equipment, etc. are excluded from the following estimates. The cost of built-in equipment that is normal and standard to construction projects such as white marker boards, projection screens, and built-in casework is included. The extensive list of owner supplied FF&E, including office furniture and private office, glass-walled furniture systems would need to be provided by Purdue University.

ALTERNATES AND OTHER ITEMS

Due to the scale of the projects, Purdue has asked that specific portions of the work be identified and broken out separately for consideration. These break-out costs are provided in order to align possible funding sources available to the University for specific items or areas within each overall project. Where appropriate, these items are identified on each estimate for each building and as follows.

ME SUMMARY

The summary cost estimate for ME is included on the following pages, including additional backup information for the Phase 1 and Phase 2 projects. As requested, following are break-out costs associated with the following items, including costs for Design Contingency, GC + OH&P, and Construction Contingency:

	Estimated Costs
Classroom Upgrades	$1,710,687
ADA Upgrades, Including Elevator	$174,621
Restroom Upgrades	$409,346

RAIL SUMMARY

The summary cost estimate for ME is included on the following pages, including additional backup information for the project.

HAMP SUMMARY

The summary cost estimate for ME is included on the following pages, including additional backup information for the Phase 1 and Phase 2 projects. As requested, following are break-out costs associated with the following items, including costs for Design Contingency, GC + OH&P, and Construction Contingency:

	Estimated Costs
Classroom Upgrades	$2,630,955
ADA Upgrades, Including Elevator	$276,736
Restroom Upgrades	$459,434
Basement Level Rsch Lab Upgrades *(Opt A and Opt B on Proposed Plans)*	$1,369,748

ME BUILDING COSTS PARAMETRIC ESTIMATE

ME Building – Summary Information

Purdue University; West Lafayette, IN — **Parametric Estimate**

Mechanical Engineering Bldg **CCS Project No.** 15.044 **Date:** 6/29/2015

Department Categories	GSF	Phase Number	Cost / SF	Base Construction Cost - 2015	Environmental removal costs - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed at 4% per year compounded	Escalated Costs
Major Renovation	43,974	One	$ 177.36	$ 7,799,070	$ -	$ -	$ -	$ 7,799,070	103.00%	$ 8,033,042
Demolition	43,606	One	$ 7.20	$ 313,963	$ -	$ -	$ -	$ 313,963	103.00%	$ 323,382
Subtotal				$ 8,113,033				$ 8,113,033		$ 8,356,424
Design Contingency	15.00%			$ 1,216,955				$ 1,216,955		$ 1,216,955
GC + OH&P	12.00%			$ 1,119,599				$ 1,119,599		$ 1,119,599
Construction Contingency	10.00%			$ 1,044,959				$ 1,044,959		$ 1,044,959
								$ -		$ -
Total Construction - Phase 1	43,974			$ 11,494,545				$ 11,494,545		**$ 11,737,936**
Major Renovation	11,138	Two	$ 213.24	$ 2,375,070	$ -	$ -	$ -	$ 2,375,070	105.50%	$ 2,505,699
Demolition	11,138	Two	$ 7.20	$ 80,194	$ -	$ -	$ -	$ 80,194	105.50%	$ 84,604
Infrastructure		Two		$ -	$ -	$ -	$ -	$ -	105.50%	$ -
Subtotal				$ 2,455,264				$ 2,455,264		$ 2,590,303
Design Contingency	15.00%			$ 368,290				$ 368,290		$ 368,290
GC + OH&P	12.00%			$ 338,826				$ 338,826		$ 338,826
Construction Contingency	10.00%			$ 316,238				$ 316,238		$ 316,238
Total Construction - Phase 2	11,138			$ 3,478,617				$ 3,478,617		**$ 3,613,657**
#										
#										
Total Construction	55,112			$ 14,973,163				$ 14,973,163		**$ 15,351,593**

ADA Upgrades	Ramps	50	$ 65.00	$ 3,250	
SUBTOTAL				$ 3,250	
Design Contingency	15.00%			$ 488	
GC + OH&P	12.00%			$ 449	
Construction Contingency	10.00%			$ 419	
TOTAL COST					$ 4,605
Restroom Upgrades	Demolition	1,856	$ 7.20	$ 13,363	
	Existing Restroom	888	$ 125.00	$ 111,000	
	New Restroom	968	$ 170.00	$ 164,560	
SUBTOTAL				$ 288,923	
Design Contingency	15.00%			$ 43,338	
GC + OH&P	12.00%			$ 39,871	
Construction Contingency	10.00%			$ 37,213	
TOTAL COST					$ 409,346

ME Building – Renovation Calculations

Purdue University; West Lafayette, IN

Mechanical Engineering Bldg **CCS Project No.** 15.044

Renovation Cost Outline Matrix

Date: 6/29/2015

Categories	Space Types	Area - GSF	Phase Number	Cost / SF - Cosmetic - Paint and flooring $/ SF	Cost / SF - Minor Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Cost / SF - Major Renovation - Total Reconfiguration - all new localized construction and MEP	Base Construction Cost - 2015	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs	Subtotal
ADA Upgrades	Elevator (4 stop)	1	One			$ 120,000.00	$ 120,000	$ 120,000.00	3.00%	$ 123,600	
	Ramps	50	One	$ 15.00	$ 50.00	$ 65.00	$ 3,250	$ 3,250.00	3.00%	$ 3,348	
	Existing Restroom	888	One	$ 15.00	$ 60.00	$ 125.00	$ 111,000	$ 111,000.00	3.00%	$ 114,330	
	New Restroom	968	One	$ 15.00	$ 55.00	$ 170.00	$ 164,560	$ 164,560.00	3.00%	$ 169,497	
											$ 410,774
Classroom	SMAS	5,138	One	$ 15.00	$ -	$ 235.00	$ 1,207,430	$ 1,207,430.00	3.00%	$ 1,243,653	
	Collaboration	946	One	$ 15.00	$ 60.00	$ 175.00	$ 165,550	$ 165,550.00	3.00%	$ 170,517	
	Tutoring Room	798	One	$ 15.00	$ 60.00	$ 235.00	$ 187,530	$ 187,530.00	3.00%	$ 193,156	
											$ 1,607,325
Offices	Office Suite	13,709	One	$ 15.00	$ 50.00	$ 150.00	$ 2,056,350	$ 2,056,350.00	3.00%	$ 2,118,041	
	Private	8,594	One	$ 15.00	$ 50.00	$ 175.00	$ 1,503,950	$ 1,503,950.00	3.00%	$ 1,549,069	
	Conf / Seminar	1,534	One	$ 15.00	$ 50.00	$ 175.00	$ 268,450	$ 268,450.00	3.00%	$ 276,504	
											$ 3,943,613
Laboratory	Teaching	3,737	One	$ 15.00	$ 75.00	$ 215.00	$ 803,455	$ 803,455.00	3.00%	$ 827,559	
	Computer	963	One	$ 15.00	$ 55.00	$ 235.00	$ 226,305	$ 226,305.00	3.00%	$ 233,094	
											$ 1,060,653
Common	Corridors	5,315	One	$ 15.00	$ 50.00	$ 135.00	$ 717,525	$ 717,525.00	3.00%	$ 739,051	
	Work Café	415	One	$ 15.00	$ 50.00	$ 145.00	$ 60,175	$ 60,175.00	3.00%	$ 61,980	
											$ 801,031
Utility	Electric	43	One	$ 15.00	$ 75.00	$ 145.00	$ 6,235	$ 6,235.00	3.00%	$ 6,422	
	Custodian	196	One	$ 15.00	$ 75.00	$ 105.00	$ 20,580	$ 20,580.00	3.00%	$ 21,197	
	IDF	311	One	$ 15.00	$ 75.00	$ 185.00	$ 57,535	$ 57,535.00	3.00%	$ 59,261	
											$ 86,881
Stair Tower	2 Flights	1	One			$ 47,000.00	$ 47,000	$ 47,000.00	3.00%	$ 48,410	
	Enclosure	364	One			$ 185.00	$ 67,340	$ 67,340.00	3.00%	$ 69,360	
											$ 117,770
Window Repacement	68" x 96" fixed	2	One			$ 675.00	$ 1,350	$ 1,350.00	3.00%	$ 1,391	
											$ 1,391
New North Entrance		1	One			$ 3,500.00	$ 3,500	$ 3,500.00	3.00%	$ 3,605	
											$ 3,605
MAJOR RENOVATION - Phase 1		**43,974**					**$ 7,799,070**	**$ 7,799,070**		**$ 8,033,042**	
Laboratory	Teaching	1,506	Two	$ 15.00	$ 75.00	$ 215.00	$ 323,790	$ 323,790.00	3.00%	$ 333,504	
	Computer	1,200	Two	$ 15.00	$ 55.00	$ 235.00	$ 282,000	$ 282,000.00	3.00%	$ 290,460	
	Electronics Shop	2,521	Two	$ 15.00	$ 55.00	$ 245.00	$ 617,645	$ 617,645.00	3.00%	$ 636,174	
	Research	3,215	Two	$ 15.00	$ 50.00	$ 245.00	$ 787,675	$ 787,675.00	3.00%	$ 811,305	
											$ 2,071,443
Common	Corridors	2,696	Two	$ 15.00	$ 50.00	$ 135.00	$ 363,960	$ 363,960.00	3.00%	$ 374,879	
											$ 374,879
MAJOR RENOVATION - Phase 2		**11,138**					**$ 2,375,070**	**$ 2,375,070**		**$ 2,446,322**	
Total Renovation Cost		55,112					$ 10,174,140	$ 10,174,140		**$ 10,479,364**	

ME Building – Demolition Calculations

Purdue University; West Lafayette, IN — **Parametric Estimate** — **Shell and Infrastructure**

Mechanical Engineering Bldg — **CCS Project No.** 15.044 — **Date:** 6/29/2015

Building Division	Scope of Work - Description	Area - GSF or CF	Phase Number	Cost / SF - Light Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Base Construction Cost - 2015	Added cost - Potential Impact issue	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs
Demolition		654,090	One	$ 0.48	$ 313,963.20	$ -	$ 313,963.20	3.00%	$ 323,382
Demolition		167,070	Two	$ 0.48	$ 80,193.60	$ -	$ 80,193.60	3.00%	$ 82,599
Total Demolition		821,160	CF		$ 394,157		**$ 394,157**		**$ 405,982**

ME Building – Clarifications

Ln #	Clarifications
1	FF&E and soft costs are excluded from estime. To be provided by Owner.

Ln #	Cost category	Scope of Work	Cost / SQFT
1	ADA Upgrade - Elevator	Lump sum allowance for major replacement & upgrade.	$120,000.00
2	ADA Upgrade - Ramps	Concrete ramps with steel handrails	$65.00
3	ADA Upgrade - Existing Restrooms	New HD partitions; new HD doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($12/sf) & Electrical work ($12/sf).	$125.00
4	ADA Upgrade - New Restrooms	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($65/sf) & electrical work ($18/sf).	$170.00
5	Classroom	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($65/sf) & electrical work ($22/sf).	$235 & $175
6	Office - Cosmetic	Minor floor finishes; wall patching and repair; minor finish upgrade; selective demo.	$30.00
7	Office - Minor renovation	Limited new partitions; door repair; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; selective demo; minor HVAC & electrical repairs.	$97.00
8	Office - Major Renovation	New partitions; new door; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; demolition. Fire protection, plumbing & HVAC ($65/sf) & electrical ($17/sf) work.	$150.00
9	Common / Circulation	New partitions; new doors; new floor finishes; wall patching & repair; new acoustical ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($65/sf) & Electrical work ($15/sf).	$135.00
10	Laboratory	New partitions; new doors; new highend floor finishes; wall patching & repair; new ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($65/sf) & Electrical work ($20/sf).	$215, $230 & $240
11	Utility - Electric	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; Mechanical ($65/sf) & electrical work ($11/sf).	$145.00
12	Utility - Custodian	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; demolition; fire protection & electrical work ($10/sf).	$105.00
13	Utility - IDF	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; Mechanical ($50/sf) & electrical work ($10/sf).	$185.00

Notes: Excludes general conditions, contingency, escalation and logistics factors.
Includes major mechanical equipment replacement work in major & minor renovation categories.

RAIL BUILDING COSTS PARAMETRIC ESTIMATE

RAIL Building – Summary Information

Purdue University; West Lafayette, IN — **Parametric Estimate**

American Railway Bldg **CCS Project No.** 15.044 **Date:** 6/29/2015

Department Categories	GSF	Phase Number	Cost / SF	Base Construction Cost - 2015	Premium Costs to Base Construction Costs: Environmental removal costs - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed at 4% per year compounded	Escalated Costs
Major Renovation	2,128	One	$ 200.28	$ 426,198	$ -	$ -	$ -	$ -	$ 426,198	103.00%	$ 438,984
Basement		One		$ -	$ -	$ -	$ -	$ -	$ -	103.00%	$ -
Demolition	30,105	One	$ 0.58	$ 14,450	$ -	$ -	$ -	$ -	$ 14,450	103.00%	$ 14,884
Subtotal	2,128			$ 440,648					$ 440,648		$ 453,868
Design Contingency	15.00%			$ 66,097					$ 66,097		$ 68,080
GC + OH&P	12.00%			$ 60,809					$ 60,809		$ 62,634
Construction Contingency	10.00%			$ 56,756					$ 56,756		$ 58,458
Total Construction	2,128			$ 624,311					$ 624,311		**$ 643,040**

RAIL Building – Renovation Information

Purdue University; West Lafayette, IN

American Railway Bldg **CCS Project No.** 15.044

Renovation Cost Outline Matrix

Date: 6/29/2015

Categories	Space Types	Area - GSF	Phase Number	Cost / SF - Cosmetic - Paint and flooring $/ SF	Cost / SF - Minor Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Cost / SF - Major Renovation - Total Reconfiguration - all new localized construction and MEP	Base Construction Cost - 2015	Environmental removal costs - Lump Sum	Added cost - Potential Impact issue: ACM - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs
									um Costs to Base Construction (				
ADA Upgrades	Ramps	119	One			$ 72.00	$ 8,568	$ -	$ -	$ -	$ 8,568.00	3.00%	$ 8,825
	New ADA Entrance	1	One	$ 15.00	$ 50.00	$ 5,925.00	$ 5,925	$ -	$ -	$ -	$ 5,925.00	3.00%	$ 6,103
	New flight of stairs	1	One			$ 23,500.00	$ 23,500	$ -	$ -	$ -	$ 23,500.00	3.00%	$ 24,205
	New Restroom		One	$ 15.00	$ 55.00	$ 175.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
Offices	Office Suite	1,719	One	$ 15.00	$ 50.00	$ 195.00	$ 335,205	$ -	$ -	$ -	$ 335,205.00	3.00%	$ 345,261
	Private		One	$ 15.00	$ 50.00	$ 155.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
	Conf / Seminar		One	$ 15.00	$ 50.00	$ 155.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
Laboratory	Teaching		One	$ 15.00	$ 75.00	$ 220.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
	Computer		One	$ 15.00	$ 55.00	$ 235.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
Common	Corridors	158	One	$ 15.00	$ 50.00	$ 175.00	$ 27,650	$ -	$ -	$ -	$ 27,650.00	3.00%	$ 28,480
	Work Café	130	One	$ 15.00	$ 50.00	$ 195.00	$ 25,350	$ -	$ -	$ -	$ 25,350.00	3.00%	$ 26,111
Utility	Electric		One	$ 15.00	$ 75.00	$ 235.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
	Custodian		One	$ 15.00	$ 75.00	$ 135.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
	IDF		One	$ 15.00	$ 75.00	$ 215.00	$ -	$ -	$ -	$ -	$ -	3.00%	$ -
MAJOR RENOVATION - Phase 1		**2,128**					**$ 426,198**				**$ 426,198**		**$ 438,984**
Total Renovation Cost		2,128					$ 426,198				$ 426,198		**$ 438,984**

RAIL Building – Demolition Information

Purdue University; West Lafayette, IN — **Parametric Estimate** — **Shell and Infrastructure**

American Railway Bldg — **CCS Project No.** 15.044 — **Date:** 6/29/2015

						mium Costs to Base Construction Co				
Building Division	Scope of Work - Description	Area - GSF or CF	Phase Number	Cost / SF - Light Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Base Construction Cost - 2015	Added cost - Potential Impact issue	Added cost - Potential Impact issue	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs
Demolition		30,105	One	$ 0.48	$ 14,450.40	$ -	$ -	$ 14,450.40	3.00%	$ 14,884
Demolition				$ 0.48	$ -	$ -	$ -	$ -		$ -
Total Demolition		30,105	CF		$ 14,450			**$ 14,450**		**$ 14,884**

RAIL Building – Additional Shell Information

Purdue University; West Lafayette, IN — **Parametric Estimate** — **Shell and Infrastructure**

American Railway Bldg — **CCS Project No.** 15.044 — **Date:** 6/29/2015

						mium Costs to Base Construction Co				
Building Division	Scope of Work - Description	Area - GSF or CF	Phase Number	Cost / SF - Light Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Base Construction Cost - 2015	Added cost - Potential Impact issue	Added cost - Potential Impact issue	Total Construction Costs 2015	Escalation to Midpoint of construction	Escalated Costs
Foundations		0		$ 100.00	$ -	$ -	$ -	$ -	0.00%	$ -
Slab on Grade		0		$ 75.00	$ -	$ -	$ -	$ -	0.00%	$ -
Basement		0		$ 65.00	$ -	$ -	$ -	$ -	0.00%	$ -
Floor Construction		0		$ 55.00	$ -	$ -	$ -	$ -	0.00%	$ -
Roof Construction		0		$ 60.00	$ -	$ -	$ -	$ -	0.00%	$ -
Stairs		0		$ 50.00	$ -	$ -	$ -	$ -	0.00%	$ -
Exterior Envelope		0		$ 65.00	$ -	$ -	$ -	$ -	0.00%	$ -
Exterior Windows		450	One	$ 50.00	$ 22,500.00	$ -	$ -	$ 22,500.00	3.00%	$ 23,175
Roofing		0		$ 75.00	$ -	$ -	$ -	$ -	0.00%	$ -
Conveying Systems		0		$ 75.00	$ -	$ -	$ -	$ -	0.00%	$ -
Plumbing		0		$ 50.00	$ -	$ -	$ -	$ -	0.00%	$ -
HVAC		0		$ -	$ -	$ -	$ -	$ -	0.00%	$ -
Fire Protection		0		$ 55.00	$ -	$ -	$ -	$ -	0.00%	$ -
Electrical		0		$ 60.00	$ -	$ -	$ -	$ -	0.00%	$ -
Environmental	Asbestos Abatement Allowance (By Owner)	0		$ -	$ -	$ -	$ -	$ -	0.00%	$ -
Landscaping		0		$ 50.00	$ -	$ -	$ -	$ -	0.00%	$ -
Utility relocation		0		$ -	$ -	$ -	$ -	$ -	0.00%	$ -
Total Infrastructure		450	SF		$ 22,500			**$ 22,500**		**$ 23,175**

RAIL Building – Clarifications

Ln #	Clarifications
1	FF&E and soft costs are excluded from estimate. Owner to provide.

Ln #	Cost category	Scope of Work	Cost / SQFT
1	ADA Upgrade - Elevator	Lump sum allowance.	$50,000.00
2	ADA Upgrade - Ramps	Concrete ramps with steel handrails	$65.00
3	ADA Upgrade - Existing Restrooms	New HD partitions; new HD doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($18/sf) & Electrical work ($12/sf).	$125.00
4	ADA Upgrade - New Restrooms	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($65/sf) & electrical work ($18/sf).	$180.00
5	Classroom	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($65/sf) & electrical work ($22/sf).	$235 & $175
6	Office - Cosmetic	Minor floor finishes; wall patching and repair; minor finish upgrade; selective demo.	$30.00
7	Office - Minor renovation	Limited new partitions; door repair; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; Mechanical ($60/sf) & electrical repairs ($17/sf).	$97.00
8	Office - Major Renovation	New partitions; new door; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; demolition. New Fire Protection, some Plumbing & Complete HVAC ($90/sf) & electrical ($17/sf) work.	$195.00
9	Common / Circulation	New partitions; new doors; new floor finishes; wall patching & repair; new acoustical ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($80/sf) & Electrical work ($15/sf).	$175.00
10	Laboratory	New partitions; new doors; new highend floor finishes; wall patching & repair; new ceiling; finish upgrades; misc. specialties; demolition; fire protection; special HVAC & electrical work.	$220, $235 & $245
11	Utility - Electric	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; Mechanical ($96/sf) & electrical work ($10/sf).	$145.00
12	Utility - Custodian	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; demolition; fire protection & electrical work.	$105.00
13	Utility - IDF	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; Mechanical ($50/sf) & electrical work ($10/sf).	$185.00

Notes: Excludes general conditions, contingency, escalation and logistics factors.
Includes major mechanical equipment replacement work in major & minor renovation categories.

HAMP BUILDING COSTS PARAMETRIC ESTIMATE

HAMP Building – Summary Information

Purdue University; West Lafayette, IN — **Parametric Estimate**

Hampton Hall — **CCS Project No.** 15.044 — **Date:** 6/29/2015

Department Categories	GSF	Phase Number	Cost / SF	Base Construction Cost - 2015	Premium Costs to Base Construction Costs: Environmental removal costs - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed at 4% per year compounded	Escalated Costs
Major Renovation	73,090	One	$ 178.62	$ 13,055,650	$ -	$ -	$ -	$ -	$ 13,055,650	103.00%	$ 13,447,320
Demolition	70,720	One	$ 7.20	$ 509,184	$ -	$ -	$ -	$ -	$ 509,184	103.00%	$ 524,460
Subtotal				$ 13,564,834					$ 13,564,834		$ 13,971,779
Design Contingency	15.00%			$ 2,034,725					$ 2,034,725		$ 2,034,725
GC + OH&P	12.00%			$ 1,871,947					$ 1,871,947		$ 1,871,947
Construction Contingency	10.00%			$ 1,747,151					$ 1,747,151		$ 1,747,151
Total Construction - Phase 1	70,720			$ 19,218,657					$ 19,218,657		**$ 19,625,602**
Major Renovation	21,201	Two	$ 181.76	$ 3,853,435	$ -	$ -	$ -	$ -	$ 3,853,435	105.50%	$ 4,065,374
Demolition	21,201	Two	$ 7.20	$ 152,647	$ -	$ -	$ -	$ -	$ 152,647	105.50%	$ 161,043
Subtotal				$ 4,006,082					$ 4,006,082		$ 4,226,417
Design Contingency	15.00%			$ 600,912					$ 600,912		$ 600,912
GC + OH&P	12.00%			$ 552,839					$ 552,839		$ 552,839
Construction Contingency	10.00%			$ 515,983					$ 515,983		$ 515,983
Total Construction - Phase 2	21,201			$ 5,675,817					$ 5,675,817		**$ 5,896,152**
Total Construction	91,921			$ 24,894,474					$ 24,894,474		**$ 25,521,754**

ADA Upgrades	Ramps	346	$ 65.00	$ 22,490.00	
SUBTOTAL				$ 22,490.00	
Design Contingency	15.00%			$ 3,373.50	
GC + OH&P	12.00%			$ 3,103.62	
Construction Contingency	10.00%			$ 2,896.71	
TOTAL COST					$ 31,864
Restroom Upgrades	Demolition	1830	$ 7.20	$ 13,176.00	
	New Restroom	1830	$ 170.00	$ 311,100.00	
SUBTOTAL				$ 324,276.00	
Design Contingency	15.00%			$ 48,641.40	
GC + OH&P	12.00%			$ 44,750.09	
Construction Contingency	10.00%			$ 41,766.75	
TOTAL COST					$ 459,434

HAMP Building – Renovation Information

Purdue University; West Lafayette, IN

Renovation Cost Outline Matrix

Hampton Hall **CCS Project No.** 15.044 **Date:** 6/29/2015

Categories	Space Types	Area - GSF	Phase Number	Cost / SF - Cosmetic - Paint and flooring $/ SF	Cost / SF - Minor Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Cost / SF - Major Renovation - Total Reconfiguration - all new localized construction and MEP	Base Construction Cost - 2015	Environmental removal costs - Lump Sum	Added cost - Potential Impact issue: ACM - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs	Subtotal
									um Costs to Base Construction (					
ADA Upgrades	Elevators	1	One			$ 168,050.00	$ 168,050	$ -	$ -	$ -	$ 168,050.00	3.00%	$ 173,092	
	Ramp w/ Handrail	346	Optional	$ 15.00	$ 50.00	$ 65.00	$ 22,490	$ -	$ -	$ -	$ 22,490.00	3.00%	$ 23,165	
	ADA Entrance	1	Optional	$ 15.00	$ 60.00	$ 4,785.00	$ 4,785	$ -	$ -	$ -	$ 4,785.00	3.00%	$ 4,929	
	New Restroom	1830	One	$ 15.00	$ 55.00	$ 170.00	$ 311,100	$ -	$ -	$ -	$ 311,100.00	3.00%	$ 320,433	
														$ 521,618
Classroom	SMAS	7,902	One	$ 15.00	$ -	$ 235.00	$ 1,856,970	$ -	$ -	$ -	$ 1,856,970.00	3.00%	$ 1,912,679	
														$ 1,912,679
Offices	Open Office Suite	24,122	One	$ 15.00	$ 50.00	$ 150.00	$ 3,618,300	$ -	$ -	$ -	$ 3,618,300.00	3.00%	$ 3,726,849	
	Private	7,545	One	$ 15.00	$ 50.00	$ 150.00	$ 1,131,750	$ -	$ -	$ -	$ 1,131,750.00	3.00%	$ 1,165,703	
	Focus	1,064	One	$ 15.00	$ 50.00	$ 150.00	$ 159,600	$ -	$ -	$ -	$ 159,600.00	3.00%	$ 164,388	
	Work Room	339	One	$ 15.00	$ 50.00	$ 150.00	$ 50,850				$ 50,850.00	103.00%	$ 103,226	
	Conf / Meeting	4,402	One	$ 15.00	$ 50.00	$ 150.00	$ 660,300	$ -	$ -	$ -	$ 660,300.00	3.00%	$ 680,109	
														$ 5,840,274
Laboratory	Teaching & CE	6,069	One	$ 15.00	$ 75.00	$ 220.00	$ 1,335,180	$ -	$ -	$ -	$ 1,335,180.00	3.00%	$ 1,375,235	
	CE Research	471	One	$ 15.00	$ 75.00	$ 235.00	$ 110,685	$ -	$ -	$ -	$ 110,685.00	3.00%	$ 114,006	
	CE Rrch (Opt a)	1,596	One	$ 15.00	$ 75.00	$ 235.00	$ 375,060	$ -	$ -	$ -	$ 375,060.00	3.00%	$ 386,312	
	CE Rrch (Opt b)	2,518	One	$ 15.00	$ 75.00	$ 235.00	$ 591,730	$ -	$ -	$ -	$ 591,730.00	3.00%	$ 609,482	
	Basem't Research	1,985	One	$ 15.00	$ 75.00	$ 220.00	$ 436,700	$ -	$ -	$ -	$ 436,700.00	3.00%	$ 449,801	
	Teaching Prep	1,389	One	$ 15.00	$ 75.00	$ 220.00	$ 305,580	$ -	$ -	$ -	$ 305,580.00	3.00%	$ 314,747	
	Computer	917	One	$ 15.00	$ 55.00	$ 235.00	$ 215,495	$ -	$ -	$ -	$ 215,495.00	3.00%	$ 221,960	
														$ 3,471,543
Common	Corridors	7,154	One	$ 15.00	$ 50.00	$ 135.00	$ 965,790	$ -	$ -	$ -	$ 965,790.00	3.00%	$ 994,764	
	Work Café	970	One	$ 15.00	$ 50.00	$ 145.00	$ 140,650	$ -	$ -	$ -	$ 140,650.00	3.00%	$ 144,870	
														$ 1,139,633
Utility	Mechanical	202	One	$ 15.00	$ 75.00	$ 145.00	$ 29,290	$ -	$ -	$ -	$ 29,290.00	3.00%	$ 30,169	
	Custodian	245	One	$ 15.00	$ 75.00	$ 105.00	$ 25,725	$ -	$ -	$ -	$ 25,725.00	3.00%	$ 26,497	
														$ 56,665
First Floor	Entry Space	2,000	One	$ 15.00	$ 75.00	$ 265.00	$ 530,000	$ -	$ -	$ -	$ 530,000.00	3.00%	$ 545,900	
														$ 545,900
Windows - each	2'-4" x 6'-7" Casem't	22	One	$ 15.00	$ 75.00	$ 435.00	$ 9,570	$ -	$ -	$ -	$ 9,570.00	3.00%	$ 9,857	
														$ 9,857
MAJOR RENOVATION - Phase 1		**73,090**					**$ 13,055,650**				**$ 13,055,650**		**$ 13,498,170**	
Classroom	SMAS	1,352	Two	$ 15.00	$ -	$ 235.00	$ 317,720	$ -	$ -	$ -	$ 317,720.00	3.00%	$ 327,252	
	Mediascape	406	Two	$ 15.00	$ -	$ 235.00	$ 95,410	$ -	$ -	$ -	$ 95,410.00	3.00%	$ 98,272	
	Collaboration	997	Two	$ 15.00	$ 60.00	$ 175.00	$ 174,475	$ -	$ -	$ -	$ 174,475.00	3.00%	$ 179,709	
														$ 605,233
Offices	Open Office Suite	3,245	Two	$ 15.00	$ 50.00	$ 155.00	$ 502,975	$ -	$ -	$ -	$ 502,975.00	3.00%	$ 518,064	
	Private	2,892	Two	$ 15.00	$ 50.00	$ 195.00	$ 563,940	$ -	$ -	$ -	$ 563,940.00	3.00%	$ 580,858	
	Office Suite (Opt a)	5,469	Two	$ 15.00	$ 50.00	$ 155.00	$ 847,695	$ -	$ -	$ -	$ 847,695.00	3.00%	$ 873,126	
	Private (Opt a)	3,240	Two	$ 15.00	$ 50.00	$ 195.00	$ 631,800	$ -	$ -	$ -	$ 631,800.00	3.00%	$ 650,754	
	Conf / Meeting	961	Two	$ 15.00	$ 50.00	$ 155.00	$ 148,955	$ -	$ -	$ -	$ 148,955.00	3.00%	$ 153,424	
														$ 2,776,226
Laboratory	Research	384	Two	$ 15.00	$ 50.00	$ 245.00	$ 94,080	$ -	$ -	$ -	$ 94,080.00	3.00%	$ 96,902	
	CE Research	742	Two	$ 15.00	$ 50.00	$ 245.00	$ 181,790	$ -	$ -	$ -	$ 181,790.00	3.00%	$ 187,244	
	Simulation	967	Two	$ 15.00	$ 50.00	$ 245.00	$ 236,915	$ -	$ -	$ -	$ 236,915.00	3.00%	$ 244,022	
														$ 528,169
Utility	Custodian	88	Two	$ 15.00	$ 75.00	$ 135.00	$ 11,880	$ -	$ -	$ -	$ 11,880.00	3.00%	$ 12,236	
														$ 12,236
Common	Corridors	458	Two	$ 15.00	$ 50.00	$ 100.00	$ 45,800	$ -	$ -	$ -	$ 45,800.00	3.00%	$ 47,174	
														$ 47,174
MAJOR RENOVATION - Phase 2		**21,201**					**$ 3,853,435**				**$ 3,853,435**		**$ 3,969,038**	
Total Renovation Cost		94,291					$ 16,909,085				$ 16,909,085		**$ 17,467,208**	

HAMP Building – Demolition Information

Purdue University; West Lafayette, IN — **Parametric Estimate** — **Shell and Infrastructure**

Hampton Hall — **CCS Project No.** 15.044 — **Date:** 6/29/2015

						Premium Costs to Base Construction Costs					
Building Division	Scope of Work - Description	Area - GSF or CF	Phase Number	Cost / SF - Light Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Base Construction Cost - 2015	Added cost - Potential Impact issue	Added cost - Potential Impact issue	Added cost - Potential Impact issue	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs
Demolition	includes all walls, ceilings and floor finishes	1,060,800	One	$ 0.48	$ 509,184.00	$ -	$ -	$ -	$ 509,184.00	3.00%	$ 524,460
Demolition	includes all walls, ceilings and floor finishes	318,015	Two	$ 0.48	$ 152,647.20	$ -	$ -	$ -	$ 152,647.20	5.50%	$ 161,043
Total Demolition		1,378,815	CF		$ 661,831				**$ 661,831**		**$ 685,502**
		91,921									

HAMP Building – Clarifications

Ln #	Clarifications
1	FF& E and project soft costs are excluded from estimate. To be provided by Owner.

Ln #	Cost category	Scope of Work	Cost / SQFT
1	ADA Upgrade - Elevator	Lump sum allowance for new elevator complete.	$168,050.00
2	ADA Upgrade - Ramps	Concrete ramps with steel handrails	$65.00
3	ADA Upgrade - Existing Restrooms	New HD partitions; new HD doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($12/sf) & Electrical work ($12/sf).	$125.00
4	ADA Upgrade - New Restrooms	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($62/sf) & Electrical work ($18/sf).	$170.00
5	Classroom	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($62/sf) & Electrical work ($22/sf).	$235 & $175
6	Office - Cosmetic	Minor floor finishes; wall patching and repair; minor finish upgrade; selective demo.	$30.00
7	Office - Minor renovation	Limited new partitions; door repair; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; selective demo; minor HVAC & electrical repairs.	$97.00
8	Office - Major Renovation	New partitions; new door; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; demolition. Mechanical ($62/sf) & Electrical work ($17/sf).	$150.00
9	Common / Circulation	New partitions; new doors; new floor finishes; wall patching & repair; new acoustical ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($62/sf) & Electrical work ($15/sf).	$135.00
10	Laboratory	New partitions; new doors; new highend floor finishes; wall patching & repair; new ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($62/sf) & Electrical work ($20/sf).	$215, $230 & $240
11	Utility - Electric / Mechanical	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; Mechanical ($62/sf) & electrical work ($11/sf).	$145.00
12	Utility - Custodian	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; demolition; fire protection & electrical work ($10/sf).	$105.00
13	Utility - IDF	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; Mechanical ($50/sf) & electrical work ($10/sf).	$185.00

Notes: Excludes general conditions, contingency, escalation and logistics factors.
Includes major mechanical equipment replacement work in major & minor renovation categories.

Projected Schedules

As indicated in the cost estimates above, the following pages provide information related to proposed project schedules utilizing a traditional design-bid-build model for each project. The following information is estimated based on an immediate design start date beginning in August 2015, but we understand that the projects may be staggered and follow a different phasing strategy. With that in mind, as requested, the following information is included for illustration purposes only.

The initial step for each project will include a program verification phase. During this time, review of the existing teaching labs and research labs is required, and discussions will need to include specifics as to what teaching and research labs will remain within each building. This phase will require meetings with the building committee / team to resolve these questions.

The conceptual study plans within this study are based on the electronic CAD files provided by Purdue University, and no field verification was incorporated as part of the study and study documentation. At the first part of the proposed Schematic Design (SD) phase in the following schedules, field verification of existing conditions will be required to develop a more accurate Building Information Model (BIM) for each project.

And finally, as described within each narrative, any abatement processes and review required by Purdue University will need to be incorporated into the overall project schedule to ensure adequate time is allocated for the actual abatement.

Projected Construction Schedule – ME Phase 1

ID	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	**Design Schedule**	**46 wks**	**Mon 8/3/15**	**Fri 6/17/16**		
2	**Program Review and Verification**	**6 wks**	**Mon 8/3/15**	**Fri 9/11/15**		
3	Programming Meetings	6 wks	Mon 8/3/15	Fri 9/11/15		
4	**Schematic Design (SD) Phase**	**12 wks**	**Mon 9/14/15**	**Fri 12/4/15**	3	
5	Schematic Design and Field Verification	10 wks	Mon 9/14/15	Fri 11/20/15		
6	Code Review	2 wks	Mon 9/14/15	Fri 9/25/15		
7	SD Review	1 wk	Mon 11/23/15	Fri 11/27/15	5	
8	SD Estimate	2 wks	Mon 11/23/15	Fri 12/4/15	5	
9	**Design Development (DD) Phase**	**14 wks**	**Mon 11/23/15**	**Fri 2/26/16**	5	
10	Design Development	12 wks	Mon 11/23/15	Fri 2/12/16		
11	Code Review	2 wks	Mon 11/23/15	Fri 12/4/15		
12	DD Review	1 wk	Mon 2/15/16	Fri 2/19/16	10	
13	DD Estimate	2 wks	Mon 2/15/16	Fri 2/26/16	10	
14	**Construction Documents (CD) Phase**	**18 wks**	**Mon 2/15/16**	**Fri 6/17/16**	10	
15	Construction Documents	15 wks	Mon 2/15/16	Fri 5/27/16		
16	Code Review	2 wks	Mon 2/15/16	Fri 2/26/16		
17	FRCD Review	1 wk	Mon 5/30/16	Fri 6/3/16	15	
18	CD Estimate	2 wks	Mon 5/30/16	Fri 6/10/16	15	
19	Bidding Documents Pkg	2 wks	Mon 6/6/16	Fri 6/17/16	17	
20	**Bidding and Award - Purdue University**	**8.2 wks**	**Mon 6/20/16**	**Mon 8/15/16**	19	
21	Bid Docs	1 day	Mon 6/20/16	Mon 6/20/16		
22	1st Ad	2 wks	Tue 6/21/16	Mon 7/4/16	21	
23	2nd Ad	2 wks	Tue 7/5/16	Mon 7/18/16	22	
24	Bid	2 wks	Tue 7/19/16	Mon 8/1/16	23	
25	Award	2 wks	Tue 8/2/16	Mon 8/15/16	24	
26	**Construction Phase**	**56 wks**	**Tue 8/16/16**	**Mon 9/11/17**		
27	ME-P1 Renovation Construction	14 mons	Tue 8/16/16	Mon 9/11/17	20	

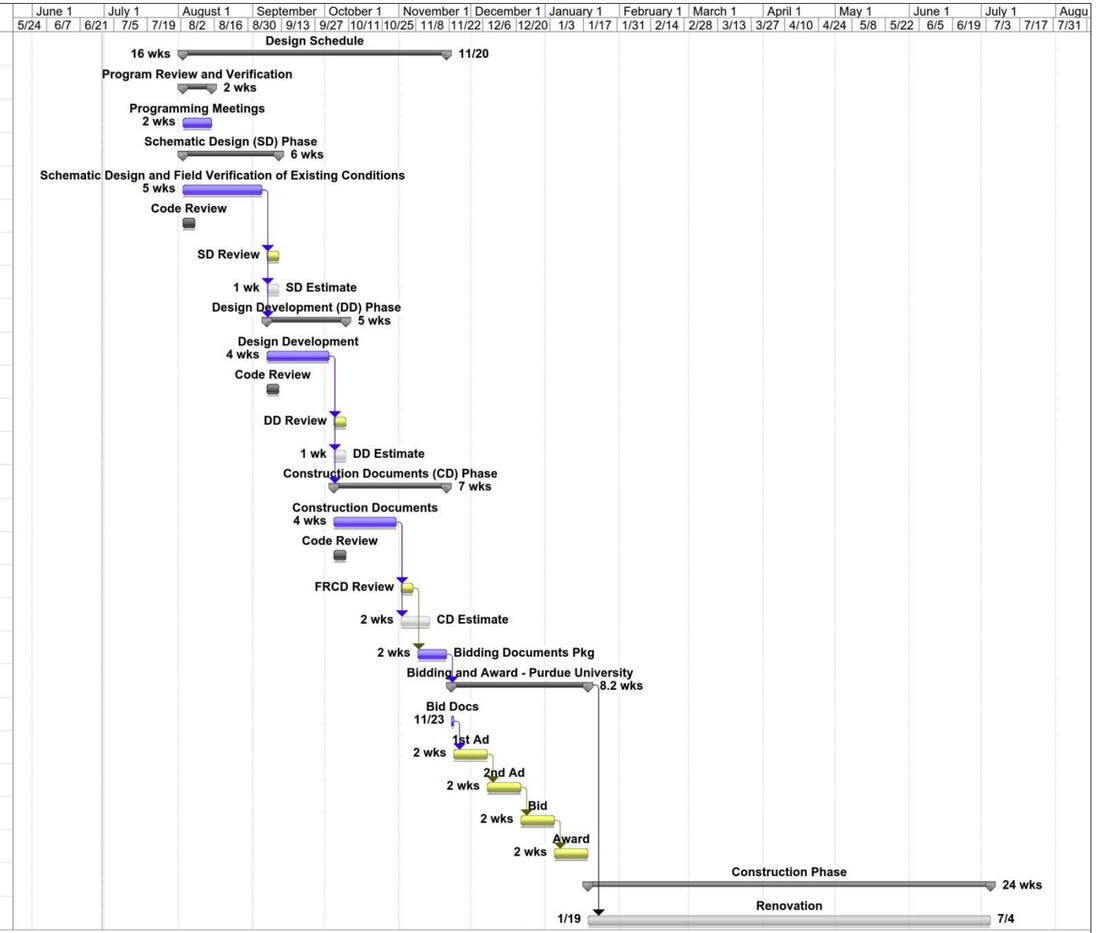
Project: Purdue University ME-P1 Ren
Date: Tue 6/30/15

Task | Split | Progress | Milestone | Summary | Project Summary | External Tasks | External Milestone | Deadline

Page 1

Projected Construction Schedule – RAIL

ID	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	**Design Schedule**	**16 wks**	**Mon 8/3/15**	**Fri 11/20/15**		
2	**Program Review and Verification**	**2 wks**	**Mon 8/3/15**	**Fri 8/14/15**		
3	Programming Meetings	2 wks	Mon 8/3/15	Fri 8/14/15		
4	**Schematic Design (SD) Phase**	**6 wks**	**Mon 8/3/15**	**Fri 9/11/15**		
5	Schematic Design and Field Verification	5 wks	Mon 8/3/15	Fri 9/4/15		
6	Code Review	1 wk	Mon 8/3/15	Fri 8/7/15		
7	SD Review	1 wk	Mon 9/7/15	Fri 9/11/15	5	
8	SD Estimate	1 wk	Mon 9/7/15	Fri 9/11/15	5	
9	**Design Development (DD) Phase**	**5 wks**	**Mon 9/7/15**	**Fri 10/9/15**	**5**	
10	Design Development	4 wks	Mon 9/7/15	Fri 10/2/15		
11	Code Review	1 wk	Mon 9/7/15	Fri 9/11/15		
12	DD Review	1 wk	Mon 10/5/15	Fri 10/9/15	10	
13	DD Estimate	1 wk	Mon 10/5/15	Fri 10/9/15	10	
14	**Construction Documents (CD) Phase**	**7 wks**	**Mon 10/5/15**	**Fri 11/20/15**	**10**	
15	Construction Documents	4 wks	Mon 10/5/15	Fri 10/30/15		
16	Code Review	1 wk	Mon 10/5/15	Fri 10/9/15		
17	FRCD Review	1 wk	Mon 11/2/15	Fri 11/6/15	15	
18	CD Estimate	2 wks	Mon 11/2/15	Fri 11/13/15	15	
19	Bidding Documents Pkg	2 wks	Mon 11/9/15	Fri 11/20/15	17	
20	**Bidding and Award - Purdue University**	**8.2 wks**	**Mon 11/23/15**	**Mon 1/18/16**	**19**	
21	Bid Docs	1 day	Mon 11/23/15	Mon 11/23/15		
22	1st Ad	2 wks	Tue 11/24/15	Mon 12/7/15	21	
23	2nd Ad	2 wks	Tue 12/8/15	Mon 12/21/15	22	
24	Bid	2 wks	Tue 12/22/15	Mon 1/4/16	23	
25	Award	2 wks	Tue 1/5/16	Mon 1/18/16	24	
26	**Construction Phase**	**24 wks**	**Tue 1/19/16**	**Mon 7/4/16**		
27	Renovation	6 mons	Tue 1/19/16	Mon 7/4/16	20	



Project: Purdue University RAIL Renov
Date: Tue 6/30/15

Task | Split | Progress | Milestone | Summary | Project Summary | External Tasks | External Milestone | Deadline

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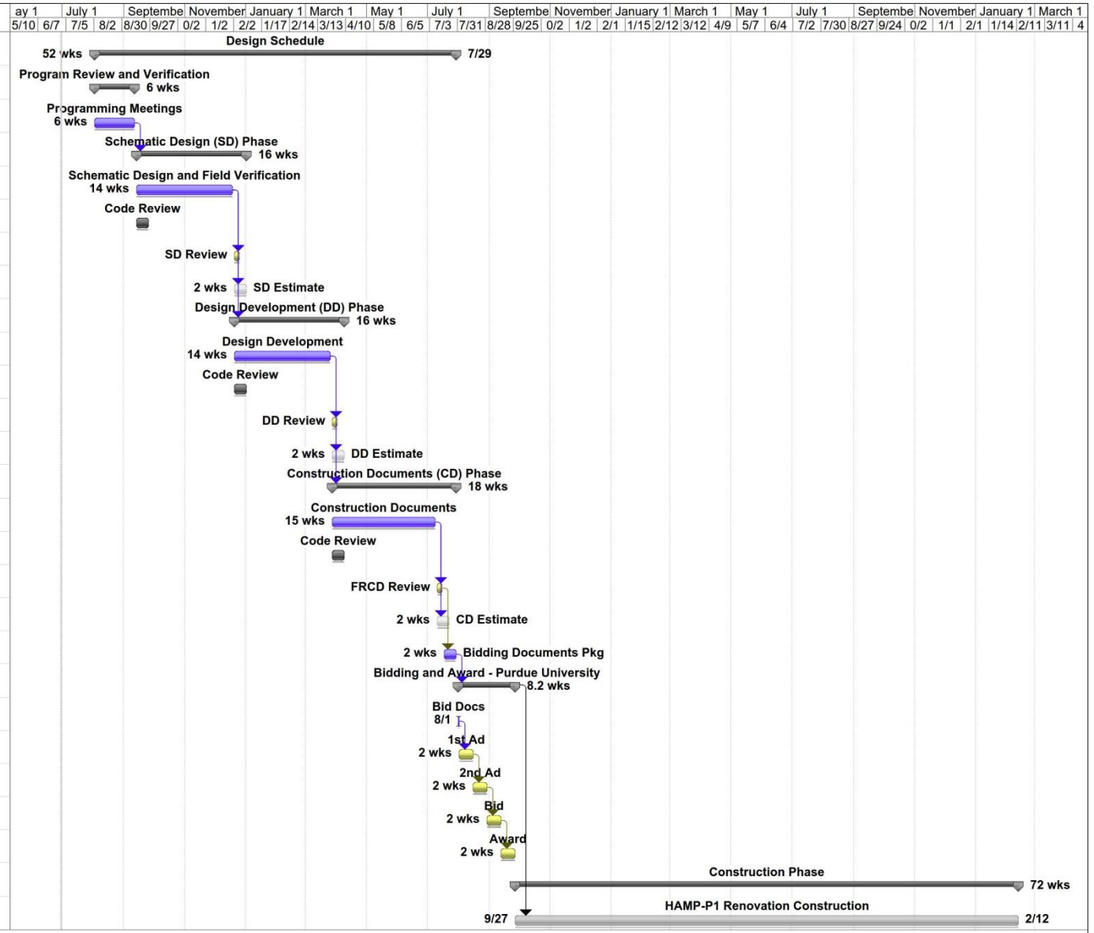
Projected Construction Schedule – HAMP Phase 1

ID	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	**Design Schedule**	**52 wks**	**Mon 8/3/15**	**Fri 7/29/16**		
2	**Program Review and Verification**	**6 wks**	**Mon 8/3/15**	**Fri 9/11/15**		
3	Programming Meetings	6 wks	Mon 8/3/15	Fri 9/11/15		
4	**Schematic Design (SD) Phase**	**16 wks**	**Mon 9/14/15**	**Fri 1/1/16**	3	
5	Schematic Design and Field Verification	14 wks	Mon 9/14/15	Fri 12/18/15		
6	Code Review	2 wks	Mon 9/14/15	Fri 9/25/15		
7	SD Review	1 wk	Mon 12/21/15	Fri 12/25/15	5	
8	SD Estimate	2 wks	Mon 12/21/15	Fri 1/1/16	5	
9	**Design Development (DD) Phase**	**16 wks**	**Mon 12/21/15**	**Fri 4/8/16**	5	
10	Design Development	14 wks	Mon 12/21/15	Fri 3/25/16		
11	Code Review	2 wks	Mon 12/21/15	Fri 1/1/16		
12	DD Review	1 wk	Mon 3/28/16	Fri 4/1/16	10	
13	DD Estimate	2 wks	Mon 3/28/16	Fri 4/8/16	10	
14	**Construction Documents (CD) Phase**	**18 wks**	**Mon 3/28/16**	**Fri 7/29/16**	10	
15	Construction Documents	15 wks	Mon 3/28/16	Fri 7/8/16		
16	Code Review	2 wks	Mon 3/28/16	Fri 4/8/16		
17	FRCD Review	1 wk	Mon 7/11/16	Fri 7/15/16	15	
18	CD Estimate	2 wks	Mon 7/11/16	Fri 7/22/16	15	
19	Bidding Documents Pkg	2 wks	Mon 7/18/16	Fri 7/29/16	17	
20	**Bidding and Award - Purdue University**	**8.2 wks**	**Mon 8/1/16**	**Mon 9/26/16**	19	
21	Bid Docs	1 day	Mon 8/1/16	Mon 8/1/16		
22	1st Ad	2 wks	Tue 8/2/16	Mon 8/15/16	21	
23	2nd Ad	2 wks	Tue 8/16/16	Mon 8/29/16	22	
24	Bid	2 wks	Tue 8/30/16	Mon 9/12/16	23	
25	Award	2 wks	Tue 9/13/16	Mon 9/26/16	24	
26	**Construction Phase**	**72 wks**	**Tue 9/27/16**	**Mon 2/12/18**		
27	HAMP-P1 Renovation Construction	18 mons	Tue 9/27/16	Mon 2/12/18	20	



Project: Purdue University HAMP-P1 F
Date: Tue 6/30/15

Task | Progress | Summary | External Tasks | Deadline
Split | Milestone | Project Summary | External Milestone

Page 1

EVALUATION – COLLEGE OF ENGINEERING PROGRAM COMPARISONS

As part of the overall College of Engineering growth plan evaluation, Purdue University requested the following summary of information for office and open-office workstation growth options and projections based on recent projects such as Wang Hall and Grissom Hall, as well as studies from the Electrical Engineering and Duncan Annex (EE) building and the ME-RAIL-HAMP study.

College of Engineering
Space Summary Information

Purdue University ME/RAIL/HAMP Study

College of Engineering Growth Plan - Program Comparisons

SUMMARY OF WORKSPACES / WORKSTATIONS PER BUILDING

	Grissom Hall (GRIS) Industrial Engineering (IE)							Electrical Engineering Bldg (EE) - P1 Electrical and Computer Engineering (ECE) and Engineering Computer Network (ECN)							Electrical Engineering Bldg (EE) - P1+2 Electrical and Computer Engineering (ECE) and Engineering Computer Network (ECN)						
	Existing Quantities 2012 (People)	Existing Configuration (Space - estimated)	MP Projections, Incl. Growth (Space)	Actual/ Proposed Config		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)	Existing Quantities 2012 (People)	Existing Configuration Space Review	MP Projections, Incl. Growth (Space)	Actual/ Proposed Config ECE+ECN		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)	Existing Quantities 2012 (People)	Existing Configuration Space Review ECE+ECN	MP Projections, Incl. Growth (Space) ECE+ECN	Actual/ Proposed Config ECE+ECN		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)
				# People	# Workspace						# People	# Workspace						# People	# Workspace		
Faculty	33	33	54	46	46	13	139%	95	45		50	50	5	111%	95	45	115	62	62	17	138%
Staff	11	11	29	26	26	15	236%	41	48		103	103	55	215%	41	48	135	80	80	32	167%
Graduate Students	79	79	120	246	172	167	218%	310	208		245	171	37	82%	310	208	381	369	258	161	124%
Subtotal	**123**	**123**	203	**318**	**244**		**198%**	**446**	**301**		**398**	**324**		**136%**	446	301	631	511	400		143%
								No. of Grads x (0.7) = No. of Wkstns			*No. of Grads x (0.7) = No. of Wkstns*							*No. of Grads x (0.7) = No. of Wkstns*			*152%*

Comments: Existing configuration information is not available. *Comments:* *Comments:*

CONTINUED

	Mechanical Engineering (ME) and Gatewood Mechanical Engineering (ME)							American Railway Building (RAIL) Mechanical Engineering (ME)							Hampton Hall (HAMP) Civil Engineering (CE) and Construction Engineering and Mgmt (CEM)						
	Existing Quantities 2012 (People)	Existing Configuration Space Review	MP Projections, Incl. Growth (Space)	Actual/ Proposed Config		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)	Existing Quantities 2012 (People)	Existing Configuration Space Review	MP Projections, Incl. Growth (Space)	Actual/ Proposed Config		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)	Existing Quantities 2012 (People)	Existing Configuration Space Review CE+CEM	MP Projections, Incl. Growth (Space)	Actual/ Proposed Config		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)
				# People	# Workspace						# People	# Workspace						# People	# Workspace		
Faculty	88	77	93	103	103	**26**	**134%**	0	0		0	0	**0**	**N/A**	62	74	90	122	122	**60**	**165%**
Staff	52	53	53	61	61	**8**	**115%**	0	0		0	0	**0**	**N/A**	50	74	72	112	112	**62**	**151%**
Graduate Students	269	167	183	205	143	**38**	**86%**	26	26		60	42	**34**	**162%**	180	243	223	440	308	**260**	**127%**
Subtotal	**409**	**297**	329	**369**	**307**		**111%**	**26**	**26**		**60**	**42**		**162%**	**292**	**391**	385	**674**	**542**		**148%**
								No. of Grads x (0.7) = No. of Wkstns			*No. of Grads x (0.7) = No. of Wkstns*							*No. of Grads x (0.7) = No. of Wkstns*			*158%*

Comments: Includes existing Gatewood quantities (existing to remain) *Comments:* *Comments:*

CONTINUED

	Wang Hall Fourth Floor (Rec'd 03/25/15)						
	Existing Quantities (People)	Existing Configuration Space Review	MP Projections, Incl. Growth (Space)	Actual/ Proposed Config (4th Floor)		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)
				# People	# Workspace		
Faculty				48	48	**48**	**N/A**
Staff				46	46	**46**	**N/A**
Graduate Students				175	122	**175**	**N/A**
Subtotal				**269**	**216**		**N/A**
				No. of Grads x (0.7) = No. of Wkstns			

Comments:

OVERALL SUMMARY

	Actual/ Proposed Config (incl Wang Hall, 4th Flr)		Additional People Accommodated	Calculated Efficiency (Planned/Existing Wk)	CoE Growth Plan (People)	Difference
	# People	# Workspace				
Faculty	369	369	**152**	**137%**	107	45
Staff	348	348	**186**	**179%**	105	81
Graduate Students	1371	958	**711**	**135%**	700	11
Subtotal	**2088**	**1675**	**1049**	**150%**		
	No. of Grads x (0.7) = No. of Wkstns			*158%*		

Comments: Growth calculations are only for buildings within the scope and do not reflect other parts of campus for growth (i.e., 0.7 factor to be applied to existing Graduate Student workstation quantities to determine approximate capacity).

MSKTD Associates, Inc.

CoE Summary (EGP)
Page 1 of 1

NARRATIVE

Structural and Architectural Narrative

The east wing of the existing ME building only extends to the first floor. Towards the end of the study and related team meetings, there were discussions about a separate option to extend the east wing of ME up to the second and third floor for additional office suites and related functions (in lieu of renovating the existing building to accommodate these growth requirements). In this scenario, a fourth floor mechanical area would also be necessary to accommodate MEP/T system requirements. While this option does not address renovation and maintenance issues within the main ME building, the design team provided a high-level review for feasibility, and summary information is included in this appendix and the MEP narratives.

STRUCTURAL

The first consideration was the review of the structural implications. A high-level preliminary review of the existing drawings was provided, and based on that review, it appears that the existing buildings were designed with forward planning for a second floor level, third floor level, attic and upper roof over the entire footprint. The drawings reviewed include the 1941 "Third Unit" building, the 1949 connector building, and the 1932 "Second Unit" building. As part of that preliminary review, there are some things to consider moving forward that may influence the usage of the proposed spaces. These items are listed by building below.

1941 Building - "Third Unit" Set (Purdue #4139):

- The construction documents show a second story roof on the northern half of the building that was apparently deleted by "Addendum No. 1." While the design team does not have this addendum, it was assumed that the second level framing was constructed as shown. This existing construction should be field verified to complete the study.
- The second level as shown on the drawings should be thoroughly evaluated to determine its load-carrying capacity. Initial calculations indicate that it is likely adequate for classroom or offices with partitions, but it is not likely to be adequate for occupancies that require higher live loads.

1949 Building - "Addition & Alternations" Set (Purdue #4818):

- The existing building columns have corbels intended to support beams for a future third level and attic level beams.
- These drawings clearly indicate the future planned levels and their anticipated live loads:
    - Present Ground Floor, Present 1st Floor = 150 PSF
    - Present Roof (Future 2nd Floor) = 100 PSF
    - Present 3rd Floor & Future Attic = 50 PSF
    - Future Roof = 30 PSF
- The live loads listed above for the 2nd floor and the 3rd floor are low (100 PSF + 50 PSF = 150 PSF accumulative). If offices with partitions are being considered for the 2nd and 3rd floors, the live loads should be 80 PSF for each floor (160 PSF accumulative). This is to show that consideration of the occupancies of the future levels should be given in order to not exceed the live loads allocated for future use.

ARCHITECTURAL

Architectural review of this option was limited to determining the quantities of faculty, staff and graduate student offices and open-office workstations that could potentially be accommodated within the addition without requiring renovations in much of the existing building. While this scenario does not address many existing building conditions, such as office configurations, space adjacencies and function, circulation, additional restroom facilities or existing MEP/T systems, the exercise provides a high-level look at the overall impact this addition may have on the College of Engineering's growth plan for additional offices and open-office workstations.

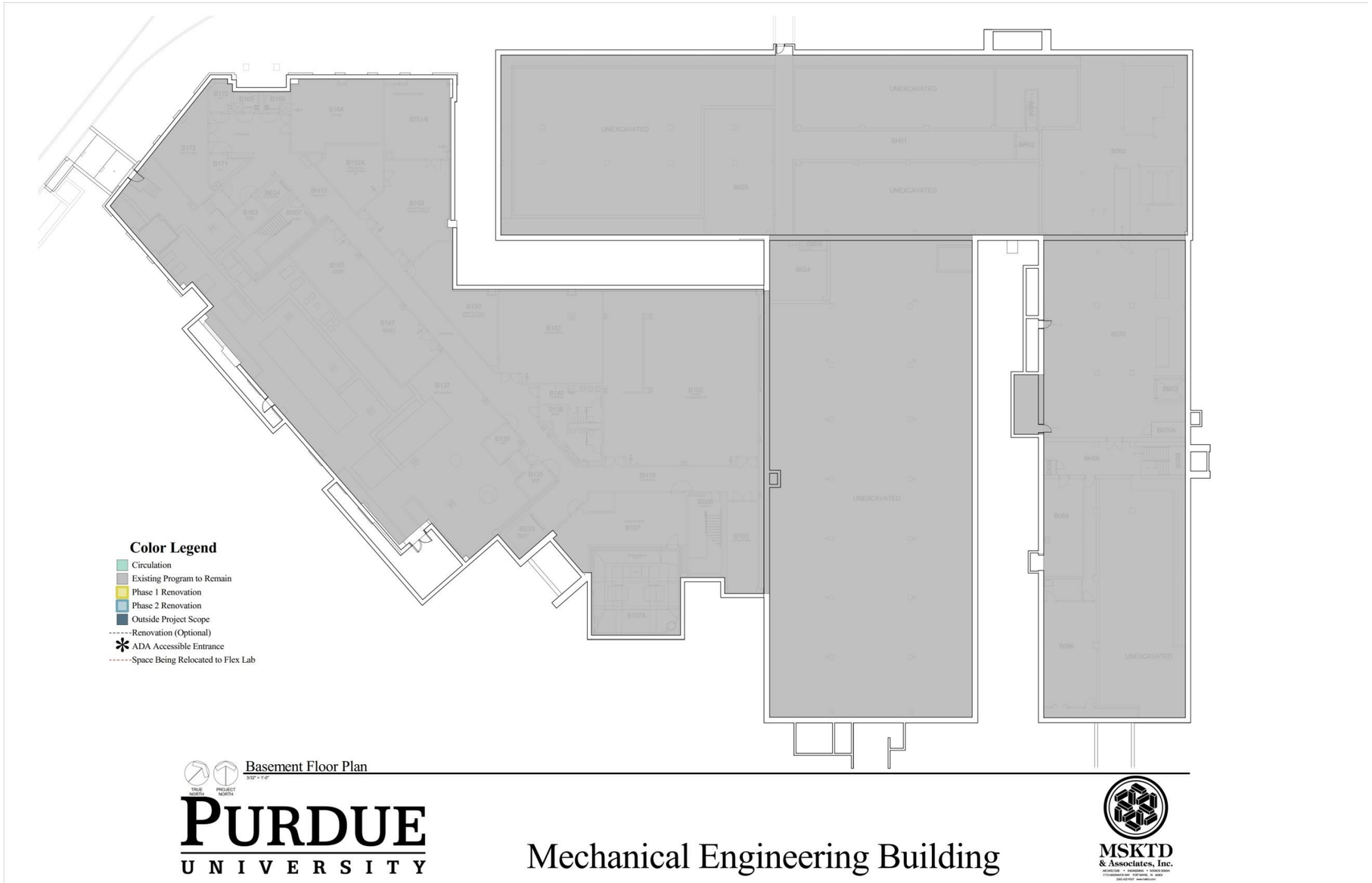
On the second and third floor, the proposed plans show additional restrooms in order to accommodate men's and women's restroom facilities on these floors where the additional offices are proposed. On the second floor, the new restroom location impacts an existing SMAS classroom, and this classroom is shown on the proposed plans as being relocated to the first floor, displacing an existing suite of offices and impacting the corridor and circulation in the east wing of the building. A larger conference room is shown on the second floor. To connect to the new addition from the north wing, an office is displaced to provide a larger circulation corridor to the new office suite.

On the third floor, the existing office suite on the east end will require reconfiguration and renovation to accommodate the corridor and circulation to access the new office suite. In the proposed plan, a new restroom is shown, as well as an informal work café, suite of offices, and meeting spaces.

In regard to circulation, a new stair tower is shown on the south end of the east wing, and the central stair tower within the east wing will need to be reconfigured to accommodate extending the stair up to serve these floors. At the second and third floor areas, the stairs may have access-controlled doors to limit access to the office suites after hours.

See the Mechanical and Electrical narratives for description of the systems impact for this option.

Architectural Narrative – Proposed Plans – ME Basement Floor - Option



Architectural Narrative – Proposed Plans – ME Ground Floor - Option

Architectural Narrative – Proposed Plans – ME First Floor - Option

FIRST FLR. TOTALS

OVERALL TOTALS
(23) Offices
(27) Workstations
- (2 Staff, 25 GA)
(2) Conference Rooms
(1) Work Rooms
(0) Teaching Labs
(1) Research Labs
(2) Restroom
(1) Reception

First Floor Plan - Proposed
3/32" = 1'-0"

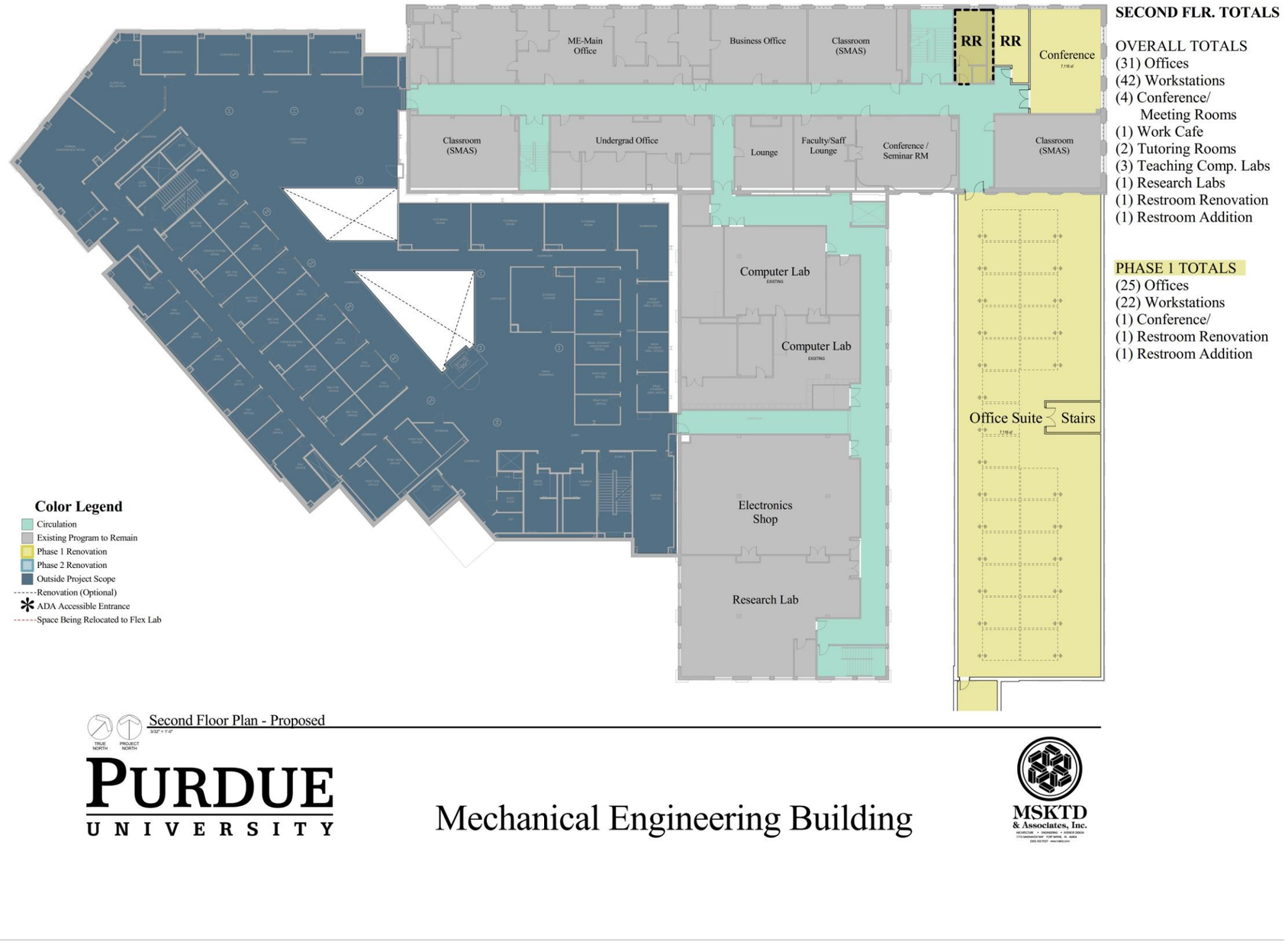
TRUE NORTH PROJECT NORTH

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

Architectural Narrative – Proposed Plans – ME Second Floor - Option



Architectural Narrative – Proposed Plans – ME Third Floor - Option

THIRD FLR. TOTALS

OVERALL TOTALS
(33) Offices
(91) Workstations
- (30 Staff, 61 GA)
(1) Conference Rooms
(2) Meeting Rooms
(1) Cafe
(1) Restroom Renovation
(1) Restroom Addition

PHASE 1 TOTALS
(25) Offices
(28) Workstations
(1) Conference Rooms
(2) Meeting Rooms
(1) Cafe
(1) Restroom Renovation
(1) Restroom Addition

Third Floor Plan - Proposed

PURDUE UNIVERSITY

Mechanical Engineering Building

MSKTD & Associates, Inc.

ME Addition Option Building Costs Parametric Estimate

ME Addition Option – Summary Information

Purdue University; West Lafayette, IN — **Parametric Estimate**

Mechanical Engineering Bldg Addition **CCS Project No.** 15.044 **Date:** **6/30/2015**

Department Categories	GSF	Phase Number	Cost / SF	Base Construction Cost - 2015	Environmental removal costs - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Added cost - Potential Impact issue - Lump Sum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed at 4% per year compounded	Escalated Costs	Cost / SQFT
Major Renovation	7,184	One	$ 169.27	$ 1,216,025	$ -	$ -	$ -	$ 1,216,025	103.00%	$ 1,252,506	
New Construction	26,058	One	$ 238.23	$ 6,207,830	$ -	$ -	$ -	$ 6,207,830	103.00%	$ 6,394,065	
Demolition	15,109	One	$ 4.22	$ 63,801	$ -	$ -	$ -	$ 63,801	103.00%	$ 65,715	
Subtotal				$ 7,487,655				$ 7,487,655		$ 7,712,285	
Design Contingency	15.00%			$ 1,123,148				$ 1,123,148		$ 1,156,843	
GC + OH&P	12.00%			$ 1,033,296				$ 1,033,296		$ 1,064,295	
Construction Contingency	10.00%			$ 964,410				$ 964,410		$ 993,342	
Total Construction	33,242			$ 10,608,510				$ 10,608,510		**$ 10,926,766**	$328.70

ME Addition Option – Renovation Information

Purdue University; West Lafayette, IN

Renovation Cost Outline Matrix

Mechanical Engineering Bldg Addition **CCS Project No.** 15.044 **Date:** 6/30/2015

Categories	Space Types	Area - GSF	Phase Number	Cost / SF - Cosmetic - Paint and flooring $/ SF	Cost / SF - Minor Renovation - Paint , Flooring , Ceiling , Lighting , Minor room reconfigurations	Cost / SF - Major Renovation - Total Reconfiguration - all new localized construction and MEP	Base Construction Cost - 2015	Environmental removal costs - Lump Sum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs	Cost / SQFT
ADA Upgrades	Existing Restroom	448.44	One	$ 15.00	$ 60.00	$ 125.00	$ 56,055	$ -	$ 56,055.00	3.00%	$ 57,737	
	New Restroom	470.66	One	$ 15.00	$ 55.00	$ 175.00	$ 82,366	$ -	$ 82,365.50	3.00%	$ 84,836	
Classroom	SMAS	1,103	One	$ 15.00	$ -	$ 235.00	$ 259,186	$ -	$ 259,186.20	3.00%	$ 266,962	
Offices	Office Suite	818	One	$ 15.00	$ 50.00	$ 160.00	$ 130,896	$ -	$ 130,896.00	3.00%	$ 134,823	
	Private	485	One	$ 15.00	$ 50.00	$ 185.00	$ 89,688	$ -	$ 89,688.00	3.00%	$ 92,379	
	Conf / Seminar	1,471	One	$ 15.00	$ 50.00	$ 185.00	$ 272,054	$ -	$ 272,053.60	3.00%	$ 280,215	
Common	Corridors	2,057	One	$ 15.00	$ 50.00	$ 135.00	$ 277,745	$ -	$ 277,744.95	3.00%	$ 286,077	
	Work Café	331	One	$ 15.00	$ 50.00	$ 145.00	$ 48,036	$ -	$ 48,035.60	3.00%	$ 49,477	
MAJOR RENOVATION		**7,184**					**$ 1,216,025**		**$ 1,216,025**		**$ 1,252,506**	$ 174.34

ME Addition Option – New Construction Information

Purdue University; West Lafayette, IN | **Parametric Estimate** | **New Construction**

Mechanical Engineering Bldg Addition | **CCS Project No.** 15.044 | **Date:** 6/30/2015

							Premium Costs to Base Construction Costs					
Categories	Space Types	Quantity	Units	Phase Number	Shell & Core - $ / SF	Interior Fit out - $ / SF - walls, doors, finishes, specialties , Mech, Elect,	Added cost - Potential Impact issue - Lumpsum	Added cost - Potential Impact issue - Lumpsum	Added cost - Potential Impact issue - Lumpsum	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed at 4% per year compounded	Escalated Costs
Gable Roof		7,988	-	One	$ 85.00	$ -	$ -	$ -	$ -	$ 678,980.00	103%	$ 699,349
Offices	Office Suite	8,280	SQFT	One	$ 235.00	$ 75.00	$ -	$ -	$ -	$ 2,566,800.00	103%	$ 2,643,804
	Private	6,240	SQFT	One	$ 235.00	$ 75.00	$ -	$ -	$ -	$ 1,934,400.00	103%	$ 1,992,432
	Conf / Seminar	1,456	SQFT	One	$ 235.00	$ 75.00	$ -	$ -	$ -	$ 451,360.00	103%	$ 464,901
	Common / Corridor	594	SQFT	One	$ 235.00	$ 50.00	$ -	$ -	$ -	$ 169,290.00	103%	$ 174,369
Stair Tower	3 Flights	4	LSUM	One	$ 225.00	$ 23,000.00	$ -	$ -	$ -	$ 92,000.00	103%	$ 94,760
	Enclosure	1,500	SQFT	One	$ 185.00	$ 25.00	$ -	$ -		$ 315,000.00	103%	$ 324,450
Total Construction		26,058								**$ 6,207,830**		**$ 6,394,065**

ME Addition Option – Demolition Information

Purdue University; West Lafayette, IN — **Parametric Estimate** — **Shell and Infrastructure**

Mechanical Engineering Bldg Addition — **CCS Project No.** 15.044 — **Date:** 6/30/2015

Building Division	Scope of Work - Description	Area - GSF or CF	Phase Number	Cost / SF -	Base Construction Cost - 2015	Added cost - Potential Impact issue	Total Construction Costs 2015	Escalation to Midpoint of construction - assumed 4% per year compounded	Escalated Costs
Demolition	Ground, Second & Third Floor	111,270	One	$ 0.48	$ 53,409.60	$ -	$ 53,409.60	3.00%	$ 55,012
Demolition	Ground Floor @ New Stair Tower	5,625	One	$ 0.48	$ 2,700.00	$ -	$ 2,700.00	3.00%	$ 2,781
Demolition	Second Floor roof	7,691	One	$ 1.00	$ 7,691.00	$ -	$ 7,691.00	3.00%	$ 7,922
Total Demolition		124,586	CF		$ 63,801		**$ 63,801**		**$ 65,715**

ME Addition Option – Clarifications

Ln #	Clarifications
1	FF&E and soft costs are excluded from estime. To be provided by Owner.

Ln #	Cost category	Scope of Work	Cost / SQFT
1	ADA Upgrade - Elevator	Lump sum allowance.	$50,000.00
2	ADA Upgrade - Ramps	Concrete ramps with steel handrails	$65.00
3	ADA Upgrade - Existing Restrooms	New HD partitions; new HD doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($18/sf) & Electrical work ($12/sf).	$125.00
4	ADA Upgrade - New Restrooms	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($54/sf) & electrical work ($18/sf).	$175.00
5	Classroom	New partitions; new doors; new floor finishes; wall patching & repair; new hard ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($54/sf) & electrical work ($22/sf).	$235 & $175
6	Office - Cosmetic	Minor floor finishes; wall patching and repair; minor finish upgrade; selective demo.	$30.00
7	Office - Minor renovation	Limited new partitions; door repair; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; selective demo; minor HVAC & electrical repairs.	$97.00
8	Office - Major Renovation	New partitions; new door; minor floor finishes; limited wall patching & repair; new std ceiling; limited finish upgrade; selective millwork; minor specialties; demolition. Fire protection, plumbing & HVAC ($54/sf) & electrical ($17/sf) work.	$160.00
9	Office - New Construction	New partitions and door; new acoustical ceiling; standard finishes; selective millwork; minor specialties. Fire protection, plumbing & HVAC ($59/sf) & Electrical ($22/sf) work.	$235.00
10	Common / Circulation	New partitions; new doors; new floor finishes; wall patching & repair; new acoustical ceiling; finish upgrades; misc. specialties; demolition; Mechanical ($54/sf) & Electrical work ($15/sf).	$135.00
11	Laboratory	New partitions; new doors; new highend floor finishes; wall patching & repair; new ceiling; finish upgrades; misc. specialties; demolition; fire protection; special HVAC & electrical work.	$220, $235 & $245
12	Utility - Electric	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; fire protection; upgraded HVAC & electrical work.	$135.00
13	Utility - Custodian	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; demolition; fire protection & electrical work.	$105.00
14	Utility - IDF	New partitions; new door; new floor finishes; wall patching & repair; finish upgrades; misc. accessories & specialties; special equipment; demolition; fire protection; upgraded HVAC & electrical work.	$185.00

Notes: Excludes general conditions, contingency, escalation and logistics factors.
Includes major mechanical equipment replacement work in major & minor renovation categories.

Grissom Hall (GRIS) and Electrical Engineering and Duncan Annex (EE) Buildings

Supplemental information was requested related to buildings outside the scope of the ME-RAIL-HAMP study, and that supplemental information is included in the following pages for Grissom Hall (GRIS) and the Electrical Engineering and Duncan Annex (EE) buildings.

Following is a square footage summary for GRIS and EE related to the gross square footage (GSF) of the existing buildings. The following chart includes the area from the original study scope (if appropriate), the renovated area, and the displaced area. Displaced areas include a summary of the square footages on each floor for spaces that were included in the existing building list of spaces but were not included in the final proposed renovation plans for the project.

GRIS AND EE
Square Footage Summary Info

BUILDING SQUARE FOOTAGE SUMMARY

GRISSOM HALL				
LEVEL	GSF	STUDY SCOPE	RENOVATED AREA	DISPLACED AREA
BASEMENT	2,275	0	0	1,416
FIRST FLOOR	18,360	0	18,360	7,511
SECOND FLOOR	15,874	0	15,874	0
THIRD FLOOR	16,053	0	16,053	2,379
TOTALS SF	52,562	0	50,287	11,306

ELECTRICAL ENGINEERING BUILDING				
LEVEL	GSF	STUDY SCOPE	PROPOSED RENO.	DISPLACED AREA
SUB-BASEMENT	503	0	0	0
BASEMENT	12,161	2,276	2,276	247
GROUND FLOOR	45,666	39,140	39,140	3,493
FIRST FLOOR	46,174	44,534	44,534	1,295
SECOND FLOOR	31,700	30,406	30,406	422
THIRD FLOOR	24,263	24,264	24,264	204
TOTALS SF	160,467	140,620	140,620	5,661

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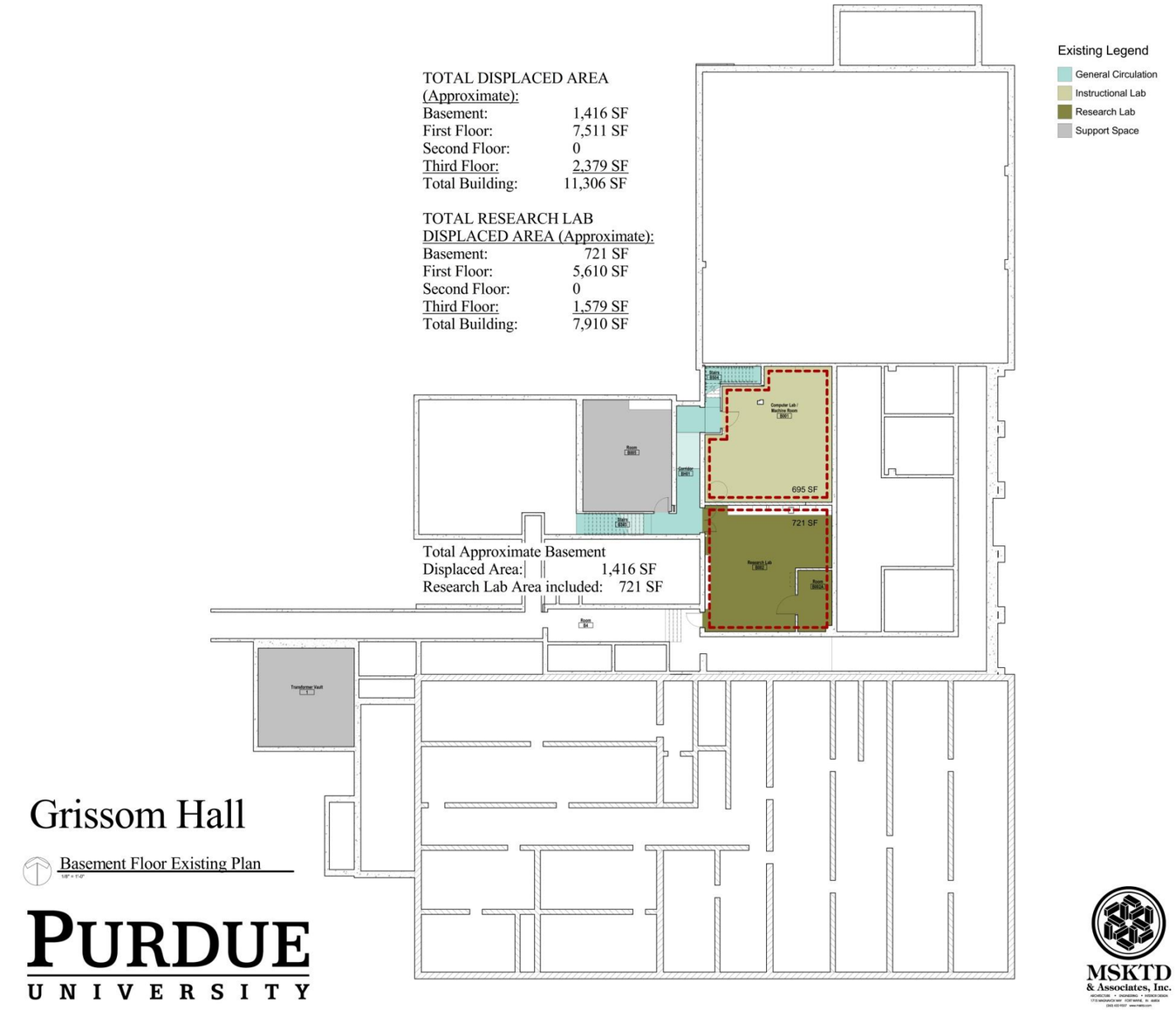
Page1 of 1

Existing Plans – GRIS Basement Floor

TOTAL DISPLACED AREA
(Approximate):
Basement: 1,416 SF
First Floor: 7,511 SF
Second Floor: 0
Third Floor: 2,379 SF
Total Building: 11,306 SF

TOTAL RESEARCH LAB
DISPLACED AREA (Approximate):
Basement: 721 SF
First Floor: 5,610 SF
Second Floor: 0
Third Floor: 1,579 SF
Total Building: 7,910 SF



Grissom Hall

Basement Floor Existing Plan

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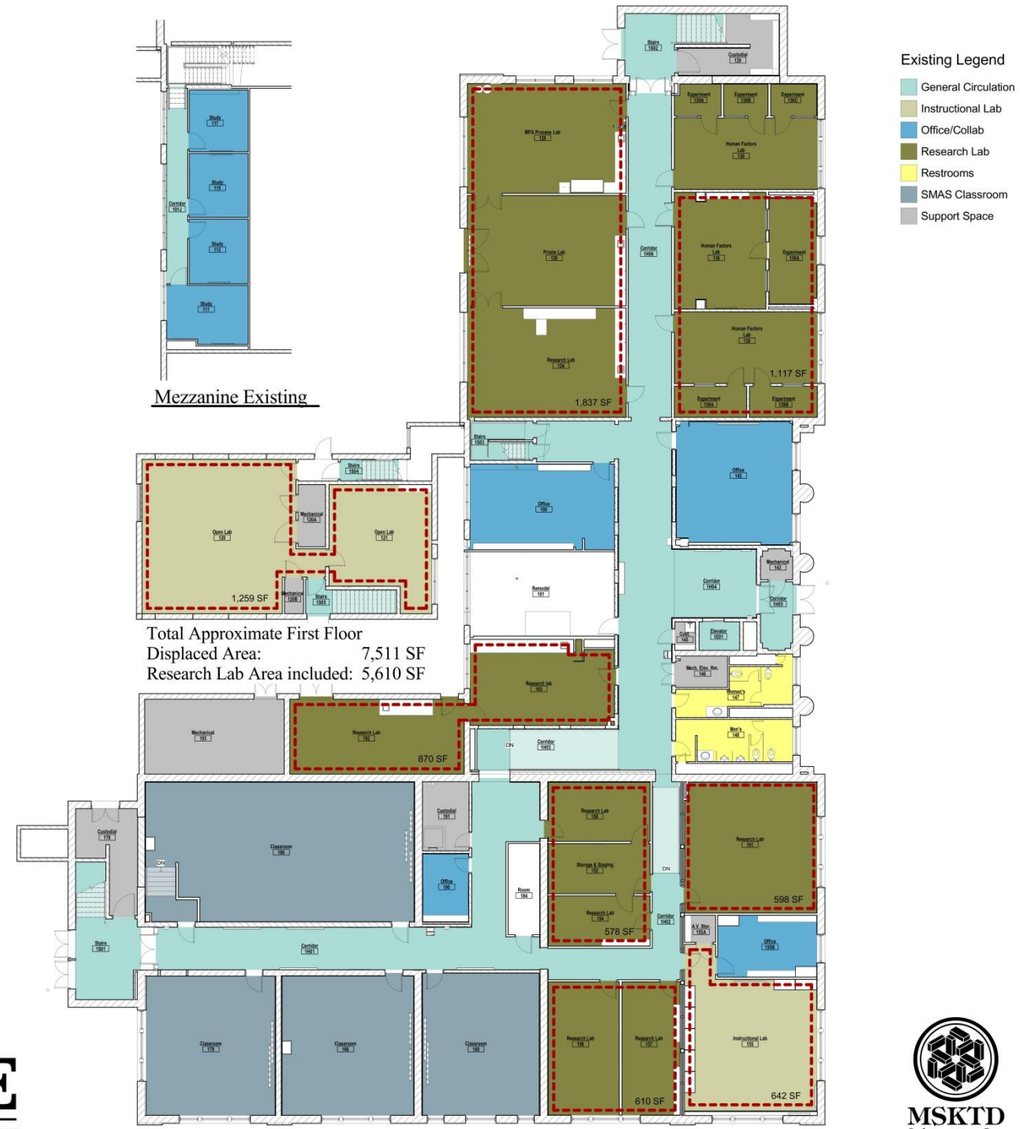
Existing Plans – GRIS First Floor



Grissom Hall

First Floor Existing Plan

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Existing Plans – GRIS Second Floor

Total Approximate Second Floor
Displaced Area: 0 SF
Research Lab Area included: 0 SF

Grissom Hall

Second Floor Existing Plan

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Existing Plans – GRIS Third Floor

Total Approximate Third Floor
Displaced Area: 1,579 SF
Research Lab Area included: 1,579 SF

Grissom Hall

Third Floor Existing Plan

PURDUE UNIVERSITY

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Existing Space Use – GRIS – Teaching Labs and Research Labs Comparison

ME/RAIL/HAMP Study
Grissom Hall (Exist)

Purdue University GRISSOM HALL

GRISSOM HALL

EXISTING SQUARE FOOTAGE SUMMARY OF TEACHING AND RESEARCH LAB SPACES							
	Basement	**First Floor**	**Second Floor**	**Third Floor**		**Current Quantities**	
TEACHING LABS	695	1,900	1,017	0		**3,612**	
RESEARCH LABS	721	5,994	0	2,795		**9,510**	

ME/RAIL/HAMP Study
Grissom Hall (Exist)

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
BASEMENT													
Computer Lab / Machine Room	B001	Eng Cpt Ntwk		CEN COMPUTER					695				
Research Lab	B002	Prvst Rsv Ac		NONCLAS LAB						627			
	B002A	Prvst Rsv Ac		NONCLS SV						94			
	B005	Prvst Rsv Ac		NONCLAS LAB									
Corridor	BH01	Gen Building		CIRCULATE									
Stairs	BS04	Gen Building		CIRCULATE									
Stairs	BS05	Gen Building		CIRCULATE									
Subtotals					0	0	0	0	695	721			

ME/RAIL/HAMP Study
Grissom Hall (Exist)

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor													
Office	100			OFFICE									
Remodel	101			REMODEL									
Research lab	103			NONCLAS LAB						418			
Study	111			STUDY									
Study	113			STUDY									
Study	115			STUDY									
Study	117			STUDY									
Open Lab	120			OPEN LAB					836				
Mechanical	120A			MECHANICAL									
Mechanical	120B			MECHANICAL									
Open Lab	121			OPEN LAB					393				
Research Lab	124			NONCLAS LAB						603			
Prisim Lab	126			NONCLAS LAB						606			
MFG Process Lab	128			NONCLAS LAB						605			
Custodial	129			BLDG SERV									
Human Factors Lab	130			NONCLAS LAB						355			
Experiment	130A			NONCLAS LAB						50			
Experiment	130B			NONCLAS LAB						49			
Experiment	130C			NONCLAS LAB						51			
Human Factors Lab	136			NONCLAS LAB						364			
Experiment	136A			NONCLAS LAB						173			
Human Factors Lab	139			NONCLAS LAB						356			
Experiment	139A			NONCLAS LAB						78			
Experiment	139B			NONCLAS LAB						73			
Student Commons	142			BLDG COMMONS									
MECHANICAL ROOM	143			MECHANICAL									
Custodial Closet	145			BLDG SERV									
ELEVATOR MACHINERY	146			MECHANICAL									
Women's Restroom Vestibule	147			REST ROOM									
Women's Restroom - 2WC,1L	147A			REST ROOM									
Men's Restroom - 3WC,4U,3L	148			REST ROOM									
Research Lab - Cheng	150			NONCLAS LAB						209			
Research Lab - Prabhu	151			NONCLAS LAB						598			
Storage + Staging	152			NONCLS SV						177			
Research Lab - R. Liu	154			NONCLAS LAB						177			
Methods Engineering Lab	155			CLASS LAB					642				
A/V Equipment + Storage	155A			CLS LB SV					29				
Methods Lab Office	155B			OFFICE									
Research Lab - Stojnic	157			NONCLAS LAB						270			
Research Lab - Lee	159			NONCLAS LAB						328			
Tablet Arm Chairs	160			CLASSROOM									
Tablet Arm Chairs	166			CLASSROOM									
Tablet Arm Chairs	170			CLASSROOM									
Custodial Closet-Beneath Stair	178			BLDG SERV									
Strip Tables	180			CLASSROOM									
TELECOMMUNICATIONS CLOSET-B	184			MECHANICAL									
AIIE & Alpha Pi Mu Office	190			OFFICE									

ME/RAIL/HAMP Study
Grissom Hall (Exist)

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor													
NW UTILITY+JANITOR	191			MECHANICAL									
Research Lab	192			NONCLAS LAB						454			
Switchgear Room+Telecomm-EDF	193			MECHANICAL									
ELEVATOR	1E01			CIRCULATE									
CORRIDOR FROM 180 TO RAMPS	1H01			CIRCULATE									
RAMP FROM 152 TO ADDITION 1	1H02			CIRCULATE									
RAMP FROM NORTH OF 191 TO 148	1H03			CIRCULATE									
CORRIDOR FROM RAMPS TO 1S03	1H04			CIRCULATE									
VESTIBULE - EAST SIDE	1H05			CIRCULATE									
CORRIDOR FROM 128 TO 139	1H06			CIRCULATE									
CORRIDOR FROM 111 TO 117	1H07			CIRCULATE									
STAIRWAY - SOUTHWEST	1S01			CIRCULATE									
STAIRWAY - NORTHEAST	1S02			CIRCULATE									
STAIRWAY TO MEZZANINE AND 2N	1S03			CIRCULATE									
STAIRWAY/CORR FROM 120 TO MZ	1S04			CIRCULATE									
STAIRWAY FROM 120 TO BASEMEN	1S05			CIRCULATE									
Subtotals					0	0	0	0	1,900	5,994			

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor													
TAP - M. Lehto	201			OFFICE									
Collaboratorium - Duket	210			NONCLAS LAB					1,017				
Faculty Collaboration Facility	218			DEPT COMMONS									
Faculty Collaboration Support	220			OFF SERV									
Faculty Collaboration Support	220A			OFF SERV									
Reception/Office	228			OFFICE									
J. Wachs	228A			OFFICE									
B. Schmeiser	228B			OFFICE									
D. Compton	228C			OFFICE									
B. Caldwell	228D			OFFICE									
Storage	229			OFF SERV									
Undergrad Counseling - P. Brunese	230			OFFICE									
Reception Area	231			OFF SERV									
Internal Corridor	231A			OFF SERV									
Graduate Office - C. Barnhart	232			OFFICE									
Supplies + Copy Machine	233			OFF SERV									
S. Duket	234			OFFICE									
Office Workroom	235			OFF SERV									
L. Sommers	236			OFFICE									
H. Faith	238			OFFICE									
Reception Area	239			OFF SERV									
Assistant to Department Head - L. Del	240			OFFICE									
Department Head - A. Deshmukh	241			OFFICE									
Teaching Assistants - Tutoring	244			STUDY									
Custodial Closet	246			BLDG SERV									
Women's Restroom Lounge	247			REST ROOM									
Women's Restroom - 3WC,1L-HCA	247A			REST ROOM									
Men's Restroom - 3WC,3U,3L	248			REST ROOM									
Reception Area	250			OFF SERV									
Internal Hall from 251 to 270	250A			OFF SERV									
Kitchenette	251			OFF SERV									
R. Liu	253			OFFICE									
Instructor	254			OFFICE									
S. Nof	255			OFFICE									
Instructor	256			OFFICE									
T. Morin	257			OFFICE									
Visiting Prof Office/Scholar	258			OFFICE									
M. Lehto	259			OFFICE									
Visiting Prof Office/Scholar	260			OFFICE									
Y. Yih	261			OFFICE									
Post Doc	262			OFFICE									
Faculty	263			OFFICE									
Visiting Professor/Scholar	264			OFFICE									
Departmental Storage	265			OFF SERV									
Visiting Professor/Scholar	266			OFFICE									
Storage	267			OFF SERV									
Visiting Professor/Scholar	268			OFFICE									

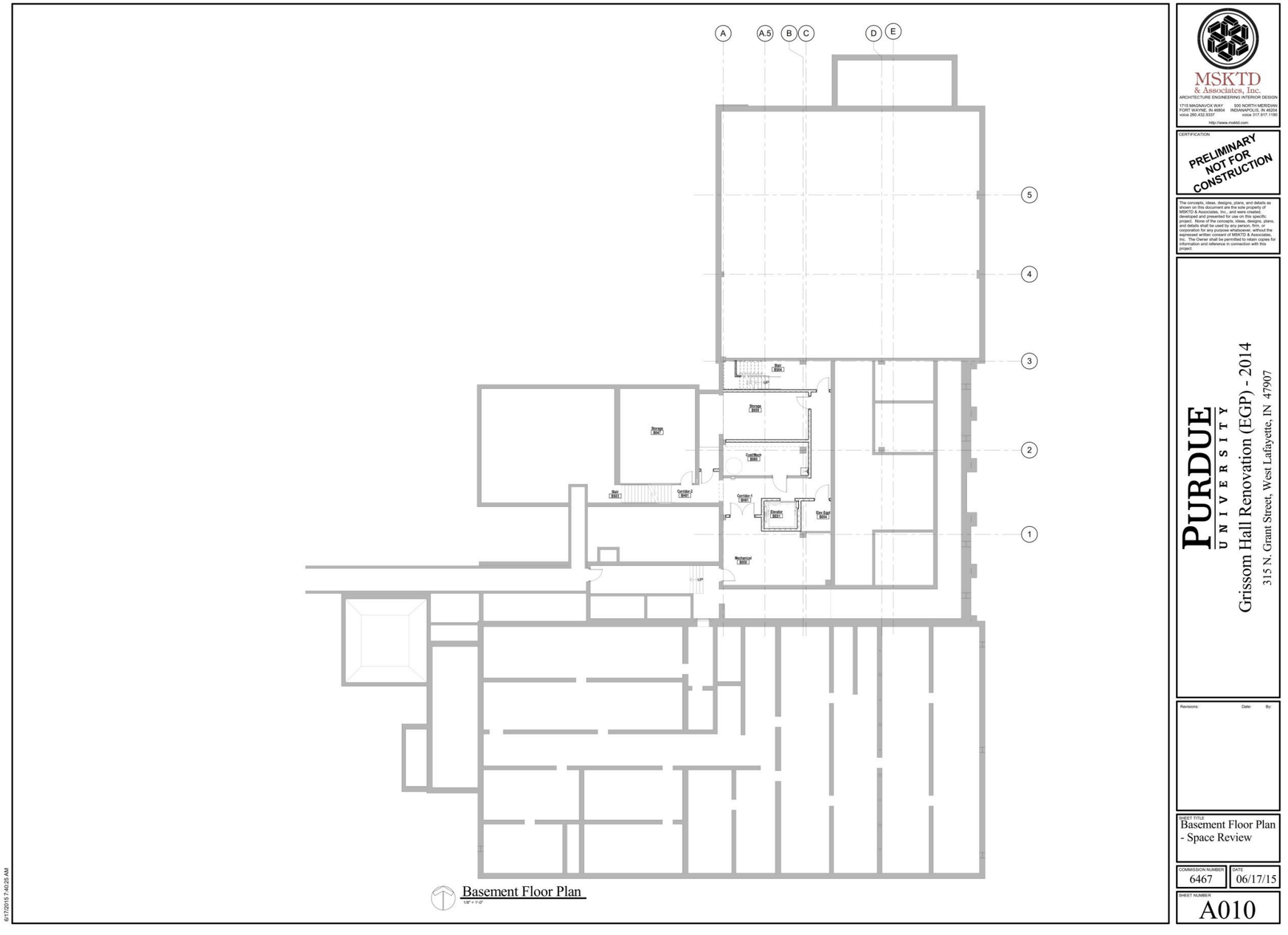
ME/RAIL/HAMP Study
Grissom Hall (Exist)

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor													
Mail/Copier Room	269			OFF SERV									
ABET Office	270			OFFICE									
Student Organization	271			OFFICE									
Student Organization	272			OFFICE									
Gilbreth Memorial Library/Conf	273			CONFERENCE									
Tables & Chairs	274			CLASSROOM									
Tablet Arm Chairs	276			CLASSROOM									
Strip Tables	280			CLASSROOM									
Storage - Thesis	290			OFF SERV									
Teaching Assistants - Tutoring	291			STUDY									
Teaching Assistants - Tutoring	292			STUDY									
ELEVATOR	2E01			CIRCULATE									
CORRIDOR - FROM 276 TO 292	2H01			CIRCULATE									
CORRIDOR - FROM 201 TO 228	2H02			CIRCULATE									
STAIRWAY S-1	2S01			CIRCULATE									
STAIRWAY S-2	2S02			CIRCULATE									
STAIRWAY S-4	2S03			CIRCULATE									
Subtotals					0	0	0	0	1,017	0			

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Third Floor													
Custodial Closet	301			BLDG SERV									
INSPIRE - Office	302			OFFICE									
INSPIRE - Undergrad Office	303			OFFICE									
INSPIRE - M. Sanger	304			OFFICE									
INSPIRE - A. Agree	305			OFFICE									
INSPIRE - E. Gajdzik	307			OFFICE									
Faculty	308			OFFICE									
INSPIRE - Office	309			OFFICE									
INSPIRE - R. Carr	310			OFFICE									
Research Lab - McComb	311			NONCLAS LAB						98			
Research Lab - McComb	312			NONCLAS LAB						96			
S. McComb	313			OFFICE									
Business Office Storage	314			OFF SERV									
Simulator	315			NONCLAS LAB						92			
Business Office Storage	316			OFF SERV									
TA Offices	317			OFFICE									
TA Offices	318			OFFICE									
J. Yi	319			OFFICE									
S. Hunter	320			OFFICE									
TA Offices	321			OFFICE									
TA Offices	322			OFFICE									
Graduate Assistants	323			OFFICE									
Research Engineer - Mann	324			OFFICE									
Corporate Visitors Office	325			OFFICE									
A. Liu	326			OFFICE									
H. Wan	327			OFFICE									
S. Landry	328			OFFICE									
Files + Storage	329			OFF SERV									
Associate Head - S. Landry	330			OFFICE									
Reception	331			OFFICE									
Internal Corridor	331A			OFF SERV									
M. Stojnic	332			OFFICE									
Conference Room	333			CONFERENCE									
V. Duffy	334			OFFICE									
Development Director - L. Sommers	336			OFFICE									
Copy Room	337			OFF SERV									
Staff Office	338			OFFICE									
Business Office	339			OFFICE									
Business Office	340			OFFICE									
Business Office	341			OFFICE									
Business Office	342			OFFICE									
Cosine Research Lab - Tanchoco	344			NONCLAS LAB						161			
Custodial Closet	346			BLDG SERV									
Women's Restroom Lounge	347			REST ROOM									
Women's Restroom - 2WC,1L	347A			REST ROOM									
Men's Restroom - 3WC,4U,3L	348			REST ROOM									
Graduate Student Commons	350			BLDG COMMONS									

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Third Floor													
INSPIRE - M. Schluttenhofer	360			OFFICE									
INSPIRE - Corridor+Files	360			OFF SERV									
INSPIRE - Storage	360A			OFF SERV									
INSPIRE - Office	360B			OFFICE									
INSPIRE - Post Docs	360C			OFFICE									
INSPIRE - Conference Room	360D			CONFERENCE									
INSPIRE - Graduate Assistants	360E			OFFICE									
Internal Corridor	370			OFF SERV									
S. Lee	372			OFFICE									
Research Lab - Nohadani	373			NONCLAS LAB						117			
Faculty	374			OFFICE									
G. Cheng	375			OFFICE									
Research Lab - Wan	376			NONCLAS LAB						95			
Research Lab - Chandrasekar	377			NONCLAS LAB						110			
Storage	378			NONCLS SV						120			
Research Lab - V. Duffy	379			NONCLAS LAB						296			
Closet	379A			NONCLS SV						10			
Research - Prabhu + Liu + Capponi	380			NONCLAS LAB						647			
Research Lab - Y. Yih	385			NONCLAS LAB						667			
Research Lab - Deshmukh	390			NONCLAS LAB						238			
Research Lab - Deshmukh	390A			NONCLS SV						48			
ELEVATOR	3E01			CIRCULATE									
CORRIDOR - FROM 379 TO 390	3H01			CIRCULATE									
CORRIDOR - FROM 301 TO 326	3H02			CIRCULATE									
STAIRWAY SOUTHWEST	3S01			CIRCULATE									
STAIRWAY NORTH EAST	3S02			CIRCULATE									
STAIRWAY S-4	3S03			CIRCULATE									
Subtotals					0	0	0	0	0	2,795			

New Renovation Plans – GRIS Basement Floor



MSKTD & Associates, Inc.
ARCHITECTURE ENGINEERING INTERIOR DESIGN
1715 MAGNAVOX WAY FORT WAYNE, IN 46804 voice 260.432.9337
930 NORTH MERIDIAN INDIANAPOLIS, IN 46204 voice 317.917.1190
http://www.msktd.com

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PURDUE UNIVERSITY

Grissom Hall Renovation (EGP) - 2014
315 N. Grant Street, West Lafayette, IN 47907

Revisions: Date: By:

SHEET TITLE
Basement Floor Plan - Space Review

COMMISSION NUMBER 6467
DATE 06/17/15

SHEET NUMBER
A010

New Renovation Plans – GRIS First Floor

MSKTD & Associates, Inc.
ARCHITECTURE ENGINEERING INTERIOR DESIGN
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PURDUE UNIVERSITY

Grissom Hall Renovation (EGP) - 2014
315 N. Grant Street, West Lafayette, IN 47907

Revisions: Date: By:

SHEET TITLE: First Floor Plan - Space Review

COMMISSION NUMBER: 6467 DATE: 06/17/15

SHEET NUMBER: A011

6/17/2015 7:40:38 AM

New Renovation Plans – GRIS Second Floor

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PURDUE UNIVERSITY
Grissom Hall Renovation (EGP) - 2014
315 N. Grant Street, West Lafayette, IN 47907

Revisions:		Date:	By:
1	Addendum D-1	08/04/14	

SHEET TITLE: Second Floor Plan

COMMISSION NUMBER	DATE
6467	06/17/15

SHEET NUMBER: A012

6/17/2015 7:40:48 AM

New Renovation Plans – GRIS Third Floor

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PURDUE UNIVERSITY

Grissom Hall Renovation (EGP) - 2014
315 N. Grant Street, West Lafayette, IN 47907

Revisions: Date: By:

SHEET TITLE
Third Floor Plan

COMMISSION NUMBER 6467
DATE 06/17/15

SHEET NUMBER
A013

6/17/2015 7:41:04 AM

Existing Plans – EE Sub-Basement and Basement Floors

TOTAL DISPLACED AREA
(Approximate):

Basement:	247 SF
Ground Floor:	3,493 SF
First Floor:	1,295 SF
Second Floor:	422 SF
Third Floor:	204 SF
Total Building:	5,661 SF

BASEMENT DISPLACED AREA
(Approximate): 247 SF

Basement Floor Plan - Existing
3/32" = 1'-0"

PURDUE UNIVERSITY

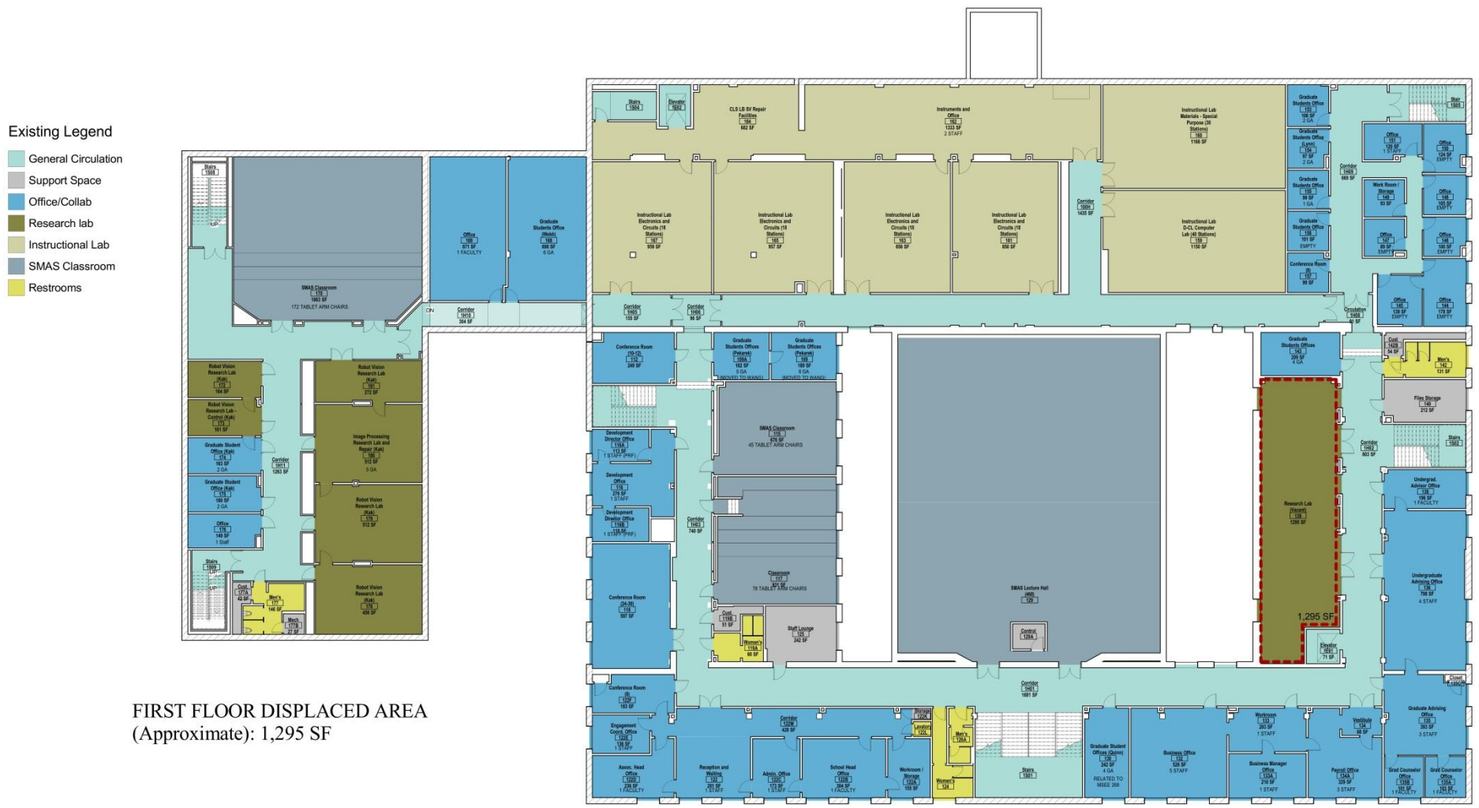
Electrical Engineering Building

MSKTD & Associates, Inc.

Existing Plans – EE Ground Floor

Existing Legend
General Circulation
Support Space
Office/Collab
Research lab
Instructional Lab
SMAS Classroom
Restrooms
OPEN TO BELOW
OPEN TO BELOW
OPEN TO BELOW
Elec Support 275 SF
Elec Support 238 SF
Elec Support 332 SF
Support 425 SF
Office 159 SF 1 Faculty
Mech 109 SF
Teaching Lab 47788 / D-CL Terminals/Multimedia 871 SF
Electrical 737 SF
Robotics Lab 468 SF
Research Lab (Polaris) 168 SF
Undergrad Teaching Lab 43780 975 SF
Teaching Lab 3620027890 938 SF
Teaching Lab 27980 / Electronics and Circuits 988 SF
Teaching Lab 47780 / Electronics and Circuits 998 SF
Lab 47788 / D-CL Terminals/Multimedia 365 SF
Teaching / Power Engineering Lab 32350 1322 SF
Research 32300 / Power Engineering Lab (Sudhoff) 1749 SF
Research 32300 / Power Engineering Lab 229 SF
Teaching Lab 401 SF
Engineering Teaching Lab 304 SF
Corridor 1448 SF
Mech 240 SF
Vestibule 55 SF
Corridor 125 SF
Stairs
Robotics Research Lab 410 SF
Ultrasonics Research Lab and Office (Furgason) 198 SF 1 Faculty
Ultrasonics Research Lab (Furgason) 387 SF
Graduate Students Office (Shiflett) 1 GA
Electronic Imaging Research Lab A/V (Kisbaugh) 165 SF
Electronic Imaging Research Lab (Kisbaugh) 168 SF
Electronic Imaging Research Lab (Kisbaugh) 168 SF
Electronic Imaging Research Lab (Kisbaugh) 168 SF
Storage/Shop 204 SF
Storage/Shop 168 SF
Storage 266 SF
Security Office 132 SF 1 Staff (Vacant)
Corridor 623 SF
Corridor 78 SF
Corridor 633 SF
Corridor 1448 SF
Corridor 1448 SF
Conference Room 1749 SF
Corridor 623 SF
Corridor 633 SF
Corridor 1448 SF
Corridor 1448 SF
Corridor 1448

Existing Plans – EE First Floor



FIRST FLOOR DISPLACED AREA
(Approximate): 1,295 SF

First Floor Plan - Existing

PURDUE UNIVERSITY

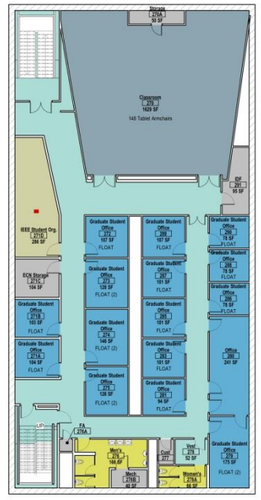
Electrical Engineering Building

MSKTD & Associates, Inc.

Existing Plans – EE Second Floor

Existing Legend

- General Circulation
- Support Space
- Office/Collab
- Research lab
- Instructional Lab
- SMAS Classroom
- Restrooms



SECOND FLOOR DISPLACED
AREA (Approximate): 422 SF

Second Floor Plan - Existing
3/32" = 1'-0"

TRUE NORTH PROJECT NORTH

PURDUE UNIVERSITY

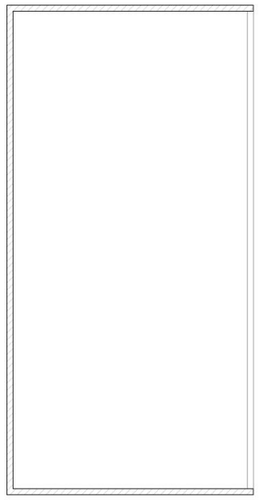
Electrical Engineering Building

MSKTD & Associates, Inc.

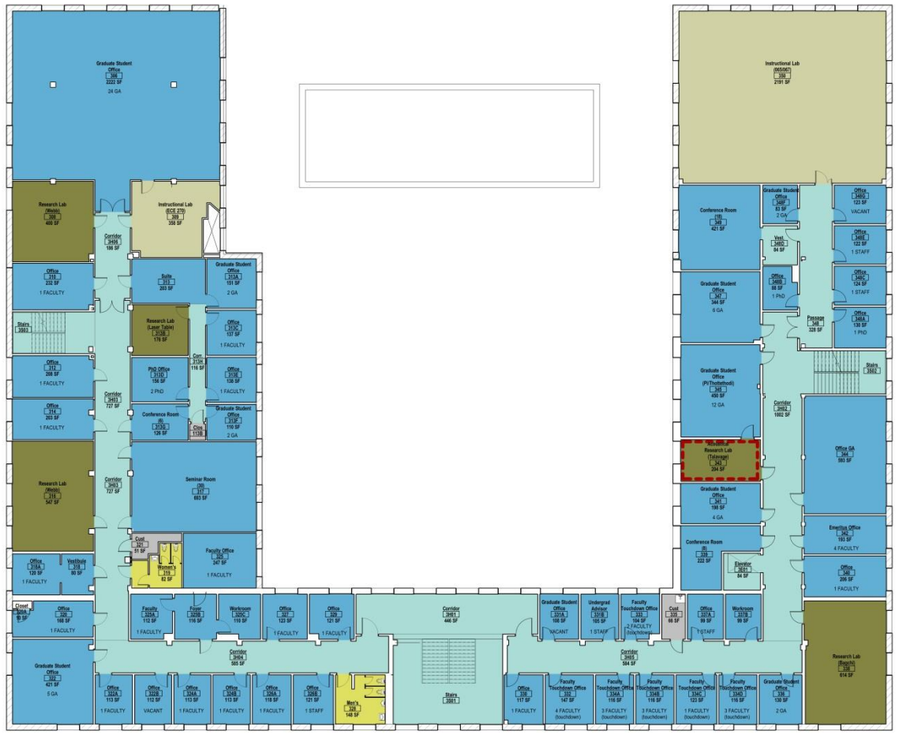
Existing Plans – EE Third Floor

Existing Legend

- General Circulation
- Support Space
- Office/Collab
- Research lab
- Instructional Lab
- Restrooms



THIRD FLOOR DISPLACED
AREA (Approximate): 204 SF



Third Floor Plan - Existing
3/32" = 1'-0"

TRUE NORTH PROJECT NORTH

PURDUE UNIVERSITY

Electrical Engineering Building

MSKTD & Associates, Inc.

Existing Space Use – EE – Teaching Labs and Research Labs Comparison

Purdue University ME/RAIL/HAMP Study

Electrical Engineering Building (EE)

EXISTING SQUARE FOOTAGE SUMMARY OF TEACHING AND RESEARCH LAB SPACES								
	Basement	**Ground Floor**	**First Floor**	**Second Floor**	**Third Floor**		**Current Quantities**	
TEACHING LABS	1,972	6,870	7,544	4,146	0		**20,532**	
RESEARCH LABS	2,572	6,131	3,429	2,394	2,364		**16,890**	

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
SUB-BASEMENT													
MECHANICAL EQUIPMENT	S066M			MECHANICAL									
Stairs from Maintenance Shop	SS06			CIRCULATE									
Subtotals					0	0	0	0	0	0			

ME/RAIL/HAMP Study
Electrical Engineering Bldg.

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
BASEMENT													
Storage	B066B			CENT STORAGE									
Maintenance Shop	B066C			SHOP									
Model Shop	B066D			SHOP									
Supervisor - C. Harrington	B066D			OFFICE									
MECHANICAL EQUIPMENT	B066N			MECHANICAL									
Reactor	B070A			NONCLAS LAB						954			
Teaching Lab	B070B			CLASS LAB					953				
TRANSFORMER VAULT	B071			MECHANICAL									
TRANSFORMER VAULT	B072			MECHANICAL									
Office	B073			OFFICE									
Instrumentation Lab	B076			CLASS LAB					468				
RADIOACTIVE MATERIAL STORAGE	B076A			CLS LB SV					95				
FUEL PROCESSING LAB	B077			NONCLAS LAB						266			
Radioactive Material Storage	B077A			NONCLS SV						72			
Men's Restroom - 1WC,1U,1L	B078			REST ROOM									
Emergency Support Center - N. Satvat	B079			NONCLS SV						120			
R. Bean	B080			OFFICE									
A. Garner	B081			OFFICE									
Custodial Closet	B082			BLDG SERV									
Women's Restroom - 1WC, 1L	B083			REST ROOM									
SUBCRITICAL ASSEMBLY	B084			NONCLAS LAB						516			
Multi-Phase Flow Research Lab	B085			NONCLAS LAB						456			
Undergrad Thermal Hydraulics Lab	B086			CLASS LAB					456				
Maintenance Shop	B087			NONCLS SV						120			
Graduate Assistants	B087			OFFICE									
SHOP STORAGE	B087A			NONCLS SV						68			
ELEVATOR	BE02			CIRCULATE									
RAMP TO SHOP	BH05			CIRCULATE									
CORRIDOR - FROM B070A TO B084	BH11			CIRCULATE									
STAIRS N.W.	BS04			CIRCULATE									
STAIRS TO SUB BASM'T	BS06			CIRCULATE									
STAIRS S.W. CORNER	BS08			CIRCULATE									
STAIRS S.E. CORNER	BS09			CIRCULATE									
Subtotals					0	0	0	0	1,972	2,572			

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Ground Floor													
Storage	002			CENT STORAGE									
TRANSFORMER ROOM	003			MECHANICAL									
D. Rowe	004			OFFICE									
Theater Seating	005			CLASSROOM									
RECEIVING AREA	006			RECEIVING									
Maintenance Shop	006A			SHOP									
Storage	006B			SHOP SERVICE									
Lab - Sudhoff	007			NONCLAS LAB						373			
Storage	007A			NONCLS SV						48			
Computer Room	007B			NONCLS SV						484			
FUME HOOD	007C			NONCLS SV						46			
Custodial Room	008			BLDG SERV									
air conditioning equipment	010			MECHANICAL									
Student Activity - IEEE	014			OPEN LAB					199				
Custodial Closet	015			BLDG SERV									
Student Activity - HKN Shop	016			OPEN LAB					325				
Unisex Restroom - 1WC,1U,1L-HCA	017			REST ROOM									
TELECOMMUNICATIONS CLOSET-I[	018			MECHANICAL									
Roy Graduate Assistants	020			OFFICE									
HKN Coffee Shop	022			MERCHANDISE									
MECHANICAL EQUIPMENT	023			MECHANICAL									
HKN Commons	024			BLDG COMMONS									
Help Room for EE 201	026			STUDY									
VENTILATION FANS	029			MECHANICAL									
Storage	029A			CENT STORAGE									
Storage	029B			CENT STORAGE									
Storage	029C			CENT STORAGE									
COMPRESSOR	029D			MECHANICAL									
graduate assistants	030			OFFICE									
GRADUATE ASSISTANTS	032			OFFICE									
Graduate Assistants	034			OFFICE									
Graduate Assistants	036			OFFICE									
CPT Network Center - Tan	038			NONCLAS LAB						660			
CHEMICAL STORAGE	039			NONCLS SV						81			
ENTRANCE	039A			NONCLS SV						17			
CPT Network Center - Tan	040			NONCLAS LAB						426			
Storage	042			NONCLS SV						126			
Electronic Imaging - Allebach	043			NONCLAS LAB						191			
TELECOMMUNICATIONS CLOSET-B	044			MECHANICAL									
Electronic Imaging - Allebach	045			NONCLAS LAB						191			
Electronic Imaging - Allebach	046			NONCLAS LAB						186			
Media Room	046A			NONCLS SV									
Ultrasonics Lab - Furgason	047			NONCLAS LAB						400			
Electronic Imaging - Allebach	048			NONCLAS LAB						186			
Ultrasonics Lab - Furgason	049			NONCLAS LAB						50			
E. Furgason	049			R LAB OFFICE									
Electronic Imaging - Allebach	050			NONCLAS LAB						186			

ME/RAIL/HAMP Study
Electrical Engineering Bldg.

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Ground Floor													
research - lee	051			NONCLAS LAB						406			
Allebach Graduate Assistants	052			OFFICE									
Men's Restroom - 2WC,3U,2L	054			REST ROOM									
Men's Restroom Entrance	054A			REST ROOM									
Custodial Closet	054B			BLDG SERV									
Engineering Teaching Lab	056			CLASS LAB					319				
Teaching Lab	057			CLASS LAB					609				
Power Engineering Lab	057			NONCLAS LAB						610			
Robotics Lab - Lee	057A			NONCLAS LAB						546			
Power Engineering Lab - Sudhoff	057B			NONCLAS LAB						492			
Power Engineering Lab	057C			NONCLAS LAB						264			
Lab - M. Johnson	058			OPEN LAB									
Research - Pekarek	060			NONCLAS LAB						162			
D-CL Terminals/Multimedia	061			OPEN LAB					352				
ELECTRICAL EQUIPMENT	062			MECHANICAL									
ELECTRONICS+CIRCUITS	063			CLASS LAB					1,010				
D-CL Terminals/Multimedia	064			CLASS LAB					1,050				
AIR CONDITIONING CONTROL	064A			MECHANICAL									
J. Bougher	064B			OFFICE									
Electronics + Circuits Lab	065			CLASS LAB					994				
Storage	066			CENT STORAGE									
Lab Workroom	067			CLS LB SV					981				
UNDERGRAD TEACHING LAB	069			CLASS LAB					1,031				
PANEL CIRCUITS	071			MECHANICAL									
air conditioning equipment	072			MECHANICAL									
ELEVATOR	GE01			CIRCULATE									
ELEVATOR	GE02			CIRCULATE									
CORRIDOR - FROM 018 TO 042	GH01			CIRCULATE									
CORRIDOR - FROM 044 TO 054	GH02			CIRCULATE									
CORRIDOR - FROM 005 TO 016	GH03			CIRCULATE									
VESTIBULE - EAST	GH04			CIRCULATE									
CORRIDOR BY 005	GH05			CIRCULATE									
CORRIDOR - FROM 057A TO 67	GH07			CIRCULATE									
CORRIDOR BY 066	GH08			CIRCULATE									
HALLWAY BY LOADING DOCK	GH10			CIRCULATE									
CORRIDOR - DUNCAN SE	GH11			CIRCULATE									
STAIRWAY - EAST ENTRY	GS01			CIRCULATE									
CORRIDOR+STAIRS - NORTH	GS02			CIRCULATE									
STAIRS - BY 008	GS03			CIRCULATE									
STAIRWAY BY 066	GS04			CIRCULATE									
CORRIDOR+STAIR - NORTHWEST	GS05			CIRCULATE									
STAIRWAY - DOWN FROM 064	GS06			CIRCULATE									
STAIRWAY TO NUCL	GS07			CIRCULATE									
STAIRWAY - DUNCAN SOUTHWEST	GS08			CIRCULATE									
STAIRWAY - DUNCAN SE	GS09			CIRCULATE									

ME/RAIL/HAMP Study
Electrical Engineering Bldg.

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Ground Floor													
Subtotals					0	0	0	0	6,870	6,131			

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor													
Switchgear Room	103			MECHANICAL									
GRADUATE ASSISTANTS	109			OFFICE									
Graduate Assistants	109A			OFFICE									
Conference Room	112			CONFERENCE									
Tablet Arm Chairs	115			CLASSROOM									
K. Mouldoon + Student	116			OFFICE									
Development Director - A. McIntyre	116A			OFFICE									
Development Director - B. Silotto	116B			OFFICE									
Tablet Arm Chairs	117			CLASSROOM									
Conference Room	118			CONFERENCE									
Women's Restroom Entrance	119			REST ROOM									
Women's Restroom - 2WC, 1L	119A			REST ROOM									
Custodial Closet	120			BLDG SERV									
Secretary - D. Starewich	122			OFFICE									
Storage	122A			OFF SERV									
R. Balakrishnan	122B			OFFICE									
K. Jurss	122C			OFFICE									
Assoc Dept Head - M. Melloch	122D			OFFICE									
M. Kissinger	122E			OFFICE									
Conference Room	122F			CONFERENCE									
STORAGE	122K			OFF SERV									
LAVATORY	122L			OFF SERV									
Internal Corridor	122M			OFF SERV									
Women's Restroom Entrance	124			OFF SERV									
Women's Restroom - 1WC,1L	124A			OFF SERV									
Faculty/Staff Lounge	125			DEPT COMMONS									
Men's Restroom Entrance	126			OFF SERV									
Men's Restroom - 1WC,1L	126A			OFF SERV									
Tablet Arm Chairs	129			CLASSROOM									
Vestibule between GH07 and EE129	129A			CIRCULATE									
Elmqvist Graduate Assistants	130			OFFICE									
Account Clerks	132			OFFICE									
Office Service	133			OFF SERV									
Business Mgr - B. Goodrick	133A			OFFICE									
134A Entrance + Waiting	134			OFF SERV									
Account Clerks	134A			OFFICE									
Counseling - M. Golden + M. Wagner	135			OFFICE									
Grad Counselor - C. Ong	135A			OFFICE									
Grad Counselor - S. Midkiff	135B			OFFICE									
Closet	135C			OFF SERV									
A. Rainwater + C. Glotzbach + C. Quil	136			OFFICE									
J. Gray	138			OFFICE									
Research Lab - Power Group	139			NONCLAS LAB						1,310			
FILES	140			OFF SERV									
Men's Restroom - 2WC,3U,3L	142			REST ROOM									
Men's Restroom Entrance	142A			REST ROOM									
Custodial Closet	142B			BLDG SERV									

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor													
GRADUATE ASSISTANTS	143			OFFICE									
O. Wasynczuk	144			OFFICE									
L. Campbell	145			OFFICE									
M. Saeedifard	146			OFFICE									
Graduate Assistants	147			OFFICE									
S. Pekarek	148			OFFICE									
Storage + Copier	149			OFF SERV									
S. Sudhoff	150			OFFICE									
B. Robinson	151			OFFICE									
GRADUATE ASSISTANTS	153			OFFICE									
Graduate Assistants	154			OFFICE									
GRADUATE ASSISTANTS	155			OFFICE									
GRADUATE ASSISTANTS	156			OFFICE									
Conference Room	157			CONFERENCE									
D-CL Computer Lab	159			CLASS LAB					1,182				
MATERIALS-SPECIAL PURPOSE LA	160			CLASS LAB					1,182				
ELECTRONICS+CIRCUITS	161			CLASS LAB					827				
Electronic Repair + Storage	162			CLS LB SV					1,134				
D. Azpell	162			OFFICE									
ELECTRONICS+CIRCUITS	163			CLASS LAB					833				
REPAIR FACILITY	164			CLS LB SV					464				
ELECTRONICS+CIRCUITS	165			CLASS LAB					980				
Electronics + Circuits Lab	167			CLASS LAB					942				
GRADUATE ASSISTANTS	168			OFFICE									
Graduate Assistants	169			OFFICE									
Tablet Arm Chairs	170			CLASSROOM									
ROBOT VISION LAB - KAK	172			NONCLAS LAB						154			
ROBOT VISION LAB - KAK	173			NONCLAS LAB						154			
Graduate Assistants	174			OFFICE									
Graduate Assistants	175			OFFICE									
H. Medeiros	176			OFFICE									
Men's Restroom - 2WC,2U,3L-HCA	177			REST ROOM									
Custodial Closet	177A			BLDG SERV									
MECHANICAL	177B			MECHANICAL									
Robot Vision Lab-Kak	178			NONCLAS LAB						487			
Robot Vision Lab-Kak	179			NONCLAS LAB						525			
Image Processing - Kak	180			NONCLAS LAB						457			
Electronics Repair	180			NONCLS SV						62			
Robot Vision Lab-Kak	181			NONCLAS LAB						280			
ELEVATOR SHAFT	1E01			CIRCULATE									
ELEVATOR	1E02			CIRCULATE									
FRONT HALL	1H01			CIRCULATE									
WEST HALL	1H02			CIRCULATE									
EAST HALL	1H03			CIRCULATE									
CORRIDOR NEAR 167	1H05			CIRCULATE									
CORRIDOR FROM 165 TO 167	1H06			CIRCULATE									
CORRIDOR - FROM 157 TO 165	1H07			CIRCULATE									

ME/RAIL/HAMP Study
Electrical Engineering Bldg.

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
First Floor													
CORRIDOR NEAR 145	1H08			CIRCULATE									
CORRIDOR-FROM 144-151-157	1H09			CIRCULATE									
CORRIDOR(RAMP)-FROM 168 TO 17	1H10			CIRCULATE									
CORRIDOR - FROM 170 TO 199U	1H11			CIRCULATE									
FRONT STAIRS	1S01			CIRCULATE									
WEST STAIRS	1S02			CIRCULATE									
EAST STAIRS	1S03			CIRCULATE									
CORRIDOR+STAIR BY 164	1S04			CIRCULATE									
STAIRWAY - NORTHWEST	1S05			CIRCULATE									
DUNC ANX SOUTH STAIRS	1S08			CIRCULATE									
DUNC ANX NORTH STAIRS	1S09			CIRCULATE									
Subtotals					0	0	0	0	7,544	3,429			

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor													
Teaching Lab	206			CLASS LAB					1,065				
Teaching Lab	207			CLASS LAB					800				
Help Room	208			STUDY									
Help Room	209			STUDY									
Conference Room	210			CONFERENCE									
mechanical room	211			MECHANICAL									
electrical room	211A			MECHANICAL									
Research Lab - Jung	214			NONCLAS LAB						447			
Class Lab	215			CLASS LAB					672				
Research Lab - V. Raghunathan	216			NONCLAS LAB						529			
Embedded Systems Lab	217			CLASS LAB					693				
air conditioning equipment	218			MECHANICAL									
Men's Restroom - 2WC,1U,1L	219			REST ROOM									
Men's Restroom Entrance	219A			REST ROOM									
Entry to Lab	220			NONCLS SV						44			
TELECOMMUNICATIONS CLOSET-ID	220A			MECHANICAL									
MECHANICAL CLOSET	220B			MECHANICAL									
Research - Lu	220C			NONCLAS LAB						398			
Custodial Closet	221			BLDG SERV									
Tablet Arm Chairs	222			CLASSROOM									
Tablet Arm Chairs	224			CLASSROOM									
GRADUATE ASSISTANTS	225			OFFICE									
Tablet Arm Chairs	226			CLASSROOM									
graduate assistants	230			OFFICE									
Office	232			OFFICE									
Tablet Arm Chairs	234			CLASSROOM									
Tablet Arm Chairs	236			CLASSROOM									
TELECOMMUNICATIONS CLOSET-ID	237			MECHANICAL									
On Loan: Research Lab - JVK	238			NONCLAS LAB						605			
air conditioning equipment	240			MECHANICAL									
RESEARCH - LEHNERT	240A			NONCLAS LAB						171			
S. Devault + N. Cope	241			OFFICE									
Audio Visual Room	241			CLS LB SV					686				
Mail Room	241A			OFF SERV									
W. Crabill	241A			OFFICE									
GRADUATE ASSISTANTS	242			OFFICE									
Women's Restroom - 2WC, 2L	243			REST ROOM									
Women's Restroom Entrance	243A			REST ROOM									
Neurocomputer-Ersoy	244			NONCLAS LAB						200			
Men's Restroom - 3WC,3U,2L	245			REST ROOM									
Men's Restroom Entrance	245A			REST ROOM									
Custodial Closet	245B			BLDG SERV									
M. Swabey	246			OFFICE									
M. Johnson	248			OFFICE									
Conference Room	249			CONFERENCE									
Research Office - Vijaykumar	249A			OFFICE									
Research Office - Eigenmann	249B			OFFICE									

ME/RAIL/HAMP Study
Electrical Engineering Bldg.

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor													
Research Office - Vijaykumar	249C			OFFICE									
Research Office - Vijaykumar	249D			OFFICE									
J. Lax	250			OFFICE									
G. Hadley	252			OFFICE									
J. Park	254			OFFICE									
A. Ault	256			OFFICE									
Research Office - Bagchi	258			OFFICE									
WALK WAY TO ROOF	260			CIRCULATE									
Meyer	262			OFFICE									
J. Nyenhuis	264			OFFICE									
A. Hughes	266			OFFICE									
M. Erickson	268			OFFICE									
Tablet Arm Chairs	270			CLASSROOM									
Projection Booth	270A			CLSRM SRV									
Vacant	271A			STUDY									
Vacant	271B			STUDY									
Computer Support	271C			COMPU SRV									
Student Activity	271D			OPEN LAB					230				
Vacant	272			STUDY									
Vacant	273			OFFICE									
Vacant	274			OFFICE									
Graduate Assistants	275			OFFICE									
Men's Restroom - 2WC,2U,3L-HCA	276			REST ROOM									
FIRE ALARM PANEL CLOSET	276A			MECHANICAL									
MECHANICAL	276B			MECHANICAL									
Custodial Closet	277			BLDG SERV									
Women's Restroom - 2WC, 1L	278			REST ROOM									
Vacant	279			OFFICE									
Student Group	280			OFFICE									
Graduate Assistants	281			OFFICE									
Graduate Assistants	283			OFFICE									
GRADUATE ASSISTANTS	285			OFFICE									
Graduate Assistant	286			OFFICE									
GRADUATE ASSISTANTS	287			OFFICE									
Graduate Assistant	288			OFFICE									
GRADUATE ASSISTANTS	289			OFFICE									
Graduate Assistant	290			OFFICE									
TELECOMMUNICATIONS CLOSET-ID	291			MECHANICAL									
ELEVATOR	2E01			CIRCULATE									
MAIN HALLWAY	2H01			CIRCULATE									
WEST HALLWAY	2H02			CIRCULATE									
east corridor from 211 to 219	2H03			CIRCULATE									
corridor from 206 to 210	2H04			CIRCULATE									
HALLWAY BY RM 270	2H10			CIRCULATE									
CORRIDOR THRU LSS OFFICES	2H11			CIRCULATE									
FRONT STAIRS	2S01			CIRCULATE									
NORTHWEST STAIRS	2S02			CIRCULATE									

ME/RAIL/HAMP Study
Electrical Engineering Bldg.

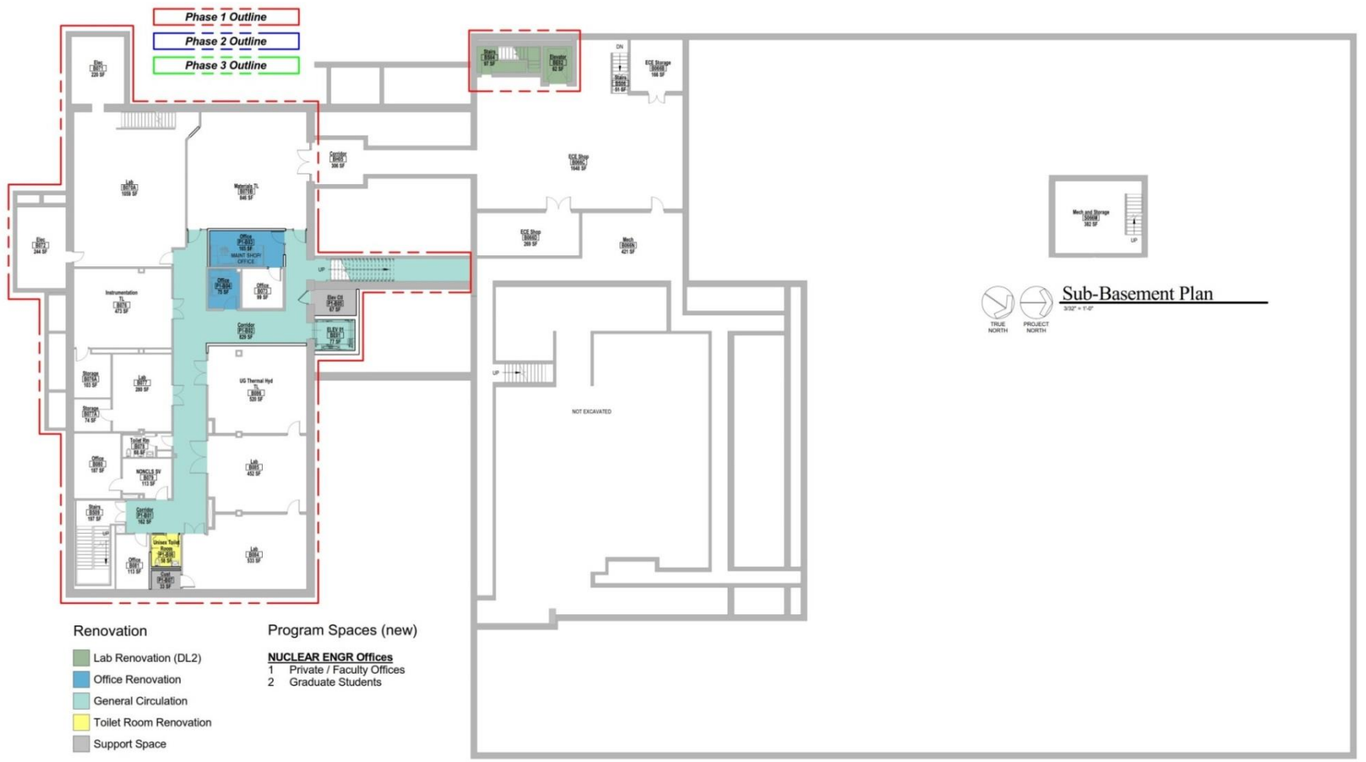
Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Second Floor													
SOUTHEAST STAIRS	2S03			CIRCULATE									
CORNER STAIRS	2S05			CIRCULATE									
DUNC ANX SOUTH STAIRS	2S08			CIRCULATE									
DUNC ANX NORTH STAIRS	2S09			CIRCULATE									
Subtotals					0	0	0	0	4,146	2,394			

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Third Floor													
Graduate Assistants	306			OFFICE									
Research Lab	308			NONCLAS LAB						400			
Conference Room	309			CONFERENCE									
S. Midkiff	310			OFFICE									
R. Pierret	312			OFFICE									
Internal Corridor	313			OFF SERV									
graduate assistants	313A			OFFICE									
graduate assistants	313B			OFFICE									
R. Givan	313C			OFFICE									
Post Docs	313D			OFFICE									
J. Siskind	313E			OFFICE									
graduate assistant	313F			OFFICE									
Conference Room	313G			CONFERENCE									
CLOSET	313H			OFF SERV									
K. Webb	314			OFFICE									
Research - Jung+Webb	316			NONCLAS LAB						536			
VIDEO-CONFERENCING ROOM	317			CONFERENCE									
M. Melloch	318			OFFICE									
M. Melloch	318A			OFFICE									
Women's Restroom - 2WC, 1L	319			REST ROOM									
Women's Restroom Entrance	319A			REST ROOM									
T. Vijaykumar	320			OFFICE									
Custodial Closet	321			BLDG SERV									
Central Storage	322			CENT STORAGE									
S. Rao	322A			OFFICE									
V. Raghunathan	322B			OFFICE									
M. Kulkarni	324A			OFFICE									
V. Pai	324B			OFFICE									
R. Eigenmann	325			OFFICE									
J. Gray	325A			OFFICE									
Entry Foyer	325B			OFF SERV									
COPY ROOM	325C			OFF SERV									
C. Brown	326A			OFFICE									
Secretary - W. Thompson	326B			OFFICE									
M. Thottethodi	327			OFFICE									
Men's Restroom - 2WC,1U,1L	328			REST ROOM									
S. Bagchi	329			OFFICE									
M. Qi	330			OFFICE									
D. Aliprantis	331A			OFFICE									
Y. Wang	331B			OFFICE									
E. Narimanov + D. Peroulis + P. Berm	332			OFFICE									
Z. Chen + A. Botasseva	333			OFFICE									
M. Capano + J. Cooper + J. Appenzell	334A			OFFICE									
D. Janes + S. Mohammadi + E. Narim	334B			OFFICE									
M. Lundstrom	334C			OFFICE									
A. Alam + G. Klimeck + P. Ye	334D			OFFICE									
Custodial Closet	335			BLDG SERV									

ME/RAIL/HAMP Study
Electrical Engineering Bldg.

Room Description / Name	Room Number	Department Using Space	Department Assigned	Room Type	Quantities				Exist Lab SF Calcs		Additional Description	Relationship of this space to others: Pick either a department, faculty or N/A	Comments: Does this space have special room needs? Examples include, fume hood, sink, 220V power, etc.
					Faculty	Staff	GAs	Other	Teaching Labs (SF)	Research Labs (SF)			
Third Floor													
Graduate Assistants	336			OFFICE									
L. Stevenson	337A			OFFICE									
copy room	337B			OFF SERV									
Research Lab - Bagchi	338			NONCLAS LAB						631			
conference room	339			CONFERENCE									
A. Kak	340			OFFICE									
Graduate Assistants	341			OFFICE									
Emeriti Faculty	342			OFFICE									
Research - Talavage	343			NONCLAS LAB						201			
research lab - jiao	344			NONCLAS LAB						596			
graduate assistants	345			OFFICE									
Graduate Assistants	347			OFFICE									
Internal Corridor	348			OFF SERV									
Office	348A			OFFICE									
Office	348B			OFFICE									
R. DeSutter	348C			OFFICE									
Conference Room	348D			CONFERENCE									
Conference Room Entrance + Coffee	348D			CONF SERV									
J. Cychosz	348E			OFFICE									
Office	348F			OFFICE									
Emeritus - C. Chen	348G			OFFICE									
Graduate Assistants	350			OFFICE									
ELEVATOR	3E01			CIRCULATE									
FRONT HALL	3H01			CIRCULATE									
WEST HALLWAY	3H02			CIRCULATE									
EAST HALLWAY	3H03			CIRCULATE									
EAST HALLWAY	3H04			CIRCULATE									
EAST HALLWAY	3H05			CIRCULATE									
EAST HALLWAY	3H06			CIRCULATE									
FRONT STAIRS	3S01			CIRCULATE									
WEST STAIRS	3S02			CIRCULATE									
EAST STAIRS	3S03			CIRCULATE									
Subtotals					0	0	0	0	0	2,364			

Proposed Plans – EE Basement Floor – Phase 1



Basement Floor Plan

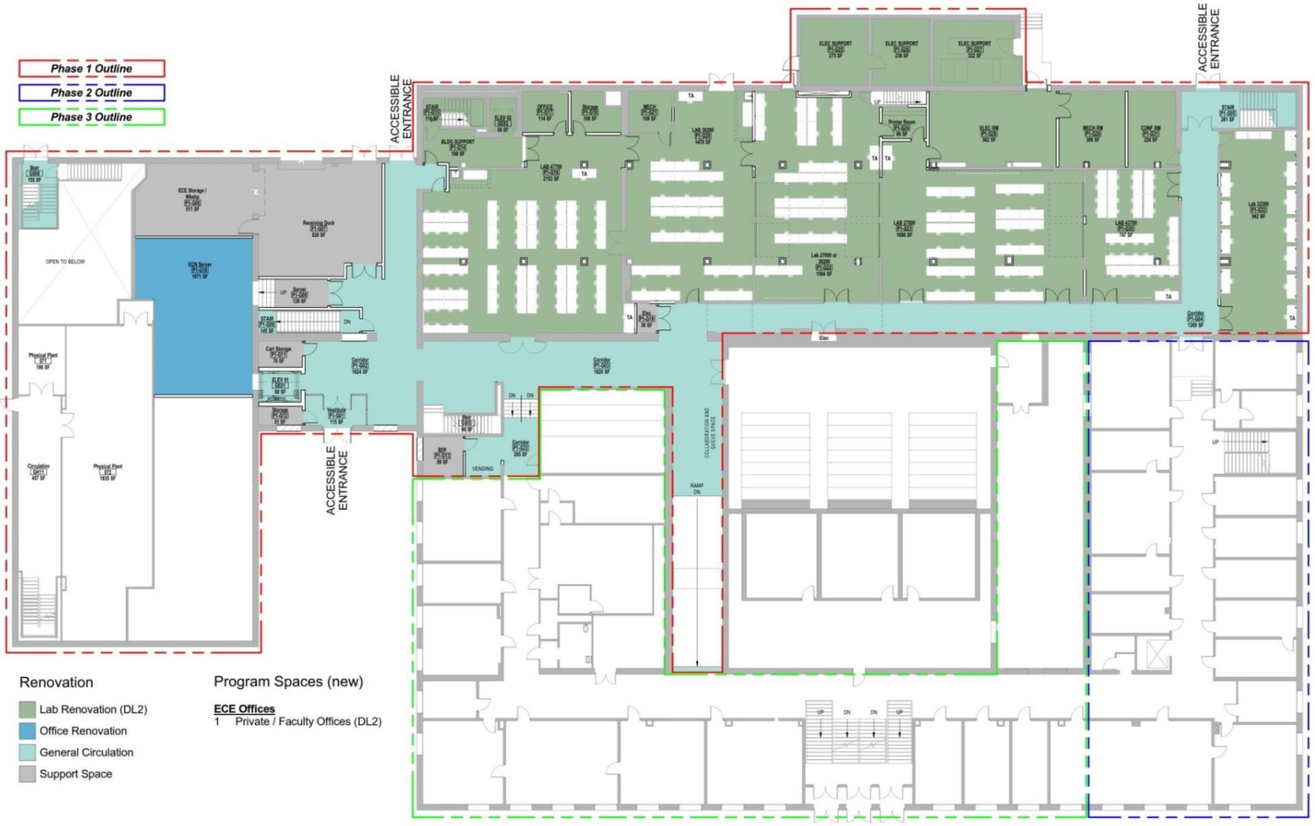
PURDUE UNIVERSITY

Electrical Engineering Building

Proposed Phase 1 Renovations

MSKTD & Associates, Inc.

Proposed Plans – EE Ground Floor – Phase 1



Ground Floor Plan

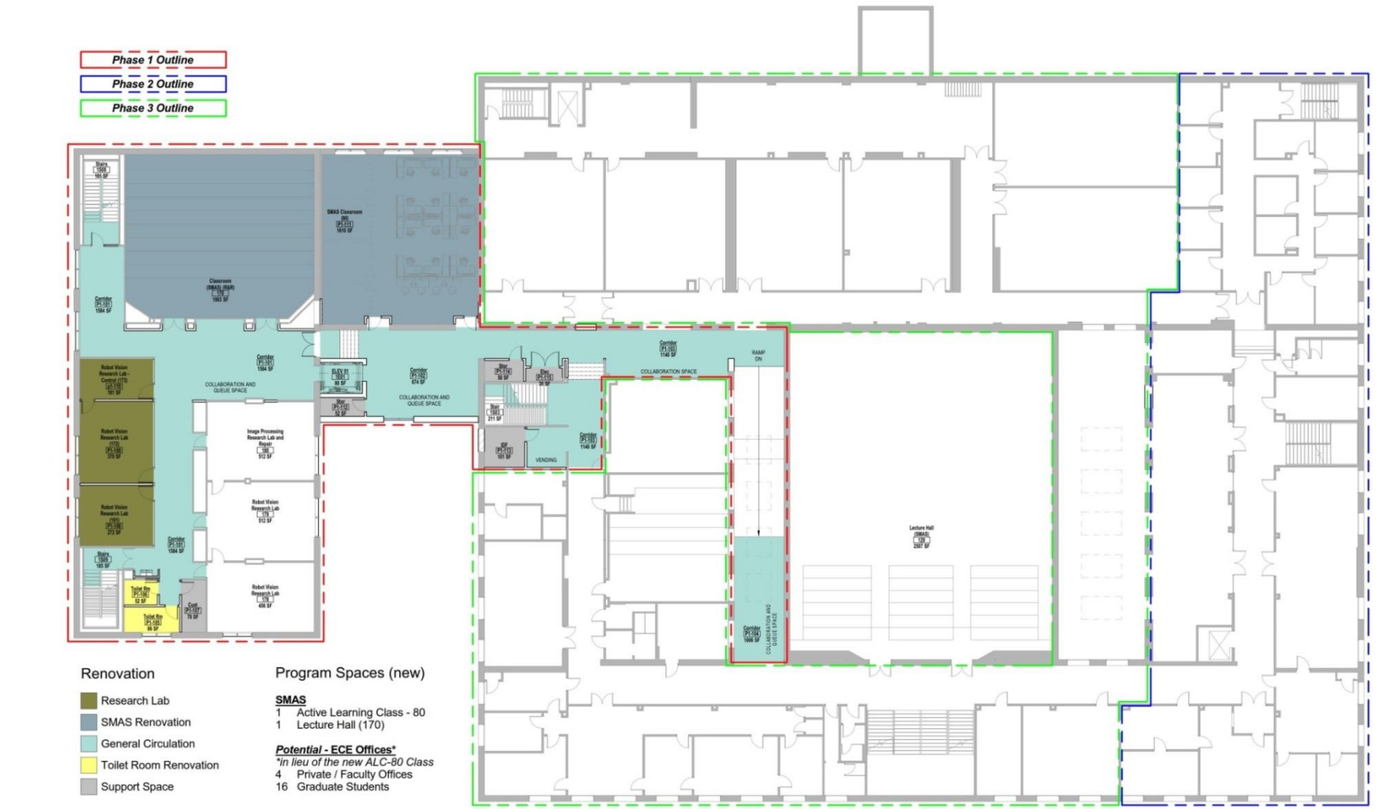
PURDUE UNIVERSITY

Electrical Engineering Building

Proposed Phase 1 Renovations

MSKTD & Associates, Inc.

Proposed Plans – EE First Floor – Phase 1



First Floor Plan

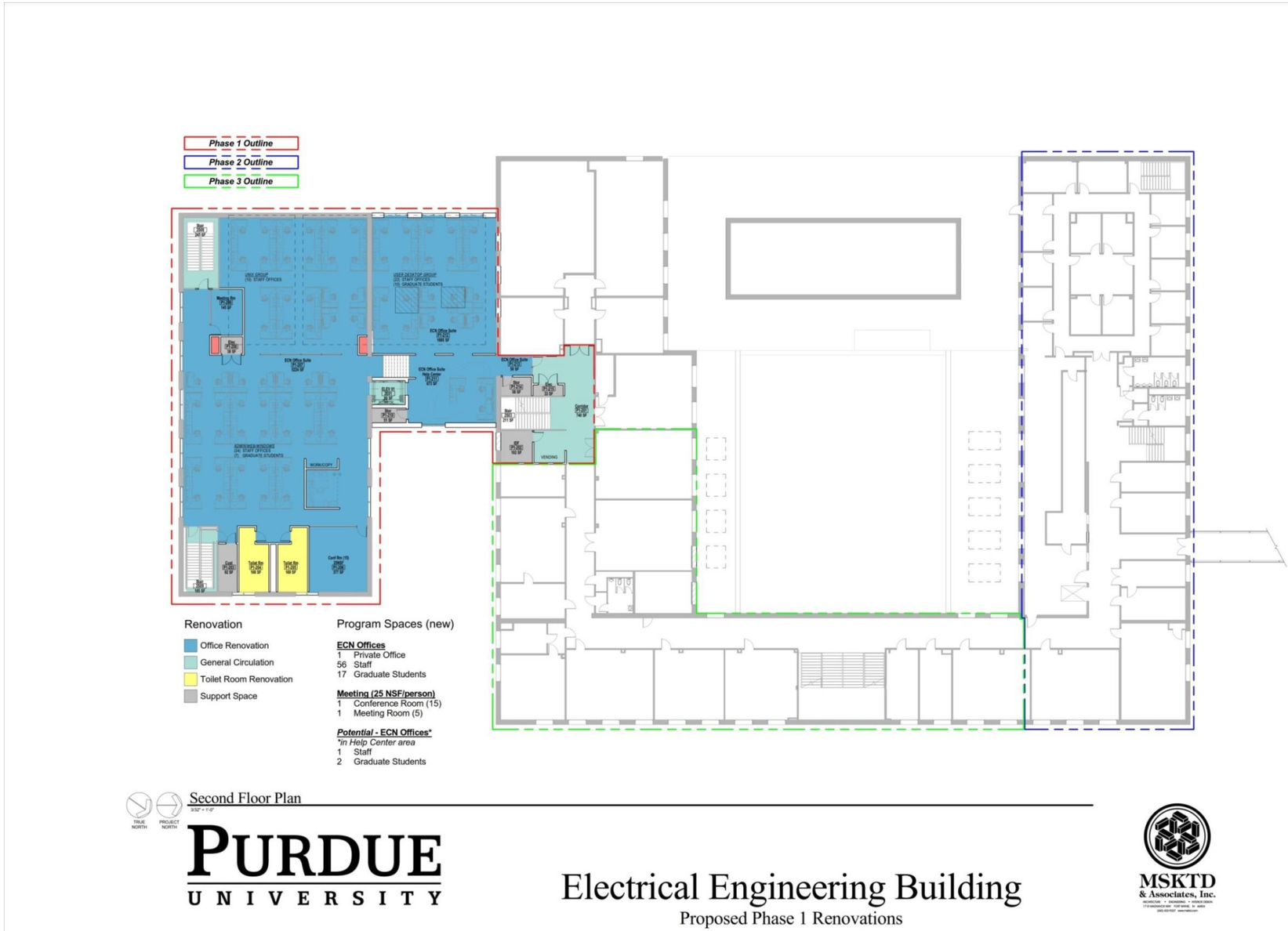
PURDUE UNIVERSITY

Electrical Engineering Building

Proposed Phase 1 Renovations

MSKTD & Associates, Inc.

Proposed Plans – EE Second Floor – Phase 1



Proposed Plans – EE Third Floor – Phase 1

Third Floor Plan

PURDUE UNIVERSITY

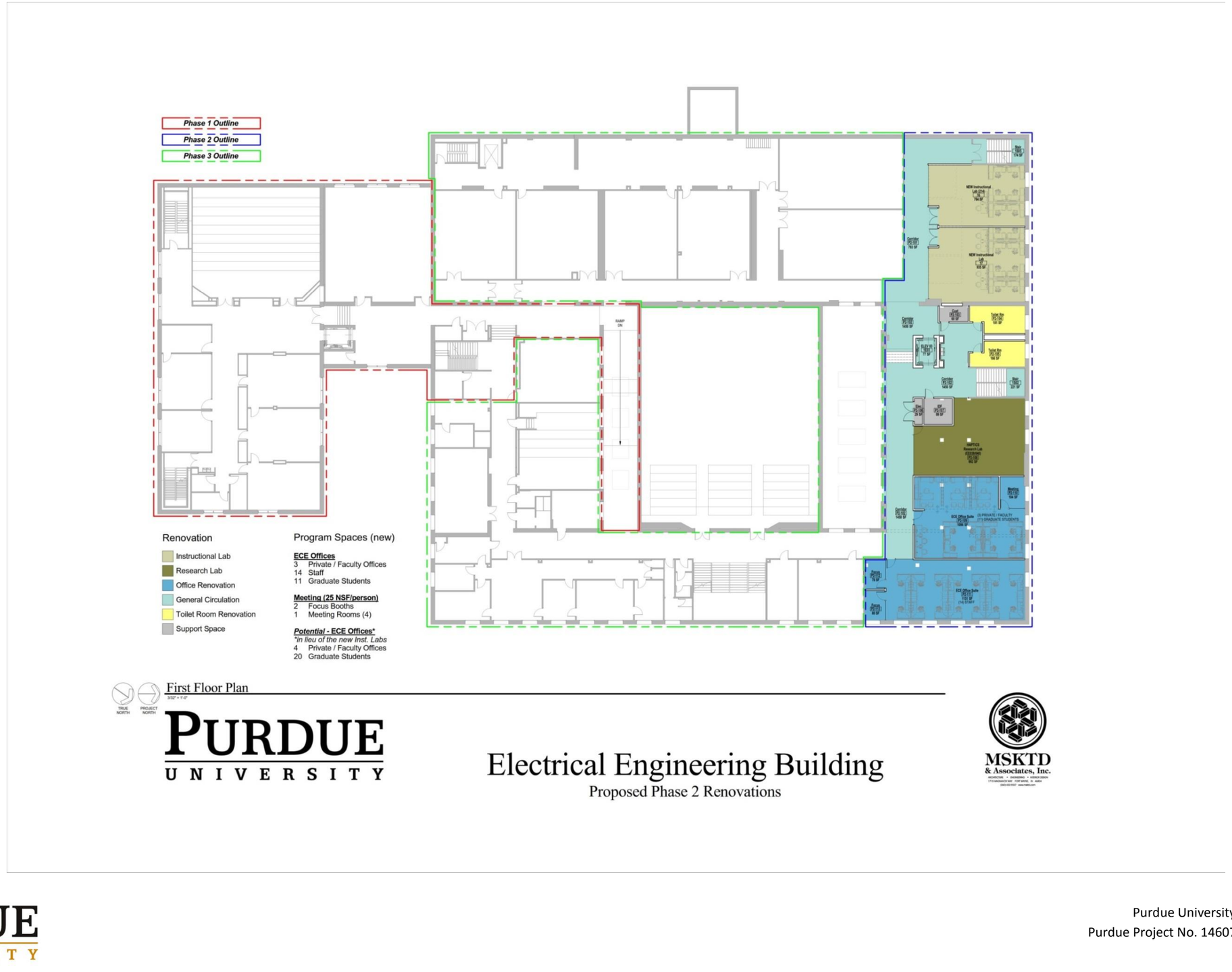
Electrical Engineering Building

Proposed Phase 1 Renovations

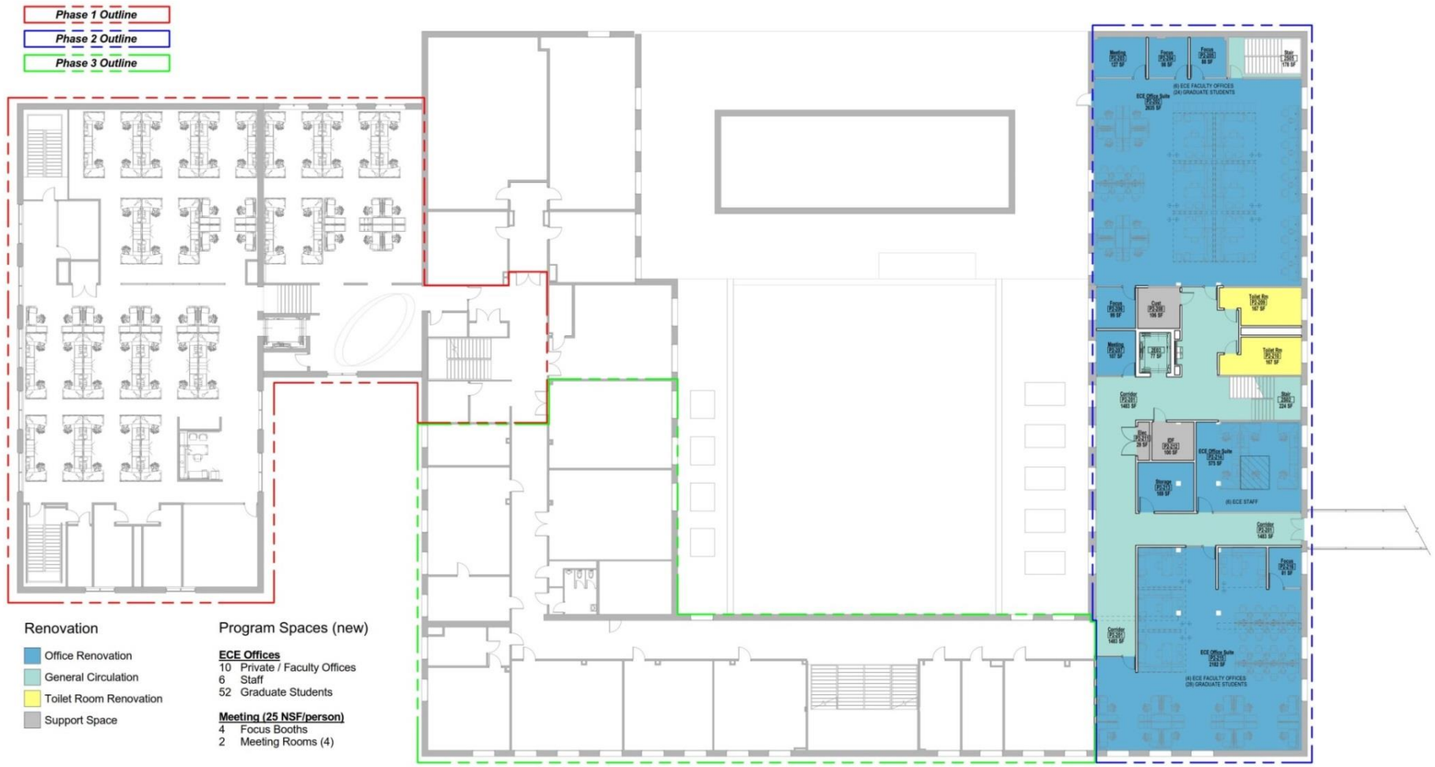
MSKTD & Associates, Inc.

Proposed Plans – EE Ground Floor – Phase 2

Proposed Plans – EE First Floor – Phase 2



Proposed Plans – EE Second Floor – Phase 2



Second Floor Plan

PURDUE UNIVERSITY

Electrical Engineering Building

Proposed Phase 2 Renovations

MSKTD & Associates, Inc.

Proposed Plans – EE Third Floor – Phase 2

Proposed Plans – EE Ground Floor – Phase 3

Ground Floor Plan

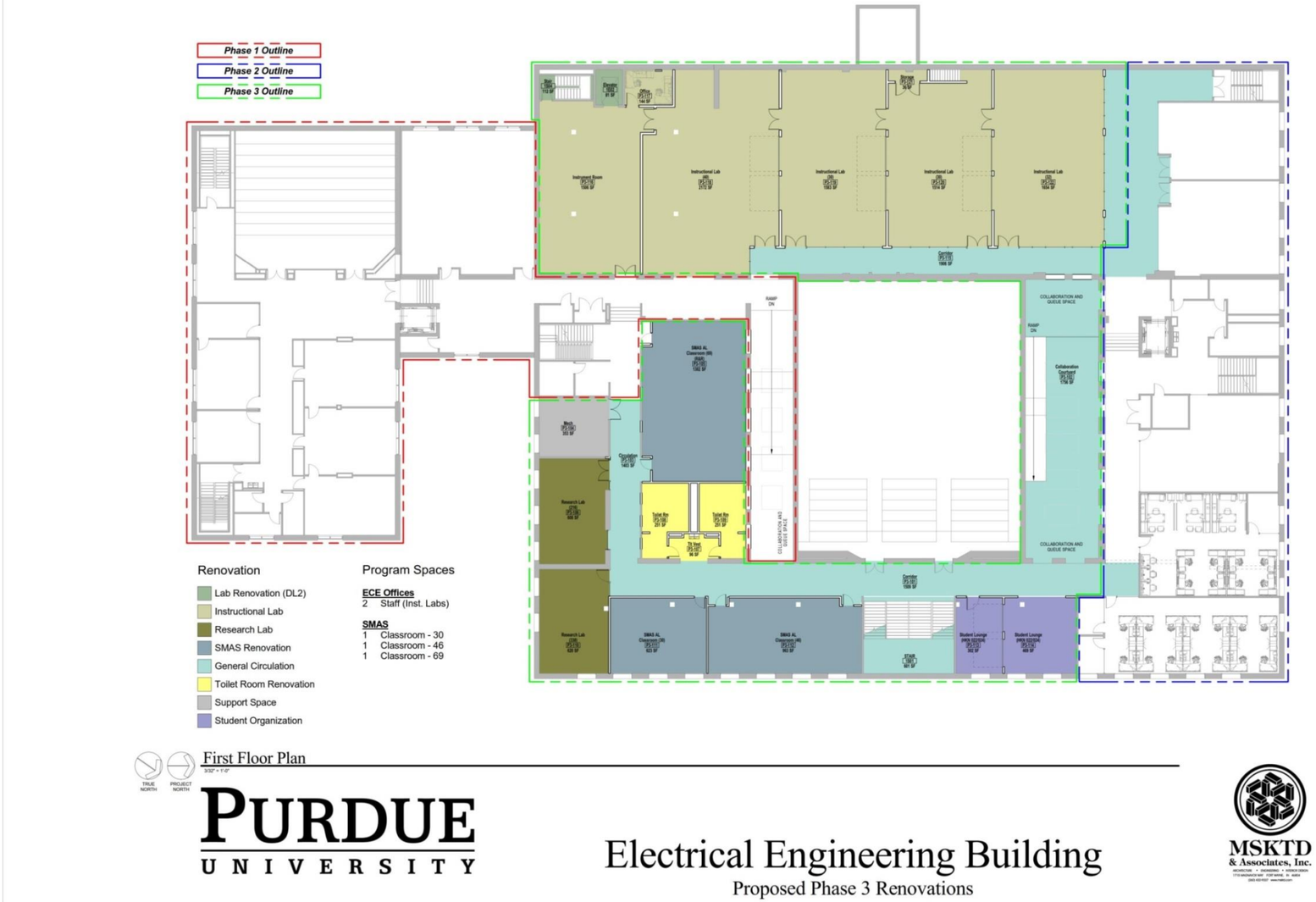
PURDUE UNIVERSITY

Electrical Engineering Building

Proposed Phase 3 Renovations

MSKTD & Associates, Inc.

Proposed Plans – EE First Floor – Phase 3



Proposed Plans – EE Second Floor – Phase 3

Second Floor Plan

PURDUE UNIVERSITY

Electrical Engineering Building

Proposed Phase 3 Renovations

MSKTD & Associates, Inc.